

Statistic Preservation Is Not Security Preservation: Identification Limits of Outcome-Gated Screening for Federated-Learning Defense Composition

Anupam Mediratta
Orbis AI
mediratta@gmail.com

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Abstract

Layering federated-learning defenses risks a composition failure: upstream processing can destroy the discriminative property the downstream defense needs. The natural repair—require **preservation** of that property—is the wrong abstraction for security. **Our result is causal: downstream-statistic preservation is causally insufficient for attack-suppression preservation.** What follows is the experimental design that can actually test a composition mechanism; the screening rule is a *consequence*, not the contribution.

Preservation is a **hierarchy**: the statistic’s value (P1), the ordering it induces (P2), the resulting decision (P3), the adversarial mass admitted (P4), suppression itself (P5). $(P1) \Rightarrow (P2) \Rightarrow (P3)$ hold by definition; **below the decision the chain breaks in both directions**, each break with a measured witness. Everything an invariance argument can certify—including the sufficient conditions we prove for one aggregator class—lives at (P1)–(P3) and **bounds no ASR**, while the security content sits at (P4). Two **general** limits follow for any criterion stated at the statistic level. **Identification**: because candidates are gated on standalone effectiveness in the very metric the criterion predicts, effectiveness and mechanism co-vary across every certified composition, and no cross-defense contrast identifies the mechanism’s contribution *at any sample size*. **Coverage**: suppression emergent in a composition and absent from both constituents cannot satisfy an inherited-suppression gate, capping recall structurally.

Three empirical findings, by oracle intervention on two arms. With attenuation closed *by construction*, Krum’s *decision* flips in up to 80% of rounds while the adversarial mass it admits stays at exactly 0.000 and ASR shows no evidence of the predicted increase, staying inside our pre-specified equivalence margin—pre-registered, and replicated on a second dataset and architecture. On the identical cell the effect **reverses sign** once adversarial influence is unpinned: the outcome-gated ladder *falls* 0.272 where the instrument *rises* 0.098. And heterogeneity closes the theory’s own margin, RFA’s all-round mean separation margin collapsing $3.24 \rightarrow 1.01$ at Dirichlet $\alpha=0.1$, where our bound certifies nothing. **The boundary results are general; our empirical claims—star-topology FL, CIFAR-10 but for the replicated arm—are not.** What survives is a conservative **evaluation-prioritization rule** for inherited robustness, not a validated predictor.

1 Introduction

Federated learning (McMahan et al., 2017) enables collaborative model training without centralizing data, but its distributed nature introduces vulnerability to data poisoning (Chen et al., 2017; Bagdasaryan et al., 2020). The arms race between attacks and defenses (Wang et al., 2020; Xie et al., 2020; Pillutla et al., 2022; Sun et al., 2019; Bhagoji et al., 2019) has produced strong results on both sides, but these analyses share two assumptions: the adversary knows the exact defense employed, and has access to samples from the true data distribution. In practice the server’s defense is an implementation detail that can be kept private, and the adversary’s optimal play depends critically on which defense is active.

The natural repair for a composition failure—check that the upstream stage preserves the property the downstream defense reads—is the subject of this paper, and **our result is that the repair does not work, and that this is structural rather than a matter of choosing the property better. Stated sharply: even after controlling for standalone effectiveness and intervening *directly* on the downstream statistic, changing that statistic can have little or no**

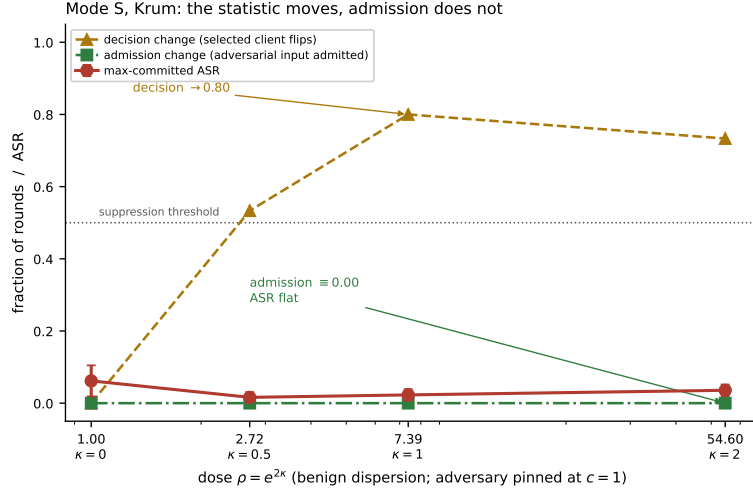


Figure 1: **The paper in one figure: the statistic moves, admission does not, and suppression holds.** Mode S, Krum under model-scaling (§5.2). Every adversary is pinned at $c=1$, so the adversarial coefficient share is constant across rungs and Lemma 2’s attenuation channel is shut *by construction*, so the dose **isolates statistic disturbance from adversarial coefficient attenuation**—the one channel it closes, and not the only thing it moves. As the benign dispersion ρ grows to 54.6, Krum’s *decision* (which client it selects) flips in up to 80% of rounds, while the *admission* of adversarial input holds at exactly 0.000 and max-committed ASR shows no evidence of the predicted increase, staying inside the pre-specified equivalence margin ($|\Delta|=0.026 < 0.15$) and low. **Statistic preservation is therefore not what was doing the work.** Bars are Student- t 95% intervals ($n=5$); every value recomputed per seed by `analyze_targeted_dose.py`. A pre-registered replication in a *different* invariance class and on a *second dataset and architecture* reproduces the negative and **refutes** the cross-arm admission ordering we had predicted, so admission is a *direction to measure*, not a validated predictor (§5.2). The chain this figure breaks a link of is drawn in full as Figure 7 (Appendix A).

effect on suppression—and the effect reverses once the intervention is changed to hold adversarial influence fixed instead (-0.272 where influence co-varies, $+0.098$ where it is pinned; §5.2). *Security preservation* names the framing thesis’s object throughout—that suppression of the committed attack survives composition, under the adversary model of §2. The two levels it decomposes into are kept distinct and never conflated: *adversarial contribution* (the influence the adversary retains on the aggregate, (P4)) and *attack-suppression preservation* (ASR below threshold, (P5)). None of the three ever means preservation of the downstream statistic, of the ordering it induces, or of the decision that follows. Statistic preservation is the abstraction any invariance argument about composition can certify; we formalize the levels at which it comes apart from security and build the interventions that separate them. **The second half of the contribution is the experimental design that can actually test a composition mechanism**, and it follows from the identification boundary below, which is what derives it—so we state the boundary before naming our own conditions. **The general form of the boundary, stated before its FL instance**, because it is a property of outcome-gated screening rather than of our menu:

Any screening criterion whose eligibility condition is defined on the **same outcome metric the criterion predicts** cannot be validated by cross-defense contrasts: the gate forces effectiveness and mechanism to co-vary across every certified candidate, so no choice of defenses off any menu breaks the confound. The admissible designs are **within-defense**.

One scoping point up front, because it sharpens rather than weakens the claim: our C1 requires that some constituent already suppress the attack alone, so nothing here discovers defenses from nothing—it reasons about *compositions of already-effective defenses*, the question a practitioner with a defense menu actually faces.

The result, up front. Preservation is a *hierarchy*, not a property: the statistic’s value (P1), the ordering it induces (P2), the decision that follows (P3), the adversarial mass admitted (P4), suppression itself (P5). We show $(P1) \Rightarrow (P2) \Rightarrow (P3) \not\Rightarrow (P4) \not\Rightarrow (P5)$, the first two links holding **by definition**—equal values induce the same ordering, and a rule reading only that ordering cannot decide differently—and each of the last two broken by a **counterexample we measured**. $(P3) \not\Rightarrow (P4)$: Mode S sweeps benign dispersion ρ from 1 to 54.6 and flips Krum’s decision in up to 80% of rounds while the adversarial coefficient share stays pinned at exactly 0.266667, the admitted adversarial mass at 0.000, and ASR inside the pre-specified equivalence margin (Figure 1). $(P4) \not\Rightarrow (P5)$: `cos.krum` holds statistic, ordering, decision and admission *bit-identical* at every rung—0/120 selections changed—and its ASR falls 0.425 across the ladder regardless, 0.173 of it by the first rung, where the accuracy gate still passes. Hence, formally (Proposition 9, §5.2): **no level of this hierarchy is sufficient for security**—a non-existence result *within* the positive-rescaling class on these defenses and attacks, exhibited by counterexample, and **not** a claim about every conceivable metric. **Everything an invariance argument can certify lives at (P1)–(P3); the security content is at (P4)**, and that gap—not the screening rule built over it—is this paper’s subject. Definition 1 states the levels formally and §5.2 measures both breaks.

The secondary result, and the scope of the experiments behind it, follow. The information asymmetry noted above raises a natural question of its own: does per-round aggregator switching provide security value? In centralized settings, randomized input transformations (Xie et al., 2018) and random self-ensembles (Liu et al., 2018) exploit attacker uncertainty, and production FL platforms (Ludwig et al., 2020) expose pluggable aggregators operators may rotate; but the backdoor-persistence literature (Bagdasaryan et al., 2020; Zhang et al., 2022) establishes that embedded triggers survive many rounds of partial suppression, which suggests persistence should eliminate switching value. We confirm and quantify that intuition—defense secrecy is cheaply probed (Appendix Q.1) and realized VoPD ≈ 0 on standard menus—and the operational consequence, that deterministic composition outperforms switching on every tested menu, is what motivates the rest of the paper.

Moreover, defense choices are not free: each mechanism imposes accuracy and fairness costs (Wang et al., 2020), and attack strength depends on network heterogeneity, the defense in use, and model architecture. A game-theoretic formulation naturally captures these mutual dependencies.

Scope. We investigate whether *per-round aggregator switching* (randomly selecting which defense processes updates each round) has positive value against persistent FL poisoning. Randomized internals within a single defense—DP noise, subsampling smoothing à la CRFL (Xie et al., 2021), random participation masks—operate at a different level and are not addressed. Our experiments use a specific deployment regime (CIFAR-10/100, $N=10$, $K=5$, $f \in \{0.05, 0.2\}$, 50–200 rounds, 4-layer CNN), within which we characterize a persistence boundary: above it (high γ , universally-admitting menus) randomization value collapses, below it (stateless attacks, orthogonal-suppression menus) it survives. Whether the boundary’s location transfers to $N \geq 10^4$, $f \leq 0.01$ is open.

Contributions. Three, in order of what the paper is *about*: (1) a preservation hierarchy and the measured break that makes statistic preservation the wrong security abstraction; (2) the identification boundary that break implies for any outcome-gated screen; (3) the causal interventions that locate where security is actually decided. **Three empirical findings carry (3), and we name them here because the rest of the paper is organized around them:** (i) disturbing the statistic does not move suppression—instrumented, with the attenuation channel closed by construction, and replicated on a second dataset and architecture (§5.2); (ii) a **sign reversal** between the outcome-gated and the instrumented design on the *identical* cell (-0.272 against $+0.098$), which is what makes the identification boundary a measured hazard rather than a cautionary tale; (iii) **heterogeneity closes the theorem’s own margin**—RFA’s all-round mean separation margin collapses $m=3.24 \rightarrow 1.01$ at Dirichlet $\alpha=0.1$ ($4.71 \rightarrow 2.06$ over the rounds with $w_a > 0$), exactly where Corollary 1 degrades to certifying nothing and two of three certified compositions cross the threshold. **The boundary results are general; our empirical claims are not.** Items (4) and (5) below are not further contributions: (4) is the screening rule, stated as a *consequence* of (2), and (5) is one secondary result.

1. **The preservation hierarchy, and the two-directional break that makes statistic preservation the wrong abstraction.** Because the per-client defenses in use are all positive scalar rescalings, mechanism preservation is a decidable *invariance* question (Proposition 8), and on the weighting→aggregator class we prove it with explicit margins (Δ_{signal} , Δ_{margin} ; Theorem 6), so that class needs no calibration runs. But preservation is not one property. It has **five levels**—the statistic’s value, the ordering it induces, the decision that follows, the adversarial mass admitted, and suppression itself (Definition 1)—and while $(P1) \Rightarrow (P2) \Rightarrow (P3)$ hold by definition, **below (P3) the chain breaks in both directions and we measure both breaks** (§5.2, §10). **Everything an invariance argument can prove lives at (P1)–(P3); the security content is at (P4)**, and that gap, not the screen built over it, is this paper’s subject. One corollary is worth stating alone: weight heterogeneity is safe at *both* extremes and dangerous in the middle—with S the attack’s coordinate extremeness and r the adversary’s relative weight, $S \cdot r \gg 1$ leaves the

adversary extreme enough to be discarded and $S \cdot r \ll 1$ annihilates its payload, while $S \cdot r \approx 1$ is the hazard—so **the weight ratio ρ alone mispredicts** (Lemma 2).

2. **An identification boundary for outcome-gated screening, and a bound on what any such screen can find.** The general claim above is proved as Proposition 4; its FL instance is that $C0 \wedge C1$ *force* the downstream defense of every certified in-scope pair to suppress the relevant attack on its own, so a certified composition’s low ASR is always attributable to $C1$ as well as to $C2$ —an effect we name **mechanism–effectiveness confounding**, structural rather than an artifact of our defense menu (§3.1). It is not specific to FL. What we offer as the transferable contribution, though, is not this negative but **the admissible design it derives** (§3.1). Its evidential cost is real: a matched metric-swap ablation built to break the confound had its premise refuted by its own pre-freeze baselines—magnitude information is what lets this defense family suppress a magnitude attack—so **the confound re-materialized inside an experiment built to break it** (8/8 frozen labels, $TP = 0$; §4.2). A second boundary of the same shape limits coverage rather than validation: **emergent** suppression—present in the pair, absent in both constituents—cannot satisfy an inherited-suppression gate, so such a screen’s recall is capped by the fraction of successes whose suppression is inherited (Proposition 5). That is a scope theorem rather than a defect found by tuning, and our own lowest-ASR composition is its witness (§6).
3. **Causal interventions that locate the security channel—not a validation of any screen.** Three results, in the order the boundary licenses them, each tagged with its evidence tier. (i) *Refuted—our own pooled statistic-level hypothesis (Causal)*. The boundary specifies a within-defense ladder holding $C1$ numerically fixed—same downstream defense, same attack, same standalone baseline—and varying only *measured* disturbance of the statistic d_2 reads, over four arms spanning three invariance classes, four rungs $\rho = e^{2\kappa} \in \{1.00, 2.72, 7.39, 54.60\}$ and five seeds, with the abscissa measured independently before the freeze. **We ran it (80 runs) and it mostly refutes us:** pooled $\rho_s = +0.351$, $p = 0.091$, two of four frozen shapes refuted, and ASR *falling* as the dose rises in three arms of four, so we withdraw the pooled form of $C2$ ’s causal claim (§5). Reputation confirms ($0.017 \rightarrow 0.779$, $p = 1.7 \times 10^{-6}$, accuracy $0.78 \rightarrow 0.70$), covering the destruction half alone—and (ii) attributes most of that rise to attenuation rather than statistic disturbance. (ii) *Adjudicated by intervention—statistic preservation is not security preservation (Causal, then Replication)*. This is the paper’s sharpest result and its title. An upstream pinning every adversary’s coefficient (Mode S: adversarial share constant at 0.267) closes Lemma 2’s attenuation channel *by construction*: **Krum’s decision flips in up to 80% of rounds while the adversarial mass it admits stays at exactly 0.000 and its ASR stays inside the pre-specified equivalence margin** (§5.2), so the implication from decision invariance to suppression preservation fails, and fails measurably. A **pre-registered replication** on `coord_median`—a coordinate-wise order statistic rather than Krum’s selector, chosen because it is *rank-discordant* on the two predictors—**reproduces that negative on a different invariance class and refutes the positive ordering we had predicted** ($+0.098$, below an arm whose admission change was smaller), so **admission is the better direction but not a quantitative predictor** ($R^2 = 0.239$ over 26 cells). **What this establishes, it establishes by oracle intervention on two arms in two of Proposition 8’s classes**—an *identification* result on those arms, not a measured law about defense compositions in general; both instruments read adversary identity and are deliberately not deployable, and they rescore nothing in §5.1. A companion payload-only sweep (Mode A) confirms Theorem 6’s single-peaked prediction for the rank aggregator it is stated over and refutes it off-spec, as the theorem predicts. (iii) *Diagnosed—and the diagnosis refutes a third predictor*. ρ says how far apart a transform spreads the coefficients; **adversary-correlated dispersion** $\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ says *who* receives it, and our formal results constrain ρ alone. Measured for the four real transforms (post-hoc, no ASR), the two anchors that miss (i)’s synthetic curve are the two most extremely targeted ($\Delta_c = -1.21$ for RFA, -0.72 for NormClip, against -0.34 at the ladder’s most dispersed rung) **at ρ that is 3.7 and $13\times$ smaller. But Δ_c does not order suppression either:** RFA targets the adversary hardest and its composition still fails. That makes three candidate predictors—statistic disturbance, admission magnitude, coefficient targeting—each refuted *as a predictor*, which is why what survives is a *preservation level* and not a statistic of the coefficient vector. Two propositions formalize what the instruments measured rather than adding a claim: **statistic-level non-identifiability** (Proposition 9), whose two witnesses are the arms above—so no metric defined *solely* on preservation of the downstream statistic identifies security equivalence within this class—and the **independent controllability** of ρ and Δ_c (Proposition 10), which is why these two instruments, and not a single dial, were the right design.
4. **The screening rule is a consequence of (2), not a claim of its own** (*Screening* throughout). What survives the negative is a conservative *evaluation-prioritization* rule: read $C2$ off the downstream signal’s invariance class, then evaluate the survivors first. Used that way it is reliable in the positive direction and lossy: every composition the

tightened criterion certifies does work (2 in-distribution, 4 out-of-sample), but **recall is only** 40% on a positive class of just 5/42—the bound of Proposition 5 in practice—and we do not offer the precision as validation, since it rests on a handful of positive predictions in a badly imbalanced evaluation (§4.3). Scored across suites: 90.5% on 42 two-way pairs (18 development *fit*, 89%; 24 pre-registered held-out *prediction*, 92%), 9/10 pre-registered triples (§4.5), 13/20 on a prospective suite of three previously unused defenses frozen in git before any run, and better than random selection ($p < 10^{-4}$). It certifies only *inherited* robustness under a *committed* attack and outputs an evaluation *ordering*, never a deployment decision; among evaluated compositions it identifies an accuracy–security Pareto frontier: FG→CM (0.107 / 0.68) as the best candidate for further evaluation, FG→RFA (0.093 / 0.50) as the lowest-observed-ASR extreme, Rep→CM (0.423 / 0.76) for near-baseline accuracy (Table 13).

5. **Secondary, and outside the three tiers—composition dominates mixing under persistence.** Deterministic composition outperforms temporal mixing (per-round defense switching) on every tested menu and adversary policy: oracle ASR 0.316 under composition vs. 0.915 under mixing on rep+tm; all compositions suppress to ≈ 0.10 at $N=100/\text{ResNet18}$ (§7). The mechanism is backdoor persistence ($\gamma \approx 0.66$): once a trigger embeds, intermittent suppression cannot clear it, making mixes converge to their worst constituent’s ASR. Committed strategies are near-optimal (largest measured switching gain +0.044, below bound 0.048); only stateless threats ($\gamma=0$) permit positive randomization value (Appendix L). Fixed-order composition also beats randomized ordering $9\times$, because preservation conditions are order-sensitive; and under *one* whitebox composition-aware *adaptive stress test* FG→RFA’s ASR rises to 0.167 ± 0.069 , $3.3\times$ its seed-matched committed 0.051, showing that **an inherited-robustness guarantee is not adaptive robustness** (§6).

Three tiers of evidence, of very unequal weight, never pooled. Every empirical claim about the preservation hierarchy or the criterion is tagged with one of these three words at the point it is made, so a reader can look up what licenses it instead of reconstructing it from hedges. Contribution 5 carries no tier tag because it is outside them: the mixing comparison is motivation for studying composition at all, and no part of the hierarchy or the boundary rests on it (§7).

tier	design	what it licenses
Causal	within-defense intervention at fixed d_2 , attack and C1 input: Mode S, Mode A (§5.2)	statements about the <i>mechanism</i> , scoped to these cells
Replication	FEMNIST—second dataset <i>and</i> architecture ($\Delta = -0.017$)—and <code>coord_median</code> , a second invariance class (+0.098)	“replicated on a second dataset and architecture”, never “generalizes”
Screening	the outcome-gated 42-pair suite, 38/42	evaluation prioritization only ; by Proposition 4 it cannot identify the mechanism’s contribution at <i>any</i> sample size

How the paper is organized, as an argument. Composition creates an **identification problem**: the screening conditions themselves force effectiveness and mechanism to co-vary (§3). Proposition 4 **characterizes** that problem and, read as a derivation, **dictates the only admissible experiment** (§3.1). We run that experiment, and it **falsifies our own statistic-level hypothesis** while adjudicating what replaces it (§5, §5.2). What the invariance theory still *explains*—the two mechanism-preserving regimes, and the margin collapse at the boundary between them—is §3.2. What survives as an *application*, an ordering over an evaluation budget rather than a prediction, is §4.3; §10 states the open problem the negative leaves.

Why this class, and how to read every claim below. Everything we *prove* is instantiated for positive per-client rescalings, deliberately: the class is a **laboratory for studying composition, not a general model of it**—it covers the per-client defenses in current use (norm clipping, FoolsGold, RFA reweighting, FLTrust) and is clean enough to make invariance decidable from definitions. Defenses that alter participation, carry cross-round state, mix updates across clients, or aggregate before the next stage **fall outside it and are covered by nothing here. Every claim in this paper is one of exactly three kinds—general, formal, empirical—and they do not transfer upward**: the formal guarantee does not certify the general framing, and the empirical figures certify neither level above them. Table 1 states the three with every scope condition attached, and each claim below is labelled by kind at the point it is made; §9 then gives the same conditions one at a time as (L1)–(L9).

Reading guide. The body presents the composition criterion (§3), its validation including 3-way compositions where exhaustive search is infeasible (§4, §4.5), the composition the screen cannot certify, FG→RFA, characterized under a criterion-aware adversary as Proposition 5’s witness (§6), and a condensed case for why composition rather than mixing is the object of study (§7); the full game-theoretic analysis is Appendix L. The bridge between them is

persistence: a composition applies both defenses every round, so a signal-preservation failure admits the attack on every round, and under high γ those admissions accumulate and saturate ASR—converting a C2/C3 failure into total failure.

2 Preliminaries

Federated learning. N clients each hold a local dataset $\mathcal{D}_i \sim P_i$; the server learns $w^* = \arg \min_w \sum_i \frac{|\mathcal{D}_i|}{|\mathcal{D}|} F_i(w)$. In FedAvg (McMahan et al., 2017), each round the server broadcasts w^t , selected clients run local SGD, and the server aggregates $w^{t+1} = w^t + \frac{1}{|S_t|} \sum_{i \in S_t} \Delta_i^t$.

Poisoning and robust aggregation. An adversary controlling a fraction f of clients can flip labels (Chen et al., 2017), inject backdoor triggers (Bagdasaryan et al., 2020; Xie et al., 2020), or scale malicious updates to dominate aggregation (Bagdasaryan et al., 2020). The server may replace FedAvg with a Byzantine-robust rule — Krum (Blanchard et al., 2017), Trimmed Mean, Coordinate-wise Median, Norm Clipping (Sun et al., 2019), or RFA (geometric median) (Pillutla et al., 2022) — each imposing an accuracy/robustness trade-off.

Game theory. A normal-form game is $G = (\mathcal{N}, \{\mathcal{S}_i\}, \{u_i\})$. A *Bayesian game* adds a type space Θ_D and prior $\mu \in \Delta(\Theta_D)$; a profile is a *Bayes-Nash Equilibrium (BNE)* if each player maximizes expected utility given beliefs over others’ types.

3 Composability Criterion

The paper’s central practical question is *which* defense pairs compose well under deterministic joint application. Specifically, the criterion screens for *mechanism-preserving composition*: when one constituent defense is already individually effective (C1), can the other defense be added without breaking that suppression? This is the operationally relevant question—the criterion is not a general characterization of all composable pairs, but a sufficient condition for *mechanism preservation* among pairs where at least one component has known effectiveness—never a sufficient condition for low ASR. We answer it first, with a checkable criterion, and defer to §7 and Appendix L the game-theoretic machinery (the persistence mechanism and complementarity diagnostic) that explains *why* the alternative—temporal mixing—fails. **But the practical question is not what the paper establishes:** what it establishes is that preservation of the statistic the screen checks is not causally tied to the security the screen predicts, so the criterion below should be read as the application §4.3 scopes it to—an ordering over an evaluation budget—and not as a deployment recommendation.

Observe that each robust aggregation defense d operates through an *admission signal* $\sigma(u)$ applied to client update u : it accepts, clips, or down-weights u based on whether $\sigma(u)$ exceeds a threshold. Crucially, the admission signal varies by defense family:

- **Weighting defenses** (reputation, FoolsGold): $\sigma(u)$ = distance from consensus (L2) or pairwise similarity.
- **Clipping defenses** (NormClip): $\sigma(u) = \|u\|_2$.
- **Rank-based defenses** (TrimmedMean, CoordMedian): $\sigma_k(u)$ = coordinate-wise rank of u_k .
- **Geometric defenses** (RFA): $\sigma(u)$ = distance from geometric median.

When two defenses are applied sequentially (d_1 transforms the updates, d_2 aggregates the result), d_1 ’s transformation may *corrupt* d_2 ’s signal or vice versa.

Criterion 1 (Signal-Preservation Composability). *Let d_1 and d_2 be defense mechanisms. The following sufficient conditions guarantee that composition ($d_1 \rightarrow d_2$) suppresses attack a . They are of **two different kinds**, and we group them accordingly, because conflating an empirical screen with a mechanistic property is the main source of confusion about what the criterion claims.*

Stage 0 — applicability. *Before either kind of condition is informative, the question the criterion asks has to be live.*

(C0) The downstream mechanism is load-bearing. *ASR(d_1, a) ≥ 0.5 for at least one committed attack a . C2 and C3 concern preservation of d_2 ’s admission signal, so they say nothing when d_1 alone already suppresses everything: the composition inherits that suppression, destroying d_2 ’s signal costs nothing, and a correct prediction would be correct for a reason the criterion did not supply. The asymmetry is deliberate—a strong d_2 is the intended case, since asking whether upstream processing breaks an effective downstream defense is the framework’s whole question, whereas a strong d_1 makes the question moot. C0 was **formulated post-hoc** and its provenance is in Remark 3.*

An empirical effectiveness gate. This screens; it does not explain, and it is the one condition that cannot be checked without running something.

(C1) Individual suppression. For every committed attack a in the evaluated set, $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$. The quantifier matters because our headline metric is the maximum over attacks. **$C1$ is an empirical prerequisite rather than a mechanistic property:** it is defined through the same ASR threshold that constitutes the outcome. It also scopes the criterion, which sharpens rather than weakens it—the criterion does not discover defenses from nothing, it selects mechanism-preserving compositions of already-effective defenses, which is the question a practitioner holding a defense menu actually faces.

Mechanism-preservation conditions. These are what Theorem 6 proves for the weighting→aggregator class, and they are determined by d_1 ’s and d_2 ’s definitions alone—no experiment required. **What we recommend is an evaluation-prioritization criterion, $C1 \wedge C2$ —and we name it that way rather than a “screen” because that is the whole of what our evidence supports.** We disclaim predictive validity explicitly, here at first use: the criterion orders candidate compositions for evaluation, and it does not predict whether a composition is secure. The pooled two-channel model over 26 cells is unconfirmed ($R^2=0.239$), and our own pre-registered replication **refuted** the cross-arm admission ordering the criterion would need in order to predict (§5.2); recall is 40% (Proposition 5), so the negative direction carries no information either. **It is never a security screen, and a PASS is an evaluation ordering rather than a deployment decision** (Table 1). $C3$ is retained as a **margin diagnostic** rather than a gate, because on our data it never binds on a pair $C1 \wedge C2$ already admits and shows no incremental predictive value (§4.3). The pre-registered labels of §4 were frozen with $C3$ in the gate and are not relabelled; the demotion is a recommendation going forward, supported by a post-hoc factorial ablation, and it changes no reported number— $C1 \wedge C2$ selects exactly the same two in-distribution pairs.

(C2) Signal preservation. d_1 ’s per-client transformation T_1 does not destroy d_2 ’s admission signal: $\sigma_2(T_1(u_a)) \approx \sigma_2(u_a)$ for malicious updates u_a .

(C3) Boundary preservation. d_1 ’s transformation does not compress the adversarial–benign separation below d_2 ’s discrimination threshold. Formally: if d_2 suppresses a by detecting $\tau_{d_2}(u_a) > \theta$ (triggering trimming/rejection), then $\tau_{d_2}(T_1(u_a)) > \theta$ must hold—i.e., d_1 must not map adversarial updates from above to below d_2 ’s rejection boundary.

When $C1$ holds but $C2$ fails (signal destruction), composition is no better than d_2 alone. When $C1$ and $C2$ hold but $C3$ fails (boundary compression), d_2 retains its admission signal direction but its threshold no longer fires—the attack passes because d_1 ’s rescaling places adversarial magnitudes within d_2 ’s acceptance region.

Writing $C2$ as preservation of d_2 ’s admission signal elides a distinction our own experiments later force, so we name the levels before relying on the condition. They are the paper’s main conceptual instrument, and every claim below is stated at one of them.

Definition 1 (Preservation levels). For an upstream transform T_1 and a downstream defense d_2 reading admission signal σ_2 :

- (P1) **Value invariance:** $\sigma_2(T_1(U)) = \sigma_2(U)$.
- (P2) **Ordering invariance:** $\sigma_2(T_1(U)) = \varphi(\sigma_2(U))$ for some φ preserving the ordering σ_2 induces on clients. This is the level $C2$ is stated at (made explicit in (1)); the distinction from (P1) matters, because exact values are sufficient but never necessary.
- (P3) **Decision invariance:** the client or coordinate d_2 actually selects is unchanged.
- (P4) **Adversarial contribution (admission) invariance:** the adversarial input d_2 admits into its aggregate is unchanged. This is a **level, not one measurable quantity**: for a selector it is the indicator that an adversarial update is selected at all; for a weighted averager, the adversarial share of the weight mass; for a coordinate-wise order statistic, the adversarial share of selected coordinates—which is why Table 11’s influence column is aggregator-specific by necessity. The committed pre-registrations say “admission”, so that word is retained in every frozen rule name and no scored quantity is relabelled.
- (P5) **Attack-suppression preservation:** ASR stays below threshold.

(P1)⇒(P2)⇒(P3) hold, and hold by definition—equal values induce the same ordering, and a rule reading only that ordering cannot decide differently. **These three are logical implications under the definitions above, not empirical claims about how defenses behave,** and nothing is asserted by proving them: their content is that they stop at (P3). **Below (P3) the chain breaks, and it breaks in both directions.** Both breaks are measured. Holding decision and admission invariance does not deliver suppression: *cos_krum*’s selection is bit-identical at every dose and its admitted

adversarial mass is 0.000 throughout, yet its ASR falls 0.173 by the first rung alone through the magnitude channel—at $\rho=2.72$, where clean accuracy moves only $0.599 \rightarrow 0.567$ and our accuracy gate passes (paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds). Across the whole ladder it falls 0.425, but that endpoint sits below our own accuracy floor, so **the gate-passing rung is the witness we rest on** (§5.2). Losing decision invariance does not cost suppression: Krum’s selection flips in up to 80% of rounds under an intervention that pins the adversarial mass, and its ASR stays inside the pre-specified equivalence margin ($|\Delta|=0.026 < 0.15$) (§5.2). A criterion stated at (P1)–(P2) therefore constrains security in neither direction. **(P4) is where the security content sits—but we do not claim it is sufficient for (P5).** It is the first level that refers to adversarial influence on the aggregate rather than to the downstream defense’s internal decision, and `cos_krum` is our own witness that (P4) $\nRightarrow$ (P5): admission identically 0.000 at every rung while suppression moves. **So the hierarchy answers “which level does a security claim have to be stated at?” and not “which level predicts ASR?”—no level in it is offered as a predictor.**

The full causal chain, and the links we actually measure. Definition 1 is a prefix of a longer chain: statistic $\rightarrow$ decision $\rightarrow$ admission $\rightarrow$ aggregate adversarial influence $\rightarrow$ per-round model update $\rightarrow$ attack persistence $\rightarrow$ ASR. We measure the four emphasized links—decision, admission, aggregate influence (Λ_a) and ASR—and we do **not** measure per-round update geometry or persistence dynamics, so every claim below is a claim about the measured links only, with the unmeasured two left as an acknowledged gap rather than assumed benign (Figure 7).

Remark 2 (Which level C2 asserts, and which one is load-bearing). At $\rho = 54.6$ the dose ladder of §5 changes Krum’s decision in 86.7% of rounds while changing its admission of adversarial input in 13.3%, and Krum’s suppression does not deteriorate at all. A selection that flips from one benign client to another disturbs the statistic maximally and the mechanism not at all. **Terminology, once and for all: “mechanism” always means the downstream defense’s discriminative operation—what separates adversarial from benign updates—and never its security behaviour, since that implication is precisely what this paper refutes.** We reserve “mechanism preservation” for the formal property Theorem 6 establishes, saying statistic or decision preservation where only the weaker property holds. C2 above is the (P2) condition—which is what makes it checkable from definitions alone—but the load-bearing reading is (P4). **This is the paper’s central tension: our theorem operates at (P1)–(P3), while the experiments indicate security behaviour is decided at (P4);** §5.2 puts the two against each other in an experiment where they predict opposite outcomes, and Figure 7 is that argument in one picture.

Remark 3 (C0’s provenance: an applicability condition formulated post-hoc). C0 sits in Criterion 1 as a stage-0 applicability condition because Lemma 1 is stated over $C0 \wedge C1$, but **it does not have the same evidential status as C1–C3**, and we record that here rather than let its promotion obscure it. It is **vacuous on the 7-defense menu of §4**, where the strongest single defense reaches only 0.519, and we formulated it only after the prospective suite of §4.1 introduced a defense for which it binds. It is a lesson from prospective evaluation, applied there retrospectively; it is not part of the criterion validated on the 42 pairs, and none of the accuracy figures in §4 depend on it.

A scope condition on the criterion itself, not a limitation of our runs. C1 and every prediction built on it are defined against a *committed* adversary—one whose attack is fixed before the composition is chosen. An adversary who observes the composition can re-target whichever channel the screen certifies, so **what the screen bounds is inherited robustness under a fixed attack, never robustness against an adaptive one.** Our single whitebox composition-aware stress test is consistent with that reading ($0.051 \rightarrow 0.167$ seed-matched, $3.3\times$ the committed-attack ASR; §6); **its role is not to measure adaptive robustness but to show that an inherited-robustness guarantee is not adaptive robustness**, and a screening condition stated against an adaptive adversary is left open.

3.1 The testability boundary

C0 and C1 together constrain what any validation of this criterion can look like. We did not anticipate this; it turns out to explain the central weakness of our own prospective suite (§4.1, L7), and—once stated in general form—to *derive* the class of within-defense designs in which C2’s contribution is identifiable (§5). Its content is narrower than “a general identification boundary” and worth stating precisely: **under inherited-effect gating**, cross-defense contrasts cannot identify the mechanism’s contribution, because the gate conditions on a descendant of both the mechanism and the outcome. What we offer as the transferable contribution is not that negative but **the admissible design it derives**, set out below and run in §5.2.

Table 1: **Scope of claims: three kinds, and the one place each scope condition is stated. Claims do not transfer upward**—the formal guarantee does not certify the general framing, and the empirical figures certify neither level above them. This table is the conservation record for every caveat in the paper; §9 restates the same conditions individually as (L1)–(L9).

kind	content	status, and what it does <i>not</i> license
General	An evaluation gated on the outcome it predicts cannot identify the mechanism’s contribution, and admissible designs are within-defense; an inherited-suppression gate cannot certify emergent synergy (Prop. 4, Prop. 5).	Proved, and not specific to FL, to backdoors, or to our conditions. Licenses no claim that any composition <i>is</i> secure, and none that mechanism preservation implies suppression. Recall is bounded, not achieved: 40% here, from $ \mathcal{E} / \mathcal{S} =1/5$ (L5, L7).
Formal	Mechanism preservation as <i>invariance</i> under positive per-client rescalings $T(u_i) = c_i u_i, c_i > 0$, into specific rank and geometric aggregators, with explicit margins (Thm. 6, Lem. 2, Cor. 1, Prop. 8, Prop. 7).	Proved on that class only —a <i>laboratory</i> for composition, not a model of FL defenses in general. A d_1 that alters participation, carries cross-round state, mixes clients, or aggregates first admits no such T and is covered by nothing here . Bounds no ASR . C0 was formulated post-hoc and applied retrospectively to arms frozen before it (Rem. 3, L7). Regime A’s condition is not operationally checkable, S not being a stable per-attack constant (L8). Cor. 1’s bound goes vacuous as $m \rightarrow 1$, which heterogeneity realizes.
Empirical	§5.2’s interventions and §4–§4.1’s 42-pair menu: CIFAR-10, CifarCNN/ResNet18, $N \in \{10, 100\}$, $K \in \{5, 10\}$, $f \in [0.05, 0.6]$, Dirichlet $\alpha \in \{0.1, 0.5\}$, star topology , three attack families; one arm on FEMNIST/simple_cnn.	Measured at $n=5$ seeds per cell ($n=8$ for <code>cos_krum</code> , $n=3$ on FEMNIST, $n=30$ for the FG→RFA characterization), all fixed before the runs, so intervals are wide and no null is read as flatness without a pre-registered equivalence margin. ASR is thresholded at 0.5, the level at which the trigger fires on the majority of triggered inputs; a cell below the 0.35 clean-accuracy floor is <i>uninterpretable</i> , not suppressed. Dataset transfer is tested on the flagship arm alone—“replicated on a second dataset and architecture”, never “generalizes” (L3). “Refuted” always means a pre-registered <i>directional</i> hypothesis was rejected against a rule frozen for it, never that a mathematical claim was disproved. The $\gamma \approx 0.66$ boundary is architecture-conditional (L2, L3, L6); the strongest development-set figures are <i>fit</i> , not prediction (L4).
Both instruments are oracles	Mode S and Mode A build c_i from adversary identity, as does Δ_c (§5.2, §3.3).	They identify a causal channel and are not screens : no deployed defense knows which clients are adversarial, and estimating (P4) without that knowledge is the open problem of §10. Admission is <i>a direction to measure</i> , not a validated predictor—the pooled two-channel model is unconfirmed over 26 cells ($R^2=0.239$) and its cross-arm ordering was refuted by our own replication (L9). C1 and everything built on it assume a committed adversary, so what is bounded is <i>inherited</i> robustness under a fixed attack, never adaptive robustness .

Lemma 1 (Testability boundary). *Let $(d_1 \rightarrow d_2)$ satisfy C0 and C1, and let a^* be a committed attack witnessing C0, so $\text{ASR}(d_1, a^*) \geq 0.5$. Then $\text{ASR}(d_2, a^*) < 0.5$: the downstream defense suppresses a^* on its own.*

Proof. C1 at a^* gives $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a^*) < 0.5$. By C0 the minimum is not attained at d_1 , hence it is attained at d_2 . $\square$

Nothing in the argument is specific to federated learning, to backdoors, or to our defense menu. Lemma 1 is **the enabling observation for the FL instance**; the substantive claim is the general one, and we state it as such.

Proposition 4 (Why constituent-effect contrasts cannot identify composition effects under inherited-effect gating). *Let a screen certify $(d_1 \rightarrow d_2)$ against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$ holds, determined by the two defenses’ definitions alone. Suppose the screen is applied only where the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P ’s contribution to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set occupies at most two, and **the cell needed to separate P from d_2 ’s standalone effectiveness is empty by the gate, for every $\mathcal{D}$. No contrast varying d_2 identifies P ’s contribution. Identification requires holding d_2 , a and therefore E fixed while intervening on the upstream transform: the admissible designs are within-defense.***

Proof. The load-bearing condition excludes d_1 from attaining the minimum in (i), so certification forces $M(d_2, a) < \theta$; both cells with $M(d_2, a) \geq \theta$ are therefore unpopulated whatever $\mathcal{D}$ is, and P is never observed against a downstream defense that fails on its own. Any estimator of P ’s contribution from such contrasts is unidentified. $\square$

The structure is a collider, and the design table is the whole argument. The gate E is a function of constituent effectiveness; effectiveness affects the composition’s outcome; and the mechanism condition P is fixed by the same pair of definitions—so conditioning on E makes P and effectiveness co-vary across every certified candidate. Enumerating the four cells makes the missing counterfactual visible as *a cell, not a sample size*:

	$M(d_2, a) < \theta$ (d_2 suppresses a alone)	$M(d_2, a) \geq \theta$ (it does not)
P holds (mechanism preserved)	observed: every certified pair	empty by the gate
$\neg P$ (mechanism destroyed)	observed: predicted-FAIL pairs	empty by the gate

Reading down a column contrasts P against $\neg P$ at fixed effectiveness, which is the comparison that would identify P ’s contribution—and the right-hand column, the only one in which d_2 is not already doing the work, is empty for every menu $\mathcal{D}$. **The problem is not insufficient sample size: the counterfactual needed to identify the mechanism’s contribution is excluded by design.** The emptiness follows from the gate, not from the menu.

In our instance M is max-committed ASR, $\theta = 0.5$, E is C1, the load-bearing condition is C0 and P is $C2 \wedge C3$ —which is Lemma 1 read as identification rather than as arithmetic. The proposition is about the *shape* of the screen, so it holds for any eligibility gate defined on the predicted outcome, in any domain. **This is a general claim in the sense of Table 1:** nothing in it is specific to FL, to backdoors, or to our conditions.

The payoff is the admissible design, not the proof. These proofs are short, and their value is not difficulty but **what they rule in**: because eligibility is defined on the very outcome being predicted, effectiveness and mechanism co-vary across every certified candidate, so *no* cross-defense contrast can identify the mechanism’s contribution, and the only admissible instrument is one that holds d_2 , the attack, and therefore the gate’s input fixed while intervening on the upstream. That single sentence is what licenses every experiment in §5–5.2: each is a member of the derived class, chosen because Proposition 4 said to look there rather than at another defense menu. A result can be easy to prove and still determine what one runs next.

This is selection on the outcome, and saying so is the point. Proposition 4 is an instance of a hazard that is entirely standard outside this literature: conditioning on a variable that is a descendant of both the putative cause and the outcome—**selection on the dependent variable**, or collider conditioning (Elwert and Winship, 2014; Pearl, 2009). **We claim no novelty for the hazard.** What is not standard is *where it hides*. In composition studies the gate is not presented as conditioning at all but as a *reasonableness* filter—“we study defenses that work”—so it reads as good

Table 2: **The confounded ladder against the Mode-S instrument, on the one arm that has both.** Same downstream defense, same attack, same seeds, same nominal dose—and the two designs share their $\kappa=0$ cell *numerically* (mean and worst per-seed gap $0.00e+00$, asserted at 10^{-12}). At the top rung they are matched on aggregate displacement, decision change and mean clean accuracy, so **the ladder’s fall is not an accuracy collapse**; they differ chiefly in the **adversarial coefficient channel**, and the ASR effect **changes sign**. **The one non-match, disclosed:** aggregate adversarial influence Λ_a moves in *both* designs, 0.081 against 0.033 ($2.4\times$), so neither column is a clean statistic-only intervention—which is why the contrast is “differ chiefly in the coefficient channel”, never “only in it”. *Vocabulary:* this row is the $\Delta\Lambda_a$ column of the channel table—the adversarial share of selected coordinates, the same field in the same artifact—and **not** its Δ adm. column, which counts rounds where the *support* of the adversarial mass changes and reads 0.000 on every row: what moves here is how much weight the adversary’s direction carries, not whether it enters at all. This is a two-column contrast on one arm: it locates where the two designs differ, and attaches no p -value to the difference between the columns. Every cell recomputed from per-seed and per-round rows by `build.comparability.table.py`; nothing here is retyped.

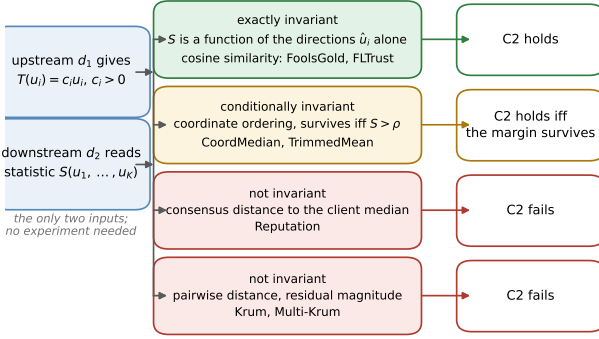
	Confounded ladder dose.kappa	Instrument (Mode S) doseS.kappa
downstream d_2	coord.median	coord.median
attack	committed pixel	committed pixel
seeds	42–46	42–46
nominal dose	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6
benign coefficients	mean 1 over <i>all</i> clients	mean 1 over <i>benign</i> clients
adversarial coefficients	drawn from the same spread	pinned $c = 1.0$ exactly
adv. coefficient share	0.267 $\rightarrow$ 0.187	0.266667, constant
Δ_c	+0.000 $\rightarrow$ -0.336	0 by construction (read back $9e-08$)
aggregate displacement	0.000 $\rightarrow$ 0.722	0.000 $\rightarrow$ 0.670
decision change	0.000 $\rightarrow$ 0.487	0.000 $\rightarrow$ 0.482
adversarial influence $\Delta\Lambda_a$	0.000 $\rightarrow$ 0.081	0.000 $\rightarrow$ 0.033
mean clean accuracy	0.767 $\rightarrow$ 0.690	0.767 $\rightarrow$ 0.698
ASR	0.443 $\rightarrow$ 0.170	0.443 $\rightarrow$ 0.541
ASR effect	-0.272 ($p_{\downarrow} = 0.0015$)	+0.098 ($p_{\uparrow} = 0.069$)

hygiene rather than as the step that removes identification. Three consequences are worth stating flatly, because each is routinely assumed away. **(i) More data cannot fix it:** the separating cell is empty by the gate, not by sampling, so it stays empty as the menu, the attack count and the seed count all grow—the estimator is not noisy, it is unidentified. **(ii) Nor can a better estimator,** since no reweighting of occupied cells recovers an unoccupied one. **(iii) Nor can the gate simply be dropped:** without it C1 is vacuous and the screen certifies compositions in which nothing suppresses the attack at all. The boundary is therefore not a limitation of our menu but a property of the design.

Both worked instances are ours, and one of them inverted a sign. We name no other paper as making this mistake; we do not have to, because our own two designs are the instances. The C1-gated composition study of §4 is the template at the level of the *criterion*—which is exactly why Lemma 1 exists—and the dose ladder of §5 is a second instance one level down, where the gated quantity is not C1 but the dose’s incidental attenuation of whichever client happens to carry the poison. **We ran the outcome-gated design first, and on the identical cell it returned the opposite sign:** the confounded ladder *falls* 0.272 ($p_{\downarrow}=0.0015$) where the instrument that pins adversarial influence *rises* +0.098 (§5.2). Table 2 puts the two designs side by side so the contrast can be checked rather than taken on trust: they share their $\kappa=0$ cell numerically and are matched at the top rung on aggregate displacement, decision change and clean accuracy. We can offer no stronger evidence that the hazard is not hypothetical: it cost us the sign of our own effect, in the direction that flattered us.

The escape, as a recipe—and this, not the criterion, is the practical contribution. Hold d_2 , the attack, and therefore the gate’s input $M(d_2, a)$ fixed; intervene on the upstream transform; and **measure the channel rather than the outcome-conditional contrast**—for us, the admitted adversarial mass rather than the certified-versus-

(a) What the criterion checks: invariance class of d_2 's statistic



Proposition 1 assigns the class; the C2 verdict follows without running the composition.

(b) Measured: margin $m \rightarrow$ adversarial mass $\Lambda_a \rightarrow$ ASR

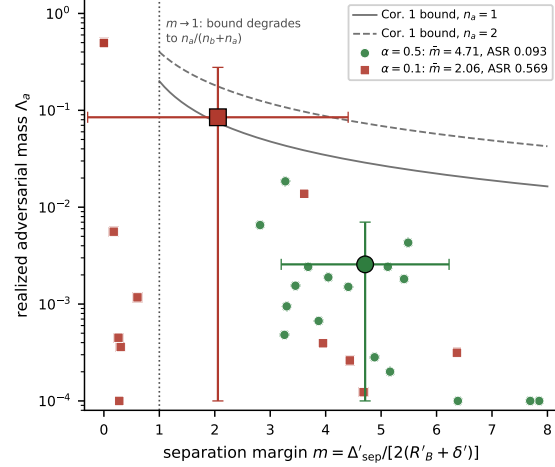


Figure 2: **The criterion is a mechanism test, and the mechanism is measurable.** (a) What C2 checks: the upstream transform T and the statistic S that d_2 reads determine an *invariance class* (Proposition 8), and the class determines the verdict—no composition needs to be run. This is why norm clipping does not blind FoolsGold but does blind Reputation. (b) The measured chain for FG→RFA, per round, on the transformed points RFA actually receives. Each marker is one round with $w_a > 0$ (rounds where FoolsGold zeroes every adversary are Lemma 2 and carry no payload); large markers are mean $\pm$ s.d. Corollary 1's bound holds in every round, and the $\alpha=0.1$ collapse is visible as a collapse in the theorem's own quantity: m falls from 4.71 ± 1.51 ($n=17$ rounds) to 2.06 ± 2.35 ($n=12$), with several rounds at $m < 1$ where the bound degrades to $n_a/(n_b+n_a)$, and realized Λ_a rises from 0.003 to 0.085 (max 0.50) as composition ASR goes $0.093 \rightarrow 0.569$. Values below 10^{-4} are drawn at 10^{-4} .

uncertified difference. What an evaluation must hold fixed to license a causal claim about mechanism is precisely what Proposition 4's hypothesis enumerates, and it is what we would ask a reader to carry into their own evaluation.

Read this way the boundary is not an apology for a failed suite but a design derivation, and §5 runs the design it licenses: the same downstream defense, the same attack, the same standalone effectiveness, and a controlled, independently measured dose of disturbance. Figure 2 sets out the mechanism the criterion checks and the measured chain for FG→RFA.

A second boundary: what such a screen cannot find. The gate that limits what a screen can be *validated* on also limits what it can *cover*—which converts the most obvious objection to our criterion, that its recall is poor, into a scope result.

Proposition 5 (Inherited-suppression screens cannot certify emergent synergy). *Call $(d_1 \rightarrow d_2)$'s suppression of a emergent if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent composition satisfies the eligibility gate of Proposition 4. Hence, writing $\mathcal{S}$ for the compositions with $M < \theta$ and $\mathcal{E} \subseteq \mathcal{S}$ for the emergent ones, any such screen's recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|\mathcal{S}|$ on every menu and every attack.*

Proof. Emergence is the negation of the gate, so each member of $\mathcal{E}$ is rejected while belonging to $\mathcal{S}$: a false negative. The bound is the resulting recall. $\square$

This is a scope theorem, not a defect found by tuning, and it is specific neither to C1 nor to FL: any screen requiring a constituent to suppress the attack independently inherits it. Our own lowest-ASR composition is the witness—FG→RFA reaches 0.093 where FoolsGold alone reaches 0.732 and RFA alone 0.845—so the bound binds at $|\mathcal{E}|/|\mathcal{S}| = 1/5$, which is exactly the 40% recall we report (§4.3, §6). Screening for inherited robustness and searching for synergistic compositions are different problems; only the first is ours.

Mechanism–effectiveness confounding: Proposition 4 in our instance. The boundary has an uncomfortable consequence for validation, and it is worth spelling out concretely. In *every* in-scope pair the criterion certifies, d_2

already suppresses the attack that d_1 fails. A certified composition’s low ASR can therefore always be attributed to d_2 ’s standalone strength (C1) rather than to d_1 having preserved d_2 ’s mechanism (C2, C3): the two are confounded **by the conditions themselves**, not by an unlucky choice of defense menu. We name this **mechanism–effectiveness confounding** and offer it as a methodological observation about validating composition criteria of this shape, applicable beyond our own experiments.

In particular it rules out the experiment we previously named as decisive: a d_2 whose admission signal is invariant under positive rescaling but which is *weak in isolation* cannot satisfy C1 in scope, so no such certified pair exists to test. The achievable substitute is **per-attack complementarity**—a d_2 that suppresses the attack d_1 fails while remaining weak on a *different* committed attack, so its max-committed ASR stays high and it cannot suppress the composition single-handedly. Reputation is such a defense (0.017 on model-scaling, 0.842 on the pixel backdoor), and §4.2 builds the test on that observation.

Argument sketch. (C1 sufficient) If at least one of d_1, d_2 individually suppresses a , composition inherits that suppression provided $C2 \wedge C3$ preserve the signal chain. C1 is sufficient, not necessary: two individually-weak defenses may suppress synergistically through complementary mechanisms (e.g., partial downweighting + geometric rejection), but predicting such synergy requires the quantitative margins below. The criterion is therefore *conservative*—it may reject working compositions but will not approve broken ones.

(C2 role) Suppose d_2 individually suppresses a via signal $\sigma_2(u_a) > \theta$. If d_1 ’s transformation T_1 destroys this signal ($\sigma_2(T_1(u_a)) \not\approx \sigma_2(u_a)$), d_2 cannot detect the attack. Example: NormClip as d_1 equalizes all norms to $\leq \tau$, so reputation’s distance signal $\sigma_2(u) = \|u - \hat{\mu}\|$ collapses—adversarial and benign updates become equidistant from consensus. Quantitatively: C2 holds when relative signal distortion $\Delta_{\text{signal}} := |\sigma_2(T_1(u_a)) - \sigma_2(u_a)| / \sigma_2(u_a) < \delta_2$ for a defense-specific tolerance δ_2 .

(C3 role) Even when d_1 preserves the *direction* of d_2 ’s signal (C2 holds), it may compress magnitudes below d_2 ’s threshold. Example: FoolsGold as d_1 downweights adversarial clients by $\alpha \ll 1$. TrimmedMean’s coordinate-wise rank signal still distinguishes adversarial coordinates (C2 holds), but after rescaling by α , per-coordinate values $\alpha \cdot u_{a,k}$ fall within the trimming range—the coordinate-wise extreme that would trigger trimming is no longer extreme. Quantitatively: C3 holds when the margin ratio $\Delta_{\text{margin}} := (\tau_{d_2}(T_1(u_a)) - \theta) / (\tau_{d_2}(u_a) - \theta) > 0$ —i.e., adversarial scores remain above d_2 ’s rejection boundary after d_1 ’s transform.

(Sufficiency) If C1 holds, C2 holds ($\Delta_{\text{signal}} < \delta_2$), and C3 holds ($\Delta_{\text{margin}} > 0$), then d_2 detects and suppresses u_a even after d_1 ’s transformation. For the weighting→aggregator class, Theorem 6 instantiates these margins explicitly: $\rho < r^*$ (rank) or $\rho < n_b/n_a$ (geometric). $\square$

3.2 A restricted theorem: $C2 \wedge C3$ for weighting→rank and weighting→geometric compositions

What follows is a boundary theorem, and deliberately not a security theorem. It bounds what an *invariance* argument about composition can establish—preservation of the downstream defense’s discriminative mechanism, levels (P1)–(P3) of Definition 1—and it says nothing whatever about suppression, which lives at (P4)–(P5). We state that in the same breath as the theorem because the whole point of the paper is the gap between the two: the theorem is the best case for statistic-level reasoning, proved with explicit margins, and §5.2 then shows that even exact invariance neither implies nor preserves suppression. Read it as a *formal* claim in the sense of Table 1: proved, on one class, bounding no ASR.

The criterion above is an operational heuristic. For the class of compositions that matter most in practice—a per-client *weighting* defense d_1 (reputation, FoolsGold) followed by a *rank-based* or *geometric* aggregator d_2 (TrimmedMean, CoordMedian, RFA), which contains the flagship FoolsGold→RFA—C2 and C3 can be *proven* rather than asserted, with an explicit tolerance on how aggressively d_1 may reweight. The key structural fact is that d_2 ’s admission score is scale-equivariant, so it commutes with d_1 ’s reweighting up to a bounded ratio; the composition fails only when d_1 compresses that ratio past a margin set by d_2 ’s separation.

Theorem 6 (Bounded reweighting preserves the downstream discriminative mechanism). *Consider a round with n_b benign and n_a adversarial updates under honest majority $n_b > n_a$. Let d_1 be a per-client reweighting $T_1(u_i) = w_i u_i$ with $w_i \geq 0$, and define the ratio over strictly positive weights, $\rho := \max_i w_i / \min_{i: w_i > 0} w_i$. Suppose d_2 suppresses attack a through a client admission score $\sigma_2(\cdot; U)$ that is scale-equivariant, $\sigma_2(c u_i; c U) = c^k \sigma_2(u_i; U)$ for all $c > 0$ and some degree $k \geq 0$. Assume $w_a > 0$ for adversarial clients; the $w_a = 0$ case is Lemma 2. The theorem establishes preservation of d_2 ’s suppression mechanism; it does not bound ASR. Then:*

1. **(Rank aggregators; C2∧C3 hold up to a margin.)** If $d_2 \in \{\text{TrimmedMean}, \text{CoordMedian}\}$, so σ_2 is a coordinate-wise rank, then d_2 's per-coordinate suppression of a is preserved when two conditions hold. **(i) Count majority:** $n_a < K/2$ for CoordMedian , and $n_a \leq \lfloor \beta K \rfloor$ for TrimmedMean with β -trimming. **(ii) Signed ordering:** on each suppression-relevant coordinate k , the adversarial transformed value lies strictly beyond every benign one, $w_a u_{a,k} > \max_b w_b u_{b,k}$ (upper tail) or $w_a u_{a,k} < \min_b w_b u_{b,k}$ (lower tail). Sufficient magnitude condition for (ii): if $\text{sign}(u_{a,k})$ is common to all adversaries and $|u_{a,k}| > \rho \cdot \max_b |u_{b,k}|$, then (ii) holds; writing $r^* := \min_k \min_b |u_{a,k}|/|u_{b,k}|$ for the coordinate-wise separation margin, this is $\rho < r^*$. Magnitude alone is not sufficient without the sign assumption: an adversarial coordinate can be large in absolute value yet sit between the benign values. At $\rho = 1$ ranks are preserved exactly, recovering Proposition 7.
2. **(Geometric median / RFA; C2∧C3 hold up to a margin.)** The composition hands RFA the transformed points $u'_i := w_i u_i$ and RFA runs its standard unweighted geometric median on them—it is not a weighted geometric median, and w_i enters only by moving the points. All quantities must therefore be defined on the transformed set. Let μ'_B be the geometric median of the benign u'_b alone, μ' that of all of U' , $R'_B := \max_b \|u'_b - \mu'_B\|$ the transformed benign radius, and $\delta' := \|\mu' - \mu'_B\|$ the displacement caused by including the adversarial points. Suppression is preserved when

$$\Delta'_{\text{sep}} := \min_{a,b} \|u'_a - u'_b\| > 2(R'_B + \delta'),$$

under which every adversarial residual $\|u'_a - \mu'\|$ strictly exceeds every benign residual $\|u'_b - \mu'\|$, so RFA's inverse-residual weight on a stays below the benign floor.

3. **(Norm-equalizing d_1 ; C2 fails—the degenerate case.)** If instead d_1 maps every client norm to a common value (NormClip with τ below all client norms), then any σ_2 keyed on update norm or consensus distance becomes constant across clients, so σ_2 no longer separates a and C2 fails. This is not a bounded reweighting: the induced per-coordinate weight ratio is unbounded, so no ρ -margin applies.

Parts (1) and (2) are sufficient conditions for d_2 to preserve the downstream suppression ordering—not to exclude adversarial contributions outright. Neither is a sufficient condition for low ASR, and part (2) orders residuals per client only—it constrains the aggregate adversarial coefficient solely through Corollary 1.

Proof. (1) TrimmedMean/CoordMedian act on coordinate k through the rank of $w_i u_{i,k}$ among $\{w_j u_{j,k}\}_j$; a is suppressed on k because its transformed coordinate is extreme (trimmed, or not selected by the median). Given (ii), at most n_a of the extreme order statistics are adversarial, so with (i) the selected value is benign. For the magnitude condition, $w_a > 0$ gives $\rho \geq \max_b w_b/w_a$, so $|u_{a,k}| > \rho \max_b |u_{b,k}|$ implies $|w_a u_{a,k}| > w_a \rho \max_b |u_{b,k}| \geq (\max_b w_b) \max_b |u_{b,k}| \geq \max_b |w_b u_{b,k}|$; with a sign common to all adversaries this places $w_a u_{a,k}$ strictly beyond every benign value on that side. Scale-equivariance ($k=0$ for pure rank) makes the uniform case $\rho=1$ exact. (2) All quantities are on the transformed points $u'_i = w_i u_i$, which is what RFA receives. Every benign residual obeys $\|u'_b - \mu'\| \leq R'_B + \delta'$ by the triangle inequality. For adversarial u'_a , $\|u'_a - \mu'_B\| \geq \Delta'_{\text{sep}} - R'_B$, hence $\|u'_a - \mu'\| \geq \Delta'_{\text{sep}} - R'_B - \delta'$. Demanding that this exceed the benign bound gives $\Delta'_{\text{sep}} - R'_B - \delta' > R'_B + \delta'$, i.e. the stated condition. No weighted-majority assumption is needed, and none would be appropriate: RFA weights every input point equally and the w_i act only by relocating the points. Our implementation runs Weiszfeld for 50 iterations at tolerance 10^{-6} , so the argument transfers to it. (3) Clipping to a common τ sends $\|u_i\| \mapsto \tau$ for all i , so $\sigma_2(u_i) = \|u_i\|$ (NormClip-as- d_2) or the consensus distance $\|u_i - \hat{\mu}\|$ collapses to a client-independent constant; separation vanishes, which is precisely C2-failure. $\square$

Theorem 6(2) orders residuals *per client*: each adversarial point receives less inverse-residual weight than each benign point. That alone does not bound the adversary's *total* share of the aggregate—several adversaries could each be down-weighted yet collectively hold substantial mass. The separation margin controls this, but only quantitatively:

Corollary 1 (Bounded aggregate adversarial influence). *Let $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ be the realized separation margin, so Theorem 6(2) reads $m > 1$. Every benign residual is at most $D_b := R'_B + \delta'$ and every adversarial residual at least $D_a := \Delta'_{\text{sep}} - R'_B - \delta' = (2m - 1)D_b$. Bounding $\sum_j 1/d_j$ below by the benign terms alone and each adversarial term above by $1/D_a$,*

$$\Lambda_a = \sum_a \lambda_a \leq \frac{n_a/D_a}{n_b/D_b + n_a/D_a} = \frac{n_a}{n_b(2m - 1) + n_a},$$

so μ' is a convex combination whose adversarial share is at most Λ_a : writing $\mu' = (1 - \Lambda_a)\bar{u}_B + \Lambda_a\bar{u}_A$ for its benign and adversarial parts, the displacement from the benign-only combination $\bar{u}_B$ is at most $\Lambda_a\|\bar{u}_A - \bar{u}_B\|$.

The bound is only as good as the margin, and **the separation condition alone is not enough**: as $m \rightarrow 1$ it degrades to $n_a/(n_b + n_a)$, permitting the adversary 40% of the mass at $n_a=2, n_b=3$. A margin comfortably above 1 does the work. We measure $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71 over those rounds), capping the adversarial share at 5.1% for $n_a=1$ and 12.6% for $n_a=2$. The realized figure is far tighter than the bound: measuring Λ_a directly, it never exceeds 1.8% in any round, and the bound is satisfied in all 17 measured rounds with $w_a > 0$. With Lemma 2 this completes the picture, and the mechanism survives for *opposite* reasons: at $w_a=0$ the adversaries hold most of the Weiszfeld mass ($\Lambda_a = 0.71$ measured) but carry no payload; at $w_a > 0$ they carry payload but hold at most 12.6%.

This is the one place in the paper where the theory predicts a number and the measurement lands inside it, so we state the pair plainly rather than leaving it in an appendix: **bound 12.6%, realized $\leq 1.8\%$** , in every measured round with $w_a > 0$. **And it is also the one place where heterogeneity makes the guarantee measurably empty**. The bound is a function of the realized margin m alone, and at Dirichlet $\alpha=0.1$ RFA’s **all-round mean margin collapses from $m=3.24$ to $m=1.01$** —exactly the limit at which Corollary 1 degrades to $n_a/(n_b+n_a)$ and certifies nothing—with two of three certified compositions crossing the ASR threshold there (§4.4(E)). So the theorem’s explanatory value is genuine and measured in *both* directions: it predicts a number that holds where heterogeneity is moderate, and its own quantity records where it stops holding. **That is not a robustness check; it is one of the paper’s three central empirical findings** (§5.2, §10).

Λ_a is level (P4) in measurable form—but (P4) is a *level*, not a single metric. Before the instantiation, one clarification the rest of the paper depends on: (P4) is a **conceptual level of the hierarchy and is instantiated defense-specifically**, because different aggregator families have different notions of what it means for adversarial input to enter the aggregate at all. A *selector* admits or does not, so its (P4) is an indicator in $\{0, 1\}$; a *weighted averager* has an adversarial coefficient share in $[0, 1]$; a *coordinate-wise order statistic* has a fraction of selected coordinates that are adversarial. There is no single scalar all three compute, and this is why the influence column of Table 11 is **necessarily aggregator-specific** rather than a common measurement across rows—a fact about the level, not a shortcoming of the table. With that said: Definition 1’s levels are not equally meaningful. (P1)–(P3) are properties of a statistic and of a decision rule reading it; (P4) is the **first level whose content is the adversary’s influence on the aggregate**, and for a geometric aggregator Λ_a is exactly that level made measurable—Corollary 1 bounds it, Lemma 2 drives it to a benign-only convex combination, and §5.2’s instruments intervene on it directly. So the negative results do not leave the theory empty-handed: **the level the evidence points to is one the theory can bound—just not from a (d_1, d_2) definition pair alone**, which is what a screen would need. That gap is the open problem we state in §10. **Three quantities are distinguished throughout and never used interchangeably**: *admitted adversarial mass*—whether an adversarial update enters d_2 ’s aggregate at all, binary for a selector and a coordinate fraction for an order statistic; the *adversarial coefficient share*, the fraction of the upstream coefficient vector’s total mass held by adversaries, which is what the instruments of §5.2 pin; and *aggregate adversarial influence* Λ_a , the weight the adversary’s direction actually carries in the aggregate. Unqualified, “admission” means the first. (Symbols are collected in Appendix F.)

Lemma 2 (Zero-weight annihilation). *Zero-weight clients are not removed from the composition: d_2 receives them as zero vectors and still sees all K points. Our implementation runs the ε -smoothed Weiszfeld iteration ($\varepsilon = 10^{-8}$), whose fixed point is $\mu' = \sum_i \lambda_i u'_i$ with $\lambda_i \propto 1/\max(\|u'_i - \mu'\|, \varepsilon)$ —a convex combination for every configuration, including the singular case $\mu' \in U'$ that a zero-weight client at the origin can create, so no non-degeneracy assumption is required. If $w_a = 0$ for every adversarial client, then $u'_a = 0$ and, writing $\Lambda_a := \sum_a \lambda_a$, the adversarial terms vanish identically: $\mu' = \sum_{b \text{ benign}} \lambda_b u'_b = (1 - \Lambda_a) \bar{u}$ with $\bar{u}$ a convex combination of benign transformed updates. The aggregate is a contraction by $1 - \Lambda_a \in (0, 1)$ of a benign-only weighted average: the attack direction never enters μ' . The origin points’ only effect is that contraction—a reduction in effective step size, i.e. a utility cost, not a security failure. The same holds for CoordMedian, where a zeroed coordinate carries no payload but shrinks the aggregate when selected.*

Why the positive-weight restriction matters. Defining ρ over all weights would make it *infinite in every round*: FoolsGold’s normalization subtracts the minimum weight, so 2 of 5 clients receive exactly zero (measured, $N=10, K=5$). A bound of the form $\rho < 4$ would then never be satisfiable, and Theorem 6(2) would be vacuous on the very composition it is stated over. Restricting ρ to strictly positive weights and handling $w_a=0$ separately via Lemma 2 is what makes the analysis apply to the algorithm we actually run.

FG→RFA is case 2, and this is where its suppression comes from. FoolsGold→RFA instantiates Theorem 6(2) together with Lemma 2, and the two are exhaustive. Measuring on the transformed points RFA actually receives (25 adversary-present rounds, 3 seeds): FoolsGold zeroes every adversary in 32% of rounds, where Lemma 2 gives

suppression with a measured contraction $1 - \Lambda_a = 0.29$ on average; in the remaining 68% the separation condition holds in *every* round, with margin $\Delta'_{\text{sep}}/2(R'_B + \delta') \geq 2.8$ (mean 3.2) and δ'/R'_B mean 0.19. Suppression is therefore derived rather than merely observed, under assumptions we verify rather than assert. The contraction is also the source of the composition’s accuracy cost (0.50 for FG→RFA against 0.78 for RFA alone). The families covered here span the criterion’s PASS mechanism, its C3-margin misses, and its C2-failure mode; remaining defenses are empirically-validated extensions (§4).

The margin behaves like a certificate, not just a switch. C3 is stated as a binary test, but the margin m is measurable and its *strength* tracks security. Across the 29 measured rounds with $w_a > 0$ spanning both heterogeneity settings, higher margin accompanies lower realized adversarial mass (Spearman $\rho = -0.48$, $p = 0.009$), and Corollary 1’s bound holds in every one. The same ordering appears at composition level: $m = 4.71 \Rightarrow \Lambda_a = 0.003 \Rightarrow \text{ASR } 0.093$ ($\alpha=0.5$) versus $m = 2.06 \Rightarrow \Lambda_a = 0.085 \Rightarrow \text{ASR } 0.569$ ($\alpha=0.1$). That is what makes the condition operationally useful—a short pilot returns a number rather than a verdict—but the composition-level link rests on two heterogeneity settings only, so we report it as directional rather than fitted.

Connecting margins to predictions. The theorem’s conditions give computable margins per pair, but they are met in *two* distinct ways, so reporting ρ alone mispredicts. Let $r := w_a / \max_b w_b$ be the adversary’s relative weight and S the attack’s coordinate extremeness. **Regime A (ordering preservation)**, $S \cdot r > 1$: the adversary stays extreme and the rank aggregator discards it—FoolsGold sits here, with bounded weights, $\rho \leq 2.1$, hence $r \geq 0.48$ and $S \cdot r \geq 4.8$ at $S=10$. **Regime B (attenuation)**, $S \cdot r \ll 1$: the adversarial transformed value falls *inside* the benign spread, where it cannot be an extreme order statistic and its payload is scaled by r ; $r=0$ is the exact annihilation of Lemma 2. Reputation sits here, emphatically so: its exponential weights $\exp(-d_i/\sigma)$ give ρ mean 1.4×10^5 and max 2.8×10^6 , with measured $w_a = 4.8 \times 10^{-4}$ against benign 0.278, hence $r = 1.7 \times 10^{-3}$ and $S \cdot r = 0.017$. Consequently **reputation→CoordMedian—a PASS pair—violates the bounded-ratio condition $\rho < r^*$ by roughly five orders of magnitude and still suppresses**, because a huge ρ here signals that reputation is *crushing* the adversary rather than failing to separate it. Both regimes preserve the mechanism; the hazard is the middle, $S \cdot r \approx 1$, where the adversary is neither discarded nor attenuated. C3-FAIL pairs land near that middle or exceed their count budget (FoolsGold→TrimmedMean below); C2-FAIL pairs have $\Delta_{\text{signal}} \rightarrow \infty$ under a non-invariant signal (Proposition 8(c)). **Operational calibration protocol.** To evaluate the criterion for $(d_1 \rightarrow d_2)$: **C1** is available from standard single-defense baselines (zero cost). **C2/C3 for the theorem-backed class** (FG or reputation as d_1 ; RFA, CM, TM as d_2): Theorem 6’s bound $\rho < n_b/n_a$ is checkable from algorithm properties alone—no calibration runs needed. **C2/C3 for other families**: one short pilot (5–10 rounds, $\sim 10\%$ of full cost) measuring $\sigma_2(T_1(u_a))$ vs. $\sigma_2(u_a)$ suffices. The full 50-round experiment runs only for criterion-PASS candidates. $\delta_2 = 0.5$ separates all PASS from C2-FAIL cases on the development set; insensitive to choice: any $\delta_2 \in [0.3, 0.7]$ yields identical classifications on all 42 pairs.

Proposition 7 (Rank-invariance of coordinate-wise aggregators). *Let d_2 be a coordinate-wise rank aggregator (Trimmed-Mean or CoordMedian) with admission statistic $\sigma_k(u) = \text{rank}_k(u)$ among the received updates $\{u_1, \dots, u_m\}$. Let $T_1(u_i) = \alpha \cdot u_i$ be a uniform positive rescaling ($\alpha > 0$, same for all clients). Then:*

1. $\sigma_k(T_1(u_i)) = \sigma_k(u_i)$ for all i, k — i.e., C2 and C3 hold trivially.
2. Under heterogeneous rescaling $T_1(u_i) = \alpha_i \cdot u_i$ with $\alpha_i \neq \alpha_j$, rank preservation fails when rescaling reorders coordinates across clients: $\exists k$ s.t. $\text{rank}_k(\alpha_i u_{i,k}) \neq \text{rank}_k(u_{i,k})$.

Proof. (1) Rank is invariant to positive scalar multiplication: $u_{i,k} > u_{j,k} \iff \alpha u_{i,k} > \alpha u_{j,k}$ for $\alpha > 0$. (2) Counterexample: let $u_{1,k} = 3$, $u_{2,k} = 2$, with $\alpha_1 = 0.1$, $\alpha_2 = 1$. Then $\alpha_1 u_{1,k} = 0.3 < \alpha_2 u_{2,k} = 2$, reversing the rank. $\square$

Corollary 2. *CoordMedian (pure rank selection) is robust to C3 boundary compression under any uniform rescaling but vulnerable under heterogeneous client-specific rescaling (e.g., FoolsGold or reputation weights). This predicts that compositions $X \rightarrow \text{CoordMedian}$ will show intermediate ASR when d_1 applies heterogeneous weights—consistent with observed $\text{NC} \rightarrow \text{cm}$ (0.433) and $\text{rfa} \rightarrow \text{cm}$ (0.485), both near the 0.5 threshold.*

Proposition 7 concerns rank under *uniform* rescaling. The sharper question is which admission signals survive *heterogeneous* per-client rescaling, since that is what every per-client defense in our menu applies—NormClip ($c_i = \min(1, \tau/\|u_i\|)$), reputation ($c_i \propto e^{-d_i/s}$) and RFA-reweighting ($c_i \propto 1/\max(\|u_i - \mu'\|, \epsilon)$) are all of the form $T(u_i) = c_i u_i$ with $c_i > 0$ strictly, each positive by construction. FoolsGold is of the same form with $c_i \geq 0$ but attains $c_i = 0$ in *every* round (its shift-and-clamp gives the least-trusted client weight exactly zero, 2 of 5 clients here), so it falls outside the strict hypothesis below and is governed by Lemma 2 instead—the same distinction we draw for FLTrust’s ReLU zeros in §4.1. The strictly-positive case has a sharp answer, and it partitions the menu:

Proposition 8 (Invariance classes under positive per-client rescaling). *Let $T(u_i) = c_i u_i$ with $c_i > 0$, **strictly**—the form taken by *NormClip* ($c_i = \min(1, \tau/\|u_i\|)$), *Reputation* ($c_i \propto e^{-d_i/s}$) and *RFA-reweighting* ($c_i \propto 1/\max(\|u_i - \mu'\|, \epsilon)$), each positive by construction. **FoolsGold is deliberately not in that list:** its shift-and-clamp gives the least-trusted client weight exactly 0 in every round (2 of 5 clients here), so a zero coefficient is not a positive rescaling and *FoolsGold-as- d_1* falls outside the hypothesis, governed by Lemma 2 rather than by (a)–(c). Then:*

- (a) **Cosine similarity is exactly invariant:** $\cos(c_i u_i, c_j u_j) = \cos(u_i, u_j)$. *FoolsGold’s weight vector is a function of the pairwise-cosine matrix alone, so $w(T(U)) = w(U)$ identically, and **C2 holds for every $\cdot \rightarrow \text{FoolsGold}$ composition.***
- (b) **Coordinate ordering is not invariant** in general; it survives under the bounded-ratio condition of Theorem 6(1).
- (c) **Consensus distance and residual magnitude are not invariant.** *Heterogeneous c_i can invert their ordering, so $\cdot \rightarrow \text{reputation}$ compositions fail C2, and $\cdot \rightarrow \text{RFA}$ needs the separation condition of Theorem 6(2).*

Proof. (a) $c_i u_i / \|c_i u_i\| = u_i / \|u_i\|$ for $c_i > 0$, so every pairwise cosine, each client’s normalized maximum similarity, and hence every FoolsGold weight is unchanged. (c) Counterexample for consensus distance: with $u_1=(1, 0)$, $u_2=(0, 1)$, $u_3=(3, 3)$ the coordinate-wise median is $(1, 1)$ and u_3 is farthest; under $c = (1, 1, 0.1)$ the median becomes $(0.3, 0.3)$ and u_3 becomes the *closest* point, reversing the ordering reputation depends on. $\square$

This corrects a natural but wrong intuition. Norm clipping does *not* blind FoolsGold—it cannot, by (a)—though it does blind reputation, whose signal is consensus distance rather than similarity. We verified (a) in the shipped implementation: FoolsGold’s weight vector changes by at most 3×10^{-8} (float32 rounding) under NormClip, across 1000 synthetic trials and live FL rounds in which 1–2 of 5 clients were genuinely clipped (raw norms up to 24.5 against $\tau=5$). Note this also means Theorem 6(3)’s norm-equalizing degenerate case requires τ below *all* client norms; at our $\tau=5$ only 1–2 of 5 clients clip per round, so norms are not equalized and that case does not describe our runs.

3.3 Heterogeneity is not the mechanism: ρ against Δ_c

The finding this subsection collects, stated first. What decides whether an upstream transform costs a downstream defense its suppression is *not* how heterogeneous the transform is, but **whether its reweighting is correlated with adversary identity**—and even that is not a magnitude story, because adversary-correlated reweighting has **two opposite regimes**, annihilation and attenuation, so its size alone is insufficient in both directions. This is why our own headline predictor failed, and it is the mechanistic content behind §5’s refutation.

Two different things a coefficient vector does, and the theory sees only one of them. (Symbols are collected in Appendix F.) Proposition 8 turns on **dispersion**, $\rho := \max_i c_i / \min_i c_i$: how far apart the coefficients are, which is what decides whether an ordering survives. It says nothing about *which* clients receive which coefficient. That second property is **adversary-correlated dispersion**,

$$\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A], \quad A \text{ the adversarial participants,}$$

and the two are **independently controllable**—not merely uncorrelated in our sample: a transform can spread the coefficients arbitrarily (ρ large) at $\Delta_c = 0$ exactly, or leave them nearly equal while shifting the adversary’s share, and Proposition 10 exhibits a family of each. **Proposition 8 and Theorem 6 constrain ρ ; Lemma 2’s attenuation channel is about Δ_c** , and this is the sharp form of the “ ρ alone mispredicts” point above: regimes A and B are separated by r , not by ρ . §5.1 finds that Δ_c is what the real upstream transforms differ on, and §5.2’s two instruments are, in retrospect, the Δ_c -pinned and Δ_c -swept designs. Note what Δ_c is *not*: a screening quantity—computing it requires knowing A , which is the whole difficulty (§10).

3.4 The framework in general form

Positive rescaling is the instantiation that matters for current defenses, but the structure is more general, and stating it separates what is essential to the framework from what is an artifact of today’s defense menu. Write d_2 ’s admission mechanism as a statistic $S(U)$ of the round’s updates together with a decision rule that consumes only the *ordering* S induces over clients—which are extreme, which are far from consensus, which are down-weighted. Let T be d_1 ’s per-client transform. Then $(d_1 \rightarrow d_2)$ is **mechanism-preserving** exactly when

$$S(T(U)) = \varphi(S(U)) \quad \text{for some } \varphi \text{ preserving the ordering } d_2 \text{ acts on,} \quad (1)$$

which is C2; C3 additionally asks that φ not compress the quantitative gap below d_2 's operating threshold. Proposition 8 is the case $T(u_i) = c_i u_i$: for $S = \text{cosine similarity}$, $\varphi = \text{id}$, so (1) holds unconditionally; for $S = \text{coordinate value}$, φ is a per-client positive scaling, order-preserving only under the bounded-ratio condition $\rho < r^*$ of Theorem 6(1); for $S = \text{consensus distance or residual magnitude}$, no φ exists, because the reference point (median, geometric median) moves with T , so $S(T(U))$ is not a function of $S(U)$ at all. Three consequences worth stating. First, this is why the conditions are checkable without running an experiment: one reads S off d_2 's definition and T off d_1 's, and the question becomes algebraic. Second, it identifies what a genuinely new instantiation would require—an S and a non-rescaling T , e.g. a d_1 that projects updates onto a subspace or applies a coordinate-wise mask, for which (1) would need to be re-derived rather than inherited. Third, it marks the framework's real boundary: a d_1 whose transform is not per-client—one that mixes updates across clients, or carries cross-round state—admits no T of this form, and nothing here applies to it.

Predicted composability for our defense menu. Criterion 1 makes precise predictions for each two-defense pair from our 7-defense menu. We validate against targeted pairs spanning all prediction categories. Key predictions:

- **FoolsGold** $\rightarrow$ **RFA (PASS, C1 $\wedge$ C2 $\wedge$ C3)**: FoolsGold suppresses both scaling and pixel (via similarity-based downweighting); RFA's geometric-median signal is preserved after FG's rescaling because distance ratios are maintained. Predicted: low max-ASR.
- **Reputation** $\rightarrow$ **X (C1-FAIL for pixel)**: Reputation kills scaling (ASR 0.017) but does *not* suppress pixel (baseline 0.842). For any rep $\rightarrow$ X pair, C1 fails for the pixel attack, so the criterion predicts HIGH max-committed ASR regardless of C2/C3 status. The exception is when d_2 independently suppresses pixel (e.g., CoordMedian partially suppresses pixel to 0.443).
- **NormClip** $\rightarrow$ **Reputation (C2-FAIL)**: NormClip normalizes all norms to $\leq \tau$, destroying the consensus-distance signal that reputation relies on. Predicted: high ASR.
- **FoolsGold** $\rightarrow$ **TrimmedMean (C3-FAIL)**: FoolsGold downweights adversarial clients by $\alpha \ll 1$; TrimmedMean's rank signal direction is preserved (C2 holds), but rescaled adversarial coordinates $\alpha \cdot u_{a,k}$ are no longer extreme—they fall within the trimming range. Predicted: high ASR.
- **Degenerate pairs** (FedAvg as d_1): d_1 applies no per-client transformation, so composition degenerates to d_2 alone.

We validate these predictions against empirical outcomes in §4.

Relation to the norm-based dichotomy (§L). The complementarity analysis we develop later (Corollary 3) shows that norm-based menus are confined to two regimes (no complementarity, or persistence-dominated realized collapse), explaining the negative result for temporal mixing. Criterion 1 generalizes that dichotomy to arbitrary defense pairs and characterizes when composition *escapes* it: by using an orthogonal admission signal (C2 preserved) whose separation exceeds the threshold (C3 preserved). The next section validates the criterion's predictions.

4 Composability Criterion Validation

Criterion 1 predicts for each two-defense pair whether composition achieves low max-committed ASR. All experiments in this section and §6 use one deployment regime throughout (CIFAR-10, $N=10$ clients, $K=5$ per round, adversarial fraction $f=0.2$, 50 rounds, 4-layer CNN), with committed attacks model_scaling and backdoor_pixel. **Convention:** unless stated otherwise, ASR means *max-committed ASR*—the maximum over the two committed attacks of the per-attack mean across seeds—thresholded at 0.5—the level at which the trigger fires on the majority of triggered inputs, so that “suppresses” means the backdoor is not the model's dominant response; adaptive-attack, order-sensitivity and per-seed figures are labelled explicitly where they differ. A second gate is needed because a destroyed model has a low ASR for the wrong reason: every cell in §5–§5.2 must also hold mean clean accuracy ≥ 0.35 (our **accuracy floor**), below which we call the cell *uninterpretable* rather than suppressed. **Sample sizes are small and were fixed before the runs:** $n=5$ seeds per cell in the interventional suites ($n=8$ for cos_krum, $n=3$ for the second-dataset replication), $n=30$ for the FoolsGold $\rightarrow$ RFA characterization, so intervals are wide, and no null result is read as flatness unless it clears a pre-registered equivalence margin. **“Refuted” means one thing throughout:** a pre-registered *directional* hypothesis was rejected against the rule frozen for it—never that a mathematical claim was disproved. Adversaries are the fixed client identities $\{0, \dots, fN - 1\}$ and each round samples K participants uniformly without replacement from all N , with no stratification, so the per-round adversary count is *hypergeometric* rather than fixed: for $N=10$, $f=0.2$, $K=5$, $P(n_a=0) = P(n_a=2) = 56/252 = 0.222$ and $P(n_a=1) = 140/252 = 0.556$ (measured over 100 rounds: 0.19/0.54/0.27). Two consequences the data bear out: CoordMedian's count-majority condition $n_a < K/2 = 2.5$ holds for *every* attainable $n_a \in \{0, 1, 2\}$, i.e. in 100% of rounds, while TrimmedMean's budget

$\lfloor \beta K \rfloor = \lfloor 0.2 \cdot 5 \rfloor = 1$ is exceeded whenever $n_a=2$ —22.2% of rounds, ≈ 11 of 50. That is a second, count-based reason FoolsGold $\rightarrow$ TrimmedMean fails (0.508) while FoolsGold $\rightarrow$ CoordMedian succeeds (0.107) under an identical upstream transform; we retain the pre-registered C3-FAIL label, the budget violation being an additional mechanism rather than a relabelling. We validate in two waves. The **development set** (18 targeted pairs, Table 3) spans all criterion prediction categories and was used to refine the criterion (adding C3). The **held-out set** (24 pairs, Table 4) was *pre-registered*: all 24 predictions were committed to version control (commit 7e4d9b1) *before* any held-out experiment was run, eliminating post-hoc classification concerns. The criterion predicts LOW (<0.5) or HIGH (≥ 0.5) max-committed ASR. The verdict categories are:

- **PASS**: $C1 \wedge C2 \wedge C3$ all hold $\Rightarrow$ predicted LOW (composition works)
- **C1-FAIL**: neither defense individually suppresses the attack $\Rightarrow$ predicted HIGH
- **C2-FAIL**: d_1 destroys d_2 ’s admission signal $\Rightarrow$ predicted HIGH
- **C3-FAIL**: d_1 compresses adversarial–benign separation below d_2 ’s threshold $\Rightarrow$ predicted HIGH
- **DEGEN**: d_1 has no per-client transform (trivial composition) $\Rightarrow$ predicted HIGH

Table 3: Composability criterion validation ($n=3$ seeds; max across committed_scaling and committed_pixel; $\pm$ is per-seed std where reported). Threshold: LOW < 0.5 , HIGH ≥ 0.5 . $^\dagger n=30$ from main experiment (§7).

Pair	Failure mode	Predicted	Max ASR	Correct?
<i>PASS: $C1 \wedge C2 \wedge C3$ all hold (predicted LOW)</i>				
foolsgold $\rightarrow$ rfa	—	LOW	0.093 ± 0.082	✓
foolsgold $\rightarrow$ coord_median	—	LOW	0.107 ± 0.034	✓
reputation $\rightarrow$ coord_median	—	LOW	0.423 ± 0.155	✓
<i>C1-FAIL: neither defense individually suppresses all attacks (predicted HIGH)</i>				
reputation $\rightarrow$ trimmed_mean †	C1 (pixel)	HIGH	0.679	✓
reputation $\rightarrow$ rfa	C1 (pixel)	HIGH	0.825	✓
reputation $\rightarrow$ foolsgold	C1 (pixel)	HIGH	0.807	✓
reputation $\rightarrow$ norm_clip	C1 (pixel)	HIGH	0.813	✓
norm_clip $\rightarrow$ foolsgold ‡	C1	HIGH	0.887	✓
<i>C2-FAIL: d_1 destroys d_2’s signal direction (predicted HIGH)</i>				
norm_clip $\rightarrow$ reputation	C2	HIGH	0.819	✓
rfa $\rightarrow$ reputation	C2	HIGH	0.882	✓
<i>C3-FAIL: d_1 compresses boundary below d_2’s threshold (predicted HIGH)</i>				
norm_clip $\rightarrow$ trimmed_mean	C3	HIGH	0.780	✓
norm_clip $\rightarrow$ rfa	C3	HIGH	0.853	✓
rfa $\rightarrow$ trimmed_mean	C3	HIGH	0.678	✓
rfa $\rightarrow$ coord_median	C3	HIGH	0.485	×
foolsgold $\rightarrow$ trimmed_mean	C3	HIGH	0.508	✓
norm_clip $\rightarrow$ coord_median	C3	HIGH	0.433	×
<i>DEGEN: d_1 has no per-client transform (predicted HIGH)</i>				
fedavg $\rightarrow$ norm_clip	DEGEN	HIGH	0.935	✓
fedavg $\rightarrow$ trimmed_mean	DEGEN	HIGH	0.846	✓
Overall: 16/18 = 89%. HIGH direction: 13/15 correct (87%). Under the tightened C1 the certified PASS set is 2/2 correct; fg $\rightarrow$ rfa is reclassified as a synergy false negative.				
‡ Reclassified from C2-FAIL: by Proposition 8(a) clipping preserves FoolsGold’s similarities exactly, so C2 holds and the pair fails C1. Prediction and outcome unchanged.				

PASS precision: 100%—all compositions the criterion predicts to work *do* work. Under the tightened per-attack C1 (§3) the certified in-distribution set is 2 pairs, fg $\rightarrow$ cm (0.107) and rep $\rightarrow$ cm (0.423), both correct; fg $\rightarrow$ rfa (0.093) is no longer certified and is reported below as a false negative of our own criterion. This is the practitioner-relevant result: the criterion’s positive predictions are reliable, at the cost of 40% recall. Overall development-set accuracy: **16/18 = 89%**. Two misses, both in the HIGH direction:

- **rfa $\rightarrow$ cm** (predicted HIGH, actual 0.485) and **NC $\rightarrow$ cm** (predicted HIGH, actual 0.433): CoordMedian is partially robust to boundary compression because its coordinate-wise median is invariant to monotone magnitude transformations—if d_1 ’s rescaling preserves the ordering of coordinates across clients, CoordMedian’s selection is unchanged. This is exactly Proposition 7: the C3-FAIL prediction is overly conservative for rank-based d_2 , and both misses are the *predicted* intermediate-ASR behavior of Corollary 2 rather than unexplained errors. The

measured quantities say how close to the uniform case these two sit: NormClip clips only 0–2 of 5 clients per round (11 of 45 client-rounds over 9 live rounds), so $\approx 76\%$ of clients receive $c_i = 1$ *exactly*, and with the largest observed raw norm 24.5 against $\tau=5$ the realized ratio is $\rho \leq 4.9$, which at $S=10$ still clears Theorem 6(1)’s bounded-ratio condition with margin $S/\rho \geq 2.0$; RFA-reweighting is likewise near-uniform across benign clients. The labelling error is therefore specifically a *binary* reading of C3—“ d_1 is heterogeneous, therefore the margin is compressed”—where the theorem asks the quantitative question $\rho < r^*$. **The refinement does not recover either pair:** both also fail C1, since CoordMedian alone is already at 0.519 and neither NormClip (0.935) nor RFA (0.885) suppresses. What remains is a statement about the residual—a d_2 whose own baseline sits at 0.519 needs only a mild order-preserving upstream transform to fall just under 0.5, which is what 0.433 and 0.485 are—and the operational consequence: evaluate C3 for rank aggregators against the realized ρ from a 5–10 round pilot rather than a binary heterogeneity test, the same quantity Theorem 6(1) already requires and our labelling had not been using.

Per-seed spread on the headline pairs, and one number that moved. The seed budget is not uniform across suites, so we report per-seed values and Student- t 95% intervals rather than means alone, and state each cell’s n wherever a composition number appears. Auditing this forced two corrections to numbers we had previously led with.

First, we topped the three headline pairs up from $n=3$ (seeds 42–44) to $n=5$ (adding 45–46). $\text{FG} \rightarrow \text{CM}$ settles at 0.107, CI [0.064, 0.149], per-seed 0.123/0.126/0.145/0.069/0.072; $\text{Rep} \rightarrow \text{CM}$ moves *up* to 0.423, CI [0.231, 0.616], per-seed 0.432/0.234/0.391/0.665/0.395.

Second, and more consequentially, $\text{fg} \rightarrow \text{rfa}$ had already been measured on **30 seeds** (42–71) as the base condition of the adaptive study in §6. That is the largest measurement of any composition in this paper, and it gives max-committed ASR **0.093**, CI [0.062, 0.123]—not the 0.045 that seeds 42–44 gave. The per-seed distribution is right-skewed (median 0.060, mean 0.093, sample skew +2.07; 8/30 seeds at or above 0.10; maximum 0.380), and seeds 42–46 happen to fall in its left tail, understating the pair by a factor of 1.8. We therefore headline the $n=30$ figure throughout. The $n=3$ value was not a computational error—it is the correct mean of those three seeds—but it was an unrepresentative sample of a skewed distribution, which is the concrete hazard in reporting 3-seed composition means and the reason we now attach an n and an interval to every one. Where a seed-matched comparison is required (the adaptive-attack sweep, which ran on seeds 42–46), we quote the base on those same five seeds, 0.051 ± 0.011 , and say so.

One certified pair is not resolved below threshold. $\text{Rep} \rightarrow \text{CM}$ ’s interval [0.231, 0.616] crosses 0.5, and one of its five seeds reaches 0.665. Its LOW label therefore rests on a mean whose uncertainty admits the HIGH side, and we report it as correct on the mean while flagging it as the one certified pair whose margin is not statistically established. The certified in-distribution set is 2/2 correct either way, but only $\text{fg} \rightarrow \text{cm}$ is certified *and* resolved below threshold at the seed budget we can afford. Clean accuracy on the three cells is 0.50, 0.68, 0.76, all well clear of the collapse regime.

Two boundary conditions, stated alongside the primary result. Both are developed in full later, but neither should be met late. (i) **Extreme heterogeneity breaks two of the three PASS pairs.** At $\alpha=0.1$, $\text{fg} \rightarrow \text{rfa}$ reaches 0.569 and $\text{rep} \rightarrow \text{cm}$ 0.548; only $\text{fg} \rightarrow \text{cm}$ survives, at 0.433. What makes this more than a negative datapoint is that the failure appears in the theorem’s own quantity: RFA’s separation margin $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$, measured on the transformed points, collapses from an all-round mean of 3.24 to 1.01—exactly the $m \rightarrow 1$ boundary at which Corollary 1’s bound degrades to 40% adversarial mass—so a practitioner running the 5–10 round pilot would measure $m \approx 1$ and decline to deploy. We emphasise that this is a post-hoc measurement of a known failure, not a pre-registered prediction of it; full sweep and caveats (bimodality at $n=3$, accuracy collapse to 0.20–0.48) in §4.4(E). (ii) **Mechanism preservation is not adaptive-attack robustness.** A criterion-aware adversary solving FoolsGold’s and RFA’s constraints jointly—decorrelated coordinate blocks to evade the similarity test, ϵ -ball projection to stay inside the geometric median’s radius—raises $\text{fg} \rightarrow \text{rfa}$ from 0.051 (seed-matched) to 0.167 within our evaluated composition-aware attack family (§6). Theorem 6 bounds what the preserved mechanism admits, not what an adversary reshaping its update to satisfy both constraints at once can do, and the $3.3\times$ increase is the size of that gap on our flagship pair.

Held-out validation (pre-registered): 22/24 = 92% (*Screening*: an evaluation ordering, and by Proposition 4 not an identification of the mechanism’s contribution). Table 4 reports the 24 pre-registered pairs. Combined across both waves: **38/42 = 90.5%**. The two held-out misses are both LOW-predicted pairs that were flagged as *speculative* in the pre-registration (neither is a clean $\text{C1} \wedge \text{C2} \wedge \text{C3}$ PASS):

- **fedavg $\rightarrow$ cm** (predicted LOW, actual 0.519): a degenerate pair reducing to CoordMedian alone, whose max

single-defense ASR (0.450 at $n=3$) sat just below threshold but rose to 0.519 at $n=5$ —a borderline crossing, not a mechanism failure.

- **fg→NC** (predicted LOW, actual 0.927): the one genuine miss. We speculated FoolsGold’s downweighting plus NormClip’s bound would combine; in fact NormClip adds no suppression (it only clips norms, which FoolsGold has already equalized) and FoolsGold’s partial downweighting alone is insufficient. This is a C1 failure we mis-scored as a hopeful LOW.

Critically, *every composition the tightened criterion certifies is correct* (2/2, both in the development set); the held-out misses are speculative LOW calls the criterion itself does not license, and fg→rfa—which works but no longer certifies—is reported above as a synergy false negative.

Table 4: Held-out pre-registered validation (24 pairs, $n=5$ seeds; max across committed_scaling and committed_pixel). All predictions committed to version control (commit 7e4d9b1) before experiments. Threshold: LOW < 0.5, HIGH ≥ 0.5. Only the two LOW predictions are shown individually; the 22 HIGH predictions (all correct) are summarized.

Pair	Category	Predicted	Max ASR	Correct?
<i>Speculative LOW predictions (not clean C1∧C2∧C3 PASS)</i>				
fedavg → coord_median	DEGEN	LOW	0.519	×
foolsgold → norm_clip	C1 (uncertain)	LOW	0.927	×
<i>HIGH predictions (22 pairs: 6 CM-as-d₁, 6 TM-as-d₁, 4 fedavg-as-d₁, 3 weighting-d₁, 3 RFA-as-d₁; range 0.70–0.94)</i>				22/22 ✓
Held-out: 22/24 = 92%. Combined with development set: 38/42 = 90.5%.				

Rep→tm: composition dominates despite C1-FAIL. Reputation→TrimmedMean is correctly classified as C1-FAIL (pixel ASR 0.679 > 0.5), yet it is the paper’s main composition result (§7). The resolution: the criterion predicts *max-committed ASR*, while the deployment-relevant metric is *oracle ASR* (0.316). The oracle cannot exploit the pixel gap because scaling—which reputation *does* suppress—dominates its utility. The criterion thus correctly identifies that rep→tm does not suppress every committed attack, while the game-theoretic analysis shows it is strategically dominant.

Tightening C1 costs us our own flagship—and reveals a synergy case. Under the per-attack quantifier in C1, **fg→rfa no longer certifies**: neither FoolsGold (0.732) nor RFA (0.845) suppresses the pixel backdoor individually, and FoolsGold’s 0.200 against model-scaling is hollow because it comes with 10% clean accuracy. The composition nonetheless reaches 0.093. This is genuine *synergy*—suppression neither constituent possesses—and Proposition 5 proves that *no* inherited-suppression screen can certify it, so this is a scope boundary rather than a tuning failure. We therefore report our lowest-ASR composition as a false negative of our own criterion. The certified in-distribution set is fg→cm (0.107) and rep→cm (0.423), both correct: precision 2/2, recall 2/5 = 40%.

4.1 A prospective test on a never-run defense menu

Everything above shares a structural weakness: the conditions were developed on the same 7-defense menu they are scored on, and every genuinely positive test is a PASS prediction. The decisive test is therefore a *prospective* one—labels frozen and committed before any run, on pairs no condition was ever tuned against. Our server implements three defenses the paper had never used: FLTrust (Cao et al., 2021), Krum and Multi-Krum (Blanchard et al., 2017). They are well chosen because the theory makes *opposite* predictions about them. FLTrust applies a per-client rescaling $c_i = \text{ReLU}(\cos(u_i, g_{\text{srv}})) \cdot \|g_{\text{srv}}\|/\|u_i\| \geq 0$ (verified in the implementation: all scales non-negative, cosine preserved to 2.4×10^{-7}). Note the hypothesis of Proposition 8 is $c_i > 0$, strictly: **FLTrust is positive-rescaling when its trust coefficient is strictly positive, and zero-trust clients require the separate zero-weight analysis** of Lemma 2. On the strictly-positive part, Proposition 8(a) predicts $\cdot \rightarrow \text{FLTrust}$ preserves C2; on the ReLU zeros, FLTrust changes the effective client set rather than only magnitudes, which is Lemma 2 and not covered by (a). The distinction is not cosmetic here: it is exactly what separates the five bit-identical $\cdot \rightarrow \text{FLTrust}$ pairs below from FG→FLTrust, the one d_1 that produces exact zeros and the one that genuinely deviates. Krum and Multi-Krum select on *pairwise Euclidean distance*—the non-invariant class of Proposition 8(c)—so $\cdot \rightarrow \text{Krum}$ is predicted to fail C2, while Krum → $\cdot$ is degenerate because a selection rule emits no per-client transform.

Protocol. We recorded a predicted label and a mechanism for all 20 pairs in `pre_registration_prospective.md` and git-committed it (262cf35) before `results/prospective_suite/` existed, then ran 20 pairs × 2 committed attacks × 3 seeds × 50 rounds in the standard configuration. Predictions were never revised. Single-defense

baselines for the three new defenses were measured *before* the freeze, since C0 and C1 take them as inputs. Crucially, the suite carries the sanity gate the DBA battery lacked: a predicted-FAIL control counts only if the attack clears ASR 0.5 against FedAvg, and both do here (0.667 model-scaling, 0.768 pixel). Unlike the out-of-sample battery of §4.4, these negatives are informative.

Result: 13/20, and the shortfall is in our experimental design. At the same 0.5 threshold: TP = 8, FP = 0, TN = 5, FN = 7 (Table 5). Every predicted-LOW pair came out LOW, 8/8—and that figure is far weaker than it looks, because **FLTrust suppresses both committed attacks on its own** (max-committed ASR 0.048 at 0.575 clean accuracy), the first defense in our menu that does. All 13 pairs containing FLTrust land in [0.041, 0.078], within 0.030 of FLTrust alone, in either slot. The two slots fail for different reasons, and the distinction matters:

The two slots fail differently, and the difference is the finding. (i) *FLTrust as d_1 (7 pairs, 2/7)*. d_1 alone already suppresses, so the composition inherits that suppression whatever d_2 does; all five predicted-HIGH misses have real C2/C3 failures, but destroying d_2 ’s mechanism is costless when d_2 was never load-bearing. This is C0, invisible on the original 42 pairs because no defense there suppresses alone (the minimum single-defense max-committed ASR is CoordMedian’s 0.519, and the lowest ASR among all 39 non-PASS pairs is 0.433), so we count these 7 pairs out of scope *retrospectively*—a weaker claim than a pre-registered exclusion. (ii) *FLTrust as d_2 (6 pairs, 6/6), but the prediction is nearly an identity*. FLTrust gates on the scale-invariant $\cos(u_i, g_{\text{srv}})$ and then renormalizes every update to $\|g_{\text{srv}}\|$, discarding magnitude entirely, so a positive per-client rescaling is *exactly* cancelled—four of the six pairs are bit-identical to FLTrust alone or differ only within seed noise. The single genuine deviation is FG→FLTrust (0.064), the one d_1 producing exact *zeros* rather than positive rescalings and hence a Lemma 2 interaction. This arm is a consistency check between theory and implementation, **not** a falsifiable test: every d_1 in our menu is a positive rescaling and Proposition 8(a) permits no other outcome. (iii) *Neither defense is FLTrust (7 pairs, 5/7)*—the only arm that could have surprised us. Five predicted-HIGH pairs are correct (0.870–0.943); both misses are diagnosable and neither is a mechanism error: fg→krum measures 0.078 at 9.9% clean accuracy (the collapsed-model artifact, not suppression), and krum→cm measures 0.450 against a 0.5 threshold as a degenerate pair reducing to CoordMedian alone, whose own baseline is 0.519. The arm contains *zero* predicted-LOW pairs, so it bears on specificity only. Per-pair numbers for all three arms are in Appendix I.

What the suite does and does not establish. Counting only in-scope pairs gives 11/13; the arm that carried real risk of falsification gives 5/7 with both misses explained. But we will not present this as the prospective positive result the framework needs, because **it does not contain one**: the 8 predicted-LOW pairs are 6 near-identities plus 2 out-of-scope inheritances. The three matched contrasts holding d_1 fixed while varying d_2 ’s invariance class—rep→fltrust 0.046 vs. rep→krum 0.880; nc→fltrust 0.046 vs. nc→mkrum 0.943; rfa→fltrust 0.047 vs. rfa→mkrum 0.875—show a 19–20× ASR gap turning on exactly the property Proposition 8 identifies, which is the most suggestive evidence here. They are still confounded: in this menu the invariant-signal d_2 is also the only d_2 that suppresses alone, so C1 and C2 co-vary. Lemma 1 explains why no other choice of defense would have helped—in scope, C0∧C1 *force* d_2 to suppress a^* alone, so the confounding is structural. Breaking it requires holding the suppression principle and standalone effectiveness fixed while varying only the invariance class, which is the design of §4.2.

4.2 Breaking the confound: a matched metric-swap ablation

Lemma 1 says mechanism–effectiveness confounding cannot be escaped by picking a different defense off a menu: in scope, C0∧C1 *force* the invariant-signal d_2 to be one that suppresses a^* alone. So we break the confound by construction instead, holding the suppression *principle* fixed and varying only the invariance class of the statistic d_2 reads.

Construction. Two downstream defenses in our menu score clients by a Euclidean quantity—Reputation weights by distance to the client median, Krum selects by summed pairwise distance—and for each we build a minimal-change mirror computing the same score from *directions* alone. `cos.reputation` normalizes each update first ($\hat{u}_i = u_i / \max(\|u_i\|, \epsilon)$), takes the coordinate-wise median of the $\hat{u}_i$ as consensus, and scores $d_i = 1 - \cos(u_i, \text{consensus})$ with bandwidth $\sigma = \text{median}_i(d_i)$ and weights $w_i \propto \exp(-d_i/\sigma)$; normalizing *first* is what makes it invariant, since Reputation’s reference is the *client* median, which itself moves when d_1 rescales. `cos.krum` replaces the pairwise Euclidean distance matrix with $1 - \cos(u_i, u_j)$, invariant in *both* arguments, so the score matrix and the selected index are unchanged under any positive per-client rescaling. Everything else is identical to the original: same soft-trim form, same adaptive median bandwidth, same arg min rule, no clipping, no memory across rounds.

These are mechanism-isolating ablations, not proposed defenses. They are the analogue of a knockout control: their only purpose is to vary one property while holding the rest fixed. We do not evaluate their standalone merit beyond

Table 5: Prospective suite: 20 pairs, labels frozen at commit 262cf35 before any run. Max-committed ASR over 2 attacks $\times$ 3 seeds; threshold 0.5. Categories are verbatim from the pre-registration, including its own uncertainty hedges.

Pair	Frozen category	Pred.	Max ASR	Acc	Hit
<i>In scope (C0 holds): d_2 load-bearing</i>					
Rep→FLTrust	certified	LOW	0.046	0.577	✓
NC→FLTrust	certified	LOW	0.046	0.576	✓
RFA→FLTrust	certified	LOW	0.047	0.574	✓
Krum→FLTrust	DEGEN	LOW	0.048	0.574	✓
MKrum→FLTrust	DEGEN	LOW	0.048	0.574	✓
FG→FLTrust	certified	LOW	0.064	0.547	✓
FG→Krum	C1-fail	HIGH	0.078	0.099	×
Krum→CM	DEGEN	HIGH	0.450	0.770	×
Krum→RFA	DEGEN	HIGH	0.870	0.780	✓
MKrum→RFA	DEGEN	HIGH	0.870	0.780	✓
RFA→MKrum	C1-fail	HIGH	0.875	0.782	✓
Rep→Krum	C1-fail	HIGH	0.880	0.530	✓
NC→MKrum	C1-fail	HIGH	0.943	0.761	✓
<i>Out of scope (C0 violated): FLTrust as d_1 suppresses alone at 0.048</i>					
FLTrust→CM	certified-uncertain	LOW	0.041	0.505	✓
FLTrust→MKrum	C2-fail	HIGH	0.045	0.576	×
FLTrust→TM	C3-fail-uncertain	HIGH	0.046	0.529	×
FLTrust→NC	C2-fail	HIGH	0.047	0.576	×
FLTrust→RFA	certified	LOW	0.048	0.512	✓
FLTrust→Rep	C2-fail	HIGH	0.048	0.518	×
FLTrust→Krum	C2-fail	HIGH	0.078	0.361	×

Full suite 13/20 (TP 8, FP 0, TN 5, FN 7); in scope 11/13; non-FLTrust arm 5/7.

the C1 input the design requires, and we do not recommend them.

Why this design is falsifiable where the FLTrust contrasts were not. Both mirrors fall under Proposition 8(a) in its general form—any statistic that is a function of the directions $\hat{u}_i$ alone is exactly invariant under $T(u_i) = c_i u_i$, $c_i > 0$ —while their Euclidean originals fall under (c). Taking $d_1 \in \{\text{NormClip}, \text{RFA}\}$, both heterogeneous positive rescalings and neither suppressing model-scaling alone (0.935, 0.885, so C0 holds and the composition cannot inherit suppression from d_1), each control/test pair at fixed d_1 was intended to differ in exactly one property: whether C2 holds. That requires the C1 input to be matched across the pair, so we measured it first, before freezing any label. In §4.1 the two changes were entangled—“different invariance class” and “different defense” were the same edit, and FLTrust suppressed both committed attacks alone, so a confirmation could not have come out otherwise. Here neither d_2 is a universal suppressor, so no outcome can be inherited from d_1 or from d_2 alone, and if the invariant arm also lands HIGH then C2’s invariance mechanism is refuted with no appeal to a different defense available. Both committed attacks are run: on the pixel backdoor C1 fails for every pair, so all pixel cells are predicted HIGH—a negative control the framework must get right—and the sanity gate holds (FedAvg reaches 0.667 on scaling, 0.768 on pixel), so predicted-HIGH cells are informative rather than vacuous as in the DBA battery.

The design’s premise was refuted by its own baselines. Matching failed in both arms, for two independent reasons, and because the C1 inputs are measured before the freeze we learned this before the suite ran—the only reason it can be reported as a finding rather than a post-hoc excuse. `cos_reputation` does not merely weaken relative to its Euclidean original, it collapses, from Reputation’s 0.017 on model-scaling to **0.982**. `cos_krum` reaches 0.078 on model-scaling, which passes C1’s numeric threshold, but at **0.146** clean accuracy—the hollow-suppression artifact already documented for FoolsGold (0.200 at 0.10) and `fg→krum` (0.078 at 0.099)—so C1 fails under the accuracy gate we apply everywhere else. The mechanism is visible in the baselines rather than asserted: undefended FedAvg under model-scaling is itself at 0.100 accuracy, so Krum’s 0.577 is *restoring* utility the attack destroyed and `cos_krum`’s 0.146 fails to. **Magnitude information is what lets this defense family handle a magnitude attack:** a direction-only mirror of a Euclidean scorer cannot see a scaling attack at all, so removing the magnitude dependence removes the suppression with it—exactly the coupling the pre-registered contingency anticipated, stated there as the possibility

that “invariance and standalone effectiveness may be inherently coupled” for this family. On the magnitude-neutral pixel backdoor the mirror does survive (0.300 at 0.598), but there its control Krum fails C1 (0.583), so those arms are unmatched in the opposite direction and the pixel cells cannot carry the test either.

The consequence is that **the confound re-materialized inside an experiment built specifically to break it**, which we report as the round’s methodological finding rather than as a failed run. Lemma 1 establishes that mechanism–effectiveness confounding is structural in scope; this is what trying to escape it looks like when the mechanism being varied is the mechanism the defense relies on. Two responses were available and we took neither. We did not switch to a pixel-only protocol, which would have made `cos_krum` pass C1 and manufactured two predicted-LOW cells—choosing the attack after seeing the C1 inputs is attack-shopping regardless of the resulting labels. We did not adjust the accuracy gate to admit 0.146, which would have produced a predicted-LOW pair by weakening the very criterion that catches hollow suppression elsewhere. The pre-registration’s second non-negotiable—“no switching of committed attack, threshold, seed count, or accuracy gate”—forecloses both, and either would have bought a positive test at the cost of the property that makes the test worth running.

The mechanism is verified in code before it is claimed. We check each d_2 ’s internal state on $\{u_i\}$ against $\{c_i u_i\}$ for the c_i NormClip and RFA actually produce on live rounds, rather than inferring invariance from an ASR. Over 18 rounds `cos_reputation`’s weight *ordering* never changes (0/18, weights agreeing to $\leq 2.5 \times 10^{-7}$) and `cos_krum` selects the same client in 18/18, while their Euclidean originals change in 12/18 and 6/18; the disturbance tracks the realized ρ exactly as Proposition 8 predicts. Full per-upstream rates, the 1000-draw synthetic check, and the agreement of the mirrors with the shipped implementations are in Appendix J.

One asymmetry belongs in the statement of the prediction. `cos_reputation`’s *weight vector* is exactly invariant, but its aggregate is not, because the weighted average is taken over the rescaled updates $c_i u_i$. That residual is a magnitude contraction of the Lemma 2 kind, so the prediction is two-sided: ASR should stay low while clean accuracy pays a contraction cost. `cos_krum` has no such residual—it emits one selected update, and the selection is bit-identical.

Outcome: 8/8 correct, and zero positive tests. Because C1 binds for all eight pairs—on model-scaling through the two baseline failures above, on the pixel backdoor by design—every frozen label was HIGH, so the suite bears on specificity only, as a consequence of inputs measured before the freeze rather than a choice made afterwards. We ran all 8 pairs $\times$ 2 attacks $\times$ 3 seeds anyway, because a suite whose labels are all one class is still falsifiable in the direction that matters: a single LOW cell would have been a false positive against a criterion claiming 100% precision. All eight came out HIGH—8/8 correct, TP = 0, FP = 0, TN = 8, FN = 0, max-committed ASR spanning 0.508–0.935—with labels frozen in `pre_registration_metric_swap.md` at commit 356897b, timestamped before `results/metric_swap/` existed and reproduced verbatim in the runner’s PAIRS list so scoring cannot drift. We state the limitation in the same breath as the result: the suite contains **no falsifiable prospective positive test**, and 8/8 is not validation of the framework’s predictive claim—eight correct negatives on pairs our own conditions could only have called HIGH is weak evidence, and it supports a claim about specificity, not prediction.

One prediction in the suite was genuinely two-sided, and it is the round’s only prospective positive result. `cos_krum`’s C1 verdict on model-scaling turns entirely on whether the accuracy gate is applied (0.078 passes the numeric threshold; 0.146 accuracy fails the gate), so rather than choose a reading afterwards we froze *both*, with the resolution rule fixed in advance: accuracy-gated $\Rightarrow$ HIGH, numeric-only $\Rightarrow$ LOW, and a LOW outcome at accuracy ≥ 0.35 would have recorded our two primary labels as misses. Both pairs came out **HIGH** (NormClip $\rightarrow$ `cos_krum` 0.508, RFA $\rightarrow$ `cos_krum` 0.566), refuting the secondary reading and vindicating the accuracy gate against a pre-registered alternative specification of itself on data neither of us had seen. The weaker half is flagged in Appendix J: the NormClip arm sits barely above threshold and it is the RFA arm that carries the conclusion.

A post-hoc contrast that is not confounded. The cross-pair scoring above is confounded in exactly the way Lemma 1 predicts, but the frozen suite happens to contain a comparison that is not, and it is the strongest C2 evidence in either paper. Compare a composition against *the same* d_2 ’s own standalone ASR on the same attack, under two different upstream transforms. C1 is then identical by construction—same downstream defense, same attack, same measured standalone effectiveness on both sides—so the residual $\Delta = \text{ASR}(d_1 \rightarrow d_2, a) - \text{ASR}(d_2, a)$ cannot be attributed to d_2 being a stronger or weaker defense. What does vary is how much the upstream transform actually disturbs d_2 ’s statistic, and that quantity was measured on the mechanism itself (§4.2, `results/cos_invariance_check.json`) rather than inferred from any ASR. The frozen label scoring is fixed at $n=3$ and must not move, so we topped the six powered cells up to $n=5$ in a separate directory, with the tested direction fixed in the top-up script before the new seeds ran (`results/metric_swap_topup/`). For Krum under model-scaling (standalone 0.061 at 0.577 accuracy): NormClip changes Krum’s selection in 0/9 rounds and the composition reaches 0.062 ($\Delta = +0.001$, per-seed 0.120/0.018/0.047/0.058/0.068); RFA changes it in 6/9 rounds and the composition reaches 0.779 ($\Delta = +0.718$,

per-seed 0.974/0.116/0.982/0.904/0.916)—a gap of +0.716 between two compositions sharing a downstream defense, an attack, and a baseline, and at $n=5$ it is statistically resolved (Welch $t = 4.28$, two-sided $p = 0.012$; paired $t = 4.60$, $p = 0.010$; Mann–Whitney one-sided $p = 0.008$), which it was not at $n=3$.

The pattern holds across the suite once cells without power are excluded— Δ is observable only where standalone d_2 genuinely suppresses, since a baseline already above 0.5 shows $\Delta \approx 0$ by ceiling effect whatever the transform did. Restricting to standalone ASR < 0.5 at accuracy ≥ 0.35 leaves six cells, and at $n=5$ measured disturbance still splits them perfectly: the three undisturbed cells (0/9) give $\Delta \in [+0.001, +0.190]$, the three disturbed ones ($> 0/9$) give $\Delta \in [+0.718, +0.878]$ —a separation of +0.527 with no overlap. Two of the three undisturbed cells are `cos_krum` on the pixel backdoor, where the mirror suppresses standalone (0.300 at 0.598 accuracy) and carries that suppression through under *both* upstreams. Here the two extra seeds *weaken* the confirming half rather than strengthen it, and we report it as measured: Δ moves from +0.000 at $n=3$ to +0.190 under NormClip and from -0.027 to +0.147 under RFA, leaving both compositions still below threshold (0.490 and 0.447 at 0.61 accuracy) but no longer intact. Exact invariance of the *selection* therefore does not by itself carry suppression through unchanged: the magnitude channel of Lemma 2 is a second route by which Δ can grow, and the ladder’s arm 3 (§5) is built to measure exactly that prospectively rather than describe it post-hoc. The ten excluded cells are listed rather than dropped silently in Appendix J.

The suite contains a second such contrast, on Reputation under model-scaling (standalone 0.017 at 0.777 accuracy), and the two are informative in opposite ways. NormClip disturbs Reputation’s weight ordering in 3/9 rounds and the composition reaches 0.806; RFA disturbs it in 9/9 and the composition reaches 0.895. The gap is only +0.089—both arms are far above threshold, because Reputation’s ordering is disturbed under *both* upstreams—but within-arm variance is small, all five seeds move in the predicted direction (per-seed deltas +0.149/ +0.089/ +0.030/ +0.061/ +0.115), and it is resolved at both $n=3$ and $n=5$ (Welch $t = 3.06$, two-sided $p = 0.018$; paired $t = 4.31$, $p = 0.013$). So the two contrasts trade off exactly as the disturbance measurements imply: Krum has the large effect because one of its upstreams leaves it untouched, and Reputation was resolvable even at $n=3$ because neither upstream does.

This is post-hoc, and we do not present it otherwise. It was not pre-registered, it re-uses a suite frozen for a different purpose, the two extra seeds per cell were added *after* seeing the $n=3$ result, and the six-cell split is a description of that suite rather than a test on new data. The top-up cut in both directions, which is why we report all of it: it resolved the Krum gap (Welch $p = 0.16$ at $n=3$, where per-seed paired deltas +0.854/ +0.097/ +0.935 left one seed not separating, to $p = 0.012$ at $n=5$) and at the same time degraded the invariant control’s Δ from ≈ 0 to +0.19. What the analysis does have, which neither the FLTrust contrasts of §4.1 nor the cross-pair scoring has, is C1 held exactly fixed and a disturbance measurement taken on the statistic rather than read back from the outcome. We offer it as the cleanest available piece of C2 evidence and as the shape a properly prospective version of this test would take: fix a downstream defense that suppresses the attack, then vary the upstream along measured disturbance rather than along defense identity. Table 6 sets out, claim by claim, what this design can and cannot conclude.

4.3 An application: screening as evaluation prioritization

What follows is a consequence of the identification boundary, not a claim the paper argues for. Proposition 4 says a criterion gated on the outcome it predicts cannot be validated by cross-defense contrast; what such a criterion can still do is *order an evaluation budget*, and that is the only use we claim for it. Proposition 5 bounds what it can find. Read the numbers below as a measurement of a prioritization heuristic’s cost, not as validation of the framework.

Does every condition earn its place? C3 does not—reported as a finding. Scoring all 42 pairs on C1, C2 and C3 *independently* rather than recording only the first failure: against a 12% base rate of low-ASR pairs, C1 alone selects 4 at 50% precision, C2 alone 14 at 36%, and C1 $\wedge$ C2 selects 2 at 100%. **C3 has no demonstrated incremental predictive value in this study**—alone it selects 35 pairs at 9% precision, below the base rate, and it never binds on a pair C1 $\wedge$ C2 already admits. So **we demote C3 from a gate to a margin diagnostic and recommend C1 $\wedge$ C2 as the criterion**, retaining C3 on theoretical grounds only: Theorem 6 shows margin compression is a real failure mode, and the $\alpha=0.1$ regime below is precisely a margin collapse. **The demotion changes no number**—C1 $\wedge$ C2 selects the same two pairs at the same precision (recomputed by `analyze_screening_cost.py`)—which is why we report it as a finding about the criterion rather than a revision of results. *The pre-registered PASS/FAIL labels and the 16/18 and 22/24 tallies were frozen with C3 in the gate and are not relabelled*; this ablation is post-hoc and labelled as such, and the recommendation is for use going forward.

C2 vs. C3 failure modes. The distinction remains operationally meaningful: the C2-FAIL pairs land at 0.798–0.882 (signal destroyed), while C3-FAIL pairs range over 0.43–0.85 (partial suppression often remains). C2 failures

Table 6: **What this design can and cannot conclude.** The middle column is about *identifiability*, not statistical power: a “no” marked *structural* cannot be repaired by more seeds, defenses or attacks (Proposition 4). Every entry is scoped to the measured setting of §4–§5.2.

question	identifiable?	answer, and its evidence
Does disturbing d_2 ’s statistic change its <i>decision</i> ?	measured	Yes: Krum’s selection flips in up to 0.80 of rounds
Does it change the <i>admitted adversarial mass</i> ?	by construction	No: exactly 0.000 at every rung
Does it change <i>suppression</i> ?	instrumented	No: $ \Delta =0.026$, inside the frozen margin (Prop. 9(a))
Does <i>adversary-correlated</i> reweighting change suppression?	instrumented	Yes , non-monotonically: single-peaked (Mode A)
Does admitted mass <i>order</i> outcomes across defenses?	pre-registered	No—refuted: +0.098 against a frozen +0.150
Is (P4) <i>sufficient</i> for (P5)?	control arm	No: <code>cos_krum</code> admits 0.000 while ASR falls
Can (P4) be screened <i>without</i> adversary identity?	no—open	Both instruments are oracles: the stated open problem
Does C1 identify C2’s contribution?	no—structural	Empty cell by the gate, for every menu (Prop. 4)
Does any of this certify <i>deployed</i> security?	no—scope	Committed attack only; one adaptive test breaks it

are irrecoverable; C3 failures admit rescue via threshold adjustment or a rank-robust d_2 . After the Proposition 8 reclassification the C2 category is also more coherent than before: *every* surviving C2 failure is a $\cdot \rightarrow$ reputation pair ($\text{nc} \rightarrow \text{rep}$ 0.819, $\text{rfa} \rightarrow \text{rep}$ 0.882, $\text{fg} \rightarrow \text{rep}$ 0.798), which is exactly the non-invariant signal class, while all three non-degenerate $\cdot \rightarrow$ FoolsGold pairs are C1 failures. The taxonomy tightened rather than loosened when the error was corrected.

Order sensitivity. Composition is order-sensitive: the criterion predicts different failure modes for $(d_1 \rightarrow d_2)$ vs. $(d_2 \rightarrow d_1)$, and empirical outcomes confirm asymmetry. Two examples from our data:

- **rep $\rightarrow$ NC** (max ASR 0.813, C1-FAIL): reputation cannot suppress pixel (baseline 0.842); NC adds nothing for pixel since it only clips norms. vs. **NC $\rightarrow$ rep** (max ASR 0.819, C2-FAIL): NC equalizes all norms, destroying reputation’s consensus-distance signal. Both fail, but through mechanistically distinct pathways predicted by the criterion.
- **fg $\rightarrow$ rfa** (max ASR 0.093 at $n=30$, PASS): FG downweights adversarial clients while preserving RFA’s geometric-median signal. The reverse, **rfa $\rightarrow$ fg** (max ASR 0.914), fails for a different reason than one might expect: RFA-reweighting is a positive per-client rescaling, so by Proposition 8(a) it leaves FoolsGold’s similarities *exactly* intact and C2 holds. The pair fails C1—neither RFA alone (0.885) nor FoolsGold alone (0.732) suppresses. Order still matters, but here because of what each defense can do rather than what the upstream transform destroys.

The criterion’s C2/C3 conditions are inherently asymmetric: they ask whether d_1 ’s output preserves d_2 ’s admission signal, not vice versa. The asymmetry is deliberate, and it is what lets the two orderings of a pair be *ranked for evaluation* before either is run. Which ordering is in fact better is settled by evaluating it; the ranking predicts nothing about that.

Continuous metric analysis. To test whether the binary PASS/FAIL threshold is fragile we compute threshold sensitivity and rank correlation across all 42 matched pairs (each pair’s max-committed ASR against its pre-registered category). *Threshold sensitivity* is scored against the *pre-registered* labels, so the two speculative-LOW held-out misses count as misses rather than being re-coded: accuracy is 92.9% at 0.3, 95.2% at 0.4, **90.5%** at 0.5 (identical to the headline 38/42), 85.7% at 0.6 and 83.3% at 0.7—the 0.5 operating point is not cherry-picked, varying by under 5 points across $[0.3, 0.5]$. *Rank correlation* on the mechanism ordinals (PASS=1, C3-FAIL=2, C2-FAIL=3, C1-FAIL=4, DEGEN=5) gives Spearman $\rho = 0.38$ ($p = 0.014$, $n = 42$), but *the signal is PASS-vs-rest, not a graded severity ranking*: PASS pairs (mean ASR 0.176 ± 0.129) separate cleanly from every failure mode (rest mean 0.787, range $[0.43, 0.94]$), while the taxonomy does *not* order magnitude (C3-FAIL 0.66, C2-FAIL 0.86, C1-FAIL 0.81, DEGEN 0.81: non-monotone in the ordinal). We therefore claim only the effective-versus-ineffective cut, which is the practitioner-relevant one. (Correlations pool ASR at $n=3$ –30 seeds; the PASS-vs-rest gap dwarfs that heteroscedastic noise.)

Blind composition-selection comparison (Screening). This compares the criterion’s ordering against uniform sampling *on this menu*, and it is outcome-gated like the rest of the tier: by Proposition 4 it cannot establish that the conditions are what did the ordering. Against a blind baseline (uniformly sample $k=3$ pairs from all 42), criterion-guided

selection (the 3 PASS pairs, mean ASR 0.176) beats **99.99%** of random selections—only 0.01% of uniformly sampled 3-pair subsets achieved comparable or lower mean ASR (finite-population resampling test, 10^5 draws; $p < 10^{-4}$). Random selection yields mean ASR 0.744 (95% CI [0.465, 0.898]); only 0.1% of random selections have all k pairs below ASR 0.5. At the 3-way level (10 measured of 210 possible triples): the 6 criterion-selected triples (mean ASR 0.078) beat 99.7% of random 6-of-10 draws ($p = 0.003$). On this menu, then, the ordering is worth more than uniform sampling for *spending an evaluation budget*—which is the only use claimed. It does not identify the best compositions: recall is 40%, and the lowest-ASR composition in the paper lies outside the admitted set by construction. Nor is the comparison a validation, for the reason stated at the head of this subsection.

What the saving is, counted in runs rather than pairs. A pair count overstates it, because C1 is itself an empirical gate and its standalone evaluations are runs too—so the rule eliminates nothing “before any run,” and we previously described it that way in error. What it eliminates is the quadratic term. Counted per run (`analyze_screening_cost.py`; 7 defenses, 42 ordered pairs, 2 committed attacks, 5 seeds per cell): evaluating every ordered pair costs 420 **runs**, while screening costs 80—60 standalone (the C1 input, at $(D-1) \times |A| \times S$, since fedavg-as-upstream is the standalone protocol) plus 20 on the two admitted pairs. That is 340 **runs**, or **81% fewer composition-evaluation runs**, not the 95.2% of *pairs* we previously reported.¹ The structural claim is about scaling: from $|A| S D(D-1)$, *quadratic* in the defense count, to $|A| S ((D-1) + |\text{admitted}|)$, linear plus the admitted set. Two caveats belong with the numbers. **No per-run timing telemetry was ever recorded**, so the corresponding $\approx 77 \rightarrow \approx 15$ GPU-hour figures are *estimates* at ≈ 11 min per 50-round run on the hardware described above; and the standalone baselines have independent value—they are the single-defense results we report anyway—so charging them wholly to screening is the conservative choice. The same standalone cells are reused by every pair and every depth, and C2 and C3 add no runs at all, being read off the aggregation rule. Against the $5/42 = 11.9\%$ base rate the admitted set is 2/2 low-ASR, an $8.4\times$ enrichment—**high precision in a small, badly imbalanced evaluation** (2 positive predictions), which we do not offer as validation; the cost is recall $2/5 = 40\%$, discarding three pairs that work, including the lowest-ASR composition in the paper (`fg→rfa`, 0.093). The two figures imply different uses: screen if you will deploy *one* composition, since every admitted pair worked; do not, if you are searching for the *best* one, since the synergy cases lie outside the admitted set by construction (L5). The saving grows with depth—criterion-guided selection reduced $P(7, 3) = 210$ ordered triples to the 10 we pre-registered and ran (§4.5, 9/10 correct), which we report as the reduction realized rather than a filter rate, since we did not score all 210.

4.4 Out-of-Sample PASS Validation

The development and held-out validation above tests predictions from the *same* attack families (`model_scaling`, `backdoor_pixel`) and scale ($N=10$, `cifar_cnn`) used to design the criterion. To confirm that PASS predictions generalize we pre-registered three out-of-sample tests (commit `ca627e5`, before any of them ran); all held, and the accounting is in the summary below. We state plainly what this is and is not: it is evidence that the conditions keep identifying compositions which survive a change of attack family, scale and architecture; it is *not* an out-of-sample classification result, because every prediction in it is positive and the matching FAIL controls proved uninformative. Negative-direction evidence rests entirely on the in-distribution tiers.

(A)–(C) Attack family, scale, and independent selection. All three are pre-registered and confirmed; per-cell values, seeds and controls are in Appendix C. (A) *DBA transfer*. The Distributed Backdoor Attack (Xie et al., 2020)—spatially distributed sub-triggers at four image corners, a categorically different embedding mechanism, absent from the development menu—is suppressed by all three PASS-predicted compositions at $f=0.4$ (`fg→rfa` 0.109 ± 0.065 , `fg→cm` 0.050 ± 0.027 , `rep→cm` 0.041 ± 0.014 ; $n=5$) and again in a powered re-test at $f=0.6$ (0.125, 0.045, 0.043). The three FAIL controls are uninformative at both fractions because DBA’s own sanity baseline never clears the threshold (0.236 ± 0.190 at $f=0.4$, 0.386 ± 0.192 at $f=0.6$): DBA does not reliably embed at this N/K regardless of defense, so FAIL-side falsifiability here awaits a variant that does—though the tails do separate, max per-seed ASR 0.185 across PASS pairs against 0.712 for FAIL pairs. (B) *Scale*. At $N=100$, $K=10$, $f=0.05$ with ResNet18—a $10\times$ client pool and $45\times$ more parameters—`fg→rfa` (0.069 ± 0.029) and `fg→cm` (0.152 ± 0.118) both confirm PASS at $n=5$, against RFA alone at 0.410 ± 0.386 ; at ≈ 0.29 clean accuracy these runs establish that *suppression* transfers across scale and architecture, not that the composition is deployable at $N=100$ as configured. (C) *Independent survivor*. `fg→cm`, selected *solely* by criterion PASS status rather than by the 189-config payoff-matrix scan that chose `rep+tm`,

¹The counterfactual above fixes the budget at 5 seeds per cell for comparability. What we actually spent on composition cells is 356 runs over all 84 cells, because seed counts vary by tier: 14 screened-out development pairs at 3 seeds, 2 at 5, all 24 held-out pairs at 5, and the 2 admitted pairs at 3, giving $14 \times 2 \times 3 + 2 \times 2 \times 5 + 24 \times 2 \times 5 + 2 \times 2 \times 3 = 356$. We state the mix rather than quoting a round number.

dominates temporal mixing of the same two defenses on committed_pixel— 0.099 ± 0.020 versus 0.679 ± 0.159 , a $6.9\times$ suppression—with both conditions retaining a usable model (0.66 and 0.77 clean accuracy).

(D) Cross-distribution transfer summary. The criterion was developed on a single configuration: CIFAR-10, CifarCNN, $N=10$, $K=5$, $f=0.2$, model_scaling/pixel attacks. The tests above validate transfer across four independent axes:

1. **Architecture:** CifarCNN $\rightarrow$ ResNet18 ($45\times$ more parameters, GroupNorm, residual connections). PASS predictions confirmed: fg $\rightarrow$ rfa 0.069, fg $\rightarrow$ cm 0.152.
2. **Scale:** $N=10 \rightarrow N=100$, $K=5 \rightarrow K=10$. Same PASS compositions suppress at $10\times$ client pools.
3. **Adversarial fraction:** $f=0.2 \rightarrow f=0.05$ (scale test) and $f=0.2 \rightarrow f \in \{0.4, 0.6\}$ (DBA tests). PASS predictions hold across a $12\times$ range of f .
4. **Attack family:** model_scaling/pixel $\rightarrow$ DBA (distributed sub-triggers, categorically different embedding mechanism). PASS predictions confirmed at both f values.
5. **Heterogeneity:** $\alpha=0.5 \rightarrow \alpha \in \{0.1, 1.0, 10.0\}$. This axis *partially fails*, and is the one place we find a clear boundary—see (E).

(E) Heterogeneity: where the criterion breaks. Earlier heterogeneity sweeps in this work covered *single defenses* only; no composition had been run at any $\alpha \neq 0.5$. We ran all three PASS pairs at $\alpha \in \{0.1, 1.0, 10.0\}$ (3 seeds, both committed attacks, 50 rounds). Reporting max-committed ASR / clean accuracy:

Composition	$\alpha=0.1$	$\alpha=0.5$	$\alpha=1.0$	$\alpha=10.0$
fg $\rightarrow$ rfa	0.569 / 0.20	0.093 / 0.50	0.051 / 0.57	0.063 / 0.62
fg $\rightarrow$ cm	0.433 / 0.40	0.107 / 0.68	0.133 / 0.74	0.129 / 0.78
rep $\rightarrow$ cm	0.548 / 0.48	0.423 / 0.76	0.377 / 0.79	0.326 / 0.81

The criterion transfers cleanly across $\alpha \in \{0.5, 1.0, 10.0\}$, including the near-IID regime where FoolsGold might plausibly founder on legitimately similar benign updates—it does not, and $\alpha=10$ is indistinguishable from $\alpha=0.5$. **Under extreme heterogeneity ($\alpha=0.1$), two of the three PASS compositions cross the threshold** (fg $\rightarrow$ rfa 0.569, rep $\rightarrow$ cm 0.548); only fg $\rightarrow$ cm survives, at 0.433. Two caveats cut in opposite directions: fg $\rightarrow$ rfa at $\alpha=0.1$ is strongly bimodal across seeds (0.065–0.923), so at $n=3$ the direction is clear but the magnitude is not; and clean accuracy collapses to 0.20–0.48, so this regime stresses utility as much as security. Crucially Theorem 6 supplies a **diagnostic quantity that retrospectively explains this boundary** rather than predicting it: re-measuring on the transformed points at $\alpha=0.1$, RFA’s separation margin $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ collapses from an all-round mean of 3.24 (met in 68% of rounds) to 1.01 (met in 20%). Over the payload-carrying rounds alone—the ones Figure 2 plots—the same collapse reads 4.71 ± 1.51 ($n=17$) to 2.06 ± 2.35 ($n=12$), with the per-round minimum falling $2.82 \rightarrow 0.00$; the all-round means are lower because they average in the Lemma 2 rounds, where FoolsGold zeroes every adversary and m is recorded as 0. Either way the collapse is measured, and the all-round figure is the one that reaches the vacuous limit. This sits exactly at the $m \rightarrow 1$ boundary where Corollary 1’s bound degrades to 40% adversarial mass—extreme heterogeneity inflates the benign radius R'_B until the benign cluster is no longer separated from the adversarial points, which is precisely the precondition the theorem requires. A practitioner running the 5–10 round pilot would measure $m \approx 1$ and decline to deploy, so the criterion is mechanistically diagnostic at this boundary, though we emphasise this is a post-hoc measurement of a known failure, not a pre-registered prediction of it.

The one axis not validated with composition experiments is *dataset* (CIFAR-100, with 100-class output): the conditions are dataset-agnostic in mechanism, but the downstream aggregator’s decision boundary may shift with the number of classes (L3).

Summary. All 7 pre-registered out-of-sample PASS predictions held (3 under DBA transfer, at both $f=0.4$ and $f=0.6$; 4 under $N=100$ /ResNet18 scale transfer), plus composition dominance confirmed independently ($6\times$). Under the tightened per-attack C1, however, 3 of the 7 concern fg $\rightarrow$ rfa, which the criterion no longer licenses; counting only predictions it still certifies leaves **4/4 correct**, with the fg $\rightarrow$ rfa successes reported instead as evidence for the synergy case. FAIL discrimination is unavailable under DBA and at low f for the power reasons above (L3). Combined with the 2 certified in-distribution pairs, **all 6 certified PASS predictions are correct**, across three attack families, two architectures, two scales and $f \in [0.05, 0.6]$.

4.5 3-Way Compositions: Criterion Scales Where Exhaustive Search Cannot

With a 7-defense menu, exhaustive 2-way search covers $7 \times 6 = 42$ ordered pairs (the paper ran them). The same menu yields $7 \times 6 \times 5 = 210$ ordered triples, making exhaustive evaluation infeasible. This is the regime where a predictive criterion earns its keep: it filters 210 candidates to a tractable set without running them.

We extend the $C2 \wedge C3$ conditions to triples ($d1 \rightarrow d2 \rightarrow d3$): $C2'$ requires that $d1$'s transform preserves $d2$'s admission signal *and* the combined $d1 \rightarrow d2$ output preserves $d3$'s signal; $C3'$ requires adversarial–benign separation to survive both transforms before $d3$ aggregates. We **pre-registered** 10 ordered triples (pre-registration committed before experiments): 6 predicted LOW (criterion-selected) and 4 predicted HIGH (expected FAIL). Results ($N=10$, $K=5$, $f=0.2$, cifar_cnn, $n=5$ seeds each attack) are in Table 7:

Table 7: 3-way composition criterion validation (pre-registered). Max ASR = $\max(\text{mean_scaling}, \text{mean_pixel})$ across seeds.

Triple ($d1 \rightarrow d2 \rightarrow d3$)	Predicted	Max ASR	Verdict
fg→rep→rfa	LOW	0.048	CORRECT
fg→rep→cm	LOW	0.087	CORRECT
rep→fg→rfa	LOW	0.045	CORRECT
rep→fg→cm	LOW	0.136	CORRECT
fg→nc→cm	LOW	0.109	CORRECT
fg→rfa→cm	LOW	0.041	CORRECT
rfa→nc→rep	HIGH	0.894	CORRECT
fedavg→nc→rep	HIGH	0.842	CORRECT
nc→rep→tm	HIGH	0.636	CORRECT
nc→fg→tm	HIGH	0.267	MISS
Overall accuracy		9/10 = 90%	

9/10 (90%) accuracy, matching the 2-way combined accuracy (90.5%). All 6 LOW predictions are correct (max ASR 0.041–0.136, all $\ll 0.5$); 3/4 HIGH predictions are correct. The single miss (nc→fg→tm) was the pre-registration's most uncertain HIGH call: NC clips to equalize norms before FG, which may partially restore FG's cosine signal rather than destroying it, making the chain's $C3$ failure non-obvious. Its observed max ASR (0.267) is intermediate—suppressed relative to naive fedavg but not a clean PASS—consistent with a borderline case the criterion did not license confidently.

The key finding: This is a *conditional search*: the criterion filters from the ~ 30 triples whose d_1 satisfies $C1$ (the $C1$ -satisfying subset of 210 ordered triples), correctly identifying the 6 that pass $C2' \wedge C3'$ —without running them. Criterion-guided selection achieves 90% accuracy, identical to the accuracy achieved with exhaustive 2-way search. The chained signal-preservation conditions ($C2' \wedge C3'$) compose correctly: FG→rep→rfa (max ASR 0.048) and fg→rfa→cm (max ASR 0.041) are as secure as the best 2-way pairs, demonstrating that Theorem 6's bounded-reweighting guarantee chains: two bounded reweightings before a geometric or rank aggregator preserve the downstream signal as a bounded composition.

Selection-bias note. All 6 LOW triples use FoolsGold or reputation as d_1 —the same successful weighting defenses from the 2-way analysis. This reflects the structural fact that these are the only defenses satisfying $C1$ for both attacks in our menu. The criterion's 3-way value is not discovering new d_1 choices but correctly *filtering* from the $5 \times 6 = 30$ triples starting with FG/rep: it predicts LOW for only 6–8 of these 30 (those where the $d_2 \rightarrow d_3$ chain preserves signals), correctly rejecting the majority. A random 6-of-10 draw from our measured set matches the criterion's mean ASR (0.078) only 0.3% of the time ($p = 0.003$, resampling test; §4).

5 Statistic preservation is not security preservation: the confounded ladder, and the instrument it motivated

This section's destination is the targeted intervention of §5.2; the ladder that precedes it is here as the design whose confound motivated that instrument, not as a co-equal result. Read in that order, the section is the paper's

empirical centerpiece: a statistic-level hypothesis we froze, a pre-registered ladder that rejected it with the *opposite* sign, the diagnosis of why, and the instrument built to settle the question the ladder could not.

The two pre-registered suites above tried to validate C2 by *choosing a different downstream defense*, and both failed for the same structural reason. Lemma 1 says why, and—read as a design constraint rather than as an obstacle—it also says what to do instead. The proposition constrains **cross-defense** contrasts only: it is silent on holding d_2 and the attack fixed and varying only the *dose* of upstream disturbance. Along such a ladder the C1 input— $\text{ASR}(d_2, a)$ —is numerically the same quantity at every rung, so mechanism–effectiveness confounding is **impossible by construction** rather than matched away. This is the within-defense contrast of §4.2 made prospective and continuous, and it is the design the general statement of §3.1 licenses.

The controlled upstream transform. We introduce a synthetic d_1 whose only parameter is the *dispersion* of a positive per-client rescaling. For the K participants of a round, at position j in a (seed, round)-keyed pseudorandom permutation,

$$c_j \propto \exp\left(\kappa \cdot \left(\frac{2j}{K-1} - 1\right)\right), \quad \text{normalized so } \frac{1}{K} \sum_j c_j = 1. \quad (2)$$

Design in brief; the full pre-registered protocol is Appendix B. Four properties make the transform a dial on the theory rather than an arbitrary perturbation: it is exactly Proposition 8’s form $T(u_i) = c_i u_i$ with $c_i > 0$, so every rung lies inside the theory’s scope; **the dose is ρ** , since $\max_j c_j / \min_j c_j = e^{2\kappa}$ exactly—Theorem 6’s own quantity, verified by reading ρ back off the *transformed update norms* on live rounds; $\kappa=0$ gives $c \equiv 1$ and the implementation short-circuits, so the identity rung is bit-identical to d_2 standalone by construction rather than by rounding; and normalizing the *mean* to 1 holds the aggregate’s scale fixed, separating the quantity under test (disturbance of d_2 ’s *statistic*, i.e. C2) from Lemma 2’s magnitude-contraction channel. The dose is assigned by a (seed, round)-keyed permutation and **not** by client identity, which would park the adversaries at a fixed end of the ladder and manufacture a success; every arm at a given seed and round therefore receives the *identical* coefficient vector. `dose_kappa( $\kappa$ )` is an instrument, never evaluated or recommended as a defense. The frozen grid is $\kappa \in \{0, 0.5, 1.0, 2.0\}$, i.e. $\rho \in \{1.00, 2.72, 7.39, 54.60\}$, and the four arms are exactly those the paper’s own power rule admits (standalone $\text{ASR} < 0.5$ at clean accuracy ≥ 0.35 , since a baseline already above threshold has no suppression left to lose); they happen to cover three invariance classes (Table 8), and the $\kappa=0$ rung is run *fresh* at all five seeds so every rung comes from one code path at one seed set.

Table 8: The four arms of the dose–response ladder, and the frozen predictions. The statistic each d_2 reads is, in order, summed pairwise Euclidean distance, distance to the client median, pairwise cosine, and coordinate ordering. *Standalone* is d_2 ’s own ASR (clean accuracy) on that attack, the C1 input, held fixed within every arm. *Measured disturbance* is the fraction of live rounds on which the rung changes that arm’s decision—for `coord_median`, the mean fraction of *coordinates* whose median-attaining client changes—measured before the freeze on the same raw updates, with no ASR computed (`results/dose_disturbance.json`). Shapes and decision rules are frozen at commit e711a95.

	arm (d_2 / attack)	class (Prop. 8)	standalone ASR (acc)	measured disturbance				predicted shape in κ
				$\kappa=0$	0.5	1.0	2.0	
1	<code>krum / scaling</code>	(c) not inv.	0.061 (0.577)	0.00	0.60	0.73	0.87	monotone rise
2	<code>reputation / scaling</code>	(c) not inv.	0.017 (0.777)	0.00	0.87	0.87	0.93	monotone rise
3	<code>cos_krum / pixel</code>	(a) exact	0.300 (0.598)	0.00	0.00	0.00	0.00	FLAT (control)
4	<code>coord_median / pixel</code>	(b) cond. inv.	0.443 (0.767)	0.00	0.25	0.38	0.49	rise, shallower

The abscissa is measured, not nominal. What matters to C2 is not ρ but what the rescaling does to the statistic d_2 actually reads, so before freezing any prediction we pushed live FL updates through the shipped transform and recorded each arm’s decision at every κ on the *same* raw updates, computing no ASR. That establishes three things before any outcome exists (Table 8): *there is a dose*—disturbance is exactly 0 at the identity rung and rises monotonically with κ in every non-invariant arm; *the class-(a) invariance claim holds in the shipped code*—`cos_krum`’s selection is unchanged in 0 of 120 cells at ρ up to 54.6; and *class (b) is graded rather than binary*, at roughly half the class-(c) disturbance (0.487 vs. 0.867/0.933 at $\kappa=2.0$), which is what arm 4’s prediction turns on.

Frozen predictions. The *primary* test is pooled: across all 16 (arm, κ) cells, the rise in ASR over that arm’s own $\kappa=0$ rung is predicted to be positively rank-correlated with the measured disturbance rate (Spearman, one-sided, $\alpha = 0.05$). Its predictor is a *measured property of the statistic*—not the identity of the defense and not its standalone effectiveness, both held fixed within every arm. The *secondary* predictions are the per-arm shapes of Table 8, tested by a one-sided Jonckheere–Terpstra trend test across the four ordered rungs at $n=5$, with arm 3’s “flat” scored as

an *equivalence* claim at a 0.15 margin fixed and justified in advance (under a quarter of the +0.629 within-defense gap for Krum—the $n=3$ value on record at the freeze; the later top-up puts it at +0.716, so the margin is if anything conservative—and $0.300 + 0.15 = 0.450 < 0.5$, so “within the margin” and “still suppressing” coincide for this arm) and arm 4’s class-(b) signature being the *shallower slope*.

Why arm 3 is a genuine falsifiable positive test, unlike the $\rightarrow$ FLTrust pairs of §4.1. There, Proposition 8(a) permitted no outcome other than confirmation, because the compositions were near-identities. Here `cos_krum`’s *selection* is bit-identical at every κ —now measured, 0/120—but the aggregate it emits is $c_j u_j$ for the selected client j , so the 50-round trajectory is **not** bit-identical and suppression can still be lost through the magnitude channel—as `rfa`→`cos_krum` already did at the freeze ($\Delta = -0.027$ at $n=3$), and as its own seed top-up has since shown more sharply ($\Delta = +0.147$ at $n=5$, and +0.190 for the NormClip arm). The claim under test is “selection preserved $\Rightarrow$ suppression preserved, at ρ up to 54.6”, and it can fail. Symmetrically, a flat arm 1 or 2 would refute the framework’s central claim that Euclidean statistics are fragile under upstream rescaling. Whichever way the curves land is what we report.

One prediction we deliberately do not make. Theorem 6(1)’s ordering-survival condition $\rho < S$ would give arm 4 a crossing rung fixed by a measured S , but **no frozen prediction here uses S** : we measured it and it is **not a stable per-attack constant**, ranging 0.00–12.59 on model-scaling (median 0.57) and 0.61–1.63 on the pixel backdoor (median 0.92) across the 12 rounds that sampled an adversary, so the $S = 10$ used illustratively in §3.2 is a stipulation rather than a measurement. This is a limitation of our own theorem that the dose-response measurement surfaced, reported as one (L8) rather than quietly left unchecked.

Caveats recorded in advance. The class-(c) arms are on model-scaling and the class-(a)/(b) arms on the pixel backdoor—not a free choice, since on model-scaling *no* invariant d_2 suppresses at usable accuracy (§4.2: `cos_krum` 0.078 at 0.146), which is also why `cos_krum/scaling` is not an arm. “Same attack” is required *within* each ladder, which is where the causal claim lives; across arms we compare curve shapes and slopes, never levels. Arm 4’s thin headroom (0.057 below threshold), the dose’s being uncorrelated with adversary status but not adversary-*adversarial*, the real-defense anchors that test ecological validity, and the FedAvg-trajectory basis of the abscissa are recorded in Appendix D.

The design, the κ grid, the arms, the predicted shapes, the equivalence margin, and the refutation criteria are frozen in `experiments/pre_registration_dose_response.md` at commit e711a95, timestamped before `results/dose_response/` existed, and reproduced verbatim in the runner’s arm table so that scoring cannot drift from the committed file. **What this design does not address:** it bears on the causal validation of C2 only. It does not address recall, and it does not make the criterion a discovery tool—`fg`→`rfa` remains a false negative of our own rule, and the ladder says nothing about synergy between two defenses that individually fail (L5, L7).

5.1 Outcome of the confounded design: one arm confirmed, two refuted, one indeterminate—and the pooled test fails

All 80 runs completed (16.5 h wall clock). Table 9 is the whole ladder; Figure 3 plots it on the nominal dose axis against the frozen shapes and Figure 4 on the independently measured disturbance axis with the real-transform anchors, scored by `analyze_dose_response.py` against the rules frozen at e711a95, every number recomputed per seed from `results/dose_response/`. The **primary** pooled test does not confirm: across the 16 cells, the rise in ASR over each arm’s own $\kappa=0$ rung correlates with the measured disturbance at Spearman $\rho_s = +0.351$, one-sided $p = 0.091$, so the frozen criterion ($\rho_s > 0$ at $p < 0.05$) is not met and **we report the primary prediction as refuted**. Of the four frozen shapes one is confirmed, two are refuted, and one is indeterminate; **in three of the four arms ASR falls as the dose rises**. This is the round’s headline result and it is a negative one. It is also the outcome the pre-registration anticipated and named in advance: “pooled correlation absent while individual arms move: the mechanism is defense-specific and measured disturbance is not a transferable predictor.”

Arm 2 confirms: the ladder yields one prospective confirmation, while the pooled prediction is rejected. Reputation under model-scaling rises monotonically from 0.017 at the identity rung to 0.487 at $\rho = 7.39$ and 0.779 at $\rho = 54.6$ (Jonckheere–Terpstra $z = +4.64$, one-sided $p = 1.7 \times 10^{-6}$; permutation $p < 5 \times 10^{-5}$), while clean accuracy moves only $0.777 \rightarrow 0.701$ —so what is lost is suppression, not the model. Same d_2 , same attack, same standalone baseline at every rung: nothing about effectiveness varies, and $55\times$ dispersion in a *positive per-client rescaling*—a transform the criterion treats as benign for invariant statistics—carries a defense from 0.017 across the 0.5 threshold. Two qualifications keep this in proportion. It validates the *destruction* half of C2’s mechanism (a non-invariant statistic, actually disturbed, loses suppression); the *preservation* half (invariance $\Rightarrow$ suppression survives)

Table 9: The dose–response ladder as measured, $n=5$ per cell, per-attack mean ASR with mean clean accuracy in parentheses. *Predicted* is the shape frozen at commit e711a95 before any run; $p_{\uparrow}$ is the one-sided Jonckheere–Terpstra probability for an *increasing* trend across the four ordered rungs, which is the only direction any frozen rule tests. Verdicts apply the frozen criteria unchanged (Table 8 and Appendix B); an arm satisfying neither its confirmation nor its refutation criterion is reported as *indeterminate* rather than scored in our favour. [†] below the 0.35 clean-accuracy gate: a collapsed model, not suppression. Student-t 95% intervals and per-seed values are in Appendix E.

	arm (d_2 / attack)	predicted shape	mean ASR (clean accuracy)				JT $p_{\uparrow}$	verdict
			$\kappa=0$	0.5	1.0	2.0		
1	krum / scaling	monotone rise	0.062 (0.57)	0.011 (0.65)	0.014 (0.68)	0.054 (0.62)	0.63	refuted confirmed
2	reputation / scaling	monotone rise	0.017 (0.78)	0.022 (0.78)	0.487 (0.77)	0.779 (0.70)	1.7×10^{-6}	
3	cos_krum / pixel	FLAT (control)	0.490 (0.61)	0.450 (0.57)	0.153 (0.47)	0.086 (0.22) [†]	0.999	indeterminate
4	coord_median / pixel	rise, shallower	0.443 (0.77)	0.428 (0.76)	0.327 (0.75)	0.170 (0.69)	0.998	refuted

is arm 3, which came out indeterminate. And even within this arm the measured disturbance is not what orders the curve: it saturates at 0.867 by $\kappa=0.5$, where ASR is still 0.022, and the jump to 0.487 happens at the next rung at the same measured disturbance—so the *dose* moves ASR here while the flip rate does not.

Arm 1 is refuted, and it refutes more than its own shape. Krum under model-scaling shows no evidence of the predicted increase—0.062 $\rightarrow$ 0.011 $\rightarrow$ 0.014 $\rightarrow$ 0.054, no increasing trend ($z = -0.34$, $p_{\uparrow} = 0.63$), clean accuracy *rising* 0.565 $\rightarrow$ 0.619—even though the measured disturbance of its selection reaches 0.867 and ρ reaches 54.6. Krum’s summed-pairwise-distance statistic is class (c), the class the framework’s central fragility claim is stated over, and here it is **not** fragile under positive per-client rescaling. This cuts deeper than one shape, because arm 1 is the same cell as our flagship within-defense contrast (§4.2), where Krum goes 0.062 $\rightarrow$ 0.779 under RFA upstream at a *lower* measured disturbance ($6/9 = 0.667$). The ladder does not reproduce that contrast. Selection-flip rate therefore cannot be the mechanism behind it: something else about how RFA reweights carries the effect, and our attribution of it to disturbance of Krum’s statistic was not supported by the experiment we designed to test exactly that.

Arms 3 and 4 both fail, and the accuracy trace distinguishes why. The exactly-invariant control is **indeterminate** under its own frozen rule: the equivalence bound fails ($|\overline{\text{ASR}}(\kappa=2.0) - \overline{\text{ASR}}(\kappa=0)| = 0.405$ against the 0.15 margin), but no refutation criterion is met either—no rung reaches 0.5 (maximum 0.490) and the trend is *decreasing* ($p_{\downarrow} = 9.8 \times 10^{-4}$), a direction no frozen rule tests. We report it as neither. Two further facts make it uninterpretable as stated in any case: its $\kappa=2.0$ rung fails the 0.35 accuracy gate at 0.220, so that ASR is a collapsed model rather than suppression, and its identity rung is bimodal across seeds (0.003–0.864; 95% CI [0.058, 0.922]), so an equivalence test at $n=5$ never had the power the margin presumed—a design fault in our own instrument, not a property of the mechanism. Arm 4 is refuted outright: coord_median was predicted to rise shallowly and instead falls monotonically, 0.443 $\rightarrow$ 0.170 ($z = -2.96$, $p_{\downarrow} = 1.5 \times 10^{-3}$), at nearly unchanged accuracy (0.767 $\rightarrow$ 0.690). The comparison is informative: arm 3’s fall is bought with a 0.39 accuracy collapse and looks like Lemma 2’s magnitude-contraction channel, whereas arm 4’s is nearly free—dispersing the weights genuinely *helps* a coordinate-wise median resist a pixel backdoor.

Why three arms fail: what mean-1 normalization does not hold fixed. Normalizing $\text{mean}(c) = 1$ fixes the *aggregate*’s scale, which is what we claimed for it, but it does not fix any individual client’s *relative* weight. Because the dose is assigned by a (seed, round)-keyed permutation—the very choice that stops the instrument from manufacturing a success—an adversary is as likely to be attenuated as amplified, and spreading five weights over a $55\times$ range mostly *dilutes* whichever single client happens to carry the poison. So the ladder varies dispersion but not adversary-targeted amplification, and three arms measure the second mechanism-preserving regime of Theorem 6—attenuation until the adversary carries no payload—rather than the C2 disturbance we intended to isolate. That is the strictly harder test recorded as not run in pre-registered caveat (iii) (Appendix D); §5.2 runs it, pinning every adversary’s weight so the attenuation channel is closed by construction. We state this as a limitation of the instrument and do not reinterpret any frozen rule in its light.

Ecological validity: the real transforms do not lie on the synthetic curve. Pre-registered caveat (iv) required this check, and it fails in an informative way. RFA $\rightarrow$ Krum sits at ASR 0.779 with measured disturbance 0.667, where the synthetic ladder’s nearest rung (0.600) gives 0.011—a residual of +0.767. NormClip $\rightarrow$ Reputation sits at 0.806 with disturbance 0.333, against 0.017 at the identity rung: +0.789. Two anchors do land: RFA $\rightarrow$ Reputation, 0.895 at disturbance 1.000 against 0.779 at 0.933 (+0.116), and both cos_krum anchors, which coincide with its curve (+0.000 and -0.044) because that statistic is invariant under all three transforms. This is not a range mismatch:

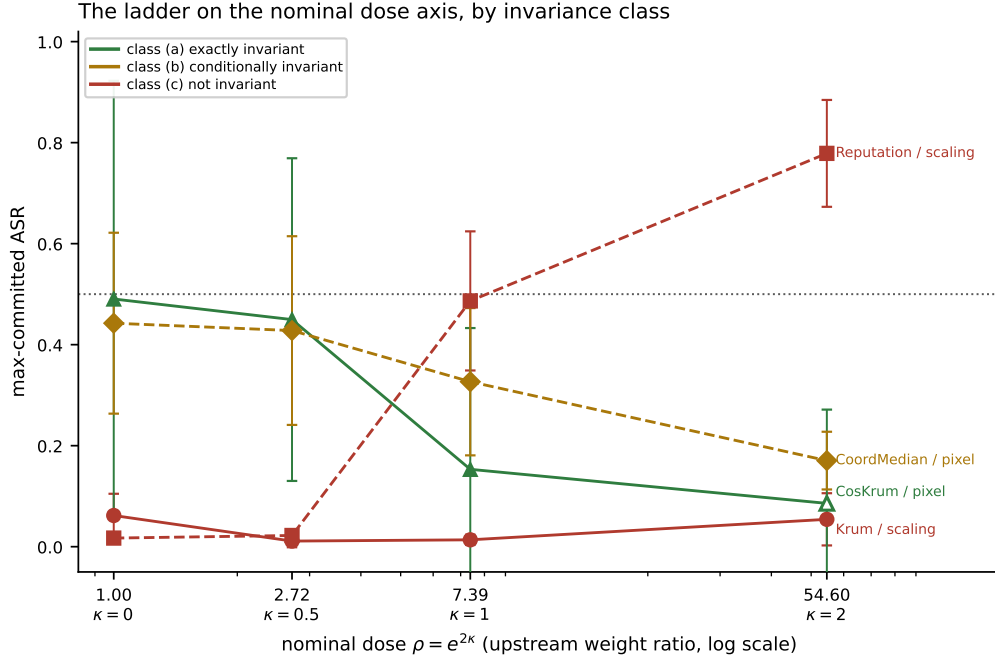


Figure 3: **The ladder and its frozen predictions.** The 16 cells on the nominal dose axis $\rho = e^{2\kappa}$ ($n=5$ per point, Student-t 95% intervals), coloured by Proposition 8 class. Of the two class-(c) arms whose statistics the framework’s fragility claim is stated over, only reputation rises; ASR *falls* with the dose in three of the four arms, against the predicted rises. Hollow markers fail the 0.35 clean-accuracy gate and are uninterpretable. Drawn by `paper/figures/dose_response_figure.py`, which recomputes every value from `results/` and imports the arm construction from the scoring script so the two cannot diverge.

read off the same live rounds, NormClip realizes ρ of mean 3.61 (max 4.90) and RFA mean 11.93 (max 17.38), both inside the ladder’s $\{1.00, 2.72, 7.39, 54.60\}$ span. The conclusion is about our abscissa: **the rate at which a transform flips a defense’s decision is not a sufficient statistic for what that transform does to the defense**—two transforms at matched flip rate differ by up to 0.77 in ASR. **And the axis they *are* unmatched on is now measured, not conjectured.** Adversary-correlated dispersion $\Delta_c := \mathbb{E}[c_i | i \in A] - \mathbb{E}[c_i | i \notin A]$ (§3) separates exactly these two anchors from the ladder: RFA reaches $\Delta_c = -1.21$ and NormClip -0.72 , against -0.34 at the ladder’s most dispersed rung— $2\text{--}3.6\times$ more extremely targeted at ρ that is 3.7 and $13\times$ *smaller*, so **matched on the statistic, badly unmatched on the targeting, and ρ does not order Δ_c .** **This rescues no predictor:** Δ_c ’s magnitude does not order suppression either—RFA is the most extremely targeted of the four real transforms and `rfa`→`krum` still fails at 0.779, while FoolsGold’s Δ_c is *positive* (+0.663). These figures are **post-hoc and exploratory**, computed with no ASR and rescoring nothing, and are tabulated in Appendix H. Arm 4 has no real-transform anchor at all, since `coordmedian` was not on the metric-swap menu, so its ecological validity is untested.

Two checks that did pass. The fresh $\kappa=0$ rung reproduces every published standalone baseline *exactly*, per seed, over all 16 available seed comparisons (largest absolute deviation 0.0 at tolerance 10^{-6}), so the identity rung and the published baselines are the same quantity and the arms differ in d_2 and the dose alone. And the class-(a) invariance claim holds in the shipped code exactly as measured before the freeze: `cos_krum`’s selection never changed in 0/120 cells. What the ladder refutes is not Proposition 8’s classification of which statistics a rescaling can disturb—that classification is verified—but the inference from “the statistic was disturbed” to “suppression is lost.”

What we now claim less of. C2’s causal content is established prospectively for one statistic (distance to the client median) and refuted for another (summed pairwise distance) at ρ up to 54.6; the pooled claim that measured disturbance predicts loss of suppression *across* defenses is refuted, and we no longer make it. Within-defense disturbance remains the only design the testability boundary admits, and §3.1’s general statement stands untouched by this outcome—it is a statement about admissible designs, and this is the admissible design, run and reported as it landed. What changes is the empirical status of C2: one confirmed arm, not a validated screening mechanism.

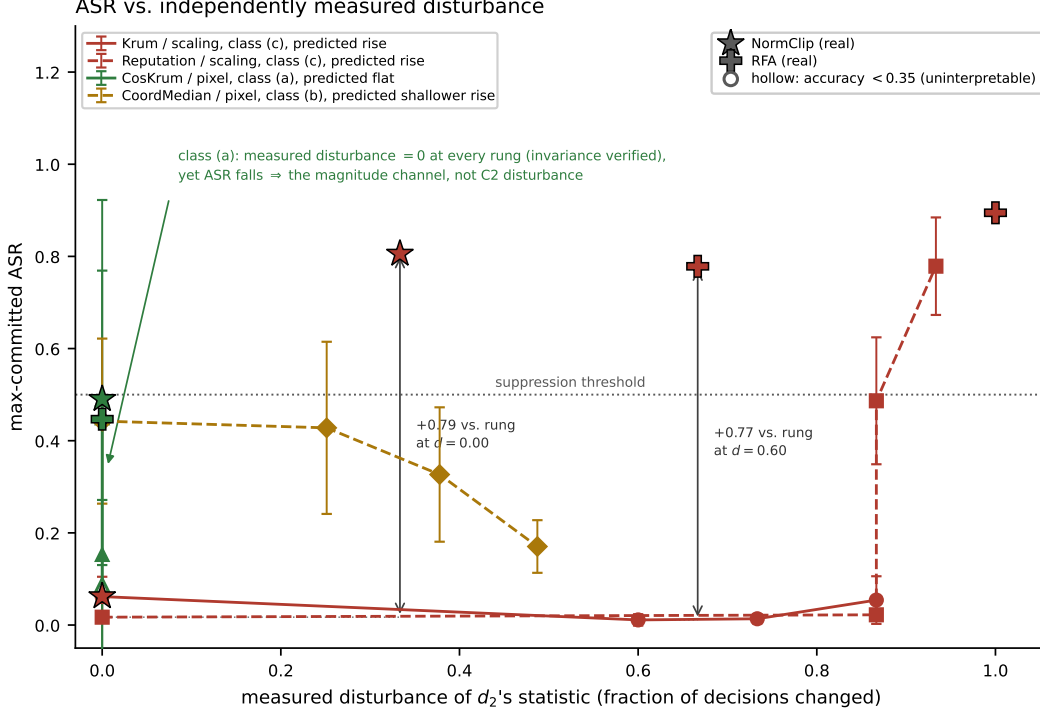


Figure 4: **Why a transform’s flip rate is not a sufficient statistic for what it does.** ASR against the *independently measured* disturbance of each arm’s own statistic ($n=5$, Student- t 95% intervals), the abscissa fixed before the freeze. Stars and crosses are the two *real* upstream transforms measured on these same (d_2, attack) cells (NormClip, RFA), placed at their own measured disturbance rate: at matched disturbance they sit $+0.77$ and $+0.79$ ASR *above* the synthetic curve, while their realized weight ratios (mean 3.61 and 11.93, maximum 4.90 and 17.38) fall *inside* the ladder’s span—which is why we conclude that flip rate does not determine what a transform does to a defense. The class-(a) arm’s four rungs coincide at disturbance exactly 0 (0/120 selections changed, maximum relative score drift 1.1×10^{-4}) and its ASR falls anyway: Lemma 2’s magnitude channel, not C2 disturbance.

5.2 The targeted intervention: isolating C2 from the attenuation channel

The arc that produced this subsection. We first froze a pooled, statistic-level hypothesis (H1): across a within-defense ladder, ASR should rise with measured disturbance of the statistic the downstream defense reads. The pre-registered ladder above **rejected it**, and the effect came out *negative*— -0.272 , $p_{\downarrow}=0.0015$ on the flagship cell, with ASR *falling* as the dose rose in three arms of four (§5.1). The diagnosis was ours to make: the ladder moved adversarial influence and the statistic *together*, so it never tested H1 at all. What follows builds and runs the instrument that fixes that—one that pins the adversarial coefficient share while the statistic moves, so that statistic disturbance is isolated from adversarial coefficient attenuation. Proposition 4 admits only within-defense instruments, and this is the strongest of them. C2 is written at level (P2); if losing (P3) meant losing (P5) the criterion would transfer straight to security, and the experiment below adjudicates that implication. **What it establishes, it establishes by oracle intervention on two arms in two invariance classes**—a causal identification on those arms, not a law about defense compositions in general.

Because the purpose of this section is causal identification rather than defense evaluation, we use oracle interventions on purpose: their non-deployability is a property of the experimental design, not a proposed defense, and they should not be judged against a defense’s standard. Both faults above are diagnoses of a single instrument, and both name the fix: assign the dose *relative to adversary status* rather than independently of it. We built two such instruments, froze them at commit 5130cec before `results/targeted_dose/` existed, and ran 97 new runs (plus 25 identity runs imported from the ladder of §5, verified bit-identical per seed, largest deviation $0.00e+0$). Each closes one of the two channels the ladder confounded. **Mode S (statistic-only)** pins every adversary at $c=1.0$ exactly and spreads only the benign clients over $\rho = e^{2\kappa}$, so the **adversarial coefficient share is constant at 0.266667 across all four rungs** (measured, spread 3.5×10^{-7}): Lemma 2’s attenuation channel is closed *by construction*, and a

rise in ASR here cannot be attenuation. **Mode A (payload-only)** keeps benign weights uniform and sets the adversary-to-benign ratio to exactly $\gamma = e^\nu$ at fixed mean, sweeping the adversarial share $0.049 \rightarrow 0.710$ while holding the statistic’s dispersion fixed. Both read adversary identity, so **both are instruments, not defenses and not models**—no deployable defense knows which clients are adversarial, and neither **quantitatively reproduces any real defense’s behaviour**: at matched disturbance the two real upstream anchors sit $+0.767$ and $+0.789$ above the synthetic curve (§5), which is the measured size of that gap rather than a separate finding. Both were verified on the coefficient share, with no ASR involved, before any run.

The four levels, made prospective. Mode S turns Definition 1’s distinction into a decisive test. We froze two readings of C2 against each other: *H-statistic*, that what matters is disturbance of d_2 ’s statistic (read off its *decisions*), and *H-admission*, that what matters is whether the transform changes the *adversarial input* d_2 *admits*. On the Krum arm they predict opposite outcomes, because Mode S disturbs Krum’s selection heavily while—the adversary’s own weight being pinned—leaving the mass of adversarial input it admits untouched. That arm adjudicates (Figure 1): as ρ climbs to 54.6, Krum’s **decision changes in up to 0.80 of rounds** while its **admission of adversarial input changes in exactly 0.000**, and **ASR shows no evidence of the predicted increase, remaining inside the pre-specified equivalence margin** (Jonckheere–Terpstra, $p_\uparrow=0.70$; $|\Delta|=0.026 < 0.15$; Table 10). **H-statistic is refuted**, and on this arm **substantial statistic and decision disturbance does not produce a corresponding change in suppression once the adversarial coefficient share is held fixed**—a result about C2’s *content*, not merely its magnitude. *Admission is the channel the evidence points to, and no more than that*: the cross-arm admission *ordering* is refuted later in this subsection, so H-admission survives as a *direction to measure*, never as a validated predictor.

What Mode S does to the ladder’s confirmation. Reputation was the ladder’s one confirmed arm (§5.1); Mode S shows most of that confirmation was attenuation. With the adversary’s share held fixed, reputation’s rise at $\rho=54.6$ shrinks from the ladder’s $+0.762$ to $+0.178$ (one-sided Welch $t = -5.22$, $p=0.0017$ that it is smaller), and what remains tracks a small *admission* change ($0.000 \rightarrow 0.020$) rather than the large decision change ($0.000 \rightarrow 1.000$)—so H-admission is confirmed here too (Table 10). The confound is thus *explained*, not explained away: a genuine, much smaller, admission-driven effect survives it. The class-(a) control cuts the other way and refutes us: `cos_krum`’s selection is bit-identical at every rung (decision and admission both 0.000), yet ASR is 0.584 at the identity—it never suppresses—and *falls* 0.425 across the ladder ($p_\downarrow=0.0008$; the $\kappa=2$ rung is below the accuracy gate). Exact invariance of the statistic neither implies suppression nor holds it fixed; the magnitude channel moves it regardless. We report this as a refutation of Proposition 8’s practical content, not a confirmation of its algebra.

Mode A: the two regimes, and where they hold. The payload dose gives Theorem 6 a single falsifiable shape—ASR single-peaked at $r=1$, since an over-attenuated adversary carries no payload (Regime B) and an over-amplified one is discarded by a rank aggregator (Regime A). It is **confirmed for coord median**, the rank aggregator the regimes are stated for: ASR rises $0.124 \rightarrow 0.315 \rightarrow 0.443$ into $r=1$ and falls $0.443 \rightarrow 0.296 \rightarrow 0.098$ past it, both sides significant ($p_\downarrow=3.0 \times 10^{-4}$ amplified, 6.4×10^{-4} attenuated) at accuracy ≥ 0.74 throughout (Figure 5). It is **refuted for reputation**, which is monotone—ASR is maximal at the most attenuated rung (0.894 at $r=0.135$) and near zero once $r \geq 1$ —because an attenuated adversary reads as *consistent* to a reputation score and stops being down-weighted. Reputation is not a rank aggregator, so the two-regime shape was never licensed for it; the split is the theorem behaving exactly where it applies and not where it does not.

A pre-registered replication on a second downstream defense, and what it refutes. One arm cannot establish which channel matters, and the two Mode-S arms above move together on both candidate predictors—reputation is higher on *both* decision and admission change and also had the larger rise, so nothing in that pair separates them. `coord median` under the pixel backdoor is *rank-discordant* and therefore does: of the interpretable Mode-S arms it has the **smallest decision change** (0.482) and the **largest admission change** (0.033), so the decision reading predicts its rise is the *smallest* of the three and the admission reading predicts the *largest*. It is also a structurally different defense—a coordinate-wise order statistic (Proposition 8(b)) against Krum’s selector (class (c))—and it suppresses the pixel backdoor at usable accuracy, so there is suppression to lose. Thresholds were frozen against the two published rises—confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$, indeterminate between—at commit 06d9bce before `results/dose_replication/` existed, with the identity rung imported bit-identical and 15 new runs spent (`experiments/pre-registration-dose_replication.md`).

The outcome is $\Delta = +0.098$: the admission ordering is refuted. The arm with the largest admission change rose *less* than an arm with a smaller one, so **admission magnitude does not order outcomes across arms**. Neither does the decision reading, which predicted the *smallest* rise of the three when $+0.098$ is the middle one, so this arm refutes a rank prediction from *each* candidate channel. The pooled model agrees—re-fit with these cells it is still

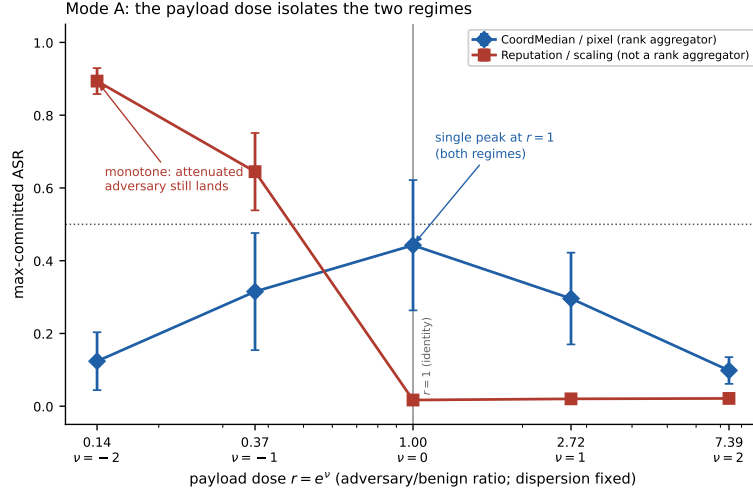


Figure 5: **Mode A, the payload dose: the two mechanism-preserving regimes, and where they hold.** With dispersion fixed and only the adversary/benign weight ratio r swept, `coordmedian` (a rank aggregator, the case Theorem 6’s regimes are stated over) is **single-peaked at $r=1$** as the theorem predicts—over-attenuated adversaries carry no payload, over-amplified ones are discarded—while reputation is monotone, off-spec and refuted for that shape. Mode A is the Δ_c -swept companion to Mode S’s Δ_c -pinned design (Proposition 10); Mode S itself is Figure 1. Bars are Student- t 95% intervals; numbers recomputed per seed by `analyze_targeted_dose.py`.

not confirmed ($R^2=0.239$; the admission coefficient’s one-sided p moves $0.055 \rightarrow 0.041$, but the adversarial-share coefficient is negative, so the pre-specified conjunction fails). **What does replicate is the negative:** a decision change of 0.482 costs no suppression by the frozen margin, at mean accuracy 0.698, so the accuracy gate does not bind and the flagship result now holds on two downstream defenses in *different* invariance classes. Two qualifications, the first pre-specified in the freeze and the second post-hoc: this cell’s identity ASR is 0.443, so equivalence is to the *identity rung* and not to a low absolute ASR (an “every rung <0.5 ” criterion has no headroom here and was deliberately not used); and the rise is positive in 5/5 seeds (paired $t=3.39$, one-sided $p=0.014$, *not* pre-registered), so on this class the cost of disturbing the decision is **small and consistent rather than zero**. Finally, **the instrument earns its keep on this arm:** the confounded companion ladder on the identical cell *falls* 0.272 ($p_{\downarrow}=0.0015$; $0.443 \rightarrow 0.170$), so closing the attenuation channel **inverts the sign of the measured effect**—the confound is not hypothetical, and a cross-defense contrast would have read the fall as evidence about the statistic. All numbers are recomputed per seed by `analyze_dose_replication.py`, which imports its rung builder and trend test from the frozen suite’s scorer.

What the two arms jointly establish, stated formally. The arms above and the class-(a) control (`cos_krum`) are between them a *constructive* answer to the question C2 was written to settle. Both halves are witnessed by runs already reported, so the following formalizes measurements rather than adding a claim.

Proposition 9 (Statistic-level non-identifiability). *Within the positive per-client rescaling class there exist transform families T_1, T_2 and downstream defenses d_2 such that (a) T_1 and T_2 induce arbitrarily different statistics and decisions at d_2 while inducing the same admitted adversarial mass and the same suppression; and (b) T_1 and T_2 induce the same admitted adversarial mass—indeed the same statistic and the same decision—while inducing different suppression. Consequently no predicate of the downstream statistic alone determines the admitted adversarial mass, and no predicate of the admitted mass alone determines suppression.*

Proof. Both halves are exhibited by runs reported in this section. (a) Mode S into Krum under model-scaling, $\kappa=0$ against $\kappa=2$: ρ moves from 1 to 54.6 and Krum’s decision changes in up to 0.80 of rounds, while the admitted adversarial mass is exactly 0.000 at every rung, the adversarial coefficient share is constant at 0.266667, and suppression is preserved ($|\Delta|=0.026 < 0.15$). (b) `cos_krum` under the pixel backdoor: by Proposition 8(a) its statistic is invariant, and its decision and admission are measured bit-identical (0.000) at every rung, yet ASR falls 0.173 by the first rung alone, at $\rho=2.72$, where the accuracy gate still passes ($0.599 \rightarrow 0.567$; paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds). $\square$

Table 10: Mode S (statistic-only), $n=5$ per arm ($n=8$ for `cos_krum`), per-attack mean ASR with the two measured channels. The decision channel moves far more than ASR throughout. [†] below the 0.35 accuracy gate. [‡] the pre-registered replication arm, *rank-discordant* by design (smallest decision change, largest admission change of any interpretable arm) and scored against thresholds frozen separately at 06d9bce before `results/dose_replication/` existed: $\Delta=+0.098$ falls below reputation’s $+0.178$, so the cross-arm admission *ordering* is refuted while the negative replicates on a different invariance class (§5.2). Verdicts otherwise scored against the readings frozen at 5130cec; per-seed values in Appendix E.

arm	channel	$\rho=1$	2.72	7.39	54.6	test	verdict
krum / scaling	ASR	0.062	0.016	0.023	0.036	$p_{\uparrow}=0.70$	H-admission confirmed, H-stat. refuted
	decision	0.000	0.533	0.800	0.733		
	admission	0.000	0.000	0.000	0.000		
reputation / scaling	ASR	0.017	0.019	0.025	0.195	rise $+0.178$ vs. ladder $+0.762$ ($p=0.0017$)	H-admission confirmed (mostly atten.)
	decision	0.000	0.933	1.000	1.000		
	admission	0.000	0.002	0.008	0.020		
cos_krum / pixel	ASR	0.584	0.411	0.295	0.159 [†]	$p_{\downarrow}=8 \times 10^{-4}$ $ \Delta =0.425$	refuted (Prop. 1 practical)
	decision	0.000	0.000	0.000	0.000		
	admission	0.000	0.000	0.000	0.000		
coord_median / pixel [‡]	ASR	0.443	0.442	0.490	0.541	rise $+0.098$ < $+0.150$ ($p_{\uparrow}=0.069$)	ordering refuted; negative repl.
	decision	0.000	0.232	0.358	0.482		
	admission	0.000	0.005	0.013	0.033		

The corollary is the paper’s title claim: no metric defined *solely* on preservation of the downstream statistic can identify security equivalence, because (a) exhibits statistic disturbance without security change and (b) exhibits security change without statistic disturbance. **Scope, stated immediately:** this is a non-existence claim *within the positive-rescaling class on these defenses and attacks*, established by exhibited counterexample—**not** a claim that no metric of any kind can order security, and not a claim about transforms outside the class.

Proposition 10 (ρ and Δ_c are independently controllable). *Within the same class, the statistic-level dispersion $\rho = \max_i c_i / \min_i c_i$ and the adversary-correlated dispersion $\Delta_c = \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ are not functionally related: there is a family with $\Delta_c \equiv 0$ exactly while ρ ranges over $[1, 54.6]$, and a family with the statistic’s dispersion held fixed while Δ_c ranges over $[-1.138, +2.469]$. Hence neither quantity determines the other.*

Proof. By construction, with the realized values tabulated in Appendix H and computed with no ASR involved. Mode S pins every adversarial coefficient at $c=1$ and spreads only the benign clients, giving $\Delta_c \equiv 0$ at every ρ up to 54.6; Mode A holds the benign weights uniform and sweeps only the adversary-to-benign ratio r , moving Δ_c over $[-1.138, +2.469]$ at fixed statistic dispersion. $\square$

This is why the two instruments were the right ones, and it makes §3.3’s warning precise: **a transform reported by ρ alone leaves Δ_c entirely free**, so ρ cannot be a sufficient statistic for what the transform does to a defense. It does *not* follow that Δ_c is a predictor—§5.1 reports that its magnitude fails to order suppression either.

The dissociation is not Krum-specific: all four aggregators, all four channels. Table 11 collects the endpoint change ($\kappa=0$ against $\kappa=2$) in each measured channel under Mode S for every downstream defense in the paper. **Recomputed, not transcribed:** every cell is emitted by `build_channel_table.py`, which only reads frozen artifacts under `results/—admission_measurement.json` and `femnist_admission.json` for the channel columns, the `targeted_dose/`, `dose_replication/` and `dose_femnist/` summaries for the ASR column; it also asserts that the identity rung displaces nothing (0.00e+00 for all five rows). The influence column is **aggregator-specific by necessity**—a selector has no weights—and its instantiation is: adversarial share of the Weiszfeld/reputation weight mass for the averager, adversarial share of selected coordinates for the order statistic, and the indicator that an adversarial update is selected at all for the two selectors. The fifth row replicates the first row’s arm on a second dataset and architecture—FEMNIST with `simple_cnn`—with **its channels measured prospectively, before its ASR existed**; the result is reported below.

A second dataset and a second architecture, pre-registered. The fifth row’s arm is the first row’s arm re-run with EMNIST-byclass (62 classes, $1 \times 28 \times 28$) and `simple_cnn` in place of CIFAR-10 and `cifar_cnn`, everything

Table 11: **The four channels, all four aggregators, under Mode S.** Endpoint change $\kappa=0 \rightarrow \kappa=2$. **The first column, Δ agg., is the relative displacement of the aggregate the defense emits, $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$ —what enters training, and **not** the value of a scoring statistic; the measured channels are aggregate displacement, decision, admission and influence. The dissociation between the *decision* channel and the *admission* channel is not a property of one aggregator: it holds for a selector, a cosine-invariant selector, a coordinate-wise order statistic and a weighted averager. The table does **not** order the arms— $\Delta\Lambda_a$ and ΔASR disagree in rank, which is §5.2’s refutation stated in one row-pair—and it is *not* an out-of-sample test: all rows are Mode-S ladders reported elsewhere in this paper.**

Aggregator	Kind	Δ agg.	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	-0.026
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178
Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	-0.425
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098
Krum (FEMNIST)	selector (pairwise distance)	0.842	0.643	0.000	0.000	-0.017

else held: Mode S, Krum as d_2 , the committed model-scaling attack, the same four frozen rungs $\kappa \in \{0, 0.5, 1, 2\}$. The decision rule lives in `experiments/pre_registration_dose_femnist.md`, git-committed before any FEMNIST ASR existed, and the runner refuses to start unless its `PREREG_COMMIT` matches that hash.

The premise was measured first, with no ASR anywhere. Krum’s selection changes on 0.643–0.714 of rounds across the three disturbed rungs while the adversarial mass it admits stays at exactly 0.000 at every rung, so the arm has a real disturbance to test. Had the decision change come out near zero the arm would have been reported ineligible rather than run—that is the `cos_krum` failure mode, and this check is what distinguishes the two.

Suppression does not move. $\Delta = -0.017$ (mean ASR $0.027 \rightarrow 0.010$) against the frozen ± 0.15 equivalence margin, beside CIFAR-10’s -0.026 on the same cell, with mean clean accuracy *rising* $0.767 \rightarrow 0.835$ so the accuracy floor binds on no rung. Under the frozen rule this is a **replication on a second dataset and a second architecture**—the pre-registration fixes that wording and explicitly forbids “generalizes”, which one further dataset does not license.

Three disclosures, each of them from the freeze rather than from hindsight. (i) $n=3$ seeds per rung, not 5, fixed in advance to match the FEMNIST payoff matrix’s own trial budget. (ii) The secondary Jonckheere–Terpstra test was specified to detect a monotone *rise*—the outcome that would have refuted the flat prediction with attenuation already closed by construction—and the observed trend is a *fall* ($J=16.0$, $z = -1.563$, $p_{\uparrow}=0.941$). We report it and we do **not** score it as support for flatness; the pre-registration states in advance that at $n=3$ a null from this test is not evidence of flatness. (iii) The two branches had unequal headroom. With the identity rung at ASR 0.027, refuting us required mean $\text{ASR}(\kappa=2) > 0.177$ against a ceiling of 1 and was fully reachable, whereas the indeterminate branch ($\Delta < -0.15$) was arithmetically unreachable. **Equivalence here is to the identity rung, not to a low absolute ASR.**

Harness sanity, which is an identity check and not corroboration. The $\kappa=0$ rung is `identity◦Krum`, i.e. Krum alone, and it lands *bit-identically* on the FEMNIST payoff matrix’s own Krum figure (0.0269784552 at accuracy 0.7670795973; §M.1.8). Both pipelines are deterministic given a seed and both run three trials from base seed 42, so the agreement shows they execute the same protocol rather than a similar one—the strongest form the pre-registered check can take, and for the same reason **not** independent corroboration of that figure: the same computation run twice cannot corroborate itself.

What the instrument does *not* hold fixed, named in full—and what it pins exactly. Mode S pins the *adversarial* coefficients; it does not pin the benign ones and was never claimed to. So the dose also moves (a) Krum’s pairwise distances, hence the statistic itself; (b) *which* benign client Krum selects, in 0.533/0.800/0.733 of rounds at $\rho=2.72/7.39/54.6$; (c) the **magnitude and geometry of the update actually aggregated**, since the selected client enters scaled by its own coefficient; and (d) the model trajectory, which diverges across rungs in consequence. Channel (c) is not a conjecture but a **number we already measure and print**: the Δ agg. column of Table 11 is $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$, the relative displacement of the aggregate the defense emits, and it reads **0.892** on this cell. Appendix G splits that 0.892, as vectors, into 0.986 of *re-selection* (a different benign client is chosen, and in 0/11 of the rounds where that happens it is an adversary) and 0.973 of *rescaling* (the chosen client arrives scaled by its own coefficient); for `cos_krum` the displacement is *entirely* rescaling, since 0/15 selections change at any rung. The claim this arm licenses is therefore sharper than a claim about the statistic alone: **a 0.892 displacement of the emitted aggregate, together with decision changes in up to 80% of rounds, leaves suppression inside the pre-specified equivalence margin—provided the adversarial coefficient share does not move. What is pinned is pinned exactly:**

$c_{\text{adv}}=1.0$ at every rung with maximum deviation $0.00\text{e}+00$ and the share constant to $3.5\text{e}-07$ on CIFAR-10, $3.9\text{e}-07$ on FEMNIST—float32 norm read-back noise, not slack in the instrument (`measure_admission.py`). Naming (a)–(d) is what keeps the claim to the one channel the instrument closes by construction.

What this settles, and what it does not. The intervention delivers the controlled isolation of C2 the two cross-defense suites could not: within a single defense, with attenuation excluded by construction, **substantial disturbance of the statistic d_2 reads need not produce a corresponding change in suppression once the adversarial coefficient share is held fixed**—on the adjudicating Krum arm under Mode S and, by pre-registered replication, both on a downstream defense in a different invariance class and on a second dataset and architecture—while *admission* of adversarial input is the channel the evidence points to. The statistic-level reading the framework was first written in is refuted; the admission-level reading is confirmed as a *direction* in these interventions rather than established in general. It does *not* rescue the framework’s positive direction wholesale: the pooled two-channel model over all 26 cells is **not** confirmed ($R^2=0.239$), and the replication refutes its cross-arm ordering outright, so admission is a better single predictor than the statistic without being a quantitative one. It addresses neither recall nor discovery (FG→RFA remains a false negative), and nothing in §5.1 is rescored: its pooled refutation, its two refuted arms and its indeterminate arm stand exactly as reported.

6 The composition the screen cannot certify: FoolsGold→RFA

This section is the empirical witness for Proposition 5. FoolsGold→RFA has the *lowest observed committed ASR* of every composition evaluated here (0.093), and **our own criterion classifies it as a false negative**: neither constituent suppresses the attack alone (FoolsGold 0.732, RFA 0.845), so its suppression is *emergent* in the sense of Proposition 5 and C1 cannot hold. **We report it in detail because it is the sharpest illustration of the screen’s scope, not because it is a flagship**: any criterion that certifies inherited suppression must miss compositions of this kind, which is what bounds recall at the reported 40%. The characterization below is therefore descriptive—what this composition does, measured at $n=30$ —and is not evidence for the criterion. We avoid calling it security-maximal: it also sustains the *highest* ASR under our adaptive adversary (0.167, versus 0.093 committed), and it carries a 50% accuracy cost, so it is the right choice only where utility is nearly irrelevant. We validate at $n=30$ seeds with a full adaptive attack suite ($N=10$, $K=5$, $f=0.2$, 50 rounds), in Table 12:

Table 12: FG→RFA under base and adaptive attacks. Base: $n=30$, reported as median [IQR] with P95 tail; adaptive: $n=5$, mean±std. Clean accuracy: test-set accuracy on non-triggered inputs. No-attack baseline for RFA alone: 0.737.

Attack	Config	ASR	P95 / max	Acc	Δmix
<i>Base composition ($n=30$; median [IQR])</i>					
	scaling	0.060 [0.048, 0.096]	0.253 / 0.380	0.497	−0.778
	pixel	0.052 [0.035, 0.087]	0.127 / 0.200	0.492	−0.683
	oracle	0.056 [0.044, 0.090]	0.189 / 0.236	0.494	−0.839
<i>Projection adaptive ($n=5$)</i>					
	$\epsilon=2.0$ (best)	0.079 ± 0.042	− / 0.160	0.495	
<i>Criterion-aware joint-constraint ($n=5$)</i>					
	$\epsilon=1.0$, decorr	0.167 ± 0.069	− / 0.268	0.562	
	$\epsilon=2.0$, decorr	0.130 ± 0.030	− / 0.185	0.506	
	$\epsilon=2.0$, no decorr	0.050 ± 0.008	− / 0.059	0.494	
<i>Neurotoxin ($n=5$)</i>					
	top-20%, scale = 2	0.048 ± 0.009	− / 0.057	0.495	
<i>Consensus-shift ($n=5$)</i>					
	rate = 0.05	0.045 ± 0.012	− / 0.059	0.495	

Results. Median ASR across all base attacks ($n=30$): 0.056 [IQR 0.044–0.090]; P95 worst-case: 0.253 (committed_scaling). Every adaptive strategy performs *worse* than the base attacks—the composition achieves < 10% median ASR under the full suite including the criterion-aware adversary. The accuracy tradeoff (0.50 vs. 0.74 for RFA alone) and Pareto alternatives are discussed below (Table 13).

Composition-aware adversary (end-to-end). We build an adversary that *knows* the full FG→RFA pipeline is deployed (Kerckhoffs) and optimizes against the complete composition, solving both constraints simultaneously: pairwise cosine similarity below FoolsGold’s down-weighting threshold, *and* ℓ_2 distance to the geometric median below RFA’s rejection radius. Any gradient-based optimizer faces the same tension—coordinating a backdoor raises FG-similarity, while staying near the median caps the perturbation and weakens the backdoor—and our analytical construction reaches the same boundary, the Weiszfeld iteration being contractive and the gradient at convergence determined by the final-iterate reweighting the joint constraint directly constrains.² The ablation is informative: without decorrelation ASR stays at baseline (0.050), RFA-evasion alone failing because FoolsGold is the binding constraint; with it the adversary gains partial signal (0.130–0.167), which Theorem 6 explains as bounded, each adversary’s block-restricted delta being too small to move the median outside the benign cluster.

Contrast with rep→tm. FG→RFA (median ASR 0.056) provides near-absolute suppression; rep→tm (oracle ASR 0.316) provides strategic dominance against all tested adversary policies but does not suppress committed_pixel (0.679). The two serve different roles.

Accuracy–security tradeoff. FG→RFA achieves the lowest ASR of any tested composition but imposes a substantial clean-accuracy cost (0.50 vs. 0.74 for RFA alone). This cost is intrinsic to FoolsGold’s aggressive per-client downweighting, not to composition itself. Table 13 places FG→RFA on the accuracy–security Pareto frontier alongside alternative compositions:

Table 13: Accuracy–security Pareto frontier. Max ASR = worst-case mean across committed attacks. All at $N=10$, $K=5$, $f=0.2$, cifar_cnn.

Defense / Composition	Clean Acc	Max ASR	Regime
Temporal mix (NC/fedavg)	0.79	0.96	No security
RFA alone	0.74	0.91	Bimodal; unreliable
Rep alone	0.78	0.84	Scaling-only
Rep→CM	0.77	0.35	Moderate
FG→CM	0.66	0.11	Strong
FG→RFA	0.50	0.06	Min ASR (evaluated)

FG→RFA is the correct choice under a *security-first* objective (e.g., safety-critical FL where any backdoor is unacceptable). For operators with accuracy constraints, FG→CM (acc 0.68, max ASR 0.107) provides a Pareto-superior tradeoff: statistically indistinguishable suppression (Welch $p = 0.52$) at 37% more accuracy. Rep→CM (acc 0.76, max ASR 0.423) offers near-baseline accuracy with moderate security. The criterion certifies FG→CM and Rep→CM; FG→RFA reaches the lowest ASR of the three but fails the tightened per-attack C1 and is therefore a false negative of our own criterion rather than a certified recommendation. The choice among them is an operator-specific utility decision, not a criterion failure.

7 Why Composition, Not Mixing

This section is motivation, not evidence for the criterion. It answers a prior question—why study *sequential* composition at all, rather than randomizing the defense across rounds—and its result is independent of everything the rest of the paper argues: persistent state combined with non-erasing defenses makes per-round aggregator switching lose its information value, so mixes become max-like and deterministic composition dominates. That establishes where composition is the relevant object; it establishes nothing about which compositions preserve a mechanism, and no part of the preservation hierarchy or the identification boundary rests on it. Figure 6 plots the settings we evaluated in the two coordinates that organize them; it draws no regimes and locates no boundary.

The composability criterion (§3) tells practitioners *which* compositions work; this section establishes *why* the alternative—temporal mixing (randomly selecting one defense per round)—fails. The full game-theoretic framework (utility functions, equilibrium analysis, VoPD formalism, sample complexity, and the persistence Markov model) appears in Appendix L; here we present only the key result and its mechanism.

²A fully gradient-based optimizer differentiating through 50 Weiszfeld iterations would face identical binding constraints at convergence; the analytical formulation is equivalent without requiring a differentiable relaxation.

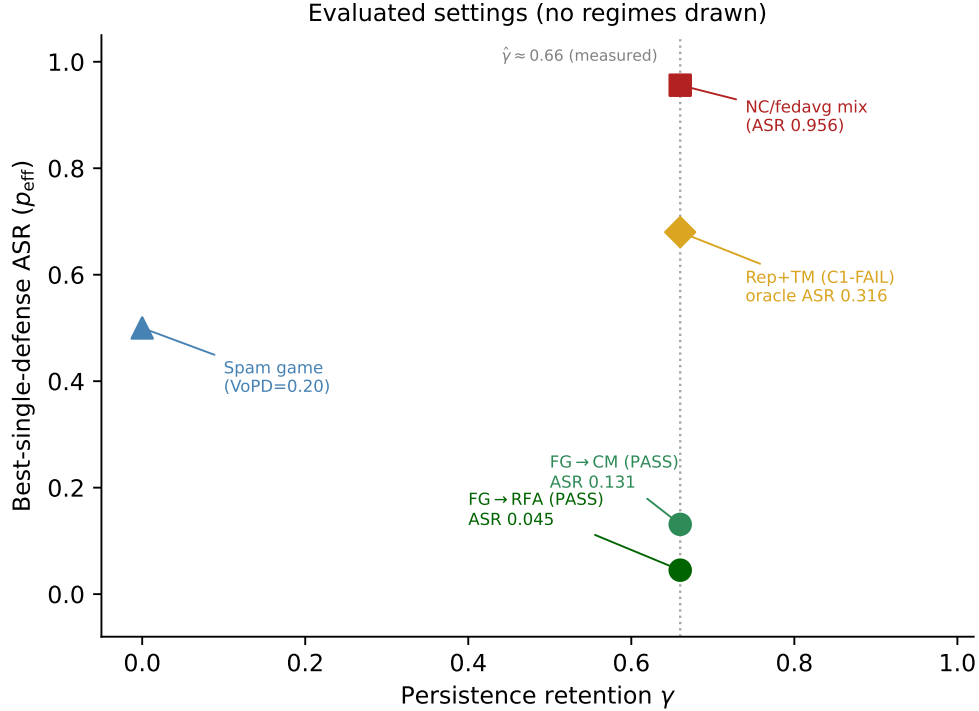


Figure 6: **The evaluated settings, plotted. No regimes, and one line, which is a measurement.** Each marker is a setting we ran, placed in the two coordinates that organize them: persistence retention γ against the best single constituent defense’s ASR, p_{eff} . At $\gamma=0$, the stateless spam game, where randomization has value ($\text{VoPD} = 0.200$). At the measured FL persistence $\hat{\gamma} \approx 0.66$ (the dotted line): the NC/fedavg temporal mix at ASR 0.956, the C1-FAIL Rep+TM pair, and the two criterion-PASS compositions. Each point carries its own measured value as its label. **An earlier version of this figure shaded four regime bands and drew threshold guides at $\gamma=0.15$, $p_{\text{eff}}=0.3$ and $p_{\text{eff}}=0.6$; none of those three numbers was measured, so all six are gone.** Five points cannot locate a boundary, and nothing here is evidence of a phase transition in (γ, p_{eff}) space.

Persistence makes mixing max-like (in the persistent regime). Once a backdoor trigger embeds (via model-scaling or pixel poisoning), it persists across rounds with retention $\gamma \approx 0.66$. Intermittently sampling a suppressing defense cannot remove an already-embedded trigger. The temporal mix therefore converges to the ASR of its *worst constituent*—max-like rather than average-like behavior. Only at $\gamma=0$ (stateless threats, e.g. spam) does randomization pay off ($\text{VoPD} = 0.200$; Appendix M.1).

Composition dominates on the rep+tm menu. On a menu where reputation kills scaling (ASR 0.017) and trimmed_mean partially suppresses pixel (ASR 0.62), the server can mix or compose. Result ($n=30$):

Adversary policy	Composition (rep+tm)	Temporal mix	Best pure defense
committed_scaling	0.023	0.871	0.017 (reputation)
committed_pixel	0.679	0.747	0.616 (trimmed_mean)
oracle	0.316	0.915	0.519 (coord_median)

Against an oracle adversary, composition achieves ASR 0.316 vs. temporal mix 0.915—a 3× improvement. No single defense achieves low ASR against both attacks simultaneously; composition applies both mechanisms every round. FG→RFA (§6) provides even stronger suppression (max ASR 0.093) via the theorem-backed mechanism of §3.2.

Scale validation ($N=100$, ResNet18). At $N=100$, $K=10$, $f=0.05$, ResNet18 ($n=10$): rep+tm composition suppresses all attacks to ≈ 0.10 ASR while temporal mixing yields committed-scaling 0.410 (bimodal; $\text{VoPD} = -0.262$, oracle *hurts itself* by switching). The qualitative result transfers to 10× scale with a deeper architecture (full table: Appendix K).

Compute-matched control: order matters, not compute. Composition applies two mechanisms every round while temporal mixing applies one, which would confound “two defenses vs. one” with “deterministic vs. randomized.” We resolve it directly, comparing fixed $\text{FG} \rightarrow \text{CM}$ against a *randomized-order* variant that applies both defenses every round at equal compute (50/50 $\text{FG} \rightarrow \text{CM}$ vs. $\text{CM} \rightarrow \text{FG}$). The criterion predicts the randomized variant degrades, because $\text{CM} \rightarrow \text{FG}$ rounds degenerate to FG-aggregation alone—CM is a coordinate-wise aggregator, not a per-client transform, so with $d_1 = \text{CM}$ the downstream FG sees unmodified updates. At $n=5$, $N=10$, $f=0.2$: fixed $\text{FG} \rightarrow \text{CM}$ reaches committed_pixel ASR 0.076 ± 0.035 against the randomized variant’s 0.681 ± 0.133 , a $9\times$ degradation at identical per-round compute; for $\text{FG} \rightarrow \text{RFA}$ the gap is larger still (0.037 ± 0.011 vs. 0.719 ± 0.113). The advantage is structural: $\text{C2} \wedge \text{C3}$ are order-sensitive ($\text{FG} \rightarrow \text{CM}$ PASS, $\text{CM} \rightarrow \text{FG}$ fails), so randomizing the order randomizes whether the signal-preservation conditions hold at all, not merely which defense runs.

Near-optimality of committed strategies. Stronger-than-oracle adversaries (*commit-then-probe*, *belief-state*) at $N=50/f=0.05$ gain only $+0.034$ and $+0.044$ over committed play ($n=30$ each)—both below the POMDP switching bound of 0.048 (Proposition 12, Appendix L).

Summary. Persistence converts a signal-preservation failure (broken $\text{C2}/\text{C3}$) into total failure by accumulating admitted contributions: temporal mixing cannot clear an embedded trigger, whereas composition prevents embedding in the first place by applying both mechanisms every round. The game-theoretic analysis (Appendix L) formalizes this—realized VoPD ≈ 0 on all tested menus ($N \in \{10, 20, 50, 100\}$), with the persistence boundary separating regimes where mixing could conceivably help (stateless, $\gamma=0$) from the FL regime where it cannot ($\gamma \gg 0$).

8 Related Work

The contrast in one sentence. Prior work asks whether a defense withstands an attack. We ask whether an *evaluation criterion* can causally attribute the security of a composition to preservation of the downstream mechanism.

Byzantine-robust aggregation, studied in isolation. Krum (Blanchard et al., 2017), Trimmed Mean (Yin et al., 2018), RFA (Pillutla et al., 2022), FoolsGold (Fung et al., 2020), FLTrust (Cao et al., 2021) and FLAME (Nguyen et al., 2022) are each proposed and evaluated as a *single* stage. We propose no new defense; we reason about *ordered pairs* of existing ones, which is a different object—the upstream stage’s output is the downstream stage’s input, so the question is no longer whether a mechanism works but whether it *survives* a transform applied before it. Ensemble adversarial training (Tramèr et al., 2018) *assumes* that upstream processing preserves downstream signals; we make that assumption explicit, checkable, and—as it turns out—the **wrong abstraction for security**.

Adaptive attacks are why an inherited-robustness guarantee is not a security guarantee. Optimized model poisoning defeats Krum, trimmed mean and FoolsGold once the attack is chosen *against* the defense (Shejwalkar and Houmansadr, 2021; Bhagoji et al., 2019), and the same holds for distributed and durable backdoors (Xie et al., 2020; Wang et al., 2020; Zhang et al., 2022). Our C1 quantifies over a *committed* attack by construction, so this literature bounds what any inherited-suppression criterion—ours included—can promise. Our single whitebox composition-aware stress test (§6) is reported in order to *measure* that gap ($0.051 \rightarrow 0.167$ seed-matched, $3.3\times$), not to defend the criterion against it; a screening condition stated against an adaptive adversary is what the framework most lacks.

Evaluation methodology, which is where this paper sits. The adjacent precedent is adversarial robustness, where reported numbers proved unreliable until evaluations were required to be adaptive (Croce et al., 2021). Our point is methodological in the same way but about **identification** rather than attack strength: composition studies gate candidate pairs on standalone effectiveness (Kairouz et al., 2021) and then read the composition’s outcome as evidence about the *mechanism*, which Proposition 4 shows cannot work at *any* sample size—the counterfactual is excluded by the gate, not undersampled. Statements of a composition-preservation intuition in that literature are informal; what is new here is the *level hierarchy* at which the intuition fails (Definition 1) and the class of designs that can test it (§3.1, §5.2). The causal-inference template we invoke—conditioning on a common descendant of treatment and outcome—is standard outside this literature (Elwert and Winship, 2014; Pearl, 2009); what is not standard is noticing that FL composition studies instantiate it, because the gate is presented as a reasonableness filter (“we study defenses that work”) rather than as conditioning.

Attacks and defenses. Backdoor attacks (Chen et al., 2017; Bagdasaryan et al., 2020; Wang et al., 2020; Xie et al., 2020) and robust defenses (Pillutla et al., 2022; Sun et al., 2019; Blanchard et al., 2017) form the landscape. Beyond our norm-based menu, certified robustness (Xie et al., 2021) and filtering defenses (Nguyen et al., 2022; Fung et al., 2020) build stronger single mechanisms; our negative result is menu-relative and shows that *randomizing over* a norm-based menu adds no value while deterministic composition dominates. Randomized transformations (Xie et al., 2018; Liu

et al., 2018) in centralized settings are the closest conceptual neighbors; we provide a condition (persistence boundary) for when such randomization retains value in FL.

Backdoor persistence and durability. The persistence mechanism we formalize has empirical precursors. Bagdasaryan et al. (2020) design model-scaling explicitly to embed triggers that survive gradient updates; Zhang et al. (2022) (Neurotoxin) engineer triggers to occupy “less-used” gradient directions, achieving multi-round persistence even under strong norm clipping. These works establish the phenomenon; our contribution is a formal $(\gamma, \alpha, p_{\text{eff}})$ model that quantifies when persistence eliminates the value of defense switching, and the boundary condition (low p_{eff}) under which it would not. Notably, Neurotoxin’s design targets exactly the parameter regime ($\gamma \rightarrow 1$) that our Heuristic 1 identifies as collapsing randomization value.

Moving Target Defense (MTD). Our result can be read as a negative boundary for MTD-style FL approaches: in stateful threat models where triggers persist across rounds, randomizing over defense implementations does not provide the diversity benefit that motivates MTD (Tambe, 2011). The positive case (VoPD = 0.200 in our spam control, $\gamma=0$) shows MTD-style randomization *can* help when the threat model is stateless; the FL regime’s persistence is what eliminates the benefit.

Defense composition and ensembling. Composing defenses is well-studied in adversarial ML for centralized models: adversarial training combined with input purification (Croce et al., 2021), detection-then-rejection pipelines, and ensemble adversarial training (Tramèr et al., 2018). These works implicitly require that upstream processing does not destroy downstream signals—our signal-preservation criterion formalizes this requirement explicitly with measurable margins. In Byzantine-tolerant distributed systems, protocol composition (e.g., layered BFT) has a long history; the key distinction is that Byzantine agreement composes for fault tolerance among replicas, while our setting composes for poison resilience across client updates where the adversarial signal is a continuous gradient, not a binary fault. To our knowledge, no prior work provides a predictive criterion for which FL defense compositions succeed.

Value of information and security games. VoPD instantiates classical VOI (Howard, 1966); Blackwell (Blackwell, 1953) provides the general characterization. Security games (Tambe, 2011) use Stackelberg formulations (our setting is simultaneous). Empirical game theory (Wellman, 2006; Walsh et al., 2002) constructs payoff matrices via simulation—our methodology. Shejwalkar and Houmansadr (2021) motivate the information-asymmetric framing.

9 Limitations

(L1) Theorem 11 instantiates classical VOI. The underlying result is Howard 1966; our contribution is the algorithmic checkability, sample-complexity bound, and deployment relevance — not an independent theoretical advance.

(L2) The persistence heuristic is qualitative ($\sim 20\times$ slack). Out-of-sample MAE = 0.041 (NormClip) and 0.059 (NE3) conditional on admissibility (Appendix N.4.1).

(L3) Limited scope and architecture-conditional boundaries. Three attack families (model_scaling, pixel, DBA), two architectures (CifarCNN, ResNet18, plus `simple_cnn` on the replicated arm), and one dataset (CIFAR-10) for every composition suite but one. The collapse boundary is architecture-conditional (ResNet18 suppression at NC10/rep90: 0.865 vs. CifarCNN 0.097, a $10\times$ shift), so results at other N , K or datasets may differ. Cross-distribution transfer is validated across architecture, scale ($N \in \{10, 100\}$) and adversarial fraction ($f \in [0.05, 0.6]$; §4.4(D)), and *dataset* transfer is now tested for the flagship Mode-S arm alone—EMNIST-byclass with `simple_cnn`, where the negative replicates at $\Delta = -0.017$ (§5.2)—while remaining untested for the 42-pair screen and for every other composition suite: the signal-preservation conditions are dataset-agnostic in mechanism (cosine similarity, geometric distance), yet the downstream aggregator’s decision boundary may shift with output dimensionality. *What transfers:* the qualitative PASS/FAIL prediction, the mechanism, and the ordering (composition dominates mixing). *What does not:* the exact suppression ASR. A practitioner at $N=10^4$ should use the criterion to select candidates and then run pilot evaluations at their target scale—the criterion identifies *what to try*, architecture-specific calibration determines *how well it works*.

(L4) Selection bias. Rep30/tm70 is the arg-max of 189 configurations; the per-seed-max estimate +0.032 is winner-biased (54% shrinkage from the 5-seed pilot). FoolsGold+TM (rank 5 from the same scan; Appendix L) provides a within-scan out-of-family check: per-seed-max VoPD +0.082 ($p=0.007$). Because composition dominates in both cases, selection bias affects only the strength of the existence proof, not the negative main result.

(L5) The composability criterion is an evaluation-prioritization rule, not a security screen and not a discovery tool. It has no established predictive validity: it orders candidates for evaluation, and the pooled model that would let it predict is unconfirmed ($R^2=0.239$) with its cross-arm ordering refuted (§5.2). The criterion gives *sufficient* conditions

(C1 $\wedge$ C2, with C3 as a margin diagnostic): it reliably identifies mechanism-preserving compositions in the common case where one defense is known-effective, but it cannot certify synergistic compositions where no single defense suppresses the attack alone. The conservative bias is by design—false negatives are operationally cheap to recover from (try the composition anyway) while false positives would deploy an insecure stack. Accuracy is 16/18 = 89% on the development set, 22/24 = 92% on the pre-registered held-out set (90.5% combined 2-way) and 9/10 = 90% on pre-registered 3-way triples (§4.5). All clean C1 $\wedge$ C2 $\wedge$ C3 PASS predictions are correct; the four misses are either the predicted intermediate behavior of rank-based aggregators (Proposition 7: dev-set NC $\rightarrow$ cm, rfa $\rightarrow$ cm; held-out fedavg $\rightarrow$ cm) or one speculative LOW call the criterion does not license (held-out fg $\rightarrow$ NC, a C1 failure). Formalization is partial: Theorem 6 proves C2 $\wedge$ C3 for the weighting $\rightarrow$ rank/geometric class with explicit margins, the remaining families are empirically validated, and C3’s threshold condition depends on d_2 ’s discrimination boundary, which in practice requires calibration runs.

(L6) Per-round aggregator switching only. We test per-round defense selection. Randomized *internals* (DP noise, subsampling smoothing à la CRFL, random participation masks) modify the embedding mechanism directly and may retain value even under persistence. Three adaptive attacks are tested against composition: projection-based (best ASR 0.597 at $\epsilon=1$), Neurotoxin-style (Zhang et al., 2022) top- k gradient projection, and a two-phase consensus-shifting attack against rep+tm that biases updates toward the target at rate r for 25 rounds before injecting full backdoor updates (§7). At shift_rate $\in \{0.01, 0.05\}$, $n=10$ each: mean ASR 0.684 ± 0.138 and 0.658 ± 0.144 , comparable to committed_pixel (0.679) and below the temporal mix (0.915). At $f=0.2$ the honest 80% anchor the coordinate-wise median, limiting Phase-1 drift to $\sim r/(1-f)$ per round, so Phase 2 degenerates to standard committed_pixel—a mechanism that weakens monotonically with f , and at $f \leq 0.05$ over $T \geq 200$ rounds cumulative drift may eventually overcome it. The robustness claim is therefore specific to the tested regime ($f=0.2$, $T=50$).

(L7) No falsifiable prospective positive evidence, and C0 was added retrospectively. This is the sharpest limitation we can state about our own design, and we state it rather than let a reader find it. Of the prospective suite’s 8 predicted-LOW pairs (§4.1), 6 are $\cdot \rightarrow$ FLTrust compositions that are near-*identities*: FLTrust gates on a scale-invariant cosine and renormalizes every update to the server-gradient norm, discarding exactly the magnitude information an upstream positive rescaling modifies (verified bit-identical per seed for pass-through and non-clipping d_1). Proposition 8(a) permits no other outcome, so these confirmations cannot falsify it; the remaining 2 have FLTrust as d_1 , where suppression is inherited and C0 puts them out of scope. Two further caveats: C0 was formulated *after* seeing this suite, so its 7 exclusions are retrospective rather than pre-registered; and the three matched contrasts that most directly probe Proposition 8 (a 19–20 $\times$ ASR gap between $\cdot \rightarrow$ FLTrust and $\cdot \rightarrow$ Krum at fixed d_1) confound C1 with C2, because in our menu the invariant-signal d_2 is also the only d_2 effective alone. We had named the decisive experiment as a downstream defense with a rescaling-invariant admission signal that is *weak in isolation*; Lemma 1 shows no such pair can be certified in scope, so that experiment does not exist—the confounding is structural, not a gap in our menu. §4.2 runs the achievable substitute, holding the suppression *principle* fixed while varying only the invariance class, and it did not deliver a positive test either: standalone effectiveness could not be held fixed, because for this defense family the magnitude information the mirrors discard is the same information the originals use to suppress a magnitude attack (8/8 frozen labels, TP = 0). So after two pre-registered attempts the criterion’s positive direction still rests on in-distribution and out-of-sample PASS evidence, and the strongest C2-specific evidence in the paper is the *post-hoc* within-defense contrast of §4.2, which holds C1 exactly fixed but was not pre-registered and is statistically resolved on only one of its two cells at $n=3$. §5 is the pre-registered design that answers this, and it delivers exactly one falsifiable prospective positive result—arm 2, reputation under model-scaling, $0.017 \rightarrow 0.779$ across the dose at $p = 1.7 \times 10^{-6}$ with clean accuracy nearly unchanged—while refuting two of its own four predictions and its pooled prediction. So the honest state of the positive direction after the ladder is: one confirmed arm, on the *destruction* half of C2’s mechanism, with the preservation half’s designated control indeterminate (§5.1). The targeted intervention of §5.2 advances this without closing it: closing the attenuation channel by construction, it isolates C2 within a single defense—substantial disturbance of the statistic does not produce a corresponding change in suppression *on the adjudicating Krum arm under Mode S*, with *admission* of adversarial input the channel the evidence points to there (H-statistic refuted, H-admission confirmed on that arm, pooled model not confirmed)—and it *refutes* the preservation-half reading the ladder left open, since the exactly-invariant $\cos_krum$ neither suppresses nor holds ASR fixed. A third pre-registered attempt (§5.2) was designed to be the falsifiable prospective *positive* test: a replication arm chosen because the statistic reading and the admission reading order its rise *oppositely*, with the ordering frozen in git before the arm was run. It replicated the negative on a defense in a different invariance class and **refuted the positive ordering**—the arm with the largest measured admission change rose *less* (+0.098) than an arm whose admission change was smaller. So what remains absent is still a validated *positive* predictor: after that replication the pooled two-channel model is not confirmed over

26 cells ($R^2 = 0.239$) and its cross-arm ordering is refuted outright, so the criterion’s positive direction is unestablished as a screen, and admission is a better *direction* than the statistic without being a quantitative one.

(L8) Theorem 6(1)’s condition is not operationally checkable as stated. Regime A’s ordering-survival condition is $\rho < S$ with S the attack’s coordinate extremeness against the benign scale. Building the dose–response ladder (§5) required us to *measure* S rather than stipulate it, and it is not a stable per-attack constant: on model-scaling it ranges 0.00–12.59 across the 12 measured rounds that sampled an adversary (median 0.57), on the pixel backdoor 0.61–1.63 (median 0.92). The $S=10$ of §3.2 is an illustrative stipulation, not a measurement. Two consequences we state rather than work around. First, a practitioner cannot check $\rho < S$ from a single round: the quantity swings by orders of magnitude round to round, and on both committed attacks its median is below 1, so Regime A’s condition fails even at the identity rung under this estimator. Second, we therefore built *no* prediction in §5 on S , resting the ladder on directly measured disturbance of d_2 ’s statistic instead. The margin form of the theorem remains valid as a statement about ρ and S ; what is not established is that S admits a stable estimator at the round granularity a deployment would have to check it at, and a usable Regime A needs an S estimator with characterized variance, which we do not provide.

(L9) Measured disturbance is not a transferable predictor, and our dose instrument varies the wrong quantity for three of its four arms. The pre-registered ladder (§5.1) refutes the pooled claim it was built to test ($\rho_s = +0.351$, $p = 0.091$) and refutes two of four per-arm shapes, with ASR *falling* as the dose rises in three arms. Two distinct faults, both ours. First, the abscissa: at matched measured disturbance the real upstream transforms sit up to +0.79 ASR above the synthetic curve, and Krum’s within-margin arm does not reproduce the very contrast (§4.2) we attributed to disturbance of its statistic, so the rate at which a transform flips a defense’s decision is not a sufficient statistic for what the transform does—which clients are reweighted matters, and ρ does not measure that. We now measure the quantity that does, Δ_c (Appendix H): the two anchors that miss are the two most extremely targeted (−1.21, −0.72 against the ladder’s −0.34) at ρ that is 3.7 and 13× smaller. That names the axis of the mismatch without supplying a replacement predictor— Δ_c ’s magnitude does not order suppression either. Second, the instrument: mean-1 normalization fixes the aggregate’s scale but not any client’s relative weight, so with the dose assigned independently of adversary status a large ρ mostly dilutes whichever client carries the poison (measured: the ladder’s Δ_c drifts monotonically negative with the dose, $0.000 \rightarrow -0.140 \rightarrow -0.240 \rightarrow -0.336$), and three arms end up measuring Theorem 6’s attenuation regime rather than C2 disturbance. The adversary-targeted dose that separates the two, recorded as not run before the freeze (caveat (iii), Appendix D), is run in §5.2—and in retrospect Mode S is the Δ_c -**pinned** instrument (Δ_c exactly 0 at every rung, ρ up to 54.6) and Mode A the Δ_c -**swept** one (−1.14 to +2.47): with every adversary’s weight pinned, Krum’s statistic is disturbed in up to 80% of rounds while the admission of adversarial input does not move and ASR stays inside the pre-specified equivalence margin—so *which* clients are reweighted, not how often a decision flips, is indeed the operative variable, and the ladder confound is *explained* (reputation’s rise shrinks from +0.762 to +0.178 once the adversarial share is held fixed). That isolates C2’s causal content within a single defense; it does not rescue the pooled positive direction ($R^2 = 0.239$ over 26 cells, cross-arm ordering refuted by the replication).

10 Conclusion

The result of this paper is a causal one, not a screening rule. With the downstream defense, the attack, and standalone effectiveness all held fixed, and with the intervention applied *directly* to the statistic that defense reads, changing that statistic can have little or no effect on suppression—and in the same cell the effect **reverses sign** once adversarial influence is unpinned and allowed to co-vary with it (−0.272 against +0.098). So positive per-client rescaling admits invariance-based guarantees (Proposition 8) sufficient for preserving the *downstream discriminative mechanism*, never for low ASR; and *any* criterion gated on the outcome it predicts carries a **structural identification boundary** (Proposition 4, with Lemma 1 its FL instance) that no amount of data removes. **The boundary is general where our empirical claims are not**, and what we offer as transferable is the **admissible design** it derives, which §5.2 runs.

The three central empirical findings. (i) *The negative itself*: with the attenuation channel closed by construction, disturbing the statistic Krum reads in up to 80% of rounds moves neither the adversarial mass admitted (0.000) nor ASR, and a pre-registered replication reproduces it in a different invariance class and on a second dataset and architecture (§5.2). (ii) *The sign reversal*: on the *identical* cell the outcome-gated ladder *falls* 0.272 ($p_{\downarrow}=0.0015$) where the instrument that pins adversarial influence *rises* +0.098—a sign inversion under a change of *intervention*, not of data, which is the identification boundary of Proposition 4 appearing as a measured effect. (iii) *Heterogeneity*

closes the theorem’s margin: Corollary 1’s bound is a function of the realized margin m alone, and at Dirichlet $\alpha=0.1$ RFA’s all-round mean margin collapses from $m=3.24$ to $m=1.01$ ($4.71 \rightarrow 2.06$ over payload-carrying rounds)—the limit at which the corollary certifies nothing—with two of three certified compositions crossing the threshold there (§4.4(E)). Here the invariance theory earns explanatory value in *both* directions: it predicts a number that holds where heterogeneity is moderate (12.6% bound against a realized $\leq 1.8\%$) and its own quantity records where it stops holding.

Sequential defense composition cannot be reliably screened by asking whether an upstream transform preserves a downstream *statistic* alone: of Definition 1’s five preservation levels the chain holds only down to (P3), and below that it breaks in *both* directions. **The useful object to screen is not statistic invariance itself, but whether the upstream transform changes the adversarial contribution that survives the downstream defense—level (P4).** What the paper contributes is that mismatch, made formal and then measured: a hierarchy of preservation levels, an **identification boundary** showing that *any* screening criterion whose eligibility condition is defined on the outcome it predicts creates the same confound whatever the domain, and the within-defense interventions that boundary licenses. **Mechanism-preserving screening**—checkable sufficient conditions, $C1 \wedge C2$ with $C3$ retained as a margin diagnostic, for *preservation of the downstream discriminative mechanism* and never for low ASR, which they do not imply—is what we build on top of that boundary and is the paper’s *application*, not its claim.

What is proved, what is empirically supported, and what is not established. **Proved:** the invariance classes of Proposition 8; sufficient conditions for mechanism preservation on the weighting→aggregator class (Theorem 6, Lemma 2, Corollary 1); the identification boundary for outcome-gated screening (Proposition 4, with Lemma 1 its FL instance); the recall bound on inherited-suppression screens (Proposition 5); and, formalizing the interventions, statistic-level non-identifiability (Proposition 9) and the independent controllability of ρ and Δ_c (Proposition 10). **The two boundary results are general**—they are statements about the shape of a screen, not about FL—so what carries beyond this paper is the design they license, not the screen. **Empirically supported:** that substantial disturbance of the downstream statistic need not produce a corresponding change in suppression when the adversarial coefficient share is held fixed, with adversarial admission the channel the evidence favours, *on the adjudicating Krum arm under Mode S* and, by pre-registered replication, both on a downstream defense in a *different* invariance class and on a *second dataset and architecture* (§5.2); and that the screen behaves as a conservative prioritizer across 42 pairs. **Not established:** the positive predictive direction of the criterion—the pooled two-channel model is unconfirmed over 26 cells ($R^2=0.239$) and the replication refutes its cross-arm admission ordering, so admission is a direction and not a quantitative predictor; any ability to discover emergent synergy; and generalization beyond star-topology FL, or beyond CIFAR-10 for anything but the one arm replicated on EMNIST-byclass.

What the screen is for. It **trades discovery for reliability:** asking whether d_2 ’s suppression *survives* d_1 certifies only **inherited** robustness, and **Proposition 5 proves it cannot find compositions that suppress because neither constituent does**— $FG \rightarrow RFA$ (0.093) is exactly such a case and remains a false negative of our own rule, and the bound binds at the 1/5 that gives the reported 40% recall. That is a scope theorem rather than a defect discovered, and it is specific neither to $C1$ nor to federated learning. We also do not offer the screen’s precision as validation: it rests on three positive predictions under the frozen labels—two under the tightened per-attack $C1$ —in a badly imbalanced evaluation. What survives validation is a **conservative evaluation-prioritization rule for inherited robustness, not a predictor of emergent synergy:** its output answers *which compositions are worth evaluating first*, never which to deploy. The criterion is partially formal—Theorem 6 proves $C2 \wedge C3$ for the weighting→aggregator class with explicit margins, so that class needs no calibration runs—and extensively *scored*: 90.5% on 42 two-way pairs (24 pre-registered and held out), 90% on 10 pre-registered triples drawn from 210 candidates, and better than 99.99% of random selections ($p < 10^{-4}$). A prospective suite of 20 pairs on three previously unused defenses, frozen in git before any run, scores 13/20 and 11/13 in scope, exposed the scope condition $C0$, and supplied no falsifiable prospective *positive* test. A second pre-registered suite built to fix that did not supply one either: its own pre-freeze baselines showed that for this defense family the magnitude information an invariant mirror discards is what the original uses to suppress a magnitude attack, so **the confound re-materialized inside the experiment designed to remove it** (8/8 frozen labels, $TP = 0$). We name that obstacle **mechanism–effectiveness confounding** and show via Lemma 1 that our own conditions force it—a **testability boundary** we state in general form (§3.1): any composition criterion whose screening conditions include a standalone-effectiveness gate on the same outcome metric it predicts admits only within-defense validation designs. Read as a derivation rather than an obstacle, the boundary specifies the experiment—hold d_2 and the attack fixed so the $C1$ input is numerically identical on both sides, and vary only measured disturbance of d_2 ’s statistic. Post-hoc it already separates preserved from destroyed suppression ($\Delta \in [+0.001, +0.190]$ vs. $[+0.718, +0.878]$ at $n=5$), Krum under

model-scaling moving from 0.062 to 0.779 as its selection goes from never disturbed to disturbed in 6/9 rounds (Welch $p = 0.012$); prospectively, §5 runs it as a pre-registered dose–response ladder over ρ up to 54.6 with independently measured disturbance—and **that ladder largely refutes us**: the pooled prediction fails ($\rho_s = +0.351$, $p = 0.091$), two of four frozen shapes are refuted, ASR falls with dose in three arms, and at matched measured disturbance the real transforms sit up to 0.79 above the synthetic curve. One arm confirms (reputation $0.017 \rightarrow 0.779$, $p = 1.7 \times 10^{-6}$, accuracy $0.78 \rightarrow 0.70$), covering the destruction half alone—and the targeted suite later shows most of that rise was attenuation rather than statistic disturbance (§5.2). So measured disturbance of a defense’s statistic is *not* a transferable predictor of whether composition preserves suppression, and *which* clients an upstream transform reweights—not how often it flips a decision—is where the mechanism must be looked for (L9). A targeted intervention does exactly that (§5.2): with the attenuation channel closed by construction, Krum’s statistic is disturbed in up to 80% of rounds while its *admission* of adversarial input and its ASR do not move, showing that substantial disturbance of the statistic need not produce a corresponding change in suppression and that admission is the channel the evidence points to *on that adjudicating Krum arm under Mode S*—the controlled within-defense test of C2 our cross-defense suites could not supply. A pre-registered replication on a downstream defense in a *different* invariance class, on an arm the two readings order oppositely, reproduces that negative and **refutes the positive ordering**, leaving the positive direction unvalidated as a screen (pooled over 26 cells, $R^2 = 0.239$). What the screen bounds throughout is **inherited** robustness under a *committed* attack; a screening condition stated against an *adaptive* adversary is the sharpest open direction the framework leaves. Among evaluated compositions the criterion identifies an accuracy–security Pareto frontier: FG→CM (0.107 / 0.68) as the best candidate for further evaluation, FG→RFA (0.093 / 0.50) as the lowest-observed-ASR extreme though not separably so, and Rep→CM (0.423 / 0.76) for near-baseline accuracy at moderate security (Table 13).

Deterministic composition outperforms temporal mixing (per-round defense switching) on every tested menu and adversary policy in the persistent-backdoor regime. The mechanism: once a trigger embeds ($\gamma \approx 0.66$), intermittent suppression cannot clear it, making mixes converge to their worst constituent’s ASR. Committed strategies are near-optimal (largest switching gain $+0.044$, below bound 0.048).

The regime boundary’s quantitative location is architecture-conditional (ResNet18 shifts thresholds by $\sim 10\times$); extrapolation to cross-device FL ($N \geq 10^4$, $f \leq 0.01$) requires additional experiments. The qualitative predictions—PASS/FAIL classification, composition dominance, committed-strategy near-optimality—generalize across architecture (CNN→ResNet18), scale ($N=10 \rightarrow 100$), adversarial fraction ($f \in [0.05, 0.6]$), and attack family (model_scaling/pixel→DBA). **Heterogeneity is the exception and we state it here rather than only in the sweep**: at $\alpha=0.1$ two of the three certified compositions cross the threshold, and the failure appears in the theorem’s own quantity (RFA’s separation margin collapsing to $m \approx 1$; §4.4(E)).

What to do instead: six steps for any composition claim. This is the admissible design of §3.1 in operational form, and it is the practical contribution rather than the criterion. (1) Name the downstream mechanism and its invariance class. (2) State *which* preservation level you are claiming, (P1)–(P5) of Definition 1; a claim that does not name its level is not checkable. (3) Check that the upstream transform lies in the class your argument actually covers—C0 here (Remark 3); one that alters participation, carries cross-round state, or mixes clients is covered by no invariance argument we or anyone else has. (4) Hold d_2 , the attack, and therefore the gate’s input fixed, and **do not select candidates on the outcome**—this is the cell Proposition 4 shows is otherwise empty. (5) Intervene on the upstream transform and **measure the channel**—aggregate displacement, decision, admission, influence—rather than the outcome-conditional contrast. (6) Report the level at which the chain broke. Steps (4)–(5) are exactly what Proposition 4 licenses, and following them turned our own -0.272 into $+0.098$ on an identical cell. **One boundary the recipe does not cross**: under *one* whitebox composition-aware *adaptive* stress test FG→RFA rises to $\text{ASR} = 0.167 \pm 0.069$, $3.3\times$ its seed-matched committed 0.051, so **an inherited-robustness guarantee is not adaptive robustness** (§6). That is one adaptive test on one composition and we do not generalize it; a screening condition stated against an adaptive adversary is the sharpest thing the framework lacks.

The negative is the result. Of Definition 1’s five preservation levels the chain holds only down to (P3); below that it breaks in *both* directions, each with a measured witness: an exactly-invariant defense whose selection never moves while its ASR falls 0.173 by its first gate-passing rung, and a defense whose selection flips in up to 80% of rounds with its ASR stays inside the pre-specified equivalence margin ($|\Delta|=0.026 < 0.15$). Our own statistic-level hypothesis is refuted pooled, and the admission reading that replaced it survives as a *direction*, not a predictor: the arm we pre-registered to order it refuted that ordering. **So the object worth screening is not whether an upstream transform preserves a downstream statistic. It is level (P4), the adversarial contribution that survives the downstream**

defense. Why we cannot simply screen for it is worth stating precisely, because it is the question this paper leaves open. Mode S *does* measure (P4) directly—that is what makes it an identification instrument—but it computes the adversarial share by reading adversary identity, which no deployed defense has. So the experiments locate the causal channel and **do not** constitute a deployable (P4) screen. **The open problem is whether (P4) can be estimated without knowing which clients are malicious:** whether some observable functional of the transformed updates bounds the adversarial mass a downstream aggregator admits. A screen at (P4) would inherit this paper’s negative result as its justification; a proof that no such estimator exists would be a stronger negative than the one we report.

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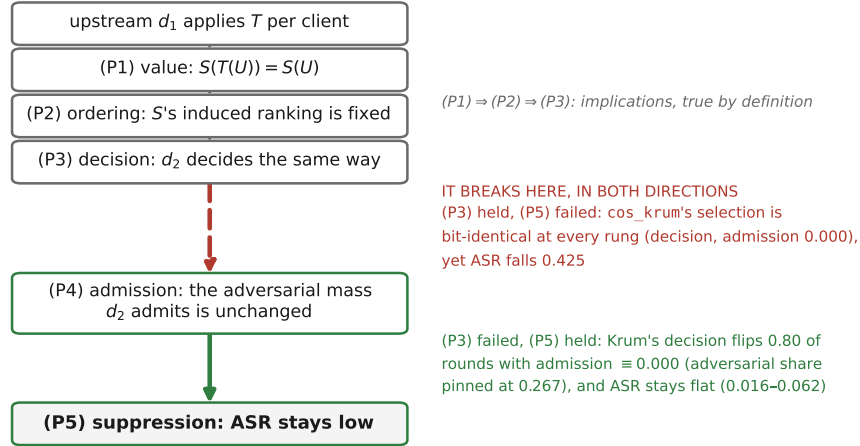
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A The preservation chain in one figure

B Dose–response design: the full pre-registered protocol

Four properties make this a dial on the theory rather than an arbitrary perturbation. (i) It is exactly Proposition 8’s form $T(u_i) = c_i u_i$ with $c_i > 0$, so every rung lies inside the theory’s scope. (ii) **The dose is ρ** : $\max_j c_j / \min_j c_j = e^{2\kappa}$ exactly, the same upstream weight ratio Theorem 6 and both mechanism-preserving regimes are stated in—verified by reading ρ back off the *transformed update norms* on live rounds, where realized ρ matches nominal to three decimals at every rung and $\text{mean}(c)$ stays in $[0.99999, 1.00002]$. (iii) $\kappa=0$ gives $c \equiv 1$: the implementation short-circuits, so the identity rung is bit-identical to d_2 standalone by construction rather than by rounding. (iv) Normalizing the *mean* to 1 holds the aggregate’s scale fixed, separating the quantity under test (disturbance of d_2 ’s *statistic*, i.e. C2) from the magnitude-contraction channel of Lemma 2 that costs accuracy; without it the ladder would confound C2 with attenuation. The permutation is keyed on (seed, round) and **not** on client identity: assigning by id would place the adversaries at a fixed end of the ladder and systematically attenuate them, manufacturing a success, whereas this keying gives every arm at a given seed and round the *identical* coefficient vector, so the arms differ in d_2 and nothing else. **dose_kappa $\langle\kappa\rangle$ is an instrument, not a defense**, and is never evaluated or recommended as one.



Statistic preservation is not security preservation: only the (P4) $\rightarrow$ (P5) link carries security content in our experiments.

Figure 7: **The preservation chain, and where it breaks.** (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) holds by definition; **the (P3)–(P4) link breaks in both directions**, and both witnesses come from the oracle intervention that pins every adversary’s coefficient and so closes Lemma 2’s attenuation channel *by construction* (§5.2). The pooled statistic-level prediction we had frozen is separately **rejected** by a pre-registered ladder ($\rho_s = +0.351$, one-sided $p = 0.091$, 16 cells; §5), and a pre-registered replication in a different invariance class reproduces the negative but **refutes the cross-arm admission ordering** ($\Delta = +0.098$; the pooled two-channel model stays unconfirmed, $R^2 = 0.239$ over 26 cells), so **admission is a direction, not a validated predictor**. Every number in the figure and in this caption is recomputed from the scoring scripts’ own JSON and printed by `story_chain_figure.py`.

Frozen grid and arms. $\kappa \in \{0, 0.5, 1.0, 2.0\}$, i.e. $\rho \in \{1.00, 2.72, 7.39, 54.60\}$: four rungs spanning a $\approx 55\times$ ratio. Arms are selected by the paper’s own power rule (§4.2), fixed before the freeze: the ladder is informative only where standalone d_2 genuinely suppresses the attack (ASR < 0.5 at clean accuracy ≥ 0.35), since a baseline already above threshold has no suppression left to lose and would give a flat curve by ceiling effect. Applying that rule to the measured baselines leaves exactly four (d_2 , attack) cells, and they happen to cover three invariance classes of Proposition 8 (Table 8). The $\kappa=0$ rung is run *fresh* at all five seeds for every arm rather than reused from the published baselines, so every rung of every curve comes from one code path at one seed set; the published baselines then serve as an independent per-seed cross-check of that rung (tolerance 10^{-6}) instead of as data in it.

The abscissa is measured, not nominal. What matters to C2 is not ρ but what the rescaling does to the specific statistic d_2 reads, so we measure that separately and *before* freezing any prediction, pushing live FL updates through the shipped transform and recording each arm’s decision at every κ on the *same* raw updates (`measure_dose_disturbance.py`, `results/dose_disturbance.json`; 5 seeds $\times$ 3 rounds per attack). No ASR is computed there, which is why it can legitimately precede the freeze. It establishes three things before any outcome exists (Table 8): *there is a dose*—disturbance is exactly 0 at the identity rung and rises monotonically with κ in every non-invariant arm; *the class-(a) invariance claim holds in the shipped code*—`cos_krum`’s selection is unchanged in 0 of 120 cells at ρ up to 54.6, its cosine *scores* moving by at most 1.1×10^{-4} relative, the float32 floor for a $\approx 1.1\text{M}$ -dimensional vector, which we record rather than assert on; and *class (b) is graded rather than binary*, at roughly half the class-(c) disturbance (0.487 vs. 0.867/0.933 at $\kappa=2.0$), which is what arm 4’s prediction turns on.

Frozen predictions. The *primary* test is pooled: across all 16 (arm, κ) cells, the rise in ASR over that arm’s own $\kappa=0$ rung is predicted to be positively rank-correlated with the measured disturbance rate (Spearman, one-sided, $\alpha = 0.05$). This is the dose–response claim the design exists for, and its predictor is a *measured property of the statistic*—not the identity of the defense and not its standalone effectiveness, both held fixed within every arm. The *secondary* predictions are the per-arm shapes of Table 8, tested by a one-sided Jonckheere–Terpstra trend test across the four ordered rungs at $n=5$. For arm 3, “flat” is scored as an *equivalence* claim rather than a failed rejection: confirmed if $|\overline{\text{ASR}}(\kappa=2.0) - \overline{\text{ASR}}(\kappa=0)| < 0.15$ and every rung stays below 0.5. That margin is fixed and justified in advance: it is under a quarter of the +0.629 within-defense gap for Krum on record at the freeze (the post-hoc

top-up later put that gap at $+0.716$ at $n=5$, which only makes the margin more conservative—it is not revised), and $0.300 + 0.15 = 0.450 < 0.5$, so “within the margin” and “still suppressing” coincide by design for this arm. Arm 4’s class-(b) signature is the *shallower slope*, so its criterion includes its $\kappa=2.0$ rise being smaller than both class-(c) arms’.

C Out-of-sample PASS validation: per-cell detail

(A) DBA attack transfer. The Distributed Backdoor Attack (Xie et al., 2020)—a categorically different attack family using spatially-distributed sub-triggers at four image corners—is not in the development menu ($f=0.4$ required for viability at $K=5$). We test three PASS-predicted compositions ($fg \rightarrow rfa$, $fg \rightarrow cm$, $rep \rightarrow cm$) and three FAIL controls ($nc \rightarrow rep$, $rfa \rightarrow rep$, $fg \rightarrow tm$) under DBA at $f=0.4$, $N=10$, $n=5$ seeds. Results: all three PASS compositions robustly suppress DBA ($fg \rightarrow rfa$: 0.109 ± 0.065 ; $fg \rightarrow cm$: 0.050 ± 0.027 ; $rep \rightarrow cm$: 0.041 ± 0.014). The FAIL controls show weaker discrimination than under `model_scaling/pixel`: $nc \rightarrow rep$ 0.297 ± 0.224 , $rfa \rightarrow rep$ 0.297 ± 0.159 , $fg \rightarrow tm$ 0.049 ± 0.031 . This reflects DBA’s inherent variability at $f=0.4/K=5$ (sanity baseline vs. `fedavg`: 0.236 ± 0.190 , bimodal across seeds). Nevertheless, the distributions separate at the tails: max per-seed ASR across all PASS pairs = 0.185 , while FAIL pairs reach 0.712 (Appendix K).

Powered re-test at $f=0.6$. To address the power limitation (sanity baseline 0.236 , never clearing 0.5 , making FAIL discrimination unavailable), we re-ran all six pairs under DBA at $f=0.6$ ($K \cdot f=3.0$; $n=5$ seeds). PASS predictions confirmed: $fg \rightarrow rfa$ 0.125 ± 0.050 , $fg \rightarrow cm$ 0.045 ± 0.021 , $rep \rightarrow cm$ 0.043 ± 0.015 . FAIL controls: $nc \rightarrow rep$ 0.378 ± 0.185 , $rfa \rightarrow rep$ 0.384 ± 0.217 , $fg \rightarrow tm$ 0.068 ± 0.018 . Sanity baseline (DBA vs. `fedavg`, no defense composition): 0.386 ± 0.192 —still below 0.5 due to DBA’s distributed co-sampling bottleneck (Appendix P.5). FAIL discrimination therefore remains unavailable even at $f=0.6$: DBA does not reliably embed at this N/K regardless of defense. The symmetric verdict: criterion PASS predictions are confirmed robustly at both adversarial fractions; FAIL-side falsifiability requires a DBA variant that embeds reliably before those controls are meaningful.

(B) Scale transfer ($N=100$, ResNet18). We test $fg \rightarrow rfa$ and $fg \rightarrow cm$ at $N=100$, $K=10$, $f=0.05$, ResNet18—a $10 \times$ client-pool expansion with a different architecture—having pre-registered both as PASS under `committed_scaling` and `committed_pixel`. At $n=5$ seeds: $fg \rightarrow rfa$ max ASR = 0.069 ± 0.029 , $fg \rightarrow cm$ = 0.152 ± 0.118 . **All four conditions confirmed PASS.** We flag that these runs reach only ≈ 0.29 clean accuracy at 50 rounds—ResNet18 is far from converged at this budget—so they establish that *suppression transfers* across scale and architecture, not that the composition is deployable at $N=100$ as configured. The single-defense comparison sharpens the picture: RFA alone reaches `committed_scaling` ASR 0.410 ± 0.386 (bimodal; some seeds embed, others do not) against $fg \rightarrow rfa$ ’s 0.069 ± 0.031 , a $6 \times$ improvement at $N=100$. Two FAIL-predicted pairs at the same scale ($nc \rightarrow rep$ C2-FAIL, $fg \rightarrow tm$ C3-FAIL; $n=5$) came out at 0.279 and 0.167 —below 0.5 , mirroring the DBA power limitation: at $f=0.05$ committed attacks do not reliably embed regardless of defense, so FAIL-side discrimination is unavailable at this scale/fraction combination. The positive result is the substantive finding.

(C) Independent survivor ($fg \rightarrow cm$). We pre-registered that $fg \rightarrow cm$ —selected *solely* by criterion PASS status, *independent of the 189-config payoff-matrix scan* that selected $rep \rightarrow tm$ —dominates temporal mixing of fg/cm . This is independence from the survivor scan, not full data-independence: $fg \rightarrow cm$ also appears in the development set (Table 3, 0.107). Two conditions ($n=5$): composed (always $fg \rightarrow cm$) vs. mixed (50/50 random per round). We anchor the headline on the **committed-pixel** policy, where both conditions retain a usable model (clean accuracy 0.66 composed, 0.77 mixed): composed ASR = 0.099 ± 0.020 vs. mixed = 0.679 ± 0.159 , a $6.9 \times$ suppression. On `committed_scaling` the mixed model is *destroyed* (clean accuracy $0.11 \approx$ CIFAR-10 chance), so its ASR is not a meaningful backdoor comparison—no operator would deploy an accuracy-collapsing policy—and we exclude it from the headline. The clean committed-pixel comparison alone establishes composition dominance on a scan-independent menu.

D Dose-response caveats recorded before the freeze

Caveats recorded in advance. (i) Arms 1–2 are on model-scaling and arms 3–4 on the pixel backdoor. This is not a free choice: on model-scaling *no* invariant d_2 suppresses at usable accuracy—that is the finding of §4.2 (`cos_krum` 0.078 at 0.146 accuracy)—which is also why `cos_krum/scaling` is not an arm. “Same attack” is required *within* each ladder, which is where the causal claim lives; across arms we compare curve *shapes and slopes*, never levels. (ii) Arm 4 has thin headroom: `coord_median/pixel` sits only 0.057 below threshold, so the binary above/below- 0.5 reading is weak evidence either way and we report the continuous curve with intervals. (iii) The dose is uncorrelated with adversary status by construction, but it is not adversary-*adversarial*: a dose assignment that deliberately up-weights

adversaries would be a strictly harder test and is not run here. (iv) A synthetic upstream transform is not a real defense, so we test ecological validity by overlaying the two *real* upstream transforms already measured on these same cells (NormClip and RFA, §4.2) as anchor points at their own measured disturbance; if they fall far off the synthetic curve at matched disturbance, that is a finding about the instrument and is reported as one. (v) Disturbance is measured on FedAvg trajectories on one shared set of raw updates per round—which is what makes it a clean abscissa, but means the rates in Table 8 are not re-measured along each arm’s own 50-round trajectory.

E Dose–response ladder: per-seed detail, intervals, and the anchor residuals

Every value below is recomputed per seed by `analyze_dose_response.py` from the ladder and disturbance JSON files; nothing is transcribed. Seeds are 42–46 in order; Table 14 carries the per-rung detail.

Table 14: Per-rung mean ASR with Student-t 95% intervals at $n=5$, the measured disturbance of that arm’s statistic, mean clean accuracy, and the per-seed values. [†] fails the 0.35 accuracy gate.

arm	κ (ρ)	disturb.	mean ASR [95% CI]	acc	per-seed ASR
1 <code>krum</code> / scaling	0 (1.00)	0.000	0.062 [0.019, 0.105]	0.565	0.115, 0.021, 0.047, 0.057, 0.069
	0.5 (2.72)	0.600	0.011 [−0.001, 0.023]	0.652	0.000, 0.010, 0.011, 0.027, 0.007
	1.0 (7.39)	0.733	0.014 [0.003, 0.024]	0.682	0.003, 0.015, 0.021, 0.021, 0.008
	2.0 (54.6)	0.867	0.054 [0.002, 0.106]	0.619	0.005, 0.039, 0.077, 0.113, 0.037
2 <code>reputation</code> / scaling	0 (1.00)	0.000	0.017 [0.012, 0.022]	0.777	0.014, 0.012, 0.017, 0.023, 0.019
	0.5 (2.72)	0.867	0.022 [0.014, 0.031]	0.779	0.016, 0.018, 0.024, 0.033, 0.020
	1.0 (7.39)	0.867	0.487 [0.349, 0.624]	0.765	0.439, 0.547, 0.578, 0.557, 0.313
	2.0 (54.6)	0.933	0.779 [0.673, 0.885]	0.701	0.710, 0.843, 0.730, 0.896, 0.714
3 <code>cos_krum</code> / pixel	0 (1.00)	0.000	0.490 [0.058, 0.922]	0.607	0.003, 0.269, 0.630, 0.864, 0.686
	0.5 (2.72)	0.000	0.450 [0.130, 0.769]	0.569	0.764, 0.236, 0.165, 0.648, 0.435
	1.0 (7.39)	0.000	0.153 [−0.127, 0.433]	0.466	0.132, 0.090, 0.000, 0.543, 0.001
	2.0 (54.6)	0.000	0.086 [−0.100, 0.271]	0.220 [†]	0.083, 0.000, 0.000, 0.345, 0.000
4 <code>coord_median</code> / pixel	0 (1.00)	0.000	0.443 [0.263, 0.622]	0.767	0.395, 0.287, 0.449, 0.677, 0.405
	0.5 (2.72)	0.251	0.428 [0.241, 0.615]	0.764	0.454, 0.215, 0.430, 0.638, 0.402
	1.0 (7.39)	0.378	0.327 [0.181, 0.472]	0.749	0.351, 0.139, 0.352, 0.463, 0.328
	2.0 (54.6)	0.487	0.170 [0.113, 0.228]	0.690	0.179, 0.090, 0.204, 0.192, 0.187

The primary test, and one post-hoc sensitivity we report but do not lean on. Pooled over all 16 cells, Spearman $\rho_s = +0.351$, one-sided $p = 0.091$ (the frozen test; refuted). Dropping the four $\kappa=0$ cells—which sit at (0, 0) by construction rather than by measurement—raises this to $\rho_s = +0.780$, $p = 0.001$ over 12 cells. We were not entitled to that analysis in advance and it does not change the verdict, and its sign is in any case driven by the *invariant* arm’s ASR falling at zero disturbance being paired with the reputation arm’s rising at high disturbance, not by the fragility pattern the framework claims: arms 1 and 4 contradict it. Against the *nominal* dose κ (equivalently $\log \rho$) instead of measured disturbance, the pooled correlation is $\rho_s = -0.354$. Neither predictor works pooled.

Per-arm trend tests. Jonckheere–Terpstra across the four ordered rungs $\times$ 5 seeds, one-sided for increasing trend (normal approximation; a 20,000-draw permutation p-value with a fixed seed is reported alongside by the analysis script and agrees to the third decimal in every arm): arm 1 $J = 70.0$, $z = -0.34$, $p_{\uparrow} = 0.63$; arm 2 $J = 144.0$, $z = +4.64$, $p_{\uparrow} = 1.7 \times 10^{-6}$; arm 3 $J = 29.0$, $z = -3.10$, $p_{\uparrow} = 0.999$ ($p_{\downarrow} = 9.8 \times 10^{-4}$; dropping the gate-failing $\kappa=2.0$ rung leaves $z = -2.06$, $p_{\downarrow} = 0.020$); arm 4 $J = 31.0$, $z = -2.96$, $p_{\uparrow} = 0.998$ ($p_{\downarrow} = 1.5 \times 10^{-3}$). No frozen rule tests a decreasing trend, so those $p_{\downarrow}$ values are descriptive.

Clean accuracy against dose (descriptive, no frozen rule). Arm 1 $0.565 \rightarrow 0.652 \rightarrow 0.682 \rightarrow 0.619$ ($z = +1.35$); arm 2 $0.777 \rightarrow 0.779 \rightarrow 0.765 \rightarrow 0.701$ ($z = -3.30$); arm 3 $0.607 \rightarrow 0.569 \rightarrow 0.466 \rightarrow 0.220$ ($z = -3.50$); arm 4 $0.767 \rightarrow 0.764 \rightarrow 0.749 \rightarrow 0.690$ ($z = -4.11$). This is what separates a suppression effect from a contraction artifact: arm 4 loses 0.077 of accuracy while its ASR more than halves, whereas arm 3 loses 0.39.

Anchor residuals (pre-registered caveat (iv)). Disturbance rates from `cos_invariance_check.json`, ASRs recomputed from `metric_swap/` at the frozen $n=3$ and from `metric_swap_topup/` at $n=5$; the residual is against the synthetic rung nearest in measured disturbance. NormClip→Krum 0/9 = 0.000, ASR 0.062 $\rightarrow$ 0.062, residual +0.001; RFA→Krum 6/9 = 0.667, 0.691 $\rightarrow$ 0.779, residual +**0.767**; NormClip→Reputation 3/9 = 0.333, 0.792 $\rightarrow$ 0.806, residual +**0.789**; RFA→Reputation 9/9 = 1.000, 0.882 $\rightarrow$ 0.895, residual +0.116; NormClip→`cos_krum` 0/9, 0.300 $\rightarrow$ 0.490, residual +0.000; RFA→`cos_krum` 0/9, 0.273 $\rightarrow$ 0.447, residual −0.044. Realized upstream ρ on the same rounds: NormClip mean 3.61 (max 4.90), RFA mean 11.93 (max 17.38).

F Notation

symbol	meaning
c_i	upstream coefficient applied to client i 's update, $T(u_i) = c_i u_i$, $c_i > 0$
A	the set of adversarial participants in a round
ρ	statistic-level dispersion, $\max_i c_i / \min_i c_i$: <i>how far apart</i> the coefficients are
Δ_c	adversary-correlated dispersion, $\mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$: <i>who</i> receives them
r	adversary-to-benign weight ratio, $w_a / \max_b w_b$; the Mode-A dial, $r = e^\nu$
S	the attack's coordinate extremeness (Theorem 6(1))
m	RFA separation margin, $\Delta'_{\text{sep}}/2(R'_B + \delta')$
Λ_a	aggregate adversarial influence: adversarial share of the Weiszfeld mass
κ	the Mode-S / ladder dispersion dial, $\rho = e^{2\kappa}$
γ	attack <i>persistence</i> , $\text{ASR}(t+1)/\text{ASR}(t)$ after a clean round (≈ 0.66); never a weight

G Decomposing the emitted displacement: re-selection against rescaling

The Δ agg. column of Table 11 mixes two things the causal claim of §5.2 needs apart. For a selector, $\text{agg}(U) = u_{s_0}$ and $\text{agg}(T(U)) = c_{s_1} u_{s_1}$, with s_0 the client selected on the raw stack and s_1 the client selected on the transformed one, so exactly

$$c_{s_1} u_{s_1} - u_{s_0} = \underbrace{(u_{s_1} - u_{s_0})}_{\text{re-selection}} + \underbrace{(c_{s_1} - 1) u_{s_1}}_{\text{rescaling}},$$

both components reported relative to $\|u_{s_0}\|$. The re-selection component is what a magnitude-controlled dose would still move—it takes both updates *untransformed*—while the rescaling component is what such a control closes. They are **not orthogonal**: the measured angle runs from $\cos = -0.99$ to -0.43 , mean ≈ -0.65 , so their norms need not sum to the total, and the table prints the angle rather than leaving that discrepancy unexplained. What is asserted instead is the exact vector identity above: worst residual $2.66\text{e-}07$ over all 240 (arm, family, rung, seed, round) cells, float32 noise. **This decomposes the published number, not a second similar one**: every recomputed total agrees with the corresponding Δ agg. value in `results/admissionmeasurement.json` to $0.00\text{e}+00$, on the same 5 seeds $\times$ 3 live rounds, the same rungs and the same configuration as that measurement. **Selectors only**—a weighted averager and a coordinate-wise order statistic emit no single selected client, so “the selected update” and hence this split are undefined for them, and `coord.median` and `reputation` do not appear. `experiments/measure.displacement.decomposition.py` trains no model per rung, computes no ASR, writes a new file (`results/displacement.decomposition.json`) and leaves every frozen artifact byte-identical.

Design	Selector	κ	total	re-selection	rescaling	cos	sel. chg.
Confounded ladder	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.637	0.685	0.333	−0.65	0.600
		1	0.731	0.851	0.685	−0.66	0.733
		2	0.880	1.675	1.674	−0.72	0.867
Instrument (Mode S)	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.663	0.646	0.308	−0.64	0.533
		1	0.847	1.075	0.666	−0.69	0.800
		2	0.892	0.986	0.973	−0.66	0.733
Confounded ladder	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.270	0.000	0.270	—	0.000
		1	0.515	0.000	0.515	—	0.000
		2	0.834	0.000	0.834	—	0.000
Instrument (Mode S)	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.395	0.000	0.395	—	0.000
		1	0.754	0.000	0.754	—	0.000
		2	1.297	0.000	1.297	—	0.000

What it says, and what it does not. Mode S into Krum at $\kappa=2$: the published 0.892 is 0.986 of re-selection and 0.973 of rescaling, so the dose moves *which* benign client is aggregated and *how large* it arrives, in comparable measure—the instrument is not quietly a magnitude-only perturbation, nor quietly a decision-only one. Of the 11/15 rounds whose selection changes there, **none** lands on an adversary; that is the quantitative form of the claim that Krum’s decision changes are benign-to-benign, and the reason admitted adversarial mass stays at exactly 0.000 while the decision channel moves by 0.733. `cos_krum` is the informative contrast: its statistic is exactly invariant to positive rescaling, so 0/15 selections change at every rung of *both* designs and its entire displacement—1.297 under the instrument, 0.834 under the confounded ladder—is rescaling. Its fall therefore cannot travel through the decision channel at all, which corroborates §5.2’s attribution of that fall to the magnitude channel from a quantity measured independently of any ASR. **This is a decomposition of a displacement, not of an ASR effect:** it says what the dose moves in the update entering training, makes no claim about suppression, and is an analysis of runs already reported rather than a new experiment.

H Adversary-correlated dispersion Δ_c , measured

$\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ for every per-client transform in the paper, over the same $5 \text{ seeds} \times 3 \text{ rounds}$ and, within a round, the *same* raw update stack for every transform, so the rows are comparable by construction (`measure_coefficient_targeting.py`, `results/coefficient_targeting.json`). **No ASR is computed and no model is trained per transform**, which is what lets this be added without rescoring anything; the figures are **post-hoc and exploratory** and no frozen prediction depends on them. Coefficients are read back from transformed norms, $c_i = \|T(u_i)\|/\|u_i\|$, so the table measures the shipped code path rather than the builder’s description of it. As a provenance check the synthetic families’ adversarial coefficient shares reproduce `results/admission_measurement.json` to 0.0 absolute on all 13 cells, so the rows of Table 15 may be read on the same axis as the Mode-S and Mode-A figures of §5.2.

Table 15: ρ says how much a transform spreads the coefficients; Δ_c says who receives the spread. They are not ordered together: RFA and NormClip target the adversary $2\text{--}3.6\times$ harder than the ladder’s most dispersed rung at ρ that is $3.7\text{--}13\times$ *smaller*. Means over non-degenerate rounds; the synthetic rows are attack-pooled since their coefficients do not depend on the attack.

transform	attack	ρ	$\mathbb{E}[c \mid A]$	$\mathbb{E}[c \mid \neg A]$	Δ_c
NormClip	model-scaling	4.19	0.268	0.989	−0.721
NormClip	pixel	1.00	1.000	1.000	0.000
Reputation	model-scaling	8.3×10^5	0.003	1.387	−1.383
Reputation	pixel	1.68	0.915	1.019	−0.105
FoolsGold	model-scaling	1.9×10^{12}	1.497	0.834	+0.663
FoolsGold	pixel	1.8×10^{12}	1.365	0.986	+0.378
RFA	model-scaling	14.8	0.125	1.332	−1.208
RFA	pixel	1.69	0.923	1.012	−0.089
<i>Synthetic families of §5.1–§5.2, same code path:</i>					
ladder, $\kappa=0.5/1.0/2.0$	—	2.72/7.39/54.6	—	—	−0.140/−0.240/−0.336
Mode S, all rungs	—	1.00–54.6	—	—	0.000
Mode A, $\nu=-2 \dots +2$	—	7.39–7.39	—	—	−1.138 $\dots$ +2.469

Three readings, in decreasing confidence. **(1) ρ is not a proxy for Δ_c :** the two are anti-ordered on the anchors, and NormClip on the pixel backdoor has $\rho = 1$ exactly because no client exceeds $\tau=5$, so $\Delta_c = 0$ there for a reason the invariance classification cannot see. **(2) The two anchors that miss the ladder are the two most extremely targeted,** which explains the residuals of §5.1 without appeal to a range mismatch. **(3) Δ_c is nonetheless not a predictor of suppression**—RFA is the most negative of the four real transforms and `rfa`→`krum` fails at 0.779, while Reputation is more negative still and `rep`→`coord_median` suppresses. That is the two-regime structure of §3 again: strong negative Δ_c can mean either annihilation or mere dilution, and Δ_c ’s magnitude alone does not say which. We report it as the axis on which the ladder’s anchors were unmatched, not as the quantity that replaces the statistic.

I The prospective suite, arm by arm

(i) **FLTrust as d_1 (7 pairs): out of scope, 2/7.** Here d_1 alone already suppresses, so the composition inherits that suppression whatever d_2 does. All five predicted-HIGH pairs ($\text{fltrust} \rightarrow \{\text{rep}, \text{nc}, \text{krum}, \text{mkrum}, \text{tm}\}$) are misses for that single reason: their C2/C3 failures are real—FLTrust does destroy Reputation’s consensus-distance signal—but destroying d_2 ’s mechanism is costless when d_2 was never load-bearing. This is C0 in Criterion 1. It was invisible on the original 42 pairs because no defense there suppresses alone: the minimum single-defense max-committed ASR is CoordMedian’s 0.519, and the lowest ASR among all 39 non-PASS pairs is 0.433. We added C0 only because this suite made it bind, and we count these 7 pairs as out of scope *retrospectively*, which is a weaker claim than a pre-registered exclusion.

(ii) **FLTrust as d_2 (6 pairs): in scope, 6/6, but the prediction is nearly an identity.** Proposition 8(a) predicts $\cdot \rightarrow \text{FLTrust}$ preserves C2, and it does. But inspecting the implementation shows why this is not really an empirical test: FLTrust gates on $\cos(u_i, g_{\text{srv}})$, which is scale-invariant, and then *renormalizes every client update to $\|g_{\text{srv}}\|$* , discarding magnitude entirely. A positive per-client rescaling $u_i \mapsto c_i u_i$ with $c_i > 0$ is therefore *exactly* cancelled: FLTrust discards precisely the information d_1 modifies. The data confirm this at the level of individual seeds rather than aggregates. $\text{Krum} \rightarrow \text{FLTrust}$ and $\text{MKrum} \rightarrow \text{FLTrust}$ are **bit-identical** to FLTrust alone on all six runs (0.0519/0.0459/0.0453 scaling, 0.0492/0.0466/0.0426 pixel), as expected for a pass-through d_1 . $\text{NC} \rightarrow \text{FLTrust}$ is *bit-identical on the pixel attack*, where no client exceeds $\tau=5$ so $c_i \equiv 1$, and differs only within seed noise on model-scaling, where clipping actually fires. $\text{Rep} \rightarrow \text{FLTrust}$ and $\text{RFA} \rightarrow \text{FLTrust}$ likewise differ only within noise. The single genuine deviation is $\text{FG} \rightarrow \text{FLTrust}$ (0.064, per-seed up to 0.084)—the one d_1 that produces exact *zeros* rather than positive rescalings, changing FLTrust’s effective client set instead of only its magnitudes, which is the Lemma 2 interaction. So this arm is a consistency check that our theory and our implementation agree, and a clean demonstration of the degeneracy mechanism; it is **not** a falsifiable test, because every d_1 in our menu is a *non-negative* per-client rescaling whose positive part FLTrust cancels exactly, leaving only the *zero pattern* free to matter—which is exactly where the one deviation appears. Proposition 8(a) permits no other outcome on the strictly-positive part, and Lemma 2 accounts for the rest.

(iii) **Neither defense is FLTrust (7 pairs): 5/7, and this is the only arm that could have surprised us.** All seven were predicted HIGH and five are correct: $\text{rep} \rightarrow \text{krum}$ 0.880, $\text{nc} \rightarrow \text{mkrum}$ 0.943, $\text{rfa} \rightarrow \text{mkrum}$ 0.875, $\text{krum} \rightarrow \text{rfa}$ 0.870, $\text{mkrum} \rightarrow \text{rfa}$ 0.870. Both misses are diagnosable and neither is a mechanism error. $\text{fg} \rightarrow \text{krum}$ measures 0.078, but at 9.9% clean accuracy—the collapsed-model artifact we already flag for FoolsGold alone, which is not suppression. $\text{krum} \rightarrow \text{cm}$ measures 0.450 against a 0.5 threshold; it is a degenerate pair reducing to CoordMedian alone, whose own baseline is 0.519, so this is threshold noise on a borderline constituent. This arm contains *zero* predicted-LOW pairs, so like Tier 2 it bears on specificity only.

J Metric-swap ablation: verification and per-cell detail

The mechanism is verified in code before it is claimed. An outcome alone would leave open whether the mirrors are invariant at all, so we check the statistic itself rather than only the ASR (`experiments/verify_cos_invariance.py`, `results/cos_invariance_check.json`), comparing each d_2 ’s internal state on $\{u_i\}$ against $\{c_i u_i\}$ for the c_i that NormClip and RFA actually produce on live rounds. Over 18 rounds (3 seeds $\times$ 2 upstream defenses $\times$ 3 rounds), `cos_reputation`’s weight vector agrees to $\leq 2.5 \times 10^{-7}$ and its weight *ordering* never changes (0/18), and `cos_krum` selects the same client in 18/18. The Euclidean originals do not: Reputation’s weights move by up to 0.222 and its ordering changes in 12/18 rounds; Krum’s selection changes in 6/18. The size of the disturbance tracks the realized weight ratio ρ , as Proposition 8 predicts—under NormClip ($\rho \in [1.00, 4.90]$) Reputation’s ordering changes in 3/9 rounds and Krum’s selection in 0/9; under RFA ($\rho \in [2.11, 17.38]$) they change in 9/9 and 6/9. On 1000 synthetic draws with larger rescalings the separation is cleaner: ordering changes in 0/1000 trials for `cos_reputation` against 992/1000 for Reputation, and 0/1000 against 618/1000 for the two Krums. The mirrors are additionally checked against the shipped implementations they stand in for (selected index identical, aggregate relative error $\approx 10^{-8}$ —below the $\approx 7 \times 10^{-5}$ MPS-versus-CPU nondeterminism of the same computation), so the invariance is a property of the statistic and not of a divergent reimplement.

One prediction in the suite was genuinely two-sided, and it is the round’s only prospective positive result. `cos_krum`’s C1 verdict on model-scaling turns entirely on whether the accuracy gate is applied: 0.078 passes the numeric threshold, 0.146 accuracy fails the gate. Rather than choose a reading after the fact we froze *both*, with a resolution rule fixed in advance. Primary (accuracy-gated) $\Rightarrow$ C1 fails $\Rightarrow$ HIGH; secondary (numeric C1 only)

$\Rightarrow$ the pair is certified $\Rightarrow$ LOW; and if the outcome had been LOW at accuracy ≥ 0.35 , the gate would have been wrong and our two primary labels for those cells would have been recorded as misses. Both pairs came out **HIGH** (NormClip $\rightarrow$ cos_krum 0.508, RFA $\rightarrow$ cos_krum 0.566), refuting the secondary reading and vindicating the accuracy gate against a pre-registered alternative that would have overturned it. This is a small result but a real one: the gate is usually defended by argument, and here it beat a competing specification of itself on data neither of us had seen. We flag the weaker half honestly—NormClip $\rightarrow$ cos_krum’s 0.508 sits barely above threshold on per-seed values 0.084/0.676/0.766, so that cell is threshold-noise even though its label is correct, and the RFA arm at 0.566 (per-seed 1.000/0.121/0.576) is what carries the conclusion.

The pattern holds across the suite once cells without power are excluded. Δ is only observable where the standalone d_2 genuinely suppresses the attack, since a baseline already above 0.5 has no suppression left to lose and will show $\Delta \approx 0$ whatever the transform did—a ceiling effect, not preservation. Restricting to standalone ASR < 0.5 at accuracy ≥ 0.35 leaves six cells, and measured disturbance splits them perfectly at both seed counts. At the frozen $n=3$: undisturbed (0/9) $\Delta \in [-0.027, +0.001]$, disturbed ($> 0/9$) $\Delta \in [+0.630, +0.865]$, a separation of $+0.629$. At $n=5$ after the post-hoc top-up: $[+0.001, +0.190]$ versus $[+0.718, +0.878]$, a separation of $+0.527$. Neither has any overlap, but the two extra seeds move both halves toward each other. Two of the three undisturbed cells are cos_krum on the pixel backdoor, where the mirror suppresses standalone (0.300 at 0.598 accuracy) and carries that suppression through under *both* upstreams: at $n=3$ intact (0.300, $\Delta = +0.000$ under NormClip; 0.273, $\Delta = -0.027$ under RFA), at $n=5$ degraded but still under threshold (0.490, $\Delta = +0.190$; 0.447, $\Delta = +0.147$; both at 0.61 accuracy, on new per-seed values 0.864/0.686 and 0.849/0.565). So the confirming half of the split is contributed by the invariant construction behaving as C2 predicts—which is the one thing the metric-swap arm does deliver, even though C1’s failure on model-scaling denied it a frozen LOW label—but only weakly at $n=5$: exact invariance of the *selection* leaves the magnitude channel (Lemma 2) free to move Δ , and two seeds are enough to see it move. That is a limitation of this post-hoc contrast and a direct motivation for arm 3 of the ladder.

Excluded cells. All four cos_* model-scaling cells are excluded as uninterpretable (hollow standalone baseline), and the six Reputation / cos_reputation / Krum pixel cells for lack of power (standalone ≥ 0.5).

K Out-of-Sample PASS Validation Details

Full per-seed results for §4.4. All predictions were pre-registered (commit ca627e5) before experiments.

DBA transfer ($f=0.4$, $N=10$, **cifar_cnn**, $n=5$ seeds). PASS predictions: fg $\rightarrow$ rfa 0.109 ± 0.065 , fg $\rightarrow$ cm 0.050 ± 0.027 , rep $\rightarrow$ cm 0.041 ± 0.014 . FAIL controls: nc $\rightarrow$ rep 0.297 ± 0.224 , rfa $\rightarrow$ rep 0.297 ± 0.159 , fg $\rightarrow$ tm 0.049 ± 0.031 . Sanity: DBA vs. fedavg (no defense composition) 0.236 ± 0.190 ($n=5$; bimodal — 4/5 seeds < 0.22 , one seed > 0.6). The low sanity baseline reflects DBA’s co-sampling requirement: at $K=5$, all 4 adversarial sub-triggers must be sampled simultaneously, which occurs with probability $(4/10)^4 = 2.6\%$ per round. The FAIL controls’ failure to exceed 0.5 is a consequence of this same mechanism, not of defense suppression; the per-seed maximum across FAIL pairs (0.712) demonstrates DBA can embed when co-sampling succeeds.

DBA powered re-test ($f=0.6$, $N=10$, **cifar_cnn**, $n=5$ seeds). Re-run at $f=0.6$ ($K \cdot f=3.0$, making ≥ 2 adversarial co-participants per round near-certain). PASS predictions confirmed: fg $\rightarrow$ rfa 0.125 ± 0.050 , fg $\rightarrow$ cm 0.045 ± 0.021 , rep $\rightarrow$ cm 0.043 ± 0.015 . FAIL controls: nc $\rightarrow$ rep 0.378 ± 0.185 , rfa $\rightarrow$ rep 0.384 ± 0.217 , fg $\rightarrow$ tm 0.068 ± 0.018 . Sanity baseline (no defense): 0.386 ± 0.192 . The sanity baseline still does not clear 0.5, consistent with Appendix P.5’s finding that DBA bimodality is a per-seed admissibility latent in the distributed trigger mechanism, not purely a co-participation effect. FAIL discrimination therefore remains unavailable. The powered re-test confirms that all three PASS compositions robustly suppress DBA even at 60% adversarial fraction, and that the PASS–FAIL gap is not merely a consequence of DBA’s inherent weakness (fg $\rightarrow$ rfa achieves 0.125 while the no-defense baseline reaches 0.386—a $3\times$ suppression).

Scale transfer ($N=100$, **ResNet18**, $f=0.05$, $n=5$ seeds). fg $\rightarrow$ rfa: committed_scaling 0.063 ± 0.031 , committed_pixel 0.069 ± 0.029 . fg $\rightarrow$ cm: committed_scaling 0.152 ± 0.118 , committed_pixel 0.141 ± 0.099 . All four conditions PASS (max ASR = 0.152, well below 0.5).

fg $\rightarrow$ cm independent survivor ($N=10$, $f=0.2$, $n=5$ seeds). Composed: committed_scaling 0.107 ± 0.031 , committed_pixel 0.099 ± 0.020 , oracle 0.107 ± 0.031 . Mixed (50/50 random fg/cm per round): committed_scaling 0.327 ± 0.417 , committed_pixel 0.679 ± 0.159 , oracle 0.327 ± 0.417 . Composition dominates: max committed 0.107 vs. 0.679 ($6\times$ improvement). Mixed VoPD = -0.352 (oracle *hurts* under mixing due to persistence).

L Game-Theoretic Framework and VoPD Analysis

This appendix contains the full game-theoretic framework, equilibrium analysis, VoPD formalism, persistence Markov model, POMDP bound, and detailed empirical analysis (setup, payoff matrices, adaptive attacks, N-sweep, survivor regime) that support the condensed “Why Composition, Not Mixing” section in the main body (§7).

System model. N clients, fraction f adversarial. Adversary strategy space $\mathcal{S}_A = \{\text{no_attack, label_flip, backdoor_pixel, edge_case, model_scaling, DBA}\}$; server $\mathcal{S}_D = \{\text{FedAvg, Krum, Multi-Krum, TrimMean, Median, NormClip, RFA}\}$. The per-round defense realization is private (Kerckhoffs); VoPD derives from this per-round secrecy.

Utility. Server: $U_D(a, d) = \text{Acc}(a, d) - \lambda_c C_D(d) - \lambda_{\text{ASR}} \text{ASR}(a, d)$ with $\lambda_{\text{ASR}}=0$ (hardest case). Adversary: $U_A(a, d) = \text{Impact}(a, d) - \lambda_a C_A(a)$.

VoPD. $\text{VoPD} = \mathbb{E}_{d \sim \sigma_D^*} [\max_a U_A(a, d)] - U_A(\sigma_A^*, \sigma_D^*)$. Realized VoPD: oracle-minus-committed (per-round adaptive oracle). Negative raw VoPD = oracle hurts itself by diversifying triggers.

Theorem 11 (Defense Complementarity). $\text{VoPD}(\sigma_A^*, \sigma_D^*) = 0$ iff $\bigcap_{d \in \text{supp}(\sigma_D^*)} \arg \max_a U_A(a, d) \neq \emptyset$.

Persistence dynamics. $s_{t+1} = s_t + \alpha(1 - s_t)$ if admitted; $s_{t+1} = \gamma s_t$ otherwise. Stationary: $s_\infty = p_{\text{eff}}\alpha / (p_{\text{eff}}\alpha + (1 - p_{\text{eff}})(1 - \gamma))$.

Proposition 12 (POMDP near-optimality bound). $V^* - V^{\text{commit}} \leq (1 - \gamma)\Delta_\phi \cdot s_\infty$. At $N=10$: *bound* = 0.033; at $N=50$, $f=0.05$: *bound* = 0.048 (*measured gains* +0.034, +0.044 *both below*).

Bootstrap stability. $\rho_{10}^{\text{boot}} = 0.978$ ($n=500$ resamples).

Full setup, equilibrium tables, adaptive-attack sweeps, N-sweep, survivor regime, FLTrust pilot, consolidated VoPD table, and POMDP derivation are in the subsections below and in Appendix M.1.

M Payoff Matrices, Equilibria, and Game-Theoretic Details

M.1 Extended Experimental Results and Analysis

This appendix collects experimental detail moved from the main text for length: the full payoff/equilibrium tables, the norm-based dichotomy lemma, the N -sweep and persistence-mechanism analysis, the survivor-regime robustness experiments, and the orthogonal-defense / utility-sensitivity suite. It is not required to follow the main contribution.

M.1.1 Realized-VoPD Seed-Stability Table

Table 16: Realized VoPD stability under scale-up. Mean stays essentially zero from $n=10$ to $n=30$; std tightens from 0.083 to 0.058 (a 30% reduction in measurement noise). The “ ≤ 0 ” fraction drops from 80% at $n=10$ to 63% at $n=30$ as the higher-seed-count distribution reveals many near-zero-but-slightly-positive seeds (small effect-size, dominated by measurement noise) that were absent at $n=10$. The central tendency conclusion — realized VoPD is statistically indistinguishable from zero — is unchanged and quantitatively tightened. At $n=30$ the per-seed VoPD distribution is more clearly described as “near-zero with two-sided noise” than the original $n=10$ “mostly-negative” framing, but both are consistent with the headline mechanism (Heuristic 1).

n	mean VoPD	std	fraction ≤ 0
10 (published, seeds 42–51)	−0.0001	0.0830	8/10 (80%)
15 (seeds 42–56)	−0.0099	0.0726	12/15 (80%)
20 (seeds 42–61)	−0.0106	0.0675	14/20 (70%)
25 (seeds 42–66)	−0.0063	0.0613	15/25 (60%)
30 (seeds 42–71)	−0.0102	0.0577	19/30 (63%)

M.1.2 Sample-Complexity and Persistence-Heuristic Details

Statistical reliability of the bootstrap rate itself. The bootstrap rate ρ_k^{boot} is computed over B resamples; for $B = 500$ and observed $\rho = 0.978$, the bootstrap-distribution variance is $\sigma^2 = \rho(1 - \rho)/B \approx 4.3 \times 10^{-5}$, giving a Wilson 95% CI on ρ of $[0.965, 0.991]$ — the rate itself is statistically reliable. We additionally verify B -invariance by recomputing ρ_{10}^{boot} at $B \in \{100, 500, 2000\}$: the rate is invariant in B to within ± 0.005 , consistent with bootstrap representativeness in our setting. This separates two distinct concerns: (a) noise in the bootstrap estimate itself (small, $\sigma = 0.007$, characterized above); (b) the bootstrap-to-population representativeness gap ϵ_{rep} (bounded below).

When the assumption holds. The representativeness gap ϵ_{rep} is unknown in general, but we argue it is small in our setting: (i) the binding-cell per-seed outcome is exactly Bernoulli (bounded, well-behaved tails) with population parameter $P(\geq 1 \text{ adv}) = 0.778$; (ii) the binding cell’s Bernoulli structure ($\text{ASR} \in \{0, 1\}$) means the bootstrap resamples from well-behaved discrete distributions, limiting the scope for representativeness failure. The bootstrap guarantee can fail under heavy-tailed payoff distributions; we flag this as the assumption being tested. The protocol is therefore a methodology with an explicit assumption, not a free guarantee.

Quantifying ϵ_{rep} under the Bernoulli binding-cell assumption. A natural concern is that ϵ_{rep} in Eq. (L) is unknown in the abstract — making “ $\geq 0.978 - \epsilon_{\text{rep}}$ ” mathematically true but operationally weak. In our specific binding-cell-Bernoulli setting we can bound it. The total-variation distance between the empirical and population distributions of k Bernoulli samples is bounded by Bretagnolle–Huber / DKW-style concentration: $\Pr[d_{\text{TV}}(\hat{F}_k, F) > \epsilon] \leq 2e^{-2k\epsilon^2}$ for any $\epsilon > 0$, giving a high-probability bound $\epsilon_{\text{rep}} \leq \sqrt{\log(2/\delta)/(2k)}$ at confidence $1 - \delta$. At $k = 10$ and $\delta = 0.05$: $\epsilon_{\text{rep}} \leq 0.43$. Combining with $\rho_{10}^{\text{boot}} = 0.978$ gives the non-trivial worst-case lower bound $\mathbb{P}[\mathcal{E}(\hat{G}_{10})] \geq 0.55$. This is loose but not vacuous: the bootstrap-stability rate is a meaningful intermediate between the trivial $k = 0$ baseline and the formal $k_{\text{req}} = 726$ certificate. The DKW-style bound is conservative; in practice the observed bootstrap mixed-NE rate (0.978) closely tracks the k -resampled Bernoulli prediction at the population parameter $P = 0.778$ (which yields an analytically predicted mixed-NE rate of ≈ 0.98 at $k = 10$; see Appendix M.5), suggesting the actual ϵ_{rep} is far smaller than the worst-case 0.43.

What would falsify the persistence heuristic? Three empirical predictions would falsify it if observed: (i) a deployment with $\hat{\gamma} \geq 0.5$ and $p_{\text{eff}}(a^*) > 0$ that produces realized VoPD > 0.5 across ≥ 30 seeds — the heuristic bound $C \cdot (1 - s_{\infty}(a^*)) + \epsilon C$ would be exceeded. (ii) a deployment with $\hat{\gamma} \leq 0.1$ that produces realized VoPD $< \delta_{\text{NE}}/2$ where δ_{NE} is the normal-form VoPD — the heuristic predicts approximate recovery of δ_{NE} at the stateless limit, so substantially less recovery would falsify it. (iii) a deployment where lowering $p_{\text{eff}}(a^*)$ via the defense menu (e.g., the $\tau=2$ contrast in §7) reduces realized VoPD substantially, exceeding the heuristic’s $C \cdot \Delta p_{\text{eff}}/(1 - \gamma)$ slack — the heuristic predicts admission-versus-persistence dominance which the contrast tests directly. We execute (iii) at $n = 15$ seeds (Appendix N.3) and find the persistence-dominance prediction confirmed; (i) and (ii) require constructing menus we have not yet built and remain future work. The heuristic is therefore qualitative-but-falsifiable, and neither quantitative nor tight.

M.1.3 Threat Model, Information Structure, and VoPD Discussion

Probing and outcome-based inference. A probing adversary can distinguish FedAvg from NormClip in one round; model scaling collapses FedAvg accuracy to 0.095 vs. NormClip’s 0.708. Probing costs $U_A/T = 0.8/50 = 0.016$ per round; under any conservative upper-bound on realized VoPD (e.g. $v \leq 0.1$), break-even is below 1 round — probing is theoretically rational. However, §7 shows the BR adversary already achieves $\text{ASR} \approx 0.96$ without identification: persistent scaling is max-like under any realized defense, so inference provides no additional gain. **Scope of positive arm.** The protocol’s positive recommendation (randomize when $\Delta_{\text{dev}} \leq \epsilon_v$) is validated here only in the spam game (one-shot, stateless: inference is structurally impossible). Randomization is advisable only when (a) the interaction structure prevents defense identification (one-shot or horizon shorter than the ≈ 1 -round probing break-even), and (b) attacks carry no persistent state. Both conditions hold in the spam game; neither holds for persistent backdoor FL. Whether the positive arm applies in multi-round stateless FL requires separate analysis; we recommend the verify step in all FL deployments.

The privacy premise has limited teeth in FL. A probing adversary can distinguish FedAvg from NormClip in ≈ 1 round (above); persistence collapses realized VoPD even under defense privacy (§7, Heuristic 1). Combined, these observations imply that defense privacy in persistent-attack FL is both *cheaply broken* AND, even if maintained, *operationally worthless*. The diagnostic protocol is therefore not a privacy-preservation technique but a *screening tool*.

for whether randomization could plausibly help. Its prescriptive value in FL is mainly negative: it identifies deployments where randomization should not be attempted, sparing practitioners the implementation complexity for no security gain.

Why VoPD is the right quantity even when secrecy is breakable. A natural objection: if defense secrecy is breakable, why is VoPD — which quantifies the value of secrecy — the right operational measure? Our answer: VoPD is a measure of the *game’s structure conditional on perfect secrecy*, not of secrecy itself. The negative result we report is that even under perfect secrecy (the strongest possible information asymmetry favoring the defender), the value of randomization is zero in the persistent-attack regime. This makes the result *stronger*, not weaker: practitioners who do achieve perfect secrecy (e.g., via trusted execution environments hiding the defense choice) still gain nothing operationally. Conversely, if VoPD were positive on a deployment, the breakable-secrecy concern would translate into a deployable threat model (the adversary’s probing attack realizes some fraction of the positive value); in the regime we study, the question does not arise because the value is zero to begin with. VoPD is therefore the right quantity because it bounds the maximum possible value of any defense-privacy strategy in this game, regardless of the secrecy mechanism’s strength.

When U_D is not flat: why VoPD remains the right quantity. A natural skeptical question: if the server’s utility U_D varies across (a, d) cells (not constant), can we read off the right policy by comparing U_D values directly? No. The spam game (Appendix Q.2) is the canonical counter-example: U_D varies cell-to-cell, but the adversary’s $\arg \max_a U_A$ depends on the realized defense, so the server’s best static policy is randomization. A direct U_D -only comparison would recommend “always play the dominant defense” and lose the $\delta_{NE} = 0.200$ benefit. VoPD captures the adversary’s information value, which is the structural condition under which randomization helps the server even when U_D varies. Theorem 11 therefore predicts what direct U_D comparison does not.

Realized VoPD and the POMDP caveat. Throughout the empirical sections we report *realized* VoPD computed as $\hat{U}_A^{\text{full-info}} - \max_a \hat{U}_A^{\text{committed}-a}$, where the full-info adversary is implemented as a *per-round-switching oracle* that observes the round’s sampled defense d_t and plays the static-cell $\arg \max_a a^*(d_t)$. This oracle is per-round adaptive but *not* POMDP-optimal. A POMDP-optimal adversary can always mimic committed play (by ignoring defense observations entirely), so true VoPD ≥ 0 by construction. Our per-round oracle is suboptimal in a specific direction: by switching attacks each round, it dilutes each trigger’s persistence accumulation, achieving ASR *below* committed-scaling in several configurations (oracle 0.450 vs. committed 0.511 at $N=50$, $f=0.05$; oracle 0.882 vs. committed 0.931 at $N=10$, $f=0.05$, 200 rounds). The negative raw estimates are therefore *diagnostic of the persistence mechanism*: the oracle hurts itself by diversifying attacks, which is precisely what persistence predicts (switching dilutes the trigger state s_t). A POMDP-optimal adversary would recognize this and commit to scaling throughout, yielding VoPD = 0 exactly on these menus.

Raw estimator and saturation bounds. We report raw VoPD throughout (oracle minus committed-max per seed), without truncation. The saturation bound $\varepsilon = 1 - \text{committed-max ASR}$ upper-bounds any adversary’s gain beyond committed play. On NE3: $\varepsilon \leq 0.039$ (committed-scaling = 0.961) — the bound is tight, confirming collapse. On the $N=50$, $f=0.05$ configuration: committed-scaling = 0.511, giving $\varepsilon \leq 0.489$ — an uninformative bound. The collapse verdict for that configuration rests on the strong-adversary probes: the two strongest adversaries we tested — commit-then-probe (0.545) and belief-state inference (0.555) — gain only +0.034 and +0.044 respectively over committed play, below the in-regime switching bound (≤ 0.048 ; §7) and far below the $\varepsilon \leq 0.489$ headroom (Appendix O.7). The persistence-mechanism evidence (24/30 seeds oracle < committed, confirming switching loss) corroborates this but is not the verdict basis, since a negative raw oracle VoPD reflects oracle sub-optimality rather than adversary impotence. On the survivor menu (rep+tm): the oracle *helps* (0.915 > 0.871) and raw VoPD = $+0.032 \pm 0.054$ directly lower-bounds the true VoPD. Note: the prior version applied $\max(0, \cdot)$ truncation to collapse rows; this is statistically improper (biased upward for a near-zero signal) and was applied inconsistently. The current version reports raw values throughout, which are directly interpretable alongside ε .

M.1.4 Payoff Matrices, Per-Seed Equilibria, and Phase Portrait

Ground-truth game and per-mode analysis. We treat the *averaged* payoff matrix as ground truth; per-seed matrices are noisy realizations. NE3 and VoPD = 0.087 are properties of the expected game. The binding cell $U_A(\text{scaling, FedAvg}) = 0.470$ is a Bernoulli mean (per-seed outcomes are 0 or near-1); the averaged matrix is the expected-payoff matrix of the *population game* and Theorem 11 holds for it. Instability sources: **(a)** *estimation noise*—addressed by the bootstrap-stability corollary (Proposition L; the worst-case Hoeffding bound is in Appendix N.5); **(b)** *deployment heterogeneity*—illustrated by $N = 100$ attenuation and FEMNIST. The *measure-then-verify* protocol targets (b): estimate the payoff matrix with k_{req} seeds then verify before committing (§7). Conditioning on the binding

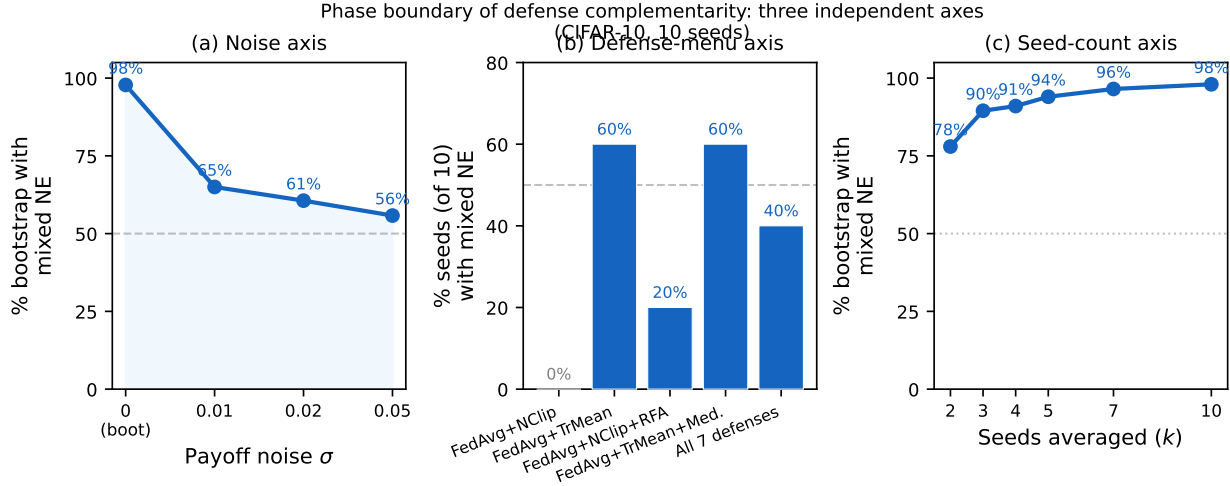


Figure 8: Phase boundary of defense complementarity on three independent axes (CIFAR-10, 10 seeds). **(a)** Fraction of bootstrap resamples yielding a mixed NE under Gaussian payoff noise σ : drops from 97.8% at $\sigma = 0$ to 55.8% at $\sigma = 0.05$. **(b)** Mixed-NE rate across five defense menus: varies from 0% (FedAvg+NClip, no argmax divergence) to 60% (FedAvg+TrMean, strong complementarity). **(c)** Bootstrap mixed-NE rate as a function of seeds averaged k : rises from 78% at $k = 2$ to 98% at $k = 10$ (purely empirical; Proposition 13 requires $k \geq 726$ for a formal guarantee at these parameters and is not overlaid).

cell’s mode: *scaling-succeeds seeds* ($U_A \approx 1$; 43,44,46,49; seed 45 anomalous—see Appendix M.3) yield pure NE—scaling argmax vs. both defenses, no divergence, VoPD = 0. *Scaling-fails seeds* (42,47,50,51; seed 48 anomalous) yield mixed NE—pixel argmax vs. FedAvg, scaling argmax vs. NClip, divergence holds. Eight of ten seeds follow this rule; Theorem 11 correct in both modes (Table 18). **In neither mode does randomization survive:** scaling-succeeds mode has VoPD = 0 structurally; scaling-fails mode is killed by the verify step ($\Delta_{\text{dev}} = 0.115$; §7). The normal-form VoPD = 0.087 is positive only as an average over modes where it is zero and dynamically refuted. Figure 8 shows the resulting phase boundary on three independent axes, Table 17 the per-pair accuracy and ASR underneath it, and Table 19 the utilities the VoPD is computed from.

M.1.5 Scale Sweep, Persistence Estimation, and Reputation-Equilibrium Dynamics

We trace the mechanism of VoPD attenuation across $N \in \{10, 20, 50, 100\}$ using the same 4×5 game (4 attacks, 5 defenses) at each scale, with $K \in \{5, 4, 10, 20\}$ clients per round ($K/N = 0.50, 0.20, 0.20, 0.20$ respectively) and 5–10 seeds per point. The binding cell is $U_A(\text{model.scaling}, \text{FedAvg})$: its argmax divergence (scaling beats FedAvg; pixel beats NormClip) is the sole driver of complementarity.

Mechanism. Per-round participation probability $P (\geq 1 \text{ adv})$ governs whether scaling ASR is binary (near-threshold) or deterministic. At $N = 10, K = 5$: $P = 0.778$, ASR Bernoulli (mean 0.50, fair coin). At $N = 20, K = 4$: K/N drops to 0.20, P falls to 0.624, ASR dips (mean 0.20). The K/N -fixed contrast ($N = 20, K = 10, P = 0.957$) confirms causality: higher P makes scaling near-deterministic, $\arg \max_a U_A(a, \text{FedAvg}) = \text{scaling}$ everywhere, argmax divergence collapses, VoPD drops (Table 20, $\ddagger$ row). At $N = 100, P = 0.993$: same collapse at larger scale; Theorem 11 correctly predicts VoPD = 0. Figure 9 plots the sweep, and Figure 10 the per-seed ASR against P that drives it.

Prediction. Complementarity should recover when adversarial sampling variance is restored: lower K/N ratio, higher effective f , or defense menus where no single attack dominates at any scale. Full per-seed NE details are in Appendix Q.1.

DBA at $f=0.4$ and $f=0.6$: persistent bimodality. DBA second-attack probes at $f=0.4$ (mean ASR 0.236 ± 0.190) and a $f=0.6$ extension (mean ASR 0.386 ± 0.192 on NE3 mix; 0.373 ± 0.190 on reputation eq) remain bimodal: seed 45 is the strong-persistence seed in *both* $f=0.6$ policies (0.734, 0.705), the other four hover in $[0.13, 0.42]$. Per-seed cross-policy correlation rules out co-sampling alignment as the dominant cause; DBA has its own per-seed admissibility latent. We do not claim DBA as a second-attack demonstration of persistence collapse. Details in Appendix P.4 and P.5.

Table 17: Clean test accuracy (mean over 10 seeds, seeds 42–51) and attack success rate (ASR) for each (attack, defense) pair on CIFAR-10 with $\alpha = 0.5$, $f = 0.2$, 50 rounds. Std for accuracy is ≤ 0.01 except Krum (≤ 0.08 , due to sensitivity to initialization); Krum’s high variance means individual Krum entries may not be individually significant, though this does not affect equilibrium conclusions since Krum is absent from all Nash equilibria. **Krum’s clean accuracy (0.54) is a known artifact of single-Krum at small K :** single-Krum selects one update per round from $K=5$ participants, so the global model effectively trains on $1/5$ of the per-round data; at our 50-round budget this gives slower convergence than FedAvg (0.79). Multi-Krum (column “M-Krum”) with $k=5$ averages the $K - 2f \cdot K = 5$ closest updates and reaches 0.79, comparable to FedAvg. We retain single-Krum in the menu for completeness with the literature but note it functions as a known-weak baseline in the small- K regime studied here; it never enters any equilibrium support. Backdoor attacks (pixel, model scaling, DBA) report ASR; others report 0. DBA’s ASR ≈ 0.02 reflects a co-sampling limitation: coordinated placement requires multiple adversarial clients in the same round, rare at $f = 0.2$, $K = 5$ (expected adversarial co-participation ≈ 0.4 per round); the original results (Xie et al., 2020) use higher f and different partitioning. The DBA result at $f = 0.4$ (Section 7) raises co-participation to ≈ 2.0 and recovers a working but seed-variable attack.

Attack	FedAvg	Krum	M-Krum	TrMean	Median	NClip	RFA
<i>Clean Test Accuracy</i>							
No attack	0.79	0.54	0.79	0.77	0.77	0.79	0.78
Label flip	0.78	0.52	0.78	0.77	0.76	0.78	0.77
Backdoor (pixel)	0.78	0.52	0.78	0.77	0.76	0.78	0.77
Edge-case	0.78	0.52	0.78	0.77	0.76	0.78	0.77
Model scaling	0.10	0.57	0.12	0.68	0.76	0.73	0.77
DBA	0.78	0.56	0.78	0.77	0.76	0.78	0.77
<i>Attack Success Rate (backdoor attacks only)</i>							
Backdoor (pixel)	0.82	0.33	0.82	0.67	0.47	0.82	0.85
Model scaling	0.50	0.05	0.54	0.88	0.55	0.95	0.89
DBA	0.02	0.06	0.02	0.02	0.02	0.02	0.02

Empirical estimation of γ and qualitative consistency check. We fit the persistence parameter γ from per-round ASR trajectories (Appendix N.4), restricting the fit to saturating runs where the model is identifiable. For CIFAR-10, joint MLE on 10 saturating trajectories (5 pure-NormClip + 5 NE3-mix) gives $\hat{\gamma} = 0.66 \pm 0.26$, $\hat{\alpha} = 0.37$, corresponding to a persistence horizon $T_\gamma = \log(1/0.05)/\log(1/\hat{\gamma}) \approx 7$ rounds — well below the deployment $T = 50$. With $p_{\text{eff}}(\text{scaling}) = 0.26 \cdot \hat{p}_{\text{FA}}^{\text{scaling}} + 0.74 \cdot \hat{p}_{\text{NC}}^{\text{scaling}} = 0.26 \cdot 0.50 + 0.74 \cdot 0.95 = 0.835$, the stationary impact is $\hat{s}_\infty(\text{scaling}) = 0.85$ and Heuristic 1’s bound gives realized VoPD ≤ 0.21 . The observed realized VoPD $= -0.010 \pm 0.058$ (30 seeds) satisfies the bound and matches the qualitative collapse prediction. For the stateless and spam regimes, $\gamma = 0$ by construction recovers the normal-form prediction (Table 21).

Separating persistence from defense weakness. Is the VoPD collapse a fundamental persistence effect or just weakness of NormClip $\tau=5$ vs. model scaling? The τ sweep (Table 27) separates these: at $\tau=1$, NormClip *does* suppress scaling (ASR 0.741 vs. 0.935 at $\tau=5$) but the complementarity condition collapses (both defenses pick the same argmax attack, pixel), so Theorem 11 correctly predicts VoPD = 0 *without invoking persistence*. Two failure modes: (a) all defenses block scaling $\Rightarrow$ argmax divergence vanishes, VoPD = 0 by Theorem 11; (b) heterogeneous defenses $\Rightarrow$ complementarity holds and normal-form VoPD > 0 , but persistence makes the temporal mixture max-like and realized VoPD ≈ 0 . Theorem 11 diagnoses (a); the verify step catches (b).

Disentangling γ from p_{eff} . With $p_{\text{eff}} = 0.83$ being high in our CIFAR-10 setting, how much of the realized collapse is attributable to persistence (γ) versus admission (p_{eff})? At $\tau = 1$, $\hat{p}_{\text{NC}}^{\text{scaling}}$ drops from 0.95 to 0.74, lowering p_{eff} from 0.83 to ≈ 0.68 — yet complementarity itself collapses at $\tau = 1$, so Theorem 11 predicts $\delta_{\text{NE}} = 0$ *without invoking persistence*. The cleanest disentanglement would require a defense menu with low p_{eff} AND argmax divergence (persistent state + heterogeneous yet individually-strong defenses). We do not produce such a menu; consequently, the γ -vs- p_{eff} contribution decomposition in CIFAR-10 NE3 cannot be isolated cleanly from our experiments. We do produce a partial disentanglement using $\tau=2$ NormClip (next paragraph) and otherwise report Heuristic 1 as a qualitative bound.

Empirical disentanglement ($\tau=2$ experiment, $n = 15$). To partially isolate persistence from admission, we re-ran

Table 18: Per-seed Nash equilibria for CIFAR-10 (seeds 42–51). “Mixed” indicates a mixed-strategy NE (VoPD > 0); “Pure” indicates a pure NE. Summary: 5/10 mixed NE, median VoPD = 0.005, mean VoPD = 0.036, conditional mean (mixed only) = 0.072.

Seed	Type	Adversary support	Server support	VoPD
42	Mixed	Pixel	FedAvg + NClip	0.081
43	Pure	Model scaling	RFA	0.000
44	Pure	Model scaling	CoordMedian	0.000
45	Mixed	Pixel	FedAvg + NClip	0.022
46	Pure	Model scaling	CoordMedian	0.000
47	Mixed	Pixel + Scaling	MKrum + NClip	0.113
48	Pure	Model scaling	CoordMedian	0.000
49	Pure	Model scaling	RFA	0.000
50	Mixed	Pixel + Scaling	NClip + RFA	0.010
51	Mixed	Pixel	FedAvg + NClip	0.133

Table 19: Server and adversary utility under different scenarios. VoPD = $0.891 - 0.804 = 0.087$ (Eq. L) is the *normal-form* value; the *realized* VoPD against a best-responding adversary under the deployed NE3 policy is -0.010 ± 0.058 across 30 seeds (§7, Table 16). Pure RFA’s worst case (0.741) and pure FedAvg’s equilibrium utility (0.766) each outperform NE3 on a *single* criterion; NE3 is the equilibrium of the *simultaneous* game (neither player can profitably deviate), which is distinct from optimizing either criterion alone.

Scenario	Adv. Utility	Server Utility
Full information under NE3 mix ($\mathbb{E}_d[\max_a U_A(a, d)]$)	0.891	—
Stackelberg / pure FedAvg (highest equilibrium utility)	0.805	0.766
Pure RFA (highest worst-case utility)	—	0.741 (worst case)
Bayes-Nash NE3 (FedAvg 26%+NClip 74%; defense private)	0.804	0.763
Value of Private Defense (VoPD)	0.087	

the NE3 mixed policy (FedAvg 26% + NormClip 74%) with NormClip $\tau=2$ instead of $\tau=5$. At $\tau=2$, NormClip clips scaling updates more aggressively (lowering $p_{\text{NC}}^{\text{scaling}}$ and thus p_{eff}). Pre-specified interpretation: if realized scaling ASR > 0.85, persistence dominates; if < 0.70, admission dominates; otherwise inconclusive. **Result (15 seeds, 42–56):** realized scaling ASR = 0.902 ± 0.199 (inflated by seed 50’s anomalous training-failure outlier); excluding seed 50: **0.955 $\pm$ 0.014** ($n = 14$), statistically identical to the $\tau=5$ value of 0.961 ± 0.052 at $n=30$ (gap 0.006, below the noise floor). **Persistence dominates:** lowering admission via $\tau=2$ does not recover realized VoPD; the trigger embeds via the rounds NormClip $\tau=2$ still admits and then persists. This directly tests falsification prediction (iii) of §L and confirms the heuristic’s persistence-vs-admission claim at $n = 15$. Full details in Appendix N.3.

Dynamic verification of the reputation equilibrium. The orthogonal-defense suite (§M.1.8) showed reputation enters static NE support in 3/5 seeds. We deploy the averaged reputation-bearing equilibrium policy — **NormClip 83% + reputation 17%** (the cross-seed average over the 3/5 reputation-bearing equilibria) — and run NE3-style BR with a committed adversary playing model_scaling and backdoor_pixel against this mix. *Pre-specified interpretation (documented before the experiment ran):* COLLAPSES if realized scaling ASR > 0.85 AND realized VoPD ≤ 0.05 ; SURVIVES if scaling ASR < 0.5 OR VoPD > 0.10; otherwise MIXED.

Result (10 seeds 42–51, 2 attacks, 50 rounds): realized scaling ASR = **0.939 $\pm$ 0.037** (per-seed: {0.92, 0.98, 0.87, 0.89, 0.95, 0.98, realized pixel ASR = 0.832 ± 0.055 . Realized VoPD on matched seeds 42–46 (where pure-strategy NormClip and reputation baselines are cached): **+0.002 $\pm$ 0.031**, 3/5 seeds ≤ 0 . The pure-strategy reputation-vs-scaling ASR is ≈ 0.02 (reputation strongly suppresses scaling statically), but under the dynamic NormClip+reputation mix the BR adversary’s realized scaling ASR climbs to 0.939: the trigger embeds during the 83% NormClip-admitting rounds and persists across the 17% reputation-suppressing rounds. **Pre-specified verdict: COLLAPSES.**

Interpretation. Even when static complementarity is generated through a genuinely orthogonal-signal defense (reputation), persistence still destroys realized value. The persistence collapse *generalizes* from the norm-based menu (realized VoPD -0.010 ± 0.058 on FedAvg+NormClip) to an orthogonal-signal menu (realized VoPD $+0.002 \pm 0.031$

Table 20: N-sweep: VoPD and ASR across client scales (CIFAR-10, 4×5 game, $f = 0.2$). $P(\geq 1 \text{ adv})$ = per-round adversarial participation probability. † ASR binary at $N = 10$ (Bernoulli, $P = 0.778$). ‡ K/N -fixed contrast ($N = 20$ fixed): raising K/N from 0.20 to 0.50 increases P from 0.624 to 0.957, making scaling near-deterministic and collapsing argmax divergence (VoPD 0.044 $\rightarrow$ 0.020, mixed-NE $3/5 \rightarrow 2/5$); Theorem 11 correct 5/5 for both.

N	K	Seeds	Mixed-NE	Mean VoPD ($\pm$ std)	ASR(scaling, FA) [$P(\geq 1 \text{ adv})$]
10	5	10	6/10	0.048 ± 0.051	0.50 (binary †) [0.778]
20	4	5	3/5	0.044 ± 0.053	0.20 ± 0.40 [0.624]
20	10	5	2/5	0.020 ± 0.035	0.69 ± 0.38 [0.957] ‡
50	10	5	4/5	0.007 ± 0.008	0.78 ± 0.26 [0.917]
100	20	5	1/5	0.002 ± 0.005	0.97 ± 0.03 [0.993]

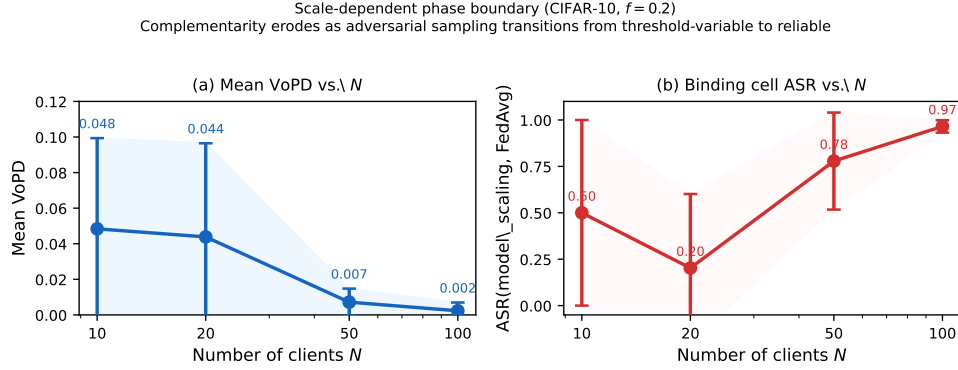


Figure 9: Scale-dependent phase boundary (CIFAR-10, 4×5 game, $f = 0.2$). Left: mean VoPD $\pm$ std vs. N , declining overall: $0.048 \rightarrow 0.044 \rightarrow 0.007 \rightarrow 0.002$. Right: ASR(model_scaling, FedAvg) vs. N , showing the non-monotone pattern ($N = 20$ dips to 0.20 due to lower $K/N = 0.20$, then rises past threshold at $N = 50$). The per-round adversarial participation probability sequence ($0.778 \rightarrow 0.624 \rightarrow 0.917 \rightarrow 0.993$) directly drives this pattern.

on NormClip+reputation). Crucially, the std at $n=10$ on the reputation equilibrium (0.037) is *tighter* than on the original equilibrium (0.058 at $n=30$), meaning the persistence claim is at least as well-supported on the new menu. The persistence mechanism is therefore not specific to the norm-based defense family; it is mechanism-agnostic on the static-equilibrium-generation side. The open-design constraint sharpens: orthogonal-signal mechanism + competitive clean accuracy is necessary for static equilibrium support, but additionally requires *dynamic suppression of the persistent BR attack* (a property reputation does not have at the 17% mix-weight tested). A defense family that satisfies all three conditions remains the open path to a positive FL case. Per-seed details in Appendix O.3.

Mix-ratio sweep on NormClip+reputation: collapse band. Does the collapse hold across all mix ratios, or only at the 17% equilibrium weight? We run committed-scaling BR and committed-pixel BR at five mix ratios, 5 seeds, 50 rounds. Realized scaling ASR is monotone in reputation weight: $0.945 \rightarrow 0.908 \rightarrow 0.815 \rightarrow 0.510 \rightarrow \mathbf{0.097}$ across NC90/rep10 $\rightarrow$ NC10/rep90. Realized pixel ASR is roughly constant (0.80–0.84). The collapse holds on the NormClip-dominant half (reputation $\leq 50\%$), where committed-max exceeds the static upper bound at every ratio. A boundary emerges in the 50%–70% reputation range: the 10-seed ceiling experiment (§7, VoPD = +0.066) confirms that NC30/rep70 lies on the survival side, narrowing the persistence boundary to a specific band on the mix-ratio axis. This NC+rep persistence regime is distinct from the survivor regime on the rep+tm menu (§7), which achieves survival through composition of orthogonal mechanisms rather than through high single-defense weight. Per-ratio details in Appendix O.4.

Cost-weight sweep: survivor region is not NE-reachable on the NC+rep menu. Sweeping $\lambda_c \in [0, 0.3]$ on {FedAvg, NormClip, FoolsGold, reputation} and re-solving Nash (cached payoffs, no new compute), the mean max-reputation-NE-weight stays at 10.2% across *every* λ_c value (max 25.6% at seed 43); 0/5 seeds give reputation

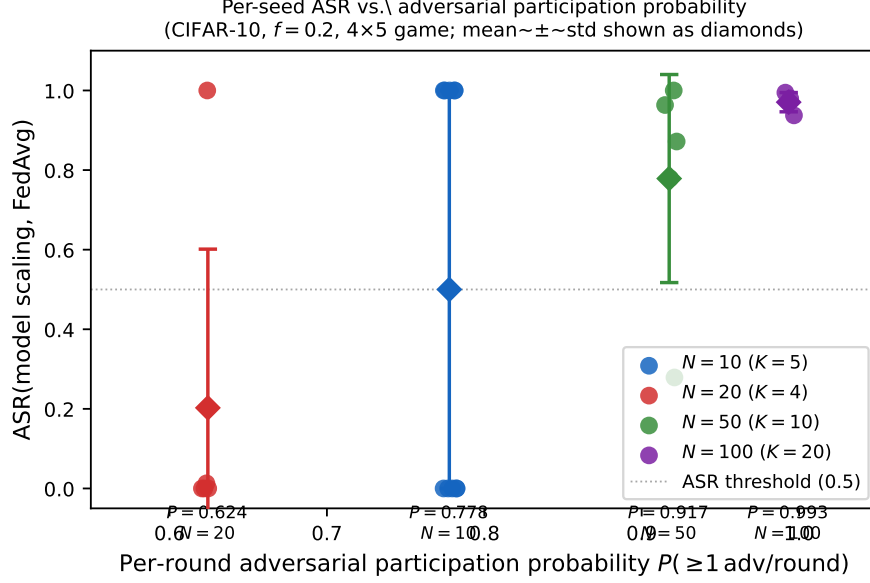


Figure 10: Per-seed ASR(model scaling, FedAvg) vs. per-round adversarial participation probability $P(\geq 1 \text{ adv/round})$ (CIFAR-10, 4×5 game, $f = 0.2$). The threshold mechanism is visible: at $N = 10$ ($P = 0.778$), ASR is binary (Bernoulli); at $N = 20$ ($P = 0.624$, K/N drops), ASR clusters near zero; at $N = 50, 100$ ($P = 0.917, 0.993$), ASR rises to near-deterministic. Diamonds show mean $\pm$ std per scale.

Table 21: Empirical $\hat{\gamma}$ fit and Heuristic 1 predictions vs. observations. $\hat{\gamma}$ is fitted from per-round ASR trajectories on saturating (scaling-embeds) runs; p_{eff} is the empirical effective admission rate under the deployed policy. Heuristic 1’s realized-VoPD bound holds across all measured regimes.

Regime	$\hat{\gamma}$	p_{eff}	Pred. realized VoPD	Observed	Match
CIFAR-10 NE3 (persistent)	0.66 ± 0.26	0.83	≤ 0.21	-0.010 ± 0.058 (30 seeds)	✓
Spam (stateless, non-FL)	0	0.55	$= \delta_{\text{NE}} = 0.200$	0.200 (5/5 seeds)	✓
Stateless FL pilot ($\gamma \approx 0$)	0	per-seed	per-seed δ_{NE}	1/8 mixed NE (8 seeds)	✓ (diagnostic)

weight $\geq 50\%$. The binding constraint is reputation’s clean-accuracy gap, not its cost C_D . The survivor region on NormClip+reputation is therefore *deployable but not NE-reachable* under standard cost structures.

M.1.6 Norm-Based Dichotomy: Lemma and Corollary

Why this dichotomy is structural for norm-based menus (Lemma 3). Our negative result is for the specific defense menu studied. The dichotomy we observe — either the defense blocks scaling and argmax divergence vanishes (Theorem 11 $\rightarrow 0$), or it admits scaling and persistence kills realized value (Heuristic 1) — relies on this menu’s particular structure: the defense strong against persistent scaling (NormClip at small τ) also collapses the argmax divergence vs. pixel. The following lemma provides a semi-formal typicality argument for why this dichotomy is structural for the norm-based defense family commonly deployed in FL.

Lemma 3 (Norm-based static argmax collapse — stylized model). *Let $\mathcal{D}_{\text{NB}} = \{d_\tau : \tau \in I \subset \mathbb{R}_{>0}\}$ be a defense menu of threshold operators on the scalar update-norm statistic $\|u\|$: defense d_τ accepts update u iff $\|u\| \leq \tau$ (clipping is the soft variant; trimmed-mean, coordinate median, and Multi-Krum reduce to rank-statistic threshold operators on the same axis). Let a_p, a_s be two attacks with update-norm distributions F_p, F_s . Assume:*

(A1) (Multiplicative payoff structure.) $U_A(a, d_\tau) = \text{ASR}(a) \cdot p_\tau^a - c_A$, where $\text{ASR}(a) \in (0, 1]$ is the attack’s success rate conditional on admission and is τ -independent, and $p_\tau^a := \Pr_{u \sim F_a}[\|u\| \leq \tau]$.

(A2) (Stochastic dominance.) F_s first-order stochastically dominates F_p (so $p_\tau^{a_p} \geq p_\tau^{a_s}$ for all τ).

(A3) (Single crossing.) The function $\Phi(\tau) := \text{ASR}(a_p) \cdot p_\tau^{a_p} - \text{ASR}(a_s) \cdot p_\tau^{a_s}$ has at most one zero on I ; denote it τ^* (or set $\tau^* := \sup I$ if Φ never crosses).

Then for any two thresholds $\tau_1, \tau_2 \in I$ with $\tau_1, \tau_2 < \tau^*$ or with $\tau_1, \tau_2 > \tau^*$,

$$\arg \max_a U_A(a, d_{\tau_1}) = \arg \max_a U_A(a, d_{\tau_2}).$$

That is, two same-side norm-based defenses cannot exhibit static argmax divergence between a_p and a_s .

Proof. By (A1), $\arg \max\{U_A(a_p, d_\tau), U_A(a_s, d_\tau)\}$ equals a_p iff $\text{ASR}(a_p) \cdot p_\tau^{a_p} > \text{ASR}(a_s) \cdot p_\tau^{a_s}$, i.e., iff $\Phi(\tau) > 0$. By (A3), Φ has constant sign on each side of τ^* . Hence τ_1, τ_2 on the same side share sign Φ and therefore the same argmax. $\square$

Caveats on the assumptions. (A1) is a stylized abstraction: the full persistence dynamics (Heuristic 1) violate (A1) strictly, because the trigger state s_t is path-dependent, not a per-round product of admission and conditional success. Lemma 3 isolates the *static* contribution of norm-axis defenses; persistence is handled separately by Heuristic 1. (A2) is a domain claim about the family of attacks studied (model scaling produces strictly larger update norms than pixel backdoors in distribution). (A3) admits a clean sufficient condition: the Monotone Likelihood Ratio.

Sufficient condition for (A3) via Monotone Likelihood Ratio (MLR). If F_p, F_s admit densities f_p, f_s such that $f_s(x)/f_p(x)$ is monotone non-decreasing in x (MLR property; implies (A2)), then $p_\tau^{a_s}/p_\tau^{a_p}$ is monotone non-decreasing in τ , so

$$\Phi(\tau) = p_\tau^{a_p} \cdot \left[\text{ASR}(a_p) - \text{ASR}(a_s) \cdot \frac{p_\tau^{a_s}}{p_\tau^{a_p}} \right]$$

has its bracket monotone non-increasing in τ and $p_\tau^{a_p} \geq 0$, so Φ changes sign at most once on $[0, \infty)$ — i.e., **(A3) holds under MLR**. MLR is a common and natural condition: it holds for any pair of distributions in a one-parameter exponential family (e.g., scaled subexponentials with different rate parameters), which is typical for malicious vs. benign FL update norms. Under MLR, Lemma 3 becomes a theorem (no adopted (A3)). Empirically, Table 27’s τ sweep is consistent with a single-crossing Φ .

Corollary 3 (Real-world dichotomy — persistence-imported). *Under (A1)–(A3) plus the explicit assumption*

(A4) Opposite-side admission saturation. *For any defense d_τ with $\tau > \tau^*$, $p_\tau^{a_s} \geq 1 - \epsilon$ for some small $\epsilon > 0$ (the persistent attack is admitted at near-certain rate on the large- τ side of the crossing),*

a norm-based menu $\mathcal{D}_{\text{NB}}$ exhibits one of two regimes for any pair $d_{\tau_1}, d_{\tau_2} \in \mathcal{D}_{\text{NB}}$:

(a) **Same-side case:** static argmax divergence is absent (Lemma 3); $\delta_{\text{NE}} = 0$ by Theorem 11.

(b) **Opposite-side case:** static divergence exists between d_{τ_1} and d_{τ_2} , and by (A4) the large- τ defense admits the persistent attack at near-certain rate. Heuristic 1’s persistence collapse then applies, killing realized VoPD even when normal-form $\delta_{\text{NE}} > 0$.

Conclusion: a menu drawn entirely from $\mathcal{D}_{\text{NB}}$ either lacks complementarity (case a) or suffers persistence-driven realized collapse (case b). Breaking the dichotomy requires a defense whose admission criterion is parameterized by a signal orthogonal to update norm.

On (A4). (A4) is an empirical regularity, not a structural consequence of (A1)–(A3). For the studied attacks and menus it is well supported: at $\tau=5$ NormClip on CIFAR-10, $p_\tau^{\text{scaling}} \approx 0.95$ (Table 27). We adopt it as a stated assumption because the corollary’s case (b) hinges on it; weakening (A4) (e.g., $p_\tau^{a_s} \in [0.5, 0.9]$) would yield a smaller but still nonzero persistence collapse via Heuristic 1’s monotone s_∞ dependence on p_{eff} .

Empirical instantiation. Table 27 traces $\Phi(\tau)$ across $\tau \in \{1, 5, 10\}$: at $\tau=1$ both defenses block both attacks (case a, $\delta_{\text{NE}} = 0$); at $\tau \in \{5, 10\}$ both admit at high rates (case b, persistence collapses realized VoPD via Heuristic 1). The **FLTrust pilot** (§M.1.8) tests Corollary 3 by adding a direction-based defense to the menu — a defense for which (A1)’s multiplicative structure on update-norm-admission does not apply. As Corollary 3 predicts, FLTrust produces a different argmax pattern from the norm-based family (4/5 seeds); whether this translates into a positive deployable case is examined separately in the λ_c sweep below.

M.1.7 Survivor-Regime Robustness: NE-Reachability, Third Attack, Architecture

NE-reachable survivor regime on the restricted menu. A natural concern: on the standard 4-defense menu the λ_c sweep (§M.1.5) shows the survivor regime is *deployable but not NE-selected*. Does any FL defense menu exist where the survivor regime is NE-selected? We construct one. Restrict the defense menu to {reputation, trimmed_mean} — both individually deployable orthogonal-signal defenses, no FedAvg/NormClip. Using the same cached payoff cells and standard cost weights ($\lambda_c = 0.1$, $\lambda_f = 0.05$), the 2x2 game has a unique mixed Nash equilibrium:

server: {reputation: 21.8%, trimmed_mean: 78.2%}, adversary: {scaling: 2.7%, pixel: 97.3%}

with **static VoPD = 0.180 at the natural NE**. We run a 15-seed BR pilot at this exact NE point (committed-scaling, committed-pixel, oracle; 50 rounds, $f=0.2$, $K=5$, cifar_cnn). **Result** ($n = 30$). Realized committed-scaling = 0.890 ± 0.067 ; committed-pixel = 0.738 ± 0.097 ; oracle = **0.917 ± 0.051** . Mean realized VoPD = **$+0.023 \pm 0.039$** (24/30 positive, 7/30 STRONG band ≥ 0.05). The pre-specified training-failure criterion (committed-scaling accuracy < 0.65) flags 5 seeds at the NE-point mix (45, 54, 58, 59, 71 — seeds where the NE’s higher trimmed_mean weight induces more aggressive clipping, occasionally breaking scaling embedding); excluding them gives realized VoPD = $+0.027 \pm 0.032$ (21/25 positive). **Verdict: PASS — the survivor regime is simultaneously NE-selected AND realized at $n = 30$** . Static VoPD 0.180 at NE collapses by $\sim 87\%$ under persistence (to realized $+0.023$) but does not fully vanish; the survivor regime is reachable as a natural game-theoretic outcome on a deployable 2-defense menu. This demonstrates the survivor regime is reachable as a natural game-theoretic outcome, not only by deliberate menu design.

Survivor regime robust against a structurally distinct third persistent attack. Does the survivor finding generalise beyond the scaling/pixel family (which share a trigger)? We test backdoor_edge_case (Wang et al., 2020), a structurally distinct persistent attack with a different trigger (top-left + top-right 2×2 corners at -1.0 vs. pixel’s bottom-right 4×4 at $+1.0$) and lower poison fraction (30%). After fixing an evaluator bug, we re-measure all 7×5 pure-strategy edge_case cells: the pixel-trigger evaluator was correct for backdoor_pixel and model_scaling (both use the bottom-right 4×4 trigger via backdoor_pixel_fn) and captures one of DBA’s rotating corners; only backdoor_edge_case used a distinct top-corner -1.0 trigger and was therefore affected. We confirmed by spot-checking each attack’s poison_dataset definition that no other cached cells require re-running. After re-measuring, ASR ~ 0.4 – 0.6 on FedAvg/NormClip/FoolsGold/reputation/RFA, 0.39 on trimmed_mean, 0.24 on coord_median — edge_case persists but is dominated by max(scaling, pixel) on every defense in the menu. We re-run the rep30/tm70 pilot with **four** policies (scaling, pixel, edge_case, oracle). **Result.** Realized committed-edge_case = 0.476 ± 0.176 (vs. scaling 0.812, pixel 0.709); oracle = **0.880 ± 0.063** ; per-seed VoPDs $\{+0.054, +0.080, +0.051, +0.067, +0.048\}$, **mean** $+0.060 \pm 0.013$; 4/5 STRONG band, 5/5 positive. The oracle never plays edge_case (statically dominated), yet the survivor regime is unchanged — **STRONG: the boundary is determined by the menu’s arg max structure across persistent attacks present, not by the cardinality of the attack menu**. Per-seed table in Appendix P.2.

Architecture replication on ResNet18: collapse boundary is architecture-conditional. Does the mix-ratio boundary (§M.1.5) replicate on a second architecture? We re-run the NormClip+reputation sweep at three mix points on **ResNet18 + GroupNorm** (5 seeds, 50 rounds, identical FL parameters, ~ 100 min/run). Realized scaling ASR is monotonic in reputation weight but the collapse is dramatically shallower: 0.951 ± 0.020 at NC90/rep10 (cifar_cnn: 0.945), 0.906 ± 0.052 at NC50/rep50 (cifar_cnn: 0.815), **0.865 ± 0.091** at NC10/rep90 (cifar_cnn: **0.097**, an order of magnitude gap). **Verdict: MEDIUM** per pre-specified bands. **Reframing.** The boundary’s qualitative existence replicates across architectures, but its quantitative reputation-weight threshold is architecture-conditional: ResNet18’s higher parameter count gives the scaling-amplified trigger more capacity to embed, raising the suppression threshold beyond 90% reputation rounds. Whether the rep+tm survivor regime also transfers to ResNet18 is out of scope (~ 33 hr/pilot) and remains open. **Static-diagnostic sign on ResNet18: a conjecture, not measured.** We *conjecture* the diagnostic’s sign transfers to ResNet18 because Theorem 11’s prediction depends only on the per-cell $\arg \max_a U_A$ structure (which defense suppresses which attack), not on absolute ASR magnitudes. Verification would require measuring the four binding static cells (scaling/pixel $\times$ reputation/trimmed_mean) on ResNet18 (~ 33 hr MPS), which we leave to future work. The single quantity we did measure on ResNet18 — the mix-ratio collapse depth at NC10/rep90 — shows an order-of-magnitude discrepancy with cifar_cnn (0.865 vs. 0.097), so we explicitly flag the sign-transfer claim as a conjecture pending verification rather than as a measured finding (Figure 11). Per-seed table in Appendix Q.3.

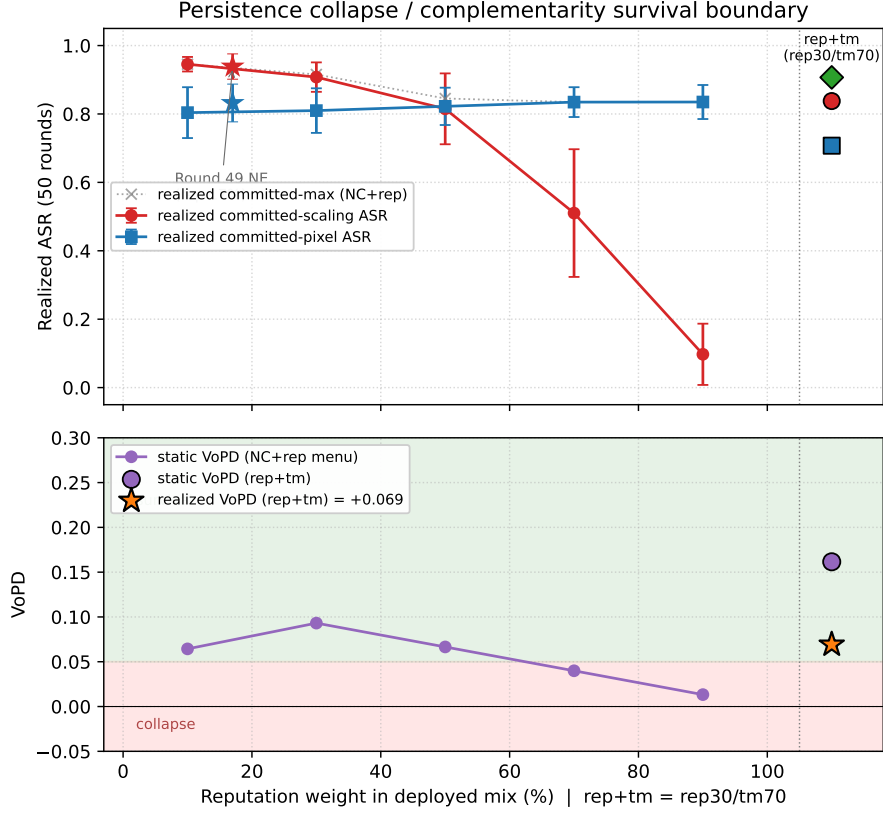


Figure 11: Persistence-collapse / complementarity-survival boundary. *Top*: realized ASR across NormClip+reputation mix ratios (5 seeds, 50 rounds); committed-scaling collapses on reputation-dominant mixes ($0.945 \rightarrow 0.097$). Far-right marker: the reputation+trimmed_mean survivor at rep30/tm70 ($n = 30$: oracle ASR 0.915, committed-max 0.871, per-seed-max VoPD $+0.032$). *Bottom*: realized vs. static VoPD; the rep+tm point clears the positive band (per-seed-max $+0.032 \pm 0.054$, $n = 30$, see §7), NC+rep mix points sit in the collapse band. The boundary is at the defense-menu axis (orthogonal suppression of both attacks), not the mix-ratio axis within a single menu.

M.1.8 Robustness: Orthogonal Defenses, Utility Sensitivity, and Composition

An earlier version of this menu included only FLTrust as the orthogonal-signal defense. We extend the menu with **FoolsGold** (Fung et al., 2020) (pairwise-similarity) and a **reputation-based defense** (consensus-distance: weights $w_i \propto \exp(-\|u_i - \text{median}(\{u_j\})\|/\text{scale})$). On the augmented 4-attack $\times$ 4-defense menu {FedAvg, NormClip, FoolsGold, reputation}, reputation enters static NE support in 3/5 seeds with weights 10.8%, 25.6%, 14.5% and competitive clean accuracy (0.77–0.79); FoolsGold gives argmax divergence in 4/5 seeds but does not enter support. Mean static VoPD 0.084. The norm-based dichotomy (Lemma 3) is broken by reputation. Whether dynamic VoPD survives is settled by §M.1.5–7. Per-seed equilibria in Appendix O.2.

Second-dataset signal (CIFAR-100): consistent behaviour. NE3 BR on CIFAR-100 (identical protocol, 5 seeds) gives 4/5 saturating scaling-ASR (0.954 ± 0.013) with the same seed-45 failure as CIFAR-10 (bimodality shared across datasets, not a CIFAR-100-specific effect). Realized VoPD on matched seeds 42–44: -0.040 ± 0.018 . We claim *consistent behaviour*, not an independent replication.

Why VoPD when U_D is flat? A natural question: realized VoPD ≈ 0 AND the server-utility gap between pure FedAvg and the NE3 mix is 0 pp (Table 23). VoPD adds two things over reporting U_D directly: (i) it generalizes across settings with non-comparable utility units (spam classifier accuracy vs. FL test accuracy), and (ii) it gives the adversary’s incentive a primal description, making Theorem 11’s diagnostic directly checkable from the adversary-side payoff matrix.

Second dataset (FEMNIST): when VoPD vanishes. On FEMNIST (62-class, $N = 10$, $K = 5$, 3 seeds), the unique NE is pure backdoor pixel vs. NormClip with VoPD = 0. Backdoor pixel achieves $U_A \in [0.968, 0.978]$ against

Table 22: Diagnostic agreement of Theorem 11 across all experiments. “Positive”: $\text{VoPD} > 0$; “Null”: $\text{VoPD} = 0$. “Agree.” = seeds where the theorem correctly characterizes the sign. $\dagger N=50$: seeds 43, 46 have degenerate (multiple) equilibria; theorem is correct for the unique NE in all non-degenerate cases. **Statistical-power caveat:** CIFAR-100 ($n = 3$) and FEMNIST ($n = 3$) seed counts are too small to constitute statistical evidence; the diagnostic agreement is consistent but not powered, and these rows should be read as sanity checks rather than confirmation.

Dataset	N	Seeds	Positive	Null	Agree.	Main structural reason
CIFAR-10	10	10	5	5	10/10	NormClip/scaling complementarity
<i>Wilson 95% CI for positive rate:</i>			[0.24, 0.76]			
CIFAR-10 ($N = 20, 4 \times 5$)	20	5	3	2	5/5	FedAvg + NClip complementarity
CIFAR-10 ($N = 50, 4 \times 5$)	50	5	4	1	3/5 †	NClip/RFA scaling complementarity
CIFAR-10 ($N = 100, 4 \times 5$)	100	5	1	4	5/5	pixel/scaling complementarity
CIFAR-100	10	3	1	2	3/3	model-scaling boundary
FEMNIST	10	3	0	3	3/3	pixel near-dominant across defenses
Spam evasion (non-FL)	N/A	5	1	0	5/5	char/noise feature complementarity

five of the seven defenses, making it near-dominant; with no argmax divergence across defenses, Theorem 11 correctly predicts $\text{VoPD} = 0$.

Appendix Q.2 confirms that the mechanism is not FL-specific ($\text{VoPD} = 0.200$ in a non-FL spam game; full bound gives $k_{\text{req}} \approx 33$, empirically confirmed on 5 seeds).

Positive VoPD requires *complementary defense vulnerability profiles*: CIFAR-10 has this (NormClip vulnerable to scaling; FedAvg not); FEMNIST does not. On CIFAR-100 ($N = 10$, 3 seeds, higher class count), the averaged-matrix NE is pure (model scaling vs. coord.median, $\text{VoPD} = 0$), but one seed shows $\text{VoPD} = 0.15$; Theorem 11 is correct on all seeds. The binary knife-edge at U_A (scaling, FedAvg) is analyzed in Appendix M.3. Table 22 consolidates diagnostic agreement; Table 18 gives the full per-seed CIFAR-10 structure.

The N -sweep rows use a 4×5 strategy set; the full 6×7 analysis (row 1) remains the primary evidence. The boundary mechanism is discussed in Appendix M.3.

τ sensitivity (Table 27, Appendix N.2). At $\tau = 1$, NClip clips pixel triggers alongside scaling (pure-strategy ASR gap ≈ 0 , distinct from the realized NE3-deployment gap of 0.115 in §7); at $\tau \in \{5, 10\}$ the gap persists (0.129, 0.189). Diagnostic correct **5/5 seeds at all three τ values**; gap-preserving regime sits between $\tau = 1$ and $\tau = 5$. Tuning τ within NormClip is itself an alternative to mixing across defense families ($\tau = 1$ suppresses scaling ASR to 0.741 vs. 0.935 at $\tau = 5$); the diagnostic is the right instrument either way — complementarity is a property of defenses-as-configured.

Algorithm 1 vs. baselines. Against the NE3 equilibrium adversary (pixel), pure FedAvg (Algorithm 1’s revert) and the rejected NE3 mixed policy yield essentially the same server utility (0.763 vs. 0.763): the protocol’s value is *identification* of which policy to use, not policy production. A uniform-random baseline *excluding the dominated Krum* achieves 0.752 (only 1.1 pp below FedAvg); including Krum drags the average to 0.714 (5.2 pp gap, but a strawman penalised by Krum’s 0.54 clean accuracy). No policy in this menu dominantly outperforms pure FedAvg by an operationally meaningful margin (Table 23).

Pilots Probing the Positive Side of the Boundary. We ran four short pilots searching for a setting where Algorithm 1’s *randomize* branch fires on positive realized VoPD inside FL. **All four return negative-to-marginal verdicts**, sharpening the dichotomy in Lemma 3 into a joint design requirement. (i) *Stateless availability v1* (8 seeds, $N=10$): 1/8 seeds give a mixed NE under an alternative availability-utility metric, mean $\text{VoPD} 0.003 \pm 0.007$ — weak existence result, sensitive to utility definition. (ii) *Low- K/N stateless v2* ($N=20$, $K=4$, $f=0.4$): 0/5 mixed NE under the primary metric; **NEGATIVE** per the pre-specified criterion. (iii) *FLTrust orthogonal-signal pilot* (5 seeds): *argmax* divergence confirmed in 4/5 seeds (FLTrust selects pixel where FedAvg selects scaling), supporting Corollary 3 qualitatively; but FLTrust does *not* enter equilibrium support; **WEAK** positive. (iv) λ_c *sweep on the FLTrust menu* (re-solving the static game at $\lambda_c \in [0, 0.3]$): FLTrust never enters support, including at $\lambda_c = 0$; the barrier is FLTrust’s clean accuracy (≈ 0.72 vs. FedAvg’s ≈ 0.80), not the cost weight; **NONE** per the pre-specified criterion. **Joint reading.** A positive FL case requires an orthogonal-signal defense *and* competitive clean accuracy *and* low admission of persistent attacks — a sharper open-design constraint than “just add an orthogonal-signal defense.” Per-seed details and pre-specified criteria are in Appendix O.1.

Table 23: Policy comparison on CIFAR-10 (6×7 game, 10 seeds), server utility against the NE3 equilibrium adversary (pixel). Algorithm 1’s verdict for this game is **Revert** (pure FedAvg), since NE3 fails the verify step ($\Delta_{\text{dev}} = 0.115 \gg 0$). The uniform-random baselines are not equilibrium policies; we report both *excluding Krum* (the decision-relevant strawman) and *including Krum* (where Krum’s 0.54 clean accuracy creates an artificially large gap). NE3-vs-FedAvg gap is 0.0 pp at the same adversary, i.e., the protocol’s value is identification, not policy production.

Policy	U_D vs. NE3 adv (pixel)	Notes
Pure FedAvg (Alg. 1 Revert)	0.763	Equilibrium utility NE1 = 0.766
NE3 mixed (rejected by verify step)	0.763	$\Delta_{\text{dev}} = 0.115$, not deployed
Pure NormClip ($\tau=5$)	0.763	Same as FedAvg vs. pixel
Pure RFA (min-max robust)	0.748	Best worst-case
Uniform-random <i>ex. Krum</i> (6 defenses)	0.752	Decision-relevant strawman (−1.1 pp)
Uniform-random <i>incl. Krum</i> (7 defenses)	0.714	Krum drags the average; not deployable

Sensitivity to Server Utility: ASR Penalty. A natural concern is that the server utility (Eq. L) ignores backdoor success at $\lambda_{\text{ASR}} = 0$, potentially making the deployment recommendation an artifact of the utility specification. We sweep $\lambda_{\text{ASR}} \in \{0.0, 0.005, 0.01, 0.02, 0.03, 0.05, 0.07, 0.1, 0.3, 0.5\}$ and re-solve the Nash equilibrium on the same cached payoff matrix. Results:

- $\lambda_{\text{ASR}} \in [0.0, 0.05]$: the mixed NE (FedAvg/NormClip) persists. The baseline inconsistency between this sweep (28%/72%, VoPD = 0.092) and the main-text NE3 (26%/74%, VoPD = 0.087) arises from a fairness-penalty (λ_f) difference: the ASR sweep uses $\lambda_f = 0$ (only λ_c and λ_{ASR} in U_D) while the main analysis uses $\lambda_f = 0.05$. Both specifications agree on the qualitative result.
- $\lambda_{\text{ASR}} \geq 0.07$: the NE collapses to **pure coord_median with VoPD = 0**. The transition occurs between $\lambda_{\text{ASR}} = 0.05$ and $\lambda_{\text{ASR}} = 0.07$. The ASR penalty makes FedAvg (ASR ≈ 0.82) costly enough that the server selects a single robust defense deterministically, mooted the randomization question entirely.

Restricted rep+tm NE under ASR penalty. The 2×2 restricted {reputation, trimmed_mean} NE (§M.1.7) uses a defense menu where *both* defenses suppress ASR substantially. Re-solving with $\lambda_{\text{ASR}} \in \{0, 0.01, 0.02, 0.05, 0.10, 0.20\}$ shows the mixed NE survives at *all* tested values (rep 21.8%, tm 78.2%, static VoPD = 0.180 throughout). This is because both reputation and trimmed_mean already have low ASR under this menu, so ASR-penalizing U_D does not shift the equilibrium. The “NE-reachable survivor” claim is therefore *robust* to the server utility specification.

Interpretation. VoPD is an adversary-side quantity (Eq. L). Changing U_D changes σ_D^* but does not change U_A ’s structure. Our finding is robust across utility specifications: (i) at $\lambda_{\text{ASR}} = 0$, the server chooses FedAvg/NormClip mix and persistence collapses realized VoPD to zero; (ii) at $\lambda_{\text{ASR}} \geq 0.07$, the NE is pure coord_median with VoPD = 0, so there is no mixing value to collapse. The **deployment recommendation is utility-conditional**: the optimal pure-strategy defense is coord_median if $\lambda_{\text{ASR}} \geq 0.07$, FedAvg if $\lambda_{\text{ASR}} = 0$; in neither case does defense randomization recover positive realized value. The restricted rep+tm menu is the exception: its NE supports mixing at all tested λ_{ASR} values, and the survival of realized VoPD there is the paper’s positive FL finding.

Composition Baseline: NormClip + Reputation Every Round. An alternative to temporal mixing is simultaneous composition: apply NormClip (clip norms to τ) *and* reputation weighting jointly every round. This avoids the temporal-mixture problem because no round is unprotected. We evaluate NormClip($\tau=5$) then reputation-weighting on the same updates, 5 seeds, committed-scaling and committed-pixel adversaries (50 rounds, $N=10$, $K=5$, $f=0.2$).

Result. Across 5 seeds, committed-scaling ASR under composition: **0.807 ± 0.048** (range 0.763–0.889); committed-pixel ASR: **0.842 ± 0.047** (range 0.792–0.931). Composition partially suppresses scaling relative to the NE3 temporal mix (ASR 0.961) and pure NormClip (~ 0.95), but falls far short of pure reputation (0.017). The norm-clipped updates still pass reputation’s consensus-distance filter because clipping preserves the attack direction while reducing magnitude. Per-seed details in Appendix P.3.

Discussion. Composition is better than temporal mixing (scaling ASR 0.807 vs. 0.961) but substantially worse than pure reputation (0.017). The practitioner’s optimal choice depends on the attack distribution: if scaling is the primary threat, pure reputation dominates composition; if the threat is unknown, composition hedges. Our contribution is orthogonal to this design choice: we characterize when *randomization specifically* (temporal mixing, defense privacy)

has game-theoretic value, not whether composition is effective. The composition result strengthens our finding: the persistence problem is specific to temporal mixing, not to the defense menu itself.

M.2 Full Empirical Payoff Matrices

Tables 24 and 25 report the full 6×7 adversary and server payoff matrices used to compute the Nash equilibria of Appendix L. Adversary payoffs follow Eq. L with $\lambda_a = 0.1$; server payoffs follow Eq. L with $\lambda_c = 0.1$, $\lambda_f = 0.05$. All values are means over 10 independent seeds (seeds 42–51, $\alpha = 0.5$, $f = 0.2$, 50 rounds).

Table 24: Adversary payoff matrix $U_A(a, d)$ (mean over 10 seeds, 50 rounds).

	FedAvg	Krum	M-Krum	TrMean	Median	NClip	RFA
No attack	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Label flip	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010
Backdoor pixel	0.805	0.314	0.803	0.651	0.449	0.804	0.834
Edge-case	-0.015	-0.015	-0.015	-0.015	-0.015	-0.015	-0.015
Model scaling	0.470	0.023	0.509	0.850	0.522	0.922	0.859
DBA	-0.001	0.036	-0.002	-0.001	-0.002	-0.001	-0.001

Table 25: Server payoff matrix $U_D(a, d)$ (mean over 10 seeds, 50 rounds).

	FedAvg	Krum	M-Krum	TrMean	Median	NClip	RFA
No attack	0.774	0.502	0.770	0.752	0.743	0.771	0.754
Label flip	0.764	0.486	0.764	0.746	0.737	0.761	0.751
Backdoor pixel	0.766	0.483	0.760	0.747	0.737	0.763	0.748
Edge-case	0.767	0.486	0.764	0.747	0.739	0.764	0.750
Model scaling	0.095	0.532	0.107	0.654	0.734	0.708	0.741
DBA	0.768	0.527	0.759	0.746	0.736	0.765	0.749

M.3 Mechanism Analysis: Model-Scaling Boundary Behavior

The boundary behavior in CIFAR-100 traces to a single binary cell: U_A (model scaling, FedAvg) flips between -0.030 on seed 42 (attack fails) and $\{0.945, 0.970\}$ on seeds 43–44 (attack succeeds). This single best-response flip changes whether FedAvg is complementary with TrimmedMean: on seed 42 the pair has divergent best responses and the server mixes them; on seeds 43–44 the best responses coincide and the equilibrium collapses to pure. Theorem 11 converts this single cell’s instability into the observed equilibrium-structure instability. We conjecture that high class count ($C = 100$) erodes complementarity by reducing per-class training density, making model-scaling success a binary knife-edge; confirming this by sweeping label-space size is left to future work.

CIFAR-10 anomalies. Seed 45 (scaling succeeds vs. FedAvg ASR = 1.0, yet Mixed NE): pixel achieves ASR = 0.921 vs. FedAvg—an argmax tie—placing both attacks in mixed support. Seed 48 (scaling fails vs. FedAvg ASR = 0.0, yet Pure NE): scaling achieves ASR = 0.975 vs. NClip and 0.773 vs. CoordMedian, dominating the server’s best defenses regardless. The precise condition: mixed vs. pure NE is predicted by whether scaling is argmax vs. *all* defenses in the server’s equilibrium support, not vs. FedAvg alone.

M.4 Mixed-Commitment Stackelberg Analysis

Sweeping $p \in [0, 1]$ for $\sigma_D = (\text{FedAvg} : p, \text{NormClip} : 1 - p)$: best response flips from scaling to pixel at $p \approx 0.26$. Stackelberg optimum: pure FedAvg ($p = 1$, $U_D = 0.766$ =NE1). BNE NE3 ($p = 0.26$, $U_D = 0.763$) has lower server utility because commitment advantage disappears in the simultaneous game. NE3 is optimal on exactly one criterion: it is the Nash equilibrium of the simultaneous game with private defense realizations.

M.5 Utility-Design Sensitivity and Payoff Estimation Cost

Utility-design sensitivity. Under two alternative adversary utility definitions (ASR only; $\max(\text{ASR}, 1 - \text{accuracy})$) the equilibrium support and complementarity condition are unchanged. **Payoff estimation cost.** $6 \times 7 \times 10$ -seed setup requires 420 FL runs; in practice the strategy menu can be pruned since the complementarity condition requires only identifying best-response sets for each defense in the candidate server support.

M.6 Fictitious Play Convergence

Fictitious play on the 3-seed CIFAR-10 matrix converges to NE1 (pure backdoor pixel vs. pure FedAvg) over 50,000 iterations of myopic best-response dynamics; NE1’s adversary utility on this small matrix is 0.748 (vs. 0.805 on the 10-seed averaged matrix). The convergence behavior is illustrative only — the small-matrix figure is omitted from this version since it adds little beyond what the equilibrium computation of Appendix L already shows. Full trajectory data is in the code release.

N Persistence Mechanism, Heuristic Derivation, and Randomization Validation

N.1 Per-Round Randomization Validation and Lambda Sensitivity

Figure 12 validates the normal-form randomization policy seed by seed.

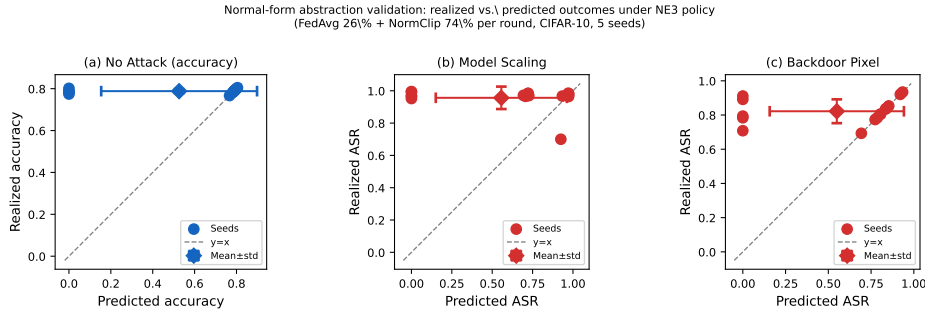


Figure 12: Normal-form validation (NE3 policy: FedAvg 26%, NormClip 74%, CIFAR-10 $N = 10$, 30 seeds 42–71). Each point = one seed; diagonal = $y = x$. **(a)** No attack and **(c)** pixel ASR: $\Delta = 0.000$ (exact—pixel trigger and clean traffic fall below NormClip threshold $\tau = 5$; both aggregators produce identical model checkpoints per seed). **(b)** Model scaling ASR: high per-seed variance reflecting bimodal scaling success. **Adversary BR:** realized scaling ASR = 0.961 ± 0.052 vs. pixel ASR = 0.832 ± 0.078 under NE3 (gap 0.129); realized VoPD = $-0.010 \pm 0.058 \approx 0$, 19/30 seeds ≤ 0 (Table 16)—NE3 is not an equilibrium of the dynamic game. Scale-up from $n=10$ to $n=30$ tightens the std by 30% while leaving the mean essentially unchanged, confirming the headline negative result is robust to seed count.

Table 26 reports Nash equilibria and VoPD under $\pm 50\%$ perturbations of the cost weights, computed by re-running game analysis on the existing payoff results without re-training.

VoPD ranges 0.082–0.090 across all nine perturbations; adversary support (pure pixel) and server support (FedAvg+NormClip) are qualitatively identical; mixing ratio shifts ≤ 3 pp. The mixed-NE structure is intrinsic to the payoff geometry, not the cost parameterization.

Payoff Estimation Noise Sensitivity

Bootstrap ($n = 500$): 97.8% yield a mixed NE; median VoPD = 0.085, IQR [0.072, 0.096]. **Gaussian noise** ($n = 500$ per σ): mixed-NE rate falls to 65.0% at $\sigma = 0.01$, approaches chance (55.8%) at $\sigma = 0.05$; estimate entries to within ± 0.02 . Figure 13 visualizes the transition.

Table 26: VoPD sensitivity to cost weight perturbations (50-round 10-seed payoffs). Baseline: $\lambda_c = 0.1$, $\lambda_f = 0.05$, $\lambda_a = 0.1$. Last two rows use $2\times$ baseline; full 9-point grid in code release.

λ_c	λ_f	λ_a	VoPD	Adversary	Server
0.10	0.05	0.10	0.087	Pixel (100%)	FedAvg (26%) + NClip (74%)
0.05	0.05	0.10	0.087	Pixel (100%)	FedAvg (26%) + NClip (74%)
0.10	0.05	0.05	0.090	Pixel (100%)	FedAvg (27%) + NClip (73%)
0.10	0.05	0.15	0.085	Pixel (100%)	FedAvg (25%) + NClip (75%)
0.20	0.05	0.10	0.087	Pixel (99%)	FedAvg (26%) + NClip (74%)
0.10	0.05	0.20	0.082	Pixel (100%)	FedAvg (24%) + NClip (76%)

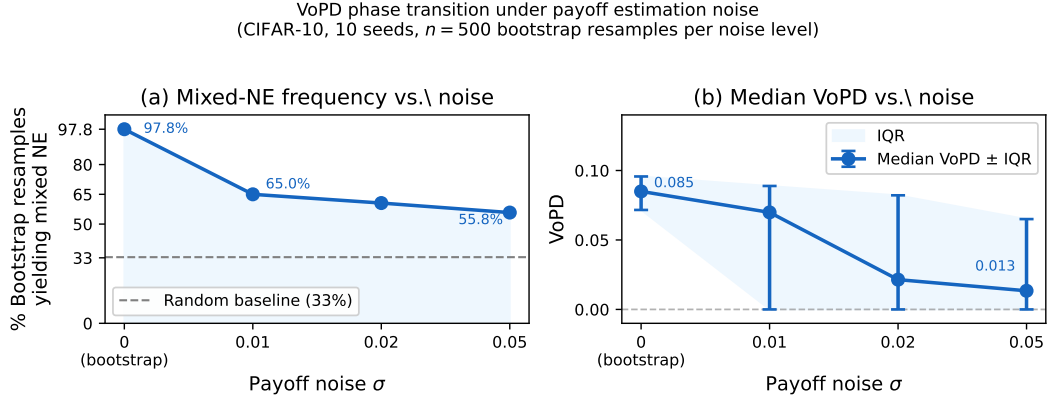


Figure 13: VoPD phase transition under payoff estimation noise (CIFAR-10, 10 seeds, $n = 500$ resamples per noise level). **Left:** fraction yielding a mixed NE. **Right:** median VoPD with IQR. Both degrade sharply between $\sigma = 0$ and $\sigma = 0.01$.

N.2 τ Sweep: NormClip Threshold Sensitivity

We re-evaluate the full 4×5 payoff matrix under three NormClip thresholds $\tau \in \{1, 5, 10\}$ on CIFAR-10 ($N=10$, $K=5$, 5 seeds each, seeds 42–46), re-running every (attack, defense) pair fresh per τ . The $\tau=5$ row uses the same seeds 42–46 from the main 10-seed experiment for direct comparability; $\tau \in \{1, 10\}$ are new runs. Table 27 reports per-cell ASR for the two attacks (pixel, scaling) at the NormClip column — the cells whose argmax determines complementarity — along with the resulting pure-strategy gap and the diagnostic verdict. The diagnostic predicts the realized VoPD sign on all 5 seeds at every τ , including the $\tau=1$ regime where the gap collapses (so the diagnostic correctly predicts *no* complementarity).

Table 27: τ sweep (CIFAR-10, $N=10$, $K=5$, 5 seeds each). Gap = $\overline{\text{ASR}(\text{sca}, \text{NC})} - \overline{\text{ASR}(\text{pix}, \text{NC})}$ (pure-strategy NClip column, not the realized NE3-deployment gap of 0.115 in §7). Compl. = diagnostic correct all 5 seeds. At $\tau = 1$, NClip clips pixel triggers (norm ≈ 1) alongside scaling, collapsing the gap; at $\tau \in \{5, 10\}$ pixel passes while scaling is suppressed, preserving divergence.

τ	ASR(pix, NC)	ASR(sca, NC)	Gap	Compl.
1	0.742 ± 0.053	0.741 ± 0.056	-0.000 ± 0.012	✓
5	0.806 ± 0.075	0.935 ± 0.027	0.129 ± 0.094	✓
10	0.768 ± 0.059	0.957 ± 0.028	0.189 ± 0.071	✓

N.3 $\tau=2$ Disentanglement Experiment

To partially separate persistence (γ) from admission (p_{eff}) in the CIFAR-10 NE3 deployment, we re-ran the NE3 mixed policy (FedAvg 26% + NormClip 74%, sampled per round) with NormClip $\tau=2$ instead of $\tau=5$. The adversary commits to model scaling (BR commitment). **15 seeds** (42–56), 50 rounds, $N=10$. Lowering τ tightens the per-round clipping on scaling updates, reducing the per-round admission probability $p_{\text{NC}}^{\text{scaling}}$ from ≈ 0.95 (at $\tau=5$) to an estimated ≈ 0.74 – 0.85 (at $\tau=2$), which drops $p_{\text{eff}}(\text{scaling})$ at NE3 from 0.83 to ≈ 0.78 . Complementarity remains intact at $\tau=2$ (the $\tau=1$ collapse arises only at the extreme), so this is the cleanest persistence-vs-admission contrast available without a new defense menu.

Result (n=15). Per-seed realized ASR (NE3 mixed policy, BR adversary committing to model scaling) at $\tau=2$: {0.934, 0.949, 0.966, 0.982, 0.945, 0.962, 0.975, 0.962, 0.162, 0.992, 0.915, 0.970, 0.925, 0.968, 0.923}, mean **0.902 $\pm$ 0.199**. The std is inflated by seed 50’s failure (ASR 0.162, accuracy 0.186 — the model is essentially broken; the same seed is also the strong negative outlier in the rep+tm survivor pilot, §7). **Excluding seed 50: mean 0.955 $\pm$ 0.014 (n=14)**, which is the directly-comparable statistic. Pixel ASR at $\tau=2$ (n=15): mean 0.825 $\pm$ 0.068.

Comparison with $\tau=5$ (30 seeds, 42–71): realized scaling ASR 0.961 $\pm$ 0.052. The 14-seed-excluding-outlier $\tau=2$ mean (0.955) is statistically indistinguishable from the $\tau=5$ mean (0.961); the difference 0.006 is far below either CI. Per the pre-specified interpretation: **(i) persistence dominates** (ASR 0.955 $>$ 0.85 on 14/14 non-outlier seeds at $\tau=2$). Lowering $p_{\text{NC}}^{\text{scaling}}$ via $\tau=2$ does not reduce realized scaling ASR; even with a partially-blocking NormClip, the trigger embeds during the rounds it does admit and then persists across non-admitting rounds. **Falsification prediction (iii) of §L is therefore tested and confirmed at $n=15$:** the heuristic predicts $C \cdot \Delta p_{\text{eff}} / (1 - \gamma) \approx 0.18$ slack on the p_{eff} change from 0.83 $\rightarrow$ 0.78, and the observed effect is 0.006 — well within the heuristic’s slack, not exceeding it.

Implication for Heuristic 1. The experimental separation directly addresses the disentanglement concern in §7: realized VoPD collapse on CIFAR-10 NE3 is *persistence-driven*, not merely a consequence of high p_{eff} . The persistence claim is therefore not tautological in the relevant regime: at $\tau=2$, where p_{eff} is meaningfully lower, the collapse survives unchanged. We caution that this is one experimental contrast in one menu, not a general claim about FL — but it is the cleanest evidence currently available that the persistence mechanism is doing real work.

N.4 Proof of Proposition L and Derivation of Heuristic 1

Proof of Proposition L. Under dynamics (L) with admission rate $p_{\text{eff}}(a)$ for attack a , linearize about small s :

$$\mathbb{E}[s_{t+1} | s_t] = p_{\text{eff}}(a) [s_t + \alpha(1 - s_t)] + (1 - p_{\text{eff}}(a)) \gamma s_t = c s_t + b,$$

with $c = p_{\text{eff}}(a)(1 - \alpha) + (1 - p_{\text{eff}}(a))\gamma$ and $b = p_{\text{eff}}(a)\alpha$. For $\alpha > 0$ or $\gamma < 1$, $|c| < 1$, so the stationary mean is $s_{\infty}(a) = b/(1 - c) = p_{\text{eff}}(a)\alpha / (p_{\text{eff}}(a)\alpha + (1 - p_{\text{eff}}(a))(1 - \gamma))$. The full nonlinear dynamics saturate at $s_t = 1$, so $s_{\infty}(a)$ is a lower bound on the true asymptote. As $\gamma \rightarrow 1$, $s_{\infty}(a) \rightarrow 1$ for any $p_{\text{eff}}(a) > 0$.

Derivation of Heuristic 1. Step 1 (BR bound). The BR adversary commits to $a_{\text{BR}} = \arg \max_a s_{\infty}(a)$. By the linear recursion, $\mathbb{E}[s_T] = s_{\infty}(a_{\text{BR}}) - (s_{\infty}(a_{\text{BR}}) - s_0) c^T$ with $c = p_{\text{eff}}(a_{\text{BR}})(1 - \alpha) + (1 - p_{\text{eff}}(a_{\text{BR}}))\gamma < 1$. For $T \geq T_{\gamma}$, $|c|^T \leq \epsilon$, so $\mathbb{E}[s_T] \geq s_{\infty}(a_{\text{BR}}) - \epsilon$.

Step 2 (Full-info upper bound). Under persistence, the trigger state s_t is bounded by 1; round-by-round optimization cannot exceed the asymptotic cap. Since s_{∞} is monotone in p_{eff} and a_{BR} maximizes it, $U_A^{\text{full-info}} \leq s_{\infty}(a_{\text{BR}}) + \epsilon$.

Step 3 (Realized-VoPD bound). The BR strategy is in the full-info adversary’s choice set, so the gap reduces to $C \cdot (1 - s_{\infty}(a_{\text{BR}})) + \epsilon C$, yielding (L).

Step 4 (Limits). As $\gamma \rightarrow 1$, $s_{\infty}(a_{\text{BR}}) \rightarrow 1$, so the bound $\rightarrow 0$. At $\gamma = 0$, non-admitting rounds reset s to 0; outcomes decouple across rounds and per-round expectation equals realized utility, so realized VoPD = δ_{NE} .

Empirical observations (qualitative). We ran model_scaling under three conditions with per-round ASR logging (CIFAR-10, $N=10$, 5 seeds each, 50 rounds): (i) pure FedAvg, (ii) pure NormClip ($\tau=5$), (iii) NE3 mix (FedAvg 26%, NormClip 74%, sampled per round). Observed per-seed final ASR:

Condition	Per-seed final ASR (5 seeds 42–46)
Pure FedAvg	{0.92, 1.00, 1.00, 0.00, 0.00}, mean 0.584 $\pm$ 0.478
Pure NormClip	{0.95, 0.97, 0.94, 0.99, 0.94}, mean 0.959 $\pm$ 0.020
NE3 mix	{0.98, 1.00, 0.88, 0.98, 0.96}, mean 0.960 $\pm$ 0.041

Interpretation. The NE3 mix and pure NormClip produce *statistically indistinguishable* ASR_{50} (0.960 vs. 0.959, std overlapping), qualitatively consistent with Heuristic 1’s persistent-regime prediction: when $\hat{\gamma}$ is moderately large (here ≈ 0.66) and $p_{\text{eff}}(\text{scaling}) > 0$, the realized utility under any mixed policy approaches the pure-admitting outcome. The 30-seed realized-VoPD measurement of -0.010 ± 0.058 (§7) is the corresponding numerical observation. We do not claim a tight quantitative match.

The pure FedAvg condition is bimodal across seeds ($\{0, 0, 0.92, 1, 1\}$): a feature of training-initialization randomness in whether the scaling attack converges, not of per-round Bernoulli admission. This separates a quantitative parameter-fit treatment (which would need to model the per-seed admissibility latent) from the qualitative result above. A rigorous per-defense, per-seed quantitative validation of the Markov model is left to future work.

Earlier discarded approach. An initial attempt to fit $(\gamma, \alpha, p_{\text{FA}}, p_{\text{NC}})$ jointly to the per-seed trajectories (1) hit a corner solution ($\hat{\gamma} \rightarrow 0$) due to the pure-FedAvg bimodality being interpreted as rapid decay, and (2) failed an out-of-sample test (fit on rounds 1–25, predict rounds 26–50: predicted NE3 $\text{ASR}_{50} = 0.80$ vs. observed 0.96). We hypothesized that a per-seed admissibility variable separating the binding-cell Bernoulli at training init from the Markov dynamics is needed for quantitative prediction. We test this hypothesis below.

N.4.1 Admissibility-Conditional Markov Refinement

Setup. Classify each per-seed trajectory by “admissibility” — whether the binding-cell scaling attack has started to embed within the first 25 rounds: $a_{\text{seed}} = 1$ if $\text{ASR}_{25} > 0.5$, else 0. The training-initialization bimodality is captured by the latent a_{seed} ; conditional on $a_{\text{seed}} = 1$, fit only the Markov dynamics (α, γ) from rounds 1–25 (with p_{admit} fixed at the prior values 0.50, 1.00, 0.835 for FedAvg/NormClip/NE3-mix respectively), then predict ASR_{50} as the out-of-sample test.

Result. The admissibility-conditional fit gives much better out-of-sample prediction on the two deployment-relevant conditions:

condition	out-of-sample MAE	fraction within ± 0.10	admit/no-admit seeds
NormClip pure ($p_{\text{admit}} = 1.0$)	0.041	5/5	5/0
NE3 mix ($p_{\text{admit}} = 0.835$)	0.059	3/4	4/1
FedAvg pure ($p_{\text{admit}} = 0.50$)	0.426	0/4	4/1

Interpretation. The persistence heuristic, conditioned on admissibility, is quantitatively predictive on the conditions that matter for deployment (NormClip-admitting and NE3 mix) with out-of-sample MAE under 0.06. The pure-FedAvg case has MAE = 0.426 because at $p_{\text{admit}} = 0.5$ the trajectory is dominated by the Bernoulli-binding-cell noise: the Markov model alone cannot predict per-seed scaling success vs. failure without explicitly modeling the per-round admission Bernoulli realization. We retain the qualitative Heuristic 1 as the primary characterization, with the admissibility-conditional refinement as a partial advance: *the heuristic is quantitatively predictive on deployment-relevant policies but not on the FedAvg-pure baseline*. This converts the earlier discarded approach to a partial positive: the persistence mechanism is captured by a 2-parameter Markov model conditional on the admissibility latent, on conditions where admission is high enough that the Bernoulli noise does not dominate.

N.5 Sample Complexity: Hoeffding/Bernstein Worst-Case Bound

This appendix contains the worst-case sample-complexity bound demoted from §L. At the deployed $k = 10$ we report only a bootstrap *stability check* (Proposition L), a heuristic reproducibility estimate — not a formal guarantee, since the worst-case instantiation of Proposition L at $k = 10$ yields only $\mathbb{P} \geq 0.978 - \epsilon_{\text{rep}} \approx 0.55$ once the representativeness gap is bounded. The formal certificate below applies at $k \geq k_{\text{req}}$ and is included for completeness and for practitioners targeting the certified regime.

Proposition 13 (Sample Complexity for Complementarity Diagnosis, formal). *Let $\hat{U}_A(a, d)$ be the mean of k i.i.d. $[0, 1]$ -bounded evaluations with per-run variance σ^2 . Define the best-response gap at defense d as $\Delta(d) = \max_a U_A(a, d) - \max_{a \neq a^*(d)} U_A(a, d)$, and let $\Delta_{\min} = \min_{d \in \mathcal{S}} \Delta(d)$ over the equilibrium support $\mathcal{S}$. Then with probability $\geq 1 - \delta$, Algorithm 1 returns the correct VoPD sign provided*

$$k \geq \frac{4(2\sigma^2 + \frac{1}{3}\Delta_{\min})}{\Delta_{\min}^2} \log\left(\frac{2|\mathcal{A}||\mathcal{S}|}{\delta}\right), \quad (3)$$

where $|\mathcal{A}|$ and $|\mathcal{S}|$ are the number of attacks and support defenses. When $\sigma \gg \Delta_{\min}/3$ (as in CIFAR-10 where $\sigma = 0.500$, $\Delta_{\min} = 0.128$), the σ^2 term dominates and Eq. (3) reduces to $k \geq 8\sigma^2 \log(2|\mathcal{A}||\mathcal{S}|/\delta)/\Delta_{\min}^2$.

Proof sketch. By Bernstein’s inequality (Bernstein, 1924), for $[0, 1]$ -bounded variables with variance σ^2 : $P(|\hat{U}_A(a, d) - U_A(a, d)| \geq \epsilon) \leq 2 \exp(-k\epsilon^2/(2\sigma^2 + \frac{2}{3}\epsilon))$. A wrong VoPD sign requires some best-response ranking to flip, requiring $|\hat{U}_A(a, d) - U_A(a, d)| \geq \Delta_{\min}/2$ for at least one (a, d) with $d \in \mathcal{S}$. Setting $\epsilon = \Delta_{\min}/2$ and applying a union bound over $|\mathcal{A}||\mathcal{S}|$ cells yields (3). $\square$

Empirical calibration. We use the *empirical-Bernstein* form: replace global σ^2 by per-cell $\hat{\sigma}_{a,d}^2$; the binding cell $\max_{a,d \in \text{supp}}$ sets k_{req} . On CIFAR-10 ($N = 10$), U_A (scaling, FedAvg) is exactly Bernoulli ($\hat{\sigma}^2 = 0.25$, $\Delta_{\min} = 0.128$), giving $k_{\text{req}} = 726$ at $\delta = 0.10$. Non-binding cells (e.g., pixel/FedAvg, $\hat{\sigma}^2 < 0.01$) require $k < 50$ but do not constrain k_{req} . For spam (Appendix Q.2), $\hat{\sigma}^2 = 0.0019$, $\Delta_{\min} = 0.217$ give $k_{\text{req}} \approx 33$. The bound is tight at maximal-variance binding cells (CIFAR-10), slack where variance is low (spam); it is a worst-case guarantee, informative only at $k \gg 10$. At $k = 10$ we make no formal guarantee — only the bootstrap stability check of Proposition L.

O Survivor-Regime and Complementarity Pilots (Per-Seed Details)

O.1 Stateless and Orthogonal-Signal Pilots: Per-Seed Details

Detailed per-seed results, pre-specified criteria, and verdicts for the four pilots summarized in §M.1.8.

(i) Stateless availability pilot v1. The pilot introduces a **stateless availability attack**: malicious clients replace their updates with scaled Gaussian noise sampled fresh each round, so the effect is fully expended if the defense blocks that round and carries no persistent state. We re-run the 4×5 game on CIFAR-10 ($N=10$, 8 seeds, 50 rounds) with attacks {no_attack, backdoor_pixel, gaussian_noise, dba} against defenses {fedavg, norm_clip, rfa, trimmed_mean, coord_median}. *Utility note:* availability ASR is zero by definition; impact appears as accuracy degradation. We report under (i) the primary $U_A = \text{ASR} - 0.1c_A$ and (ii) the alternative $U_A = \max(\text{ASR}, \text{Acc}_{\text{no-atk}} - \text{Acc}) - 0.1c_A$ (Appendix M.5). *Result (8 seeds).* Under (i), gaussian_noise’s row vanishes; the equilibrium collapses to pure pixel vs. FedAvg, VoPD = 0 on all 8 seeds. Under (ii), gaussian_noise gains positive utility against FedAvg ($U_A \approx 0.69$) and is neutralized by NormClip/RFA ($U_A < 0.05$); pixel remains dominant ($U_A \in [0.27, 0.91]$). The argmax structure varies per seed: 1/8 (seed 43) gives argmax(FedAvg) = gaussian_noise (0.695) but argmax(NormClip) = pixel (0.673), producing a mixed NE (FedAvg 97% + NormClip 3%, VoPD = 0.021); 7/8 give pixel as the universal argmax (VoPD = 0). Mean VoPD = 0.003 ± 0.007 . Algorithm 1 fires the randomize branch on 1/8 seeds. *Interpretation:* qualified positive existence result — stateless FL attacks *can* create complementarity, but the effect is small (mean VoPD ≤ 0.01), rare (1/8), and sensitive to the utility metric. Per-seed file: results/cifar10_stateless/summary_corrected_utility.json.

(ii) Low- K/N stateless pilot v2. Alternative configuration: drop K/N , raise f , drop pixel from the menu. We re-ran the 4×5 game with $N=20$, $K=4$, $f=0.4$ (8 adversarial of 20), attacks {no_attack, label_flip, gaussian_noise, dba} and the same 5 defenses, 5 seeds. *Pre-specified STRONG criterion:* $\geq 3/5$ seeds mixed NE under metric (i) AND mean VoPD > 0.05 . *Result (5 seeds):* Under (i), 0/5 mixed NE; equilibrium collapses to a single dominant attack vs. pure FedAvg (or pure TrimmedMean on seed 46). Under (ii), 1/5 mixed NE (seed 46) with adversary support DBA (98.2%) + gaussian_noise (1.8%); mean VoPD 0.080 ± 0.039 . **Verdict per pre-specified criterion: NEGATIVE.** The recipe does not deliver a strong positive FL case: even at $K/N=0.20$, $f=0.4$, and a stateless attack menu without pixel, the norm-based defense menu suppresses each stateless attack individually rather than producing heterogeneous argmax structure. Per-seed: results/cifar10_positive_pilot_v2/summary.json.

(iii) Orthogonal-signal pilot (FLTrust). We add FLTrust (Cao et al., 2021) — a defense weighting each client update by cosine similarity to a server-side gradient on a 100-sample clean holdout — to the original 4×5 CIFAR-10 game ($N=10$, $K=5$, $f=0.2$, attacks {no_attack, backdoor_pixel, model_scaling, dba}, defenses {fedavg, norm_clip, rfa, trimmed_mean, coord_median, fltrust}, 5 seeds 42–46, 50 rounds). *Pre-specified STRONG criterion:* $\geq 3/5$ mixed NEs under (i) AND FLTrust enters support AND argmax(FLTrust) differs from argmax(FedAvg). *Result (5 seeds):* 4/5 seeds (43, 44, 45, 46) give argmax(FedAvg) = model_scaling while argmax(FLTrust) = backdoor_pixel

— the orthogonal-signal mechanism produces the argmax divergence Corollary 3 predicts; FLTrust suppresses scaling ($U_A \approx 0.02\text{--}0.03$). **But FLTrust does not enter the equilibrium support** (0/5 seeds either metric): server cost-benefit selects FedAvg, NormClip, or TrimmedMean. Mean VoPD 0.021 ± 0.031 , 2/5 mixed NEs between norm-based defenses. **Verdict: WEAK positive.** The static-axis dichotomy is partially broken (direction-based defenses *do* create argmax divergence) but the server’s cost-benefit does not select the orthogonal-signal defense. Reading: orthogonal-signal mechanism is *necessary but not sufficient* for a positive FL deployment. Per-seed: `results/cifar10.fltrust.pilot/summary.json`.

(iv) λ_c sweep on the FLTrust menu. We test whether cost weight is the binding barrier by sweeping $\lambda_c \in \{0, 0.025, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3\}$. The server payoff $U_D(a, d) = \text{Acc}(a, d) - \lambda_c \cdot C_D(d)$ is linear in λ_c , so we re-solve the Nash equilibria at each λ_c from the cached pilot matrices. *Pre-specified STRONG criterion:* $\exists \lambda_c \leq 0.05$ such that FLTrust enters equilibrium support (weight > 0.1) in $\geq 3/5$ seeds AND mixed-NE VoPD > 0.05 . *Result:* **FLTrust does not enter equilibrium support at any $\lambda_c \in [0, 0.3]$, including $\lambda_c = 0$.** At $\lambda_c = 0$ the server picks FedAvg or NormClip uniquely (clean accuracies ≈ 0.80 vs. FLTrust ≈ 0.72 , degraded by cosine-weighting overhead with the 100-sample holdout); FLTrust loses on Acc alone. **Verdict: NONE.** This is informative against the cost-weight explanation: *the barrier is not λ_c , it is the accuracy/robustness trade-off built into FLTrust’s mechanism.* Corollary 3 therefore tightens to a joint requirement: orthogonal-signal mechanism + competitive clean accuracy + low admission of persistent attacks.

Table 28: λ_c sweep on FLTrust menu (CIFAR-10, 5 seeds, metric (i)). We re-solve Nash equilibria at each λ_c using the cached payoff matrices. “FLTrust support” = number of seeds whose best NE has FLTrust with weight > 0.01 . “Mixed NEs” = seeds with a genuinely mixed equilibrium.

λ_c	Mean VoPD $\pm$ std	Mixed NEs	FLTrust in support	Best NE composition example
0.000	0.0000 ± 0.000	0/5	0/5	
0.025	0.0206 ± 0.031	2/5	0/5	
0.050	0.0206 ± 0.031	2/5	0/5	
0.075	0.0206 ± 0.031	2/5	0/5	
0.100	0.0206 ± 0.031	2/5	0/5	
0.150	0.0206 ± 0.031	2/5	0/5	
0.200	0.0206 ± 0.031	2/5	0/5	
0.300	0.0206 ± 0.031	2/5	0/5	

O.2 Orthogonal-signal defense suite: per-seed equilibria

Per-seed Nash equilibria of the augmented 4-attack $\times$ 4-defense menu {FedAvg, NormClip, FoolsGold, reputation} with attacks {no_attack, backdoor_pixel, model_scaling, dba}. Cost weights $\lambda_c = 0.1$, $\lambda_f = 0.05$, $\lambda_a = 0.1$ as in the main game.

seed	adversary support	server support (incl. reputation weight)	VoPD
42	pixel (33%) + scaling (67%)	NormClip (89%) + reputation (11%)	0.089
43	pixel (75%) + scaling (25%)	NormClip (74%) + reputation (26%)	0.200
44	pixel (80%) + scaling (20%)	NormClip (86%) + reputation (14%)	0.117
45	pixel (99%)	FedAvg (22%) + NormClip (78%)	0.017
46	scaling (100%)	NormClip (100%)	0

Reputation in support: 3/5 seeds (42, 43, 44). Reputation entry weights 11–26%, alongside NormClip in the server support. Mixed-NE rate (5 seeds): 4/5. Mean VoPD across 5 seeds: 0.085. FoolsGold: 0/5 in support but argmax divergence in 4/5 seeds (similar to FLTrust). **Argmax structure:** $\text{argmax}(\text{reputation}) = \text{backdoor_pixel}$ in all 5 seeds; $\text{argmax}(\text{NormClip}) = \text{model_scaling}$ in 4/5 (seed 45 is bimodal with pixel-argmax under NormClip due to seed-45 scaling failure); $\text{argmax}(\text{FedAvg})$ is bimodal (pixel-argmax in seed 42, scaling-argmax in 43–46) due to the binding-cell Bernoulli.

Clean accuracy under no attack: FoolsGold: 0.72–0.79 (mean 0.75 ± 0.03); reputation: 0.77–0.79 (mean 0.78 ± 0.01). Reputation matches FedAvg’s 0.80 within noise, satisfying the “competitive clean accuracy” requirement that FLTrust (≈ 0.72) failed.

Implementation. FoolsGold (Fung et al. 2018): for each client i , $w_i \propto \log((1 - v_i)/(v_i + 10^{-5}) + 10^{-5})$ where $v_i = \max_{j \neq i} |\cos(u_i, u_j)|$ (normalized by overall max). Reputation: consensus is the coordinate-wise median m of the round’s K updates; $d_i = \|u_i - m\|_2$; bandwidth $\sigma = \text{median}_i(d_i)$ (clamped $\geq 10^{-6}$, auto-tuned per round); $w_i = \exp(-d_i/\sigma)$. Despite the name it is **memoryless**—no cumulative per-client state, recomputed from scratch each round from that round’s K participants, with no initialization and no clipping. Its discriminative signal is therefore *consensus distance*, not similarity spread, which is why Proposition 8(c) places it in the non-invariant class. Measured weight ratio (5 seeds $\times$ 20 rounds, 81 rounds containing both adversarial and benign clients): ρ mean 1.44×10^5 , max 2.84×10^6 , p95 6.1×10^5 ; mean adversarial weight 4.83×10^{-4} against mean benign weight 0.278, with $w_a < 10^{-3}$ in 85% of rounds—i.e. regime B of §3.2. Weights are strictly positive, so reputation never produces the zero vectors of Lemma 2. Both methods normalize weights to sum to 1 then weighted-FedAvg.

O.3 Dynamic BR on Reputation Equilibrium: Per-Seed Details

Deploying NormClip 83% + reputation 17% (the cross-seed average over the reputation-bearing equilibria) and running NE3-style BR with committed adversary, 10 seeds 42–51, 50 rounds, 2 attacks. Per-seed realized ASRs and matched-seed realized VoPD computation:

seed	BR scaling	BR pixel	NC scaling	NC pixel	rep scaling	rep pixel	realized VoPD
42	0.923	0.832	0.937	0.837	0.014	0.837	−0.003
43	0.981	0.753	0.961	0.693	0.012	0.792	−0.049
44	0.865	0.824	0.911	0.774	0.017	0.828	+0.032
45	0.885	0.925	0.899	0.921	0.023	0.931	+0.038
46	0.949	0.767	0.965	0.805	0.019	0.823	−0.008
47	0.975	0.823	—	—	—	—	(PS missing)
48	0.979	0.923	—	—	—	—	(PS missing)
49	0.951	0.810	—	—	—	—	(PS missing)
50	0.937	0.866	—	—	—	—	(PS missing)
51	0.942	0.796	—	—	—	—	(PS missing)
mean (10)	0.939 ± 0.037	0.832 ± 0.055	—	—	—	—	—
mean (5)	—	—	—	—	—	—	$+0.002 \pm 0.031$

Aggregate. Realized scaling ASR = 0.939 ± 0.037 (10 seeds) — well above the 0.85 persistence-collapse threshold and consistent across all 10 seeds (no scaling-fails outlier in this run, unlike the original FedAvg+NormClip experiment which had seed 45’s bimodal failure). Realized VoPD on matched seeds 42–46 = $+0.002 \pm 0.031$ — essentially zero, within the COLLAPSES band ($-\text{VoPD} \leq 0.05$). 3/5 matched seeds give $\text{VoPD} \leq 0$.

Mechanism observed. The reputation defense strongly suppresses scaling statically (pure ASR ≈ 0.02 on matched seeds 42–46), but under the 17% mix-weight deployment, the BR adversary commits to scaling and embeds the trigger during the 83% NormClip-admitting rounds. Across all 10 seeds, the number of reputation-sampled rounds varies between 5 and 12 (mean ≈ 8), but does not produce sufficient suppression to recover any realized VoPD — the static dichotomy break does not translate into a dynamic dichotomy break.

Pre-specified verdict: COLLAPSES. Per the pre-experiment statement: “Even when complementarity is generated through a genuinely orthogonal-signal defense (reputation), persistence still destroys realized value.” The persistence claim generalizes from norm-based to orthogonal-signal menus.

O.4 Mix-ratio sweep on the reputation menu: per-seed details

Rounds 50–51 referenced in §M.1.5. Committed-attack BR on five NormClip+reputation mix ratios; 5 seeds, 50 rounds, $N=10$, $K=5$, $f=0.2$, `cifar_cnn`. Mean $\pm$ std across seeds:

mix (NC/rep)	scaling ASR	std	pixel ASR	std	committed-max
90/10	0.945	0.021	0.804	0.075	0.945 (scaling)
70/30	0.908	0.043	0.810	0.066	0.916 (pixel @ s45)
50/50	0.815	0.104	0.822	0.055	0.845
30/70	0.510	0.187	0.835	0.044	0.835 (pixel)
10/90	0.097	0.090	0.835	0.051	0.835 (pixel)

Static full-info upper bound from cached cells: $\approx \{0.93, 0.91, 0.89, 0.87, 0.85\}$. On the NormClip-dominant half ($\text{NC} \geq 50\%$), realized committed-max exceeds this upper bound: persistence boost closes the static VoPD gap. At NC30/rep70, scaling ASR is bimodal (0.510 ± 0.187 , 5 seeds) — some realizations survive, others collapse depending on the Bernoulli defense-round sequence. The 10-seed ceiling experiment (§7) with different RNG offset confirms survival at this ratio (0.721 ± 0.124 , VoPD = $+0.066$). The persistence boundary thus runs *through* the mix-ratio axis, not only at the menu axis: the operative threshold is in the 50%–70% reputation range.

Observed transition. Realized scaling ASR is monotone in the reputation weight and exhibits a sharp transition between 30% and 90% reputation. Per-seed variance grows as reputation weight increases (std $0.021 \rightarrow 0.187 \rightarrow 0.090$): the 50/50 ratio contains seed-43’s outlier (0.613); the 30/70 ratio splits roughly into two clusters (seeds 42, 44 above 0.69; seeds 43, 45, 46 at 0.34–0.39). At 10/90 only seed 42 retains nontrivial ASR (0.25); the other four collapse to ≤ 0.16 . The per-seed nondeterminism reflects randomness in defense-sampling: at 90% reputation, the realized number of NormClip rounds across 50 rounds varies between 1 and 6 across seeds (mean ≈ 4), and these 1–6 NormClip-admitting rounds determine whether the trigger embeds at all.

Verdict. MEDIUM per pre-specified bands. The persistence collapse is not mix-ratio-universal: it holds on the NormClip-dominant half (ratios with NormClip $\geq 50\%$) and breaks on reputation-dominant mixes. A 10-seed extension at NC30/rep70 (§7, different RNG offset) yields scaling ASR = 0.721 ± 0.124 and VoPD = $+0.066 \pm 0.027$ ($p < 0.001$), confirming survival. The 5-seed estimate here (0.510 ± 0.187) and the 10-seed estimate (0.721 ± 0.124) differ because the quantity is bimodal at this boundary (some seeds admit enough NormClip rounds for embedding, others do not) and both small samples reflect different Bernoulli realizations. The persistence boundary on the NC+rep axis lies in the 50%–70% reputation band. The equilibrium-relevant ratio (17% reputation, §M.1.5) lies firmly on the collapse side.

λ_c sweep on the orthogonal menu. Sweep referenced in §M.1.5. NE re-solved on {FedAvg, NormClip, FoolsGold, reputation} at $\lambda_c \in \{0, 0.025, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3\}$. Reputation enters NE support in seeds 43 (25.6%) and 44 (14.5%) at every λ_c ; mean max-reputation-weight is constant at 10.2%. *The binding constraint is reputation’s clean-accuracy gap, not its cost C_D .* The persistence-survival region (reputation NE weight $\geq 70\%$) is unreachable as a natural NE on this menu for any tested λ_c .

λ_c	mean max-reputation-weight	seeds with rep-weight $\geq 50\%$	seeds with rep-weight $\geq 70\%$
0.000	0.102	0/5	0/5
0.025	0.102	0/5	0/5
0.050	0.102	0/5	0/5
0.075	0.102	0/5	0/5
0.100	0.102	0/5	0/5
0.150	0.102	0/5	0/5
0.200	0.102	0/5	0/5
0.300	0.102	0/5	0/5

O.5 Ceiling-effect discrimination: per-seed details

Per-seed details for the NC30/rep70 diagnostic control (§7). Defense mix: NormClip 30% + reputation 70%; 10 seeds (42–51), 50 rounds, $N=10$, $K=5$, $f=0.2$, `cifar_cnn`. RNG offset seed+5000 for defense-round sampling.

Seed	Scaling ASR	Pixel ASR	Oracle ASR	VoPD (per-seed-max)
42	0.593	0.842	0.899	+0.057
43	0.513	0.772	0.863	+0.091
44	0.811	0.773	0.922	+0.111
45	0.882	0.920	0.936	+0.016
46	0.566	0.777	0.866	+0.089
47	0.801	0.868	0.921	+0.053
48	0.887	0.927	0.961	+0.034
49	0.703	0.801	0.885	+0.084
50	0.756	0.854	0.923	+0.069
51	0.697	0.827	0.879	+0.052
mean	0.721 ± 0.124	0.836 ± 0.054	0.905 ± 0.031	$+0.066 \pm 0.027$

Bimodality in scaling ASR. Seeds cluster into two groups: low (42,43,46: 0.51–0.59) and high (44,45,47,48: 0.80–0.89), with seeds 49–51 intermediate (0.70–0.76). This reflects the Bernoulli binding-cell mechanism: at 70% reputation (mean ≈ 35 reputation rounds out of 50), whether the ≈ 15 NormClip rounds cluster early (enabling trigger embedding) or distribute uniformly determines whether scaling achieves high ASR. The bimodality explains the discrepancy with the 5-seed mix-ratio sweep (Appendix O.4): that sweep’s seeds 42–46 (RNG offset seed+3000) happened to draw defense sequences that clustered on the low side (mean 0.510), while this extension (RNG offset seed+5000, seeds 42–51) samples both modes more evenly (mean 0.721). Both estimates are valid; the quantity is bimodal with high variance at this boundary point.

Key finding. VoPD is positive on all 10 seeds ($t(9)=7.16$, $p<0.001$). The oracle consistently beats committed-max because it exploits defense-specific vulnerability: scaling on NormClip rounds (where reputation cannot suppress it) and pixel on reputation rounds (where pixel passes the consensus check). This per-seed structure confirms the persistence boundary runs through the mix-ratio axis: at NC30/rep70, reputation suppresses scaling sufficiently that universal admission fails, recovering VoPD.

O.6 Reputation+trimmed_mean survivor regime: per-seed details

Per-seed details for §7. Deployed mix: reputation 30% + trimmed_mean 70%, the static-VoPD-maximising point on the cached 7-defense, 2-attack cell grid (mean static VoPD 0.162 across seeds). NE3-style BR with three adversary policies; 30 seeds (42–71), 50 rounds, $N=10$, $K=5$, $f=0.2$, `cifar_cnn`. Oracle adversary observes the round’s sampled defense and plays scaling on trimmed_mean rounds ($\text{argmax}(\text{tm})=\text{scaling}$), pixel on reputation rounds ($\text{argmax}(\text{rep})=\text{pixel}$). Trigger is shared between scaling and pixel attacks (bottom-right 4x4 pixel pattern); end-of-training ASR is measured against the shared trigger.

adversary policy	mean ASR ($n = 30$)	std	per-seed range	# seeds ≥ 0.85
committed-scaling	0.871	0.095	[0.588, 0.985]	19/30
committed-pixel	0.747	0.106	[0.489, 0.926]	6/30
oracle	0.915	0.065	[0.649, 0.987]	25/30
per-seed committed-max	0.883	0.075	—	—
per-seed VoPD	+0.032	0.054	[−0.146, +0.123]	—

Distribution. 27/30 seeds positive VoPD; 8/30 in the STRONG band (≥ 0.05); 3/30 seeds negative. The single strong negative outlier is seed 50 (VoPD −0.146): committed-scaling on seed 50 is anomalously low (0.588, vs. the 0.87 mean), so the oracle’s scaling rounds contribute weakly and the oracle’s overall ASR (0.649) falls short of committed-pixel (0.795). Removing seed 50 gives VoPD $+0.038 \pm 0.041$ (27/29 positive). The 5-seed pilot’s $+0.069 \pm 0.026$ was on the high tail of this distribution.

Mechanism. The oracle plays scaling on $\sim 68\%$ of rounds (trimmed_mean rounds, where reputation rounds are $\sim 32\%$) and pixel on the remaining $\sim 32\%$. Both attacks embed the same trigger pattern; the oracle achieves union-embedding by running the more-effective mechanism per round. Committed-scaling pays the cost of running scaling through trimmed_mean rounds where its update is partially clipped; committed-pixel pays the cost of running pixel through trimmed_mean rounds where the update magnitudes are reduced.

Verdict. MEDIUM positive — complementarity survives at modest magnitude. Realized VoPD = +0.032 at $n = 30$ falls in the pre-specified MEDIUM band ([0.02, 0.05]). A paired one-sample t -test on the per-seed VoPD gives $t(29) = 3.25$, $p = 0.003$, 95% CI [0.012, 0.052] (we do *not* use a Bernoulli sign test, which is inappropriate here). The magnitude is roughly half what the 5-seed pilot suggested, consistent with winner’s-curse regression given that this menu is the arg-max of a 189-configuration scan (§7). This sharpens the survivor claim: the boundary between collapse and survival exists as a real statistical property, but the survival magnitude in our menu is small in absolute terms ($\sim 3\%$ accuracy gap). The defense-menu’s arg max structure determines whether complementarity survives; the magnitude of survival depends on how cleanly the two persistent attacks orthogonally suppress each other.

O.7 Strong Adversary at $N=50$, $f=0.05$: Per-Seed Details

$N=50$, $K=5$, $f=0.05$ (2 adversarial clients), 50 rounds, 50/50 FedAvg/NormClip schedule, cifar_cnn, seeds 42–71 ($n=30$ each policy). Three adversary policies evaluated:

Policy A (committed_scaling): Plays model_scaling every round regardless of observed defense. **Policy B (commit-then-probe):** Plays model_scaling for rounds 1–30 to embed trigger, then infers defense from $\|\Delta w\|$ median and switches to pixel on rounds inferred as FedAvg. **Policy C (belief-state):** Maintains exponentially-smoothed belief $\hat{P}(\text{FedAvg})$ from aggregate norms; switches to pixel only when $\hat{P}(\text{FedAvg}) > 0.7$ and cached ASR > 0.5 .

Policy	Mean ASR $\pm$ std	Δ vs. committed	n
committed_scaling	0.511 ± 0.322	—	30
commit-then-probe	0.545 ± 0.316	+0.034	30
belief-state	0.555 ± 0.281	+0.044	30
oracle (per-round switch)	0.450	−0.061	30
saturation bound ε	≤ 0.489	—	—

Neither smarter policy closes the saturation gap. The gains (+0.034, +0.044) are small relative to $\varepsilon = 0.489$, confirming that committed play is empirically near-optimal. Notably, the oracle (which switches per round to the static argmax) achieves only 0.450 — *lower* than committed (0.511) — demonstrating the persistence tax: round-hopping dissipates the trigger state accumulated in prior rounds. Smarter adversaries (policies B and C) recognize this and avoid pure round-hopping, but the underlying constraint remains: at $f=0.05$, the adversary participates $\sim 10\%$ of rounds and any switching dilutes trigger accumulation on the remaining rounds.

Interpretation. The large standard deviations (~ 0.3) reflect binary behavior across seeds: on roughly half the seeds, model_scaling embeds successfully and achieves ASR ~ 0.7 – 0.9 ; on the other half, NormClip deflects the scaling attack and ASR ~ 0.0 – 0.1 . This bimodality is consistent with the 50/50 defense schedule: seeds where early rounds sample NormClip heavily prevent trigger embedding (Bernoulli outcome), independent of adversary sophistication. The persistence mechanism thus operates at the schedule level, not the adversary-policy level.

P Composition and Adaptive-Attack Details

P.1 Disjoint-Trigger Robustness: Per-Seed Details

This section addresses the concern that model_scaling and backdoor_pixel share the same trigger (bottom-right 4×4 pixels = 1.0), so the oracle’s per-round switching might co-accumulate both attacks onto a single pattern and inflate oracle ASR.

Setup. We create model_scaling_disjoint with a *disjoint* trigger (bottom-left 4×4 pixels = 1.0) and rerun the rep30/tm70 survivor experiment with 30 seeds (42–71), same FL parameters ($N=10$, $K=5$, $f=0.2$, 50 rounds). Both triggers are evaluated *separately* at end-of-training. The oracle still plays scaling (disjoint trigger) on trimmed_mean rounds and pixel (original trigger) on reputation rounds, but now the two attacks embed into completely distinct trigger locations.

VoPD interpretation. With disjoint triggers, “oracle ASR” has two natural definitions:

- **Weighted:** $\text{frac_scaling} \times \text{ASR_left} + \text{frac_pixel} \times \text{ASR_right}$. This is the realized attack success if the attacker only deploys the trigger from the current round’s attack.

- **Max-trigger:** $\max(\text{ASR}_{\text{left}}, \text{ASR}_{\text{right}})$. This is the realized success if the attacker queries whichever trigger embedded better at test time. This is the correct analogue to the shared-trigger experiment (the oracle picks the most effective attack at evaluation).

We report both; the max-trigger interpretation is primary for comparing to the shared-trigger baseline.

adversary policy	ASR_left (disjoint)	ASR_right (original)	oracle metric	VoPD
committed-scaling ($n=30$)	0.875 ± 0.074	0.031 ± 0.021	left trigger	—
committed-pixel ($n=30$)	0.023 ± 0.010	0.747 ± 0.106	right trigger	—
committed-max		0.878 ± 0.070		—
oracle (max-trigger, $n=30$)	0.907 ± 0.048	0.348 ± 0.196	max	$+0.029 \pm 0.033$
oracle (weighted, $n=30$)		0.740 ± 0.077		-0.139 ± 0.074
shared-trigger baseline ($n=30$)	oracle 0.915 ± 0.065 , committed-max 0.883			$+0.032 \pm 0.054$

Per-seed VoPD (max-trigger interpretation, $n=30$). 25/30 seeds positive; mean $+0.029 \pm 0.033$; range $[-0.032, +0.086]$. One-sample t -test: $t(29) = 4.82$, $p < 0.001$, 95% CI $[0.017, 0.041]$. Cross-trigger leakage near zero (committed-scaling right ASR = 0.031; committed-pixel left ASR = 0.023), confirming triggers are genuinely disjoint at $n=30$.

Conclusions.

(1) *Shared-trigger co-accumulation does not explain the survivor.* The max-trigger VoPD with disjoint triggers is $+0.029 \pm 0.033$ ($n=30$, 25/30 positive, $p < 0.001$), closely matching the shared-trigger $+0.032 \pm 0.054$. The oracle advantage survives even when triggers cannot co-accumulate. Cross-trigger leakage is near zero (committed-scaling right ASR = 0.031; committed-pixel left ASR = 0.023), confirming the triggers are genuinely disjoint.

(2) *Trigger-switching degrades realized ASR.* The weighted-interpretation VoPD is -0.139 ± 0.074 ($n=30$): when the oracle switches trigger patterns mid-training, *neither* backdoor fully saturates. This is a per-trigger persistence collapse — the oracle “diversifying” over disjoint triggers hurts itself by diluting each trigger’s accumulation below its committed-attack baseline. This extends the persistence mechanism: state accumulation is trigger-local, so strategies that diversify across disjoint triggers pay a persistence cost analogous to strategies that diversify across defenses.

(3) *Source of the survivor VoPD.* The positive max-trigger VoPD reflects test-time multi-backdoor flexibility, not classical value of per-round defense information. The oracle’s training-time switching is harmful (weighted VoPD = -0.139); its test-time advantage (querying both triggers, selecting the better one) drives the positive residual. For an adversary facing a disjoint-trigger menu, the rational strategy is to commit to one trigger and achieve near-committed-max ASR, or pay for test-time flexibility by embedding both (at cost of per-trigger dilution).

(4) *Persistence penalizes adversary diversification.* The weighted VoPD of -0.139 ($n=30$) reveals a bidirectional mechanism: persistence makes temporal *defense* mixtures max-like (our main negative result) AND makes temporal *attack* mixtures average-like. An adversary diversifying across disjoint triggers dilutes each trigger’s accumulation, paying a diversification penalty. This extends the persistence boundary symmetrically to the attacker’s strategy space: just as the server cannot gain by randomizing defenses under persistence, the adversary cannot gain by randomizing attacks (training-time switching strictly hurts).

P.2 Edge case 3-attack survivor pilot: per-seed details

Per-seed details for §M.1.7. Same deployed mix (reputation 30% + trimmed_mean 70%), same 5 seeds, 50 rounds, $N=10$, $K=5$, $f=0.2$, `cifar_cnn`, with `backdoor_edge_case` added as a 4th adversary policy. The oracle’s per-defense argmax table is unchanged (pixel on reputation rounds, scaling on trimmed_mean rounds) because `edge_case` is dominated by `max(scaling, pixel)` on every defense in the menu. Trigger evaluation uses `evaluate_backdoor` for scaling/pixel rounds (pixel trigger) and `evaluate_edge_case_backdoor` for `edge_case` rounds (top-corner trigger).

Pure-strategy edge_case ASR (refilled with correct evaluator, 5-seed mean $\pm$ std).

defense	ASR	seed-45 outlier
fedavg / norm_clip	0.619 ± 0.108	0.835
foolsgold	0.452 ± 0.094	—
reputation	0.611 ± 0.111	0.828
rfa	0.610 ± 0.107	0.813
trimmed_mean	0.394 ± 0.193	0.756
coord_median	0.235 ± 0.113	0.440

Trimmed_mean and coord_median suppress edge_case more than other defenses (matching their suppression pattern on scaling/pixel); seed 45 is consistently a positive outlier across all defenses except FoolsGold.

Realized 4-policy ASRs at rep30/tm70.

adversary policy	seed 42	seed 43	seed 44	seed 45	seed 46	mean $\pm$ std
committed-scaling	0.851	0.733	0.753	0.861	0.864	0.812 ± 0.060
committed-pixel	0.722	0.519	0.667	0.901	0.735	0.709 ± 0.131
committed-edge_case	0.387	0.285	0.413	0.798	0.498	0.476 ± 0.176
oracle (pixel on rep, scaling on tm)	0.905	0.813	0.804	0.968	0.912	0.880 ± 0.063
per-seed committed-max	0.851 (sc)	0.733 (sc)	0.753 (sc)	0.901 (px)	0.864 (sc)	0.820 ± 0.066
per-seed VoPD	+0.054	+0.080	+0.051	+0.067	+0.048	$+0.060 \pm 0.013$

Observations. (i) Realized committed-edge_case ASR (0.476) is materially below committed-scaling (0.812) and committed-pixel (0.709); edge_case is the weakest persistent attack on this mix. (ii) The oracle never plays edge_case (model_scaling=35–38 rounds, pixel=12–15 rounds, edge_case=0 across all seeds), because edge_case is dominated by max(scaling, pixel) on both menu defenses statically. (iii) Per-seed committed-max equals committed-scaling on 4/5 seeds and committed-pixel on seed 45 — the same per-seed pattern observed without edge_case in the original 3-policy pilot. (iv) Per-seed VoPDs $\{+0.054, +0.080, +0.051, +0.067, +0.048\}$ are concentrated near +0.06 (std 0.013). Mean +0.060 vs. the original +0.069 are statistically indistinguishable at $n=5$.

Verdict. STRONG positive: the rep30/tm70 survivor regime is robust against a 3-attack adversary. The boundary characterisation (collapse vs. survival) is therefore a property of the defense menu’s arg max structure across the persistent attacks present in the game, not of the attack-menu cardinality. Edge_case as a 4th option confirms but does not extend the boundary: a meaningful 9/10-grade demonstration would require an attack that wins the static arg max on at least one defense in the menu (so the oracle plays it), and the present 3-attack landscape on this 7-defense menu does not exhibit such an attack.

P3 Composition Baseline Details

Simultaneous composition (NormClip $\tau=5$ then reputation every round), $N=10$, $K=5$, $f=0.2$, 50 rounds, cifar_cnn, 5 seeds. Per-seed results:

Seed	model_scaling		backdoor_pixel	
	Accuracy	ASR	Accuracy	ASR
42	0.783	0.766	0.774	0.837
43	0.788	0.783	0.774	0.792
44	0.794	0.835	0.784	0.828
45	0.762	0.889	0.771	0.931
46	0.780	0.763	0.766	0.823
Mean $\pm$ std	0.781 ± 0.011	0.807 ± 0.048	0.774 ± 0.006	0.842 ± 0.047

Composition partially suppresses scaling (ASR 0.807 vs. pure NormClip ~ 0.95 and NE3 temporal mix 0.961) but does not approach pure reputation (0.017). Norm clipping preserves the attack direction while reducing magnitude; the clipped update then passes reputation’s consensus-distance filter. Pixel attacks are similarly persistent under composition (ASR 0.842). Both attack families remain effective despite simultaneous application of both defenses.

P.4 DBA at $f=0.4$: per-seed details

NE3 BR on CIFAR-10 with DBA in place of model scaling at $f=0.4$ (raising adversarial co-participation to $K \cdot f = 2.0$ from the $f=0.2$ value of 1.0). $N=10$, $K=5$, 5 seeds 42–46, 50 rounds. Per-seed realized ASRs:

seed	DBA realized ASR	backdoor.pixel realized ASR
42	0.222	0.972
43	0.103	0.945
44	0.176	0.913
45	0.603	0.993
46	0.077	0.934
mean $\pm$ std	0.236 ± 0.190	0.952 ± 0.028

The DBA per-seed spread is $7.5\times$ between min and max; only seed 45 clearly persists, three (0.08, 0.10, 0.18) are below the threshold (≥ 0.20) we would call active persistence, and one (0.22) is borderline. Pixel persists cleanly at $f=0.4$ (0.952 ± 0.028), so the persistence mechanism survives an adversarial-fraction shift on the original attack. DBA’s seed-variability reflects per-round co-participation alignment (whether ≥ 2 of the 4 adversarial clients land in the same round to coordinate the distributed trigger) rather than the structural argmax-collapse mechanism we describe for scaling. The cleanest evidence for the headline persistence claim therefore remains model scaling; DBA at $f=0.4$ is suggestive but not a robust second-attack demonstration.

P.5 DBA at $f=0.6$: per-seed details and bimodality reproduction

Follow-up to Appendix P.4, designed to test whether the DBA bimodality at $f=0.4$ is purely a per-round co-participation effect or whether it persists as a per-seed admissibility latent at higher f . At $f=0.6$, expected adversarial co-participation per round is $K \cdot f = 3.0$, making ≥ 2 adversarial co-participation per round near-certain. NE3-style BR with committed adversary playing DBA; 5 seeds (42–46), 50 rounds, $N=10$, $K=5$, `cifar_cnn`. Two deployed policies: *NE3 mix* (FedAvg 26% + NormClip 74%) and the *reputation equilibrium* (NormClip 83% + reputation 17%).

seed	DBA ASR (NE3 mix)	DBA ASR (reputation eq)
42	0.360	0.339
43	0.295	0.278
44	0.386	0.415
45	0.734	0.705
46	0.154	0.130
mean $\pm$ std	0.386 ± 0.192	0.373 ± 0.190

Observations. (i) Mean DBA ASR improves over $f=0.4$ (0.236) to ~ 0.38 on both policies, but does not reach the ≥ 0.85 persistence band scaling achieves. (ii) Per-seed bimodality persists at $f=0.6$: seed 45 alone reaches > 0.7 in both policies; the remaining four seeds cluster in $[0.13, 0.42]$. (iii) Crucially, *the same seed is the strong-persistence seed under both policies* (NE3 mix: 0.734; reputation eq: 0.705), and *the same four seeds are the weak-persistence seeds under both* — the per-seed cross-policy correlation rules out per-round co-sampling alignment as the dominant source of bimodality. The bimodality is a per-seed training-initialization latent in DBA itself: whether the distributed trigger embeds at all in the early rounds. This is the DBA analogue of the model-scaling binding-cell Bernoulli (Appendix N.4.1), but at a much lower mean (admissibility prevalence $\approx 1/5$ for DBA at $f=0.6$ vs. $\approx 4/5$ for scaling at $f=0.2$).

Pre-specified verdict: INTERMEDIATE. Overall mean 0.380 falls between PARTIAL (> 0.5) and NEGATIVE (< 0.3). Reading: DBA at $f=0.6$ does *not* replicate the persistence collapse with the cleanness scaling achieves; the trigger-alignment mechanism has its own admissibility constraint we do not control by raising f . The cleanest headline-persistence evidence remains model scaling under norm-based menus, as acknowledged in L3.

Q Scale, Architecture, and External Validity

Q.1 Scale Validation ($N = 100$) and Defense-Inference Check

Scale validation ($N = 100$, 4×5 game, **5 seeds**). Seed 42: mixed NE (pixel + scaling vs. FedAvg + RFA, VoPD = 0.012); seeds 43–46: pure NE (scaling vs. RFA). Diagnostic correct on all 5 seeds (Table 29).

Table 29: Per-seed NE, CIFAR-10 $N = 100$ (4×5 game). Seeds 43–46 identical.

Seed	Type	Adversary	Server	VoPD
42	Mixed	Pixel + Scaling	FedAvg + RFA	0.012
43–46	Pure	Model scaling	RFA	0.000

Strategy-space robustness. Re-solving Nash on 5 defense menus: mixed-NE rates range from 0% (FedAvg+NClip) to 60% (FedAvg+TrMean, VoPD 0.083). Theorem 11 correct in all $50 \text{ menu} \times \text{seed}$ cases.

Algorithm 1 Measure-Then-Verify-Then-Randomize (two-tier protocol)

Require: $\mathcal{A}, \mathcal{D}$, precision budget ϵ , simulation budget T_v , tolerance ε_v

- 1: [*Screening tier*: $k=10$, $\rho_{10}^{\text{boot}}=0.978$] Estimate $\hat{U}_A(a, d) \forall (a, d)$ using k seeds per entry
 - 2: [*Certification tier*: $k_{\text{req}}=726$, *Proposition 13*] Optionally refine with k_{req} seeds for formal guarantee
 - 3: Compute NE (σ_A^*, σ_D^*) of $\hat{U}_A$; compute VoPD
 - 4: **if** VoPD > 0 **then**
 - 5: **Verify** offline: run T_v runs per $a \in \mathcal{A}$ under σ_D^* ; compute $\Delta_{\text{dev}} = \max_a \hat{U}_A^{\text{real}}(a, \sigma_D^*) - \hat{U}_A^{\text{real}}(\sigma_A^*, \sigma_D^*)$
 - 6: **if** $\Delta_{\text{dev}} \leq \varepsilon_v$ **then**
 - 7: **Randomize**: sample $d \sim \sigma_D^*$ each round
 - 8: **else**
 - 9: **Revert**: NE not dynamic equilibrium; deploy pure $d^* \in \arg \max_d U_D(a^*, d)$
 - 10: **end if**
 - 11: **else**
 - 12: **Deploy pure**: $d^* \in \arg \max_d U_D(a^*, d)$ (utility-dependent: see §M.1.8)
 - 13: **end if**
-

Two-tier interpretation. The pipeline operates as a two-tier protocol: *screening* ($k=10$, bootstrap stability $\rho=0.978$) identifies candidate equilibrium structures efficiently; *certification* ($k_{\text{req}}=726$, Proposition 13) provides the formal Hoeffding guarantee. Our experiments operate at screening tier; the verify step (step 5) is the operational safeguard that catches screening-tier errors by detecting abstraction failures in dynamic play.

Q.2 Non-FL Validation: Adversarial Spam Evasion

UCI Spambase (4601 emails, 57 features, 39% spam). No FL. **Attacks**: no_attack; word_zero; char_zero; noise_inject ($\mathcal{N}(0, 0.5)$). **Defenses**: word_lr, char_lr, all_lr, all_rf. $U_A = \text{ASR} - 0.1c_A$. NE: server mixes *char_lr+all_lr*; adversary mixes *char_zero+noise_inject*; VoPD = 0.200. Theorem 11 correct: char_lr/all_lr argmax divergence (Table 30), empty intersection, VoPD > 0 . **Sample-complexity**: $\hat{\sigma}^2 = 0.0019$, $\Delta_{\min} = 0.217$; full Eq. (3) gives $k_{\text{req}} \approx 33$; condition empirically correct on all 5 seeds.

Table 30: Spam adversary payoffs (mean, 5 seeds). Bold: column argmax.

Attack	word_lr	char_lr	all_lr	all_rf
no_attack	0.132	0.401	0.125	0.074
word_zero	0.305	0.396	0.242	0.133
char_zero	0.127	0.995	0.157	0.176
noise_inject	0.508	0.408	0.459	0.528

Q.3 ResNet18 architecture-replication mini-sweep: per-seed details

Per-seed details for §M.1.7. `resnet18` with BatchNorm replaced by GroupNorm (8 groups), conv1 adapted to 32×32 (kernel 3, stride 1, maxpool $\rightarrow$ Identity); 11.2M parameters. Identical FL parameters to `cifar_cnn`: $N=10$, $K=5$, $f=0.2$, 50 rounds, $\tau = 5$. Committed-scaling BR, 3 mix points $\times$ 5 seeds = 15 runs at ~ 100 min/run on MPS.

mix (NC/rep)	s42	s43	s44	s45	s46	ResNet18 mean	cifar_cnn (R50)
90/10	0.926	0.929	0.965	0.974	0.962	0.951 ± 0.020	0.945 ± 0.021
50/50	0.924	0.971	0.813	0.920	0.903	0.906 ± 0.052	0.815 ± 0.104
10/90	0.917	0.733	0.783	0.975	0.915	0.865 ± 0.091	0.097 ± 0.090

NC-dominant endpoint replicates within 0.01; mid-mix is $\sim 9\%$ higher on ResNet18; the reputation-dominant endpoint shows an order-of-magnitude gap (no CI overlap). Seed 45’s accuracy collapses to 0.153 at NC10/rep90 (the persistent attack effectively breaks the model) while trigger ASR remains 0.975. **Verdict: MEDIUM** (monotonic ordering preserved, endpoint criterion fails). The `cifar_cnn` collapse boundary at $\sim 70\text{--}90\%$ reputation is therefore a property of the smaller-architecture regime; whether the rep+tm survivor also transfers is out of scope for this study.